

# **ST7703 Data Sheet**

TFT Mobile Single Chip Driver  
For 720RGB x 1280 dot, a-Si TFT LCD, 16.7M color  
With MIPI Interface

*Version 01 Preliminary*  
*November 2016*

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## Revision History

| Version | Date       | Description of Changes                             |
|---------|------------|----------------------------------------------------|
| 01      | 2015/10/05 | 1. New setup                                       |
|         | 2016/09/19 | 1. P189 Modify Using External Power IC Application |
|         | 2016/11/29 | 1. P62 Modify 04h Register ID1 , ID2 , ID3         |

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## 1. General Description

The ST7703 is a single-chip solution for a-Si TFT LCD that combines a source driver, power supply circuit to drive a-Si TFT dot matrix LCD with 720RGBx1280 dots at maximum. The ST7703 supports MIPI Interface.

The ST7703 incorporates step-up and voltage follower circuits to generate drive voltage required for a-Si TFT and gate in panel (GIP) which including a dynamic backlight control function to control backlight brightness depending on image data, reducing power consumption at the backlight with the slightest influence on image quality.

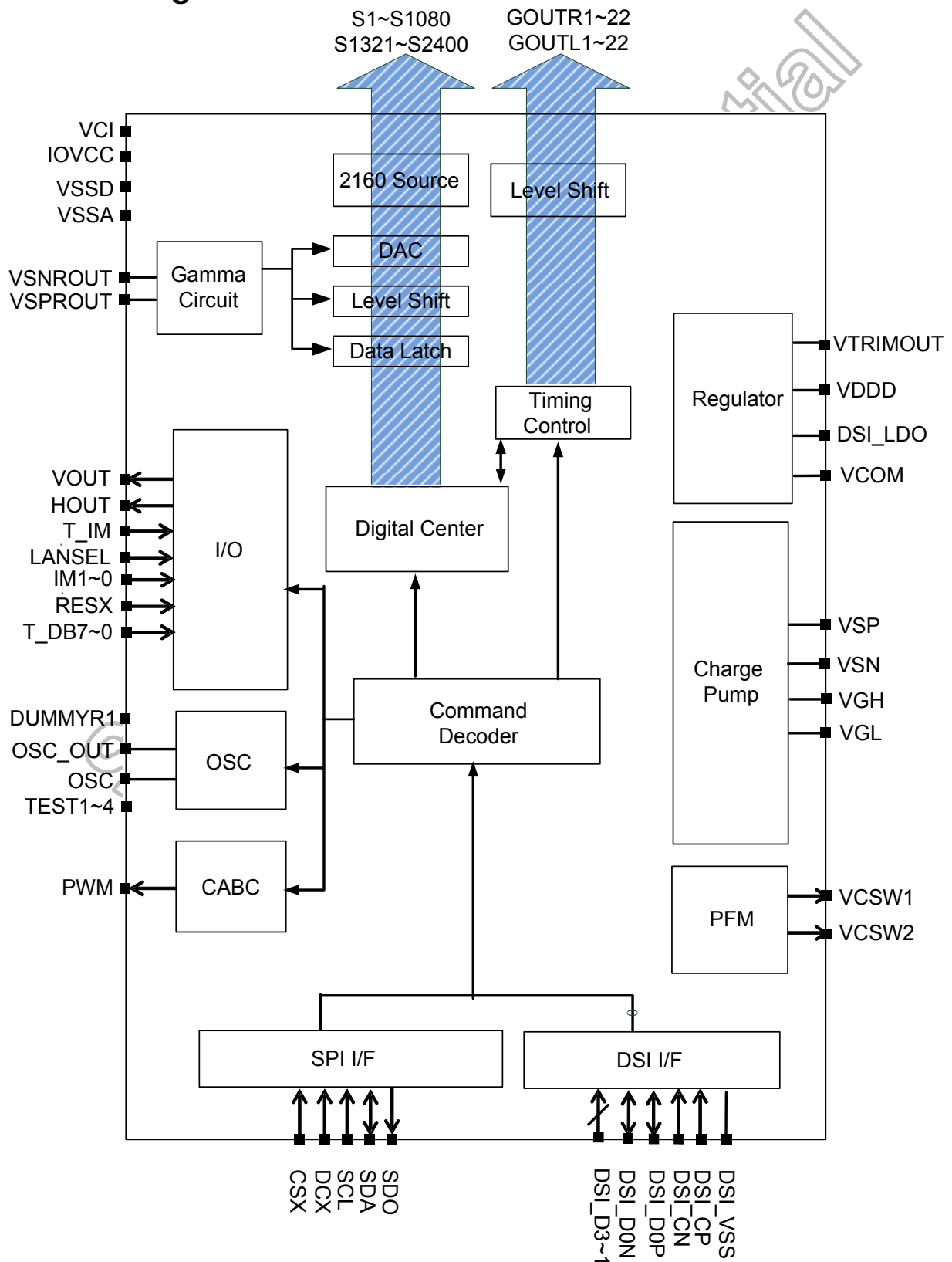
The ST7703 is suitable for small or medium-sized portable mobile solutions requiring long-term driving capabilities, such as digital mobile phones, PDAs and Smartphones.



## 2. Features

- Single chip solution for a HD720 GIP (Gate In Panel) type TFT LCD display
- Display Resolution :
  - 720RGB x (480 + 4xNL + RES\_V\_LSB)
  - 640RGB x (480 + 4xNL + RES\_V\_LSB)
  - 600RGB x (480 + 4xNL + RES\_V\_LSB)
  - 540RGB x (480 + 4xNL + RES\_V\_LSB)
- Display Color Modes :
  - Full color mode: 16.7M colors
  - Reduce color mode: 262k colors
  - Reduce color mode: 65k colors
  - Idle mode: 8 colors
- Display Interface :
  - MIPI Display Serial Interface (DSI V1.01)
- On chip :
  - VGH/VGL for GIP gate control
  - 1-dot / 1+2-dot / 2-dot / 3-dot / 4-dot / 8-dot / column / Zig-Zag inversion
  - Gamma correction function
  - Oscillator for display clock generation
  - Internal OTP programming voltage generator
  - 4 times MTP for VCOM setting · ID setting
  - CABC (Content Adaptive Brightness Control) function
  - DGC (Digital Gamma Correction) function
  - Color Enhancement function
  - 3 temperature sensing voltage
- Input power :
  - External power IC and PFM :
    - I/O and interface power supply (IOVCC) : 1.65V to 2.0V
    - Analog power supply (VCI) : 2.5V to 3.3V
  - Three-Power Mode :
    - I/O and interface power supply (IOVCC) : 1.65V to 2.0V
    - Analog power supply (VSP) : 4.5V to 6.0V
    - Analog power supply (VSN) : -4.5V to -6.0V
- Output voltage :
  - Positive source output voltage level: VSPR=3.3V to 5.6V
  - Negative source output voltage level: VSNR=-5.6V to -3.3V
  - Positive gate driver output voltage level: VGH=+10V to +20V
  - Negative gate driver output voltage level: VGL=-7.5V to -15V
  - VCOM=-3.5V to 0V

### 3. Block Diagram



## 4. Pin Description

### 4.1 Power Supply Pins

| Symbol  | Name         | Description                      | Connection when not used |
|---------|--------------|----------------------------------|--------------------------|
| IOVCC   | I/O Power    | Power supply for logic circuit.  | -                        |
| VCI     | Analog Power | Power supply for analog circuit. | -                        |
| VSSA    | Analog GND   | System ground for analog system. | -                        |
| VSSD    | I/O GND      | System ground for logic circuit. | -                        |
| DSI_VSS | MIPI GND     | DSI interface analog ground.     | -                        |

### 4.2 Interface Logic Pins

| Symbol          | I/O | Description                                                                                                                | Connection when not used |     |        |       |       |      |       |       |
|-----------------|-----|----------------------------------------------------------------------------------------------------------------------------|--------------------------|-----|--------|-------|-------|------|-------|-------|
| IM1~0<br>LANSEL | I   | Interface mode select pins.<br>IM1~IM0 and LANSEL are used for the combination of polarity swap and data lane swap of DSI. | GND                      |     |        |       |       |      |       |       |
|                 |     | IM1                                                                                                                        |                          | IM0 | LANSEL | D0P/N | D1P/N | CP/N | D2P/N | D3P/N |
|                 |     | 0                                                                                                                          |                          | 0   | 0      | D3P/N | D2P/N | CP/N | D1P/N | D0P/N |
|                 |     | 0                                                                                                                          |                          | 1   | 0      | D3N/P | D2N/P | CN/P | D1N/P | D0N/P |
|                 |     | 1                                                                                                                          |                          | 0   | 0      | D0P/N | D1P/N | CP/N | D2P/N | D3P/N |
|                 |     | 1                                                                                                                          |                          | 1   | 0      | D0N/P | D1N/P | CN/P | D2N/P | D3N/P |
|                 |     | 0                                                                                                                          |                          | 0   | 1      | D2P/N | D1P/N | CP/N | D0P/N | D3P/N |
|                 |     | 0                                                                                                                          |                          | 1   | 1      | D2N/P | D1N/P | CN/P | D0N/P | D3N/P |
|                 |     | 1                                                                                                                          |                          | 0   | 1      | D3P/N | D0P/N | CP/N | D1P/N | D2P/N |
|                 |     | 1                                                                                                                          |                          | 1   | 1      | D3N/P | D0N/P | CN/P | D1N/P | D2N/P |
| RESX            | I   | Reset pin. This signal will reset the device and must be applied to properly initialize the chip.                          | -                        |     |        |       |       |      |       |       |
| VOUT            | O   | Frame synchronization output signal pin.                                                                                   | OPEN                     |     |        |       |       |      |       |       |
| HOUT            | O   | Scan line synchronization output signal pin.                                                                               | OPEN                     |     |        |       |       |      |       |       |

Note: "1" = IOVCC Level, "0" = VSSD Level.

### 4.3 MIPI Interface Pins

| Symbol                    | I/O | Description                                                      | Connection when not used |
|---------------------------|-----|------------------------------------------------------------------|--------------------------|
| DSI_D0P,<br>DSI_D0P       | I/O | High speed interface data differential signal input/output pins. | OPEN/GND                 |
| DSI_CP,<br>DSI_CN         | I   | High speed interface clock differential signal input pins.       | OPEN/GND                 |
| DSI_D3P~1P,<br>DSI_D3N~1N | I   | High speed interface data differential signal input pins.        | OPEN/GND                 |

#### 4.4 Driver Output Pins

| Symbol                        | I/O | Description                                  | Connection when not used |
|-------------------------------|-----|----------------------------------------------|--------------------------|
| S1 to S1080<br>S1321 to S2400 | O   | Pixel electrode driving output               | Open                     |
|                               |     | Display Resolution                           |                          |
|                               |     | Used Source Pins                             |                          |
|                               |     | 720RGB                                       |                          |
|                               |     | 640RGB                                       |                          |
| CGOUTL_1~22<br>CGOUTR_1~22    | O   | Gate control signals for panel.              | Open                     |
|                               |     |                                              |                          |
| VCOM                          | O   | Regulator output for common voltage of panel | -                        |

#### 4.5 DC/DC Converter Pins

| Symbol   | I/O | Description                                                                        | Connection when not used |
|----------|-----|------------------------------------------------------------------------------------|--------------------------|
| VSP      | I   | Input voltage from the set-up circuit.                                             | -                        |
| VSN      | I   | Input voltage from the set-up circuit.                                             | -                        |
| VGH      | O   | Output voltage from the step-up circuit.                                           | -                        |
| VGL      | O   | Output voltage from the step-up circuit.                                           | -                        |
| DSI_LDO  | O   | High speed interface regulator output pin.                                         | Open                     |
| VSPROUT  | O   | Output voltage generated from VSP.<br>Output for positive gamma voltage generator. | -                        |
| VSNROUT  | O   | Output voltage generated from VSN.<br>Output for negative gamma voltage generator. | -                        |
| VDDD     | O   | Reference voltage for internal logic.                                              | -                        |
| VTRIMOUT | O   | Reference voltage for internal voltage generating circuit.                         | -                        |
| VCSW1    | O   | PFM and Power IC control output for DC/DC converter to generate VSP.               | -                        |
| VCSW2    | O   | PFM and Power IC control output for DC/DC converter to generate VSN.               | -                        |

#### 4.6 CABC Control Pins

| Symbol | I/O | Description                                          | Connection when not used |
|--------|-----|------------------------------------------------------|--------------------------|
| PWM    | O   | This pin is connecting with the external LED driver. | Open                     |

#### 4.7 Test Pins

| Symbol                 | I/O | Description                                                                                     | Connection when not used |
|------------------------|-----|-------------------------------------------------------------------------------------------------|--------------------------|
| OSC                    | I   | Oscillator test pin                                                                             | Open                     |
| CSX                    | I   | Chip select input pin ("Low" enable) in RGB I/F test mode.                                      | OPEN                     |
| DCX                    | I   | Command/parameter selection in RGB I/F test mode.<br>Low: Select command.<br>High: Select data. | IOVCC/GND/<br>OPEN       |
| SCL                    | I   | A synchronous clock signals in RGB I/F test mode.                                               | IOVCC/GND/<br>OPEN       |
| SDA                    | I/O | Serial data input/output pin in RGB I/F test mode.                                              | IOVCC/GND/<br>OPEN       |
| SDO                    | O   | A serial data output pin in RGB I/F test mode.                                                  | OPEN                     |
| T_DB7~0                | I   | 8-bit input data bus in RGB I/F test mode.                                                      | IOVCC/GND/<br>OPEN       |
| T_IM                   | I   | Let driver IC into RGB I/F test mode if set T_IM= high.                                         | Open                     |
| OSC_OUT                | O   | Test pin.                                                                                       | Open                     |
| NVDDDOUT               | O   | Test pin.                                                                                       | Open                     |
| NVTRIMOUT              | O   | Test pin.                                                                                       | Open                     |
| VREFOUT                | O   | Test pin.                                                                                       | Open                     |
| NVREFOUT               | O   | Test pin.                                                                                       | Open                     |
| VTESTOUTP              | O   | Test pin.                                                                                       | Open                     |
| VTESTOUTN              | O   | Test pin.                                                                                       | Open                     |
| DUMMYR1                | -   | Test pin for bonding quality test. They are short-circuited within the chip.                    | Open                     |
| DUMMY406~1             | -   | Not used. Let it open.                                                                          | Open                     |
| DUMMY_PM0<br>DUMMY_PM1 | -   | No use. Let them open.                                                                          | Open                     |
| DUMMY_RS0<br>DUMMY_RS1 | -   | No use. Let them open.                                                                          | Open                     |
| DUMMY_YEMP             | -   | No use. Let them open.                                                                          | Open                     |

Note: "1" = IOVCC Level, "0" = VSSD Level.

Note: "X" = Don't care.

## 4.8 BOM

### Power IC Mode

| NO. | IC Pin Name | Capacitance (uF) | Permissible Voltage (V) | Note                   |
|-----|-------------|------------------|-------------------------|------------------------|
| 1   | VCI         | -                | -                       | Don't need Capacitance |
| 2   | IOVCC       | -                | -                       | Don't need Capacitance |
| 3   | VSP         | 2.2              | 10                      | -                      |
| 4   | VSN         | 2.2              | 10                      | -                      |
| 5   | VGH         | -                | -                       | Don't need Capacitance |
| 6   | VGL         | -                | -                       | Don't need Capacitance |
| 7   | VSPROUT     | -                | -                       | Don't need Capacitance |
| 8   | VSNROUT     | -                | -                       | Don't need Capacitance |
| 9   | VTRIMOUT    | -                | -                       | Don't need Capacitance |
| 10  | VDDD        | -                | -                       | Don't need Capacitance |
| 11  | DSI_LDO     | -                | -                       | Don't need Capacitance |
| 12  | VCOM        | -                | -                       | Don't need Capacitance |

Table 4-1: BOM of Power IC Mode

### 3 Power Mode

| NO. | IC Pin Name | Capacitance (uF) | Permissible Voltage (V) | Note                   |
|-----|-------------|------------------|-------------------------|------------------------|
| 1   | VCI         | -                | -                       | Don't need Capacitance |
| 2   | IOVCC       | -                | -                       | Don't need Capacitance |
| 3   | VSP         | -                | -                       | Don't need Capacitance |
| 4   | VSN         | -                | -                       | Don't need Capacitance |
| 5   | VGH         | -                | -                       | Don't need Capacitance |
| 6   | VGL         | -                | -                       | Don't need Capacitance |
| 7   | VSPROUT     | -                | -                       | Don't need Capacitance |
| 8   | VSNROUT     | -                | -                       | Don't need Capacitance |
| 9   | VTRIMOUT    | -                | -                       | Don't need Capacitance |
| 10  | VDDD        | -                | -                       | Don't need Capacitance |
| 11  | DSI_LDO     | -                | -                       | Don't need Capacitance |
| 12  | VCOM        | -                | -                       | Don't need Capacitance |

Table 4-2: BOM of 3 Power Mode

## 5. Function Description

### 5.1 SPI Interface

The selection of serial interface is by T\_IM. ST7703 can support DPI/DBI type-C Option 1.

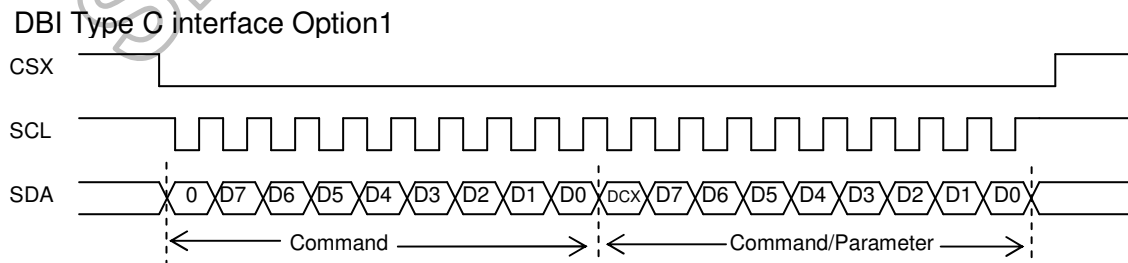
| T_IM | Interface mode                      |
|------|-------------------------------------|
| 0    | MIPI Interface                      |
| 1    | DPI/DBI type-C Option 1 (9-bit SPI) |

**Table 5.1: DBI TYPE-C Serial Interface Selection Table**

The serial interface is used to communication between the MPU and the LCD driver chip. It contains CSX (chip select), SCL (serial clock), SDA (serial data input/output). Serial clock (SCL) can be stopped when no communication is necessary.

#### 5.1.1 SPI Interface Write Mode

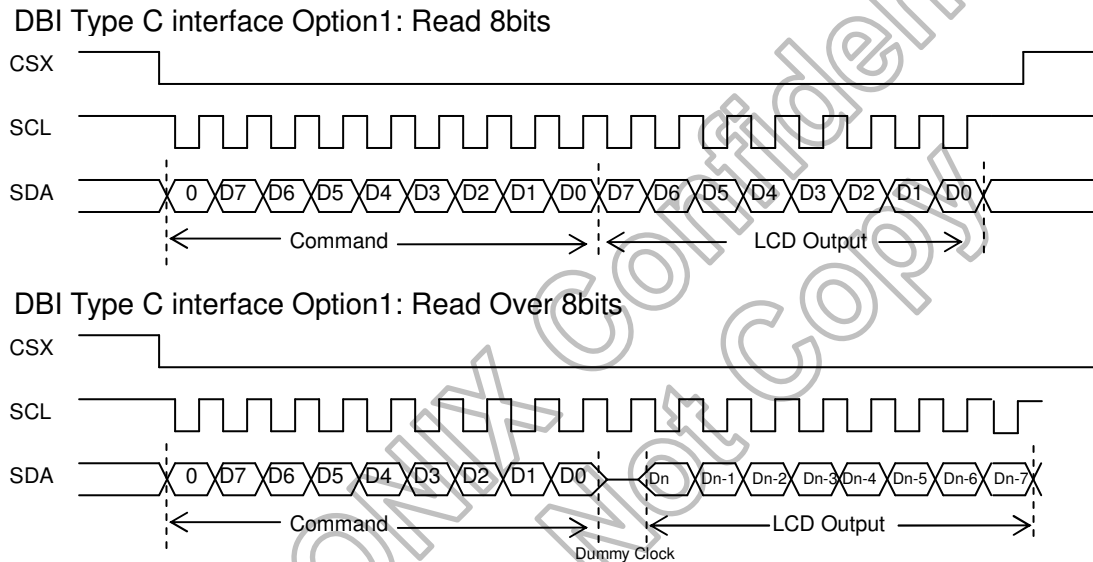
In SPI I/F Write Mode, the serial data packet contains a control bit DCX and a transmission byte. If DCX is low, the transmission byte is command byte. If DCX is high, the transmission byte is stored in to command register. The MSB is transmitted first. The serial interface is initialized when CSX is high. In this state, SCL clock pulse or serial input/output data (SDA/SDO) have no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission.



**Figure 5.1: DBI Type C interface Option 1 write mode**

## 5.1.2 SPI Interface Read Mode

In SPI I/F Read Mode, the host controller first has to send a command and then the following byte is transmitted to host controller in the SDA. The read mode has two type : one is command data 8-bits read and one is over 8-bits read.

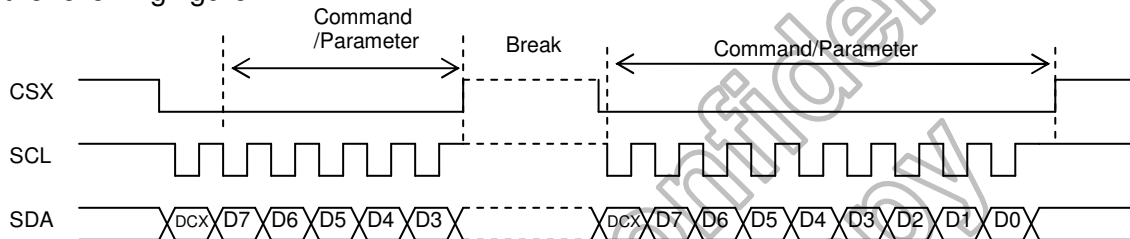


**Figure 5.2: DBI Type C interface Option 1 read mode**



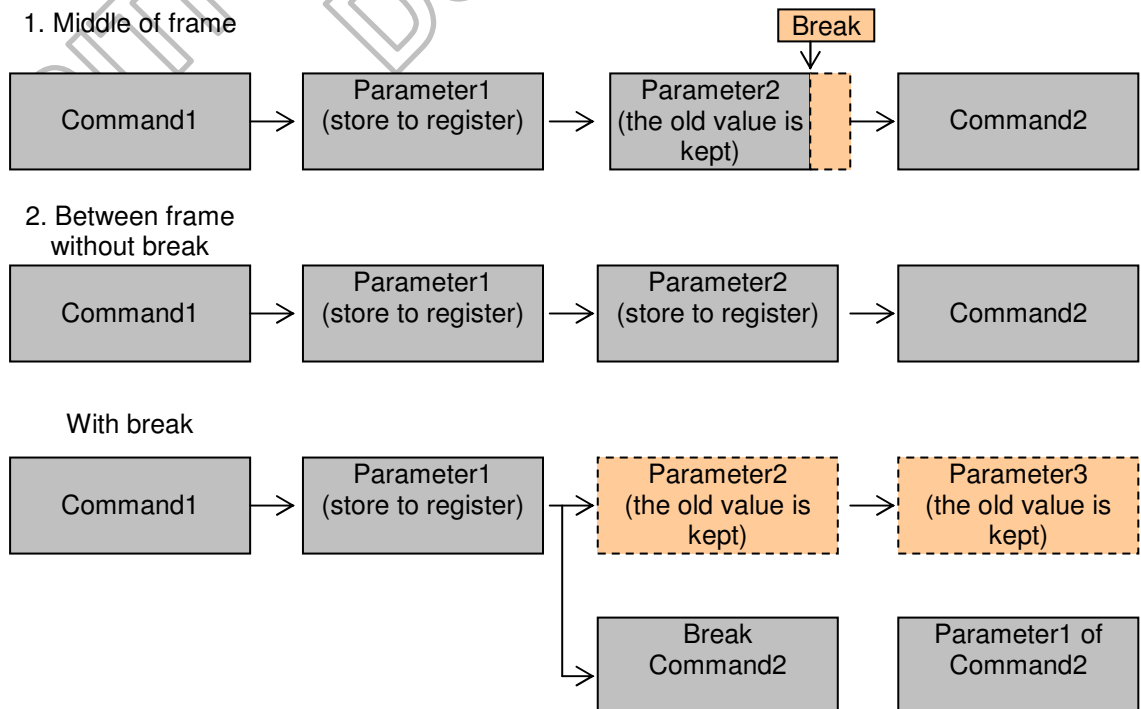
### 5.1.3 Break and Pause Sequences

If there is a break on data transmission when transmit a command before a whole byte has been completed, then the driver IC will have reset the interface such that it will be ready to receive the same byte re-transmitted when the chip select line (CSX) is next activated. See the following figure.



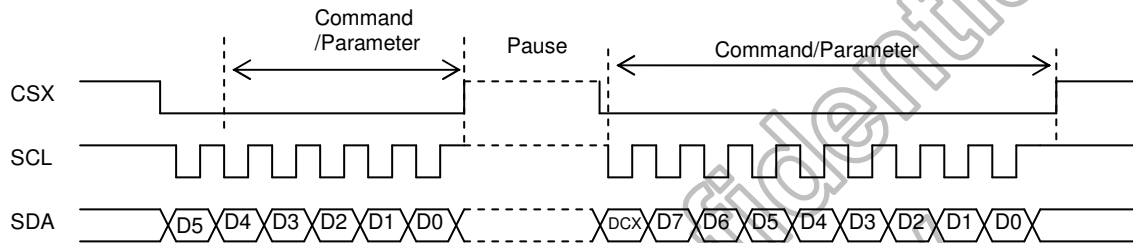
**Figure 5.3: Driver IC data transfer recovery**

If a one or more parameter command is being sent and a break occurs while sending any parameter before the last one and if the host then sends a new command rather than retransmitting the parameter that was interrupted, then the parameters that were successfully sent are stored and the parameter where the break occurred is rejected. The interface is ready to receive next byte as shown:



**Figure 5.4: Break during parameter**

The host processor can pause a write sequence by pulling the CSX signal high between command or data bytes. The driver IC shall wait for the host processor to drive CSX low before continuing the write sequence at the point where the sequence was paused.



**Figure 5.5: Driver IC data transfer pause**

There are 4 cases where there is possible to see this kind of pause:

1. Command – Pause – Command
2. Command – Pause – Parameter
3. Parameter – Pause – Command
4. Parameter – Pause – Parameter

## 5.2 MIPI Interface

The Display Serial Interface standard defines protocols between a host processor and peripheral devices that adhere to MIPI Alliance standards for mobile device interfaces. The DSI standard builds on existing standards by adopting pixel formats and command set defined in MIPI Alliance standards.

DSI-compliant peripherals support either of two basic modes of operation: Command Mode and Video Mode. Which mode is used depends on the architecture and capabilities of the peripheral. The ST7703 only support Video mode.

Video Mode refers to operation in which transfers from the host processor to the peripheral take the form of a real-time pixel stream. In normal operation, the driver IC relies on the host processor to provide image data at sufficient bandwidth to avoid flicker or other visible artifacts in the displayed image. Video information should only be transmitted using High Speed Mode.

| Lane Pair                                 | HOST(Master)/ Driver IC(Slave)                                                                                                                                             |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clock Lane                                | <ul style="list-style-type: none"> <li>- Unidirectional Lane</li> <li>- Clock Only</li> <li>- Escape mode (ULPS only)</li> </ul>                                           |
| Data Lane 0                               | <ul style="list-style-type: none"> <li>- Bi-directional Lane</li> <li>- Forward High Speed</li> <li>- Bi-directional Escape Mode</li> <li>- Bi-directional LPDT</li> </ul> |
| Data Lane 1<br>Data Lane 2<br>Data Lane 3 | <ul style="list-style-type: none"> <li>- Unidirectional Lane</li> <li>- Forward High Speed</li> <li>- Escape mode (ULPS only)</li> <li>- NO LPDT</li> </ul>                |

**Table 5.2: MIPI Interface Configuration**

## 5.2.1 DSI General Interface Communication

The driver IC uses data and clock lane differential pairs for DSI. Both differential lane pairs can be driven Low Power (LP) or High Speed (HS) mode. Low Power mode means that each line of the differential pair is used in single end mode and a differential receiver is disable (A termination resistor of the receiver is disable) and it can be driven into a low power mode.

High Speed mode means that differential pairs (The termination resistor of the receiver is enable) are not used in the single end mode. There are used different modes and protocol in each mode when there is wanted to transfer information from the HOST to the driver IC and vice versa.

| State code | Line voltage Levels |         | High speed     | Low power    |             |
|------------|---------------------|---------|----------------|--------------|-------------|
|            | DP                  | DN      |                | Control mode | Escape mode |
| HS-0       | HS Low              | HS High | Differential-0 | Note 1       | Note 1      |
| HS-1       | HS High             | HS Low  | Differential-1 | Note 1       | Note 1      |
| LP-00      | LP Low              | LP Low  | N/A            | Bridge       | Space       |
| LP-01      | LP Low              | LP High | N/A            | HS-Rqst      | Mark-0      |
| LP-10      | LP High             | LP Low  | N/A            | LP-Rqst      | Mark-1      |
| LP-11      | LP High             | LP High | N/A            | Stop         | Note 2      |

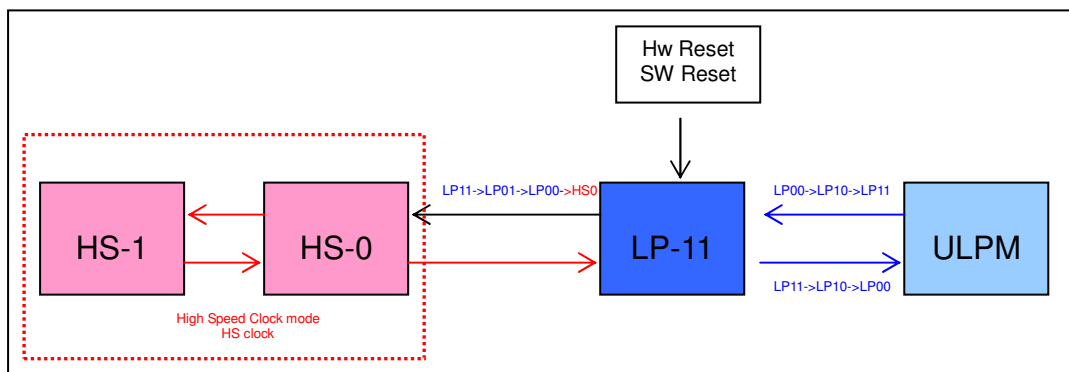
Note 1 : During high-speed transmission, the low power receivers observe LP-00 on the lines.

Note 2 :If LP-11 occurs during Escape mode, the lane returns to Stop state (Control mode LP-11)

**Table 5.3: Lane State Description**

### 5.2.1.1 DSI Clock Lane

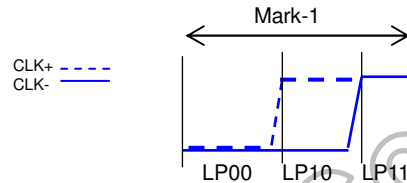
The principle flow chart of the different clock lanes power modes is illustrated below.



**Figure 5.6: DSI Clock Lane State Diagram**

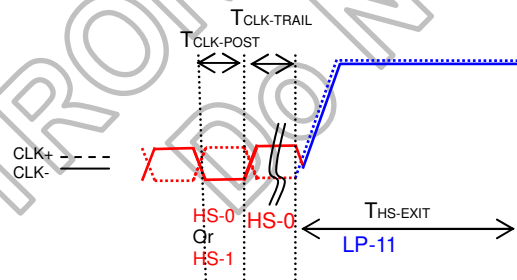
LK+/- lanes can be driven to the Low Power Mode (LPM), when DSI-CLK lanes are entering LP-11:

1. After SW Reset, HW Reset or Power On Sequence =>LP-11
2. After DSI-CLK+/- lanes are leaving Ultra Low Power Mode (ULPM, LP-00) =>LP-10 =>LP-11 (LPM). This sequence is illustrated below.



**Figure 5.7: From ULPM to LPM**

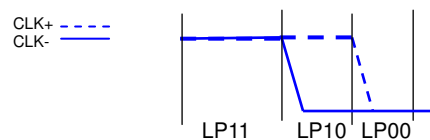
3. After DSI-CLK+/- lanes are leaving High Speed Clock Mode (HSCM, HS-0 or HS-1 State Code) =>HS-0 =>LP-11 (LPM). This sequence and all three mode changes are illustrated below.



**Figure 5.8: From High Speed Clock mode to LPM**

### Ultra Low Power Mode (LP-00: ULPM)

DSI-CLK+/- lanes can be driven to the Ultra Low power Mode (ULPM), when DSI-CLK lanes are entering LP-00 State. The entering way is from the Low Power Mode (LPM, LP-11 State) =>LP-10 =>LP-00 (ULPM). This sequence is illustrated below.



**Figure 5.9: From LPM mode to ULPM**

### High Speed Clock Mode (HSCM)

DSI-CLK+/- lanes can be driven to the High Speed Clock Mode (HSCM), when DSI-CLK lanes are starting to work between HS-0 and HS-1 State. The entering way is from the Low Power Mode (LPM, LP-11 State) =>LP-01 =>LP-00 =>HS-0 =>HS-0/1 (HSCM). This sequence is illustrated below.

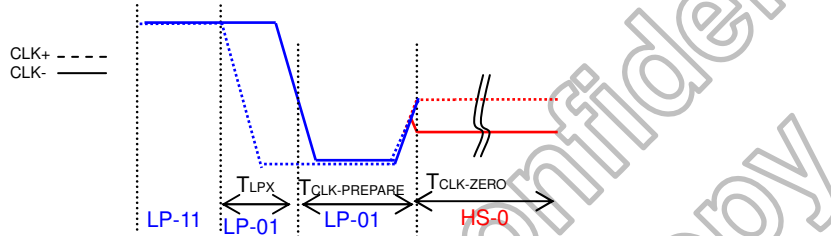


Figure 5.10: From LPM mode to HSCM

### High Speed Clock Burst

The high speed clock (DSI-CLK+/-) is started before high speed data is sent via DSI-Dn+/- lanes. The high speed clock continues clocking after the high speed data sending has been stopped.

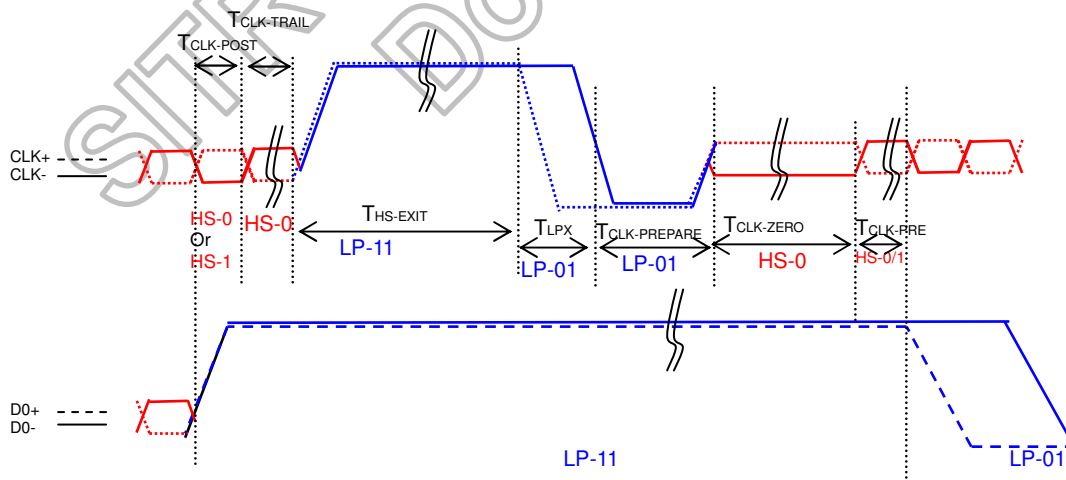


Figure 5.11: Switching the Clock Lane between HSCM and LPM

### 5.2.1.2 DSI-D0 Data Lane

DSI-D0+/- Data Lanes can be driven in different modes which are: Escape Mode, High-Speed Data Transmission and Bus Turnaround Request. The flow chart of the D0 data lanes is illustrated below.

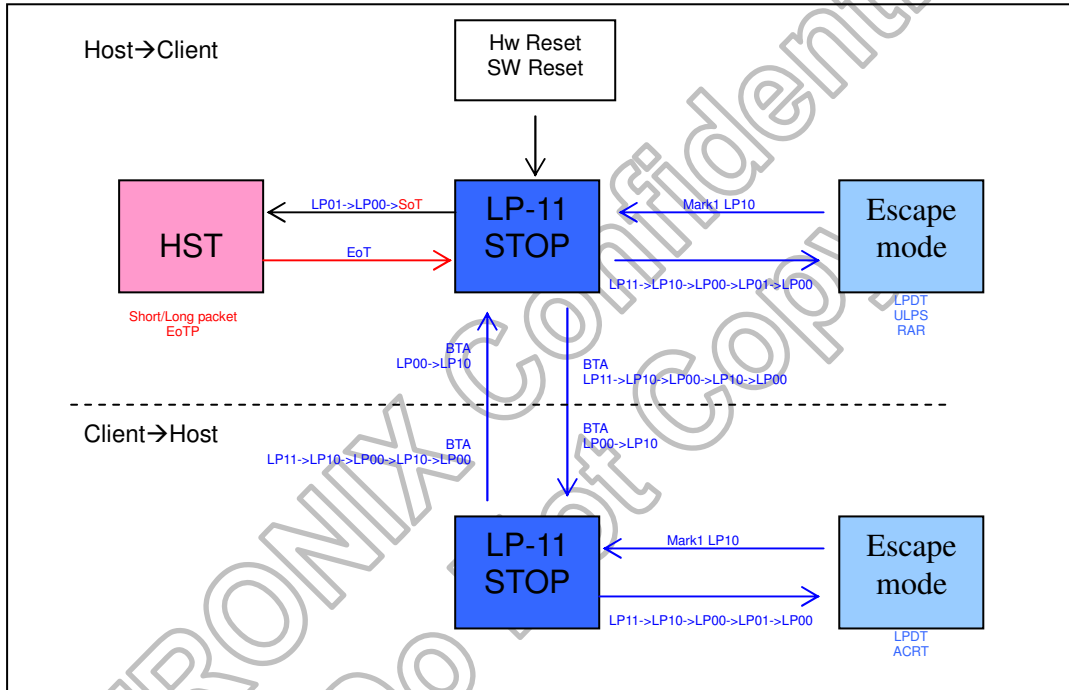


Figure 5.12: DSI Data Lane D0 State Diagram

| Description                        | Operation Code               |
|------------------------------------|------------------------------|
| High Speed Data Transmission Burst | LP11->LP01->LP00             |
| Escape mode entry                  | LP11->LP10->LP00->LP01->LP00 |
| Bus turn around                    | LP11->LP10->LP00->LP10->LP00 |
| Exit Escape mode (Mark-1)          | LP00->LP10->LP11             |

Table 5.4: Data Lane D0 Operation Modes

### ESCAPE MODE

Data lanes (DSI-D0+/-) can be used in different Escape Modes when data lanes are in Low Power (LP) mode.

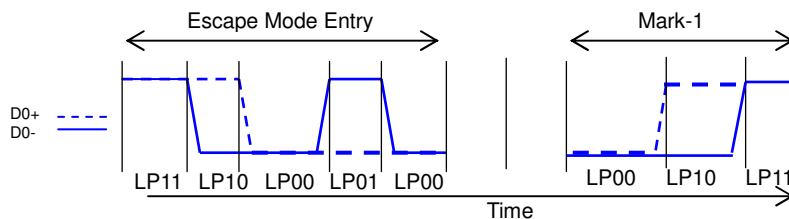


Figure 5.13: DSI Data Lane D0 general escape mode sequence

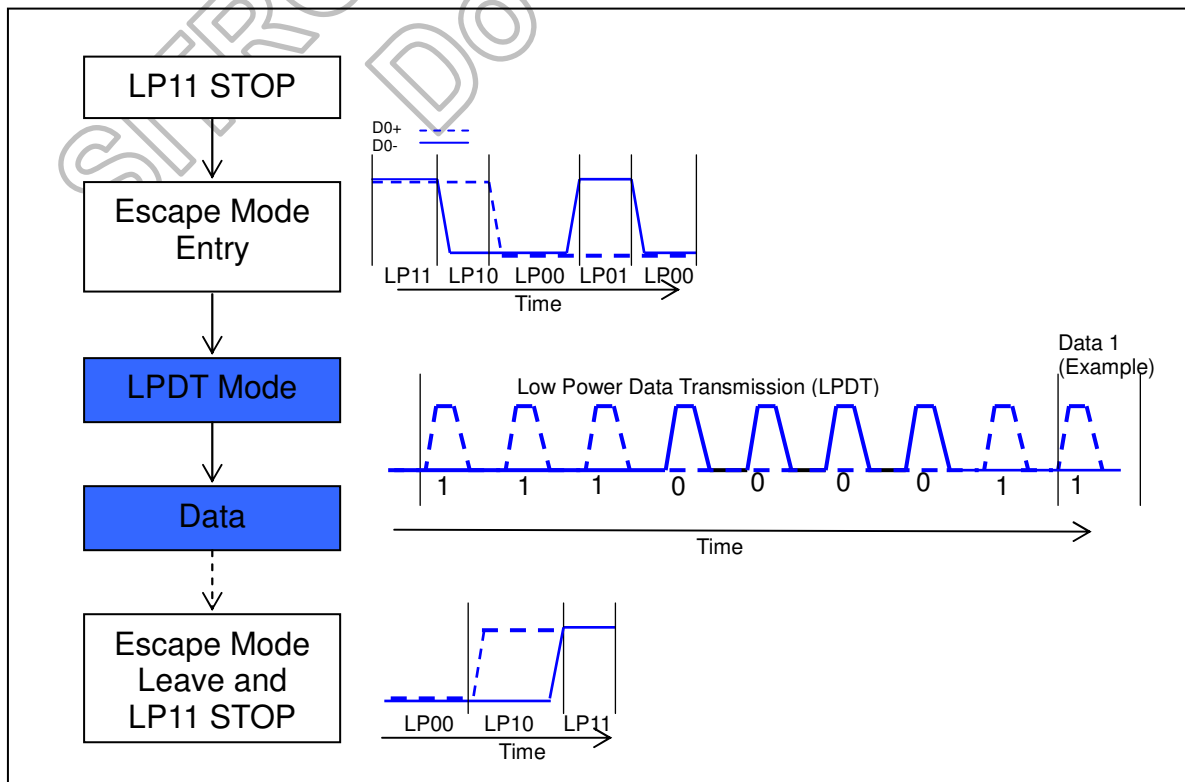
ST7703 can support three different Escape Commands. The commands (EC) can be divided 2 different groups: Mode or Trigger. The HOST is informing to the driver IC that it is controlling data lanes (DSI-D0+/-) with the mode. Escape commands are defined as below table.

| Escape Command              | Command Type | Entry Command Pattern<br>(First Bit→Last Bit Transmitted) |
|-----------------------------|--------------|-----------------------------------------------------------|
| Low Power Data Transmission | Mode         | 1110 0001                                                 |
| Ultra-Low Power mode        | Mode         | 0001 1110                                                 |
| Remote Application Reset    | Trigger      | 0110 0010                                                 |
| Tearing Effect              | Trigger      | 0101 1101                                                 |
| Acknowledge                 | Trigger      | 0010 0001                                                 |

**Table 5.5: Escape Mode Commands**

### Low-Power Data Transmission (LPDT)

The HOST can send data to the driver IC in Low-Power Data Transmission (LPDT) mode when data lanes are entering in Escape Mode and Low-Power Data Transmission (LPDT) command will be sent to this driver IC. The driver IC is also using the same sequence when it is sending data to the HOST. The Low Power Data Transmission (LPDT) is using a following sequence:

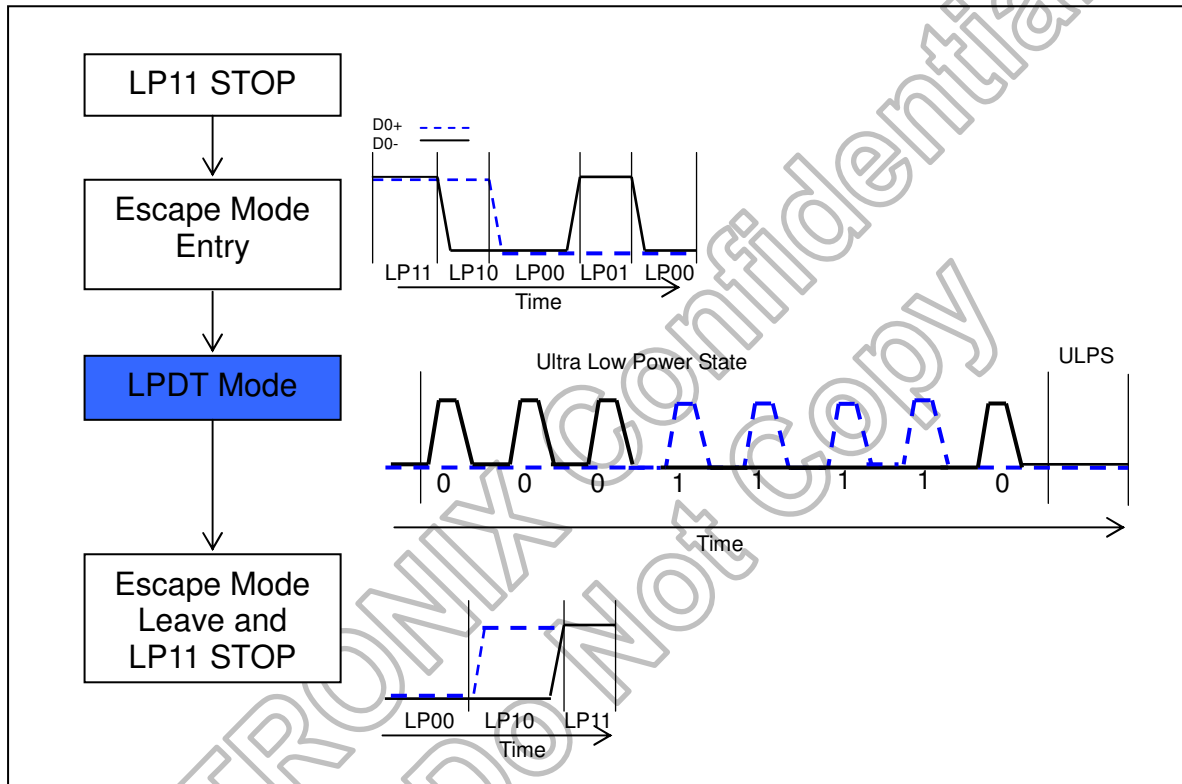


**Figure 5.14: DSI Data Lane D0 LPDT sequence**



### Ultra Low Power State (ULPS)

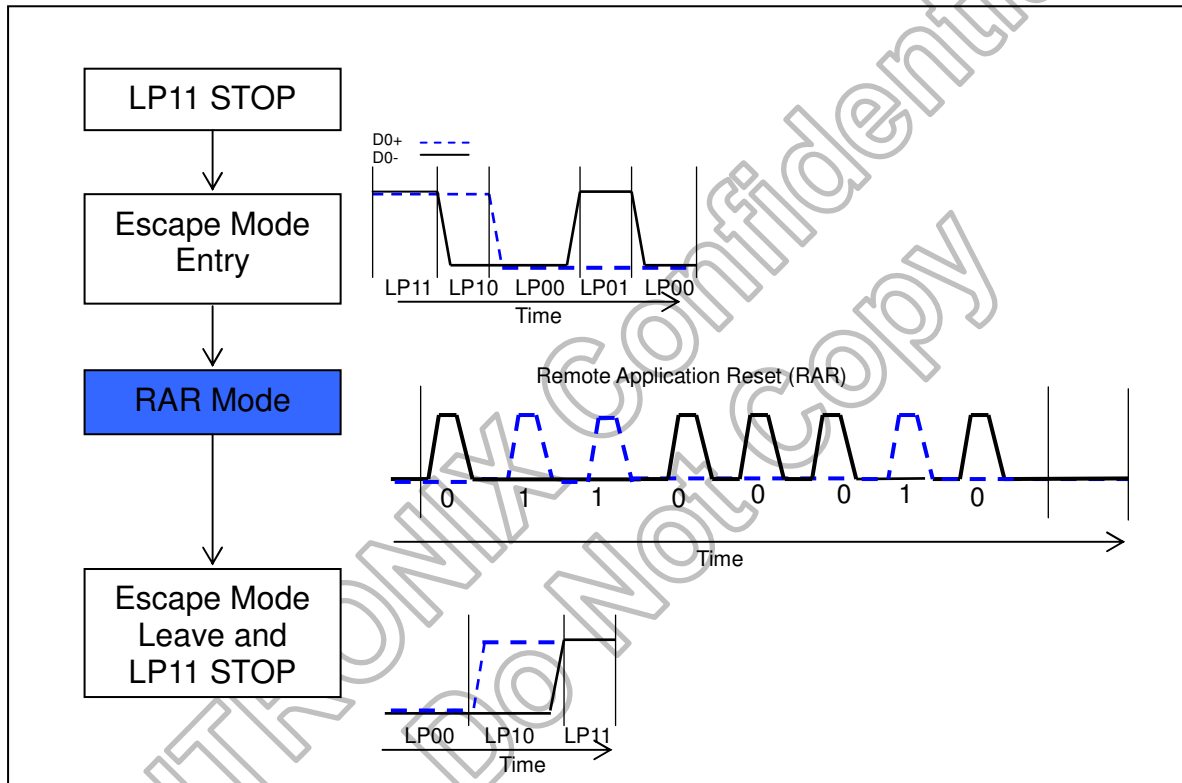
The driver IC can enter this Ultra Low Power State to save power consumption when HOST send this ULPS command. Ultra Low Power State is using below sequence:



**Figure 5.15: DSI Data Lane D0 ULPS sequence**

### Remote Application Reset (RAR)

The HOST can inform to the driver IC that it should be reset in Remote Application Reset (RAR) trigger when data lanes are entering in Escape Mode. The Remote Application Reset (RAR) is using a following sequence:



**Figure 5.16: DSI Data Lane D0 RAR sequence**

### Tearing Effect (TEE)

The driver IC can inform to the HOST when a tearing effect event (New V-synch) has been happen on the driver IC by Tearing Effect (TEE). The Tearing Effect (TEE) is using a following sequence:

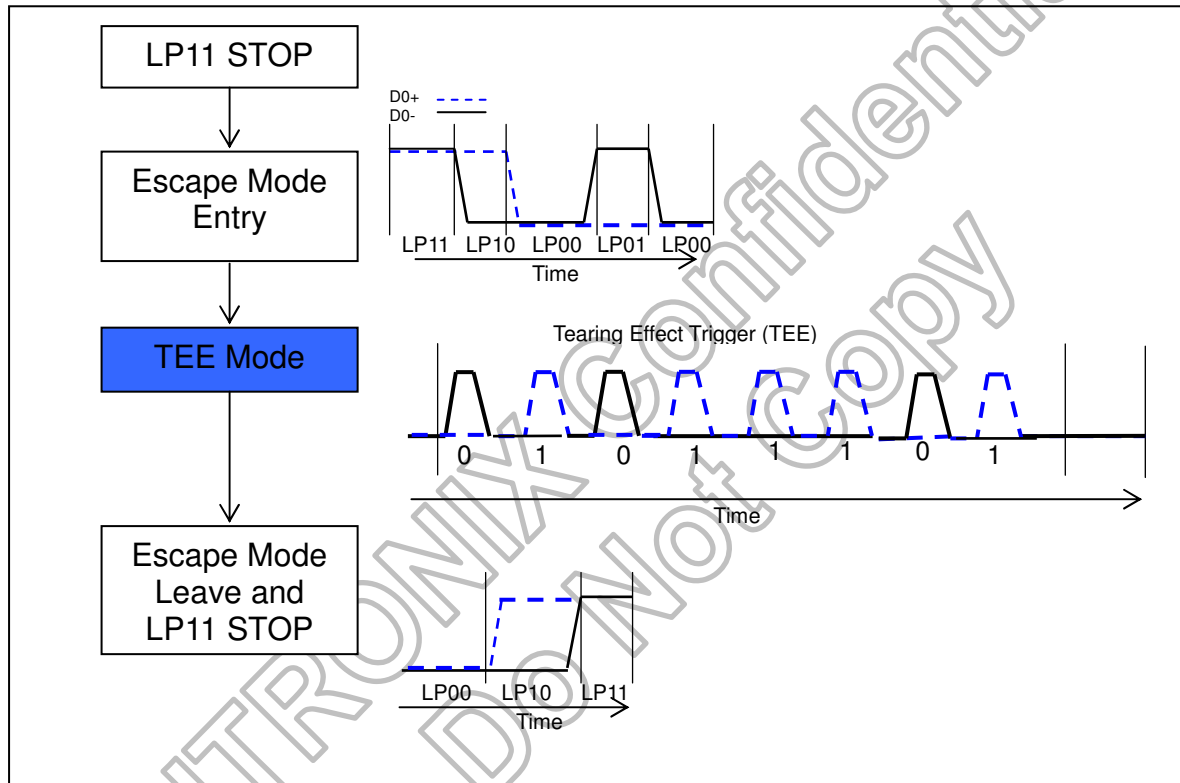


Figure 5.17: DSI Data Lane D0 TEE sequence

### Acknowledge (ACK)

The driver IC can inform to the HOST when an error has not recognized on it by Acknowledge (ACK). The Acknowledge (ACK) is using a following sequence:

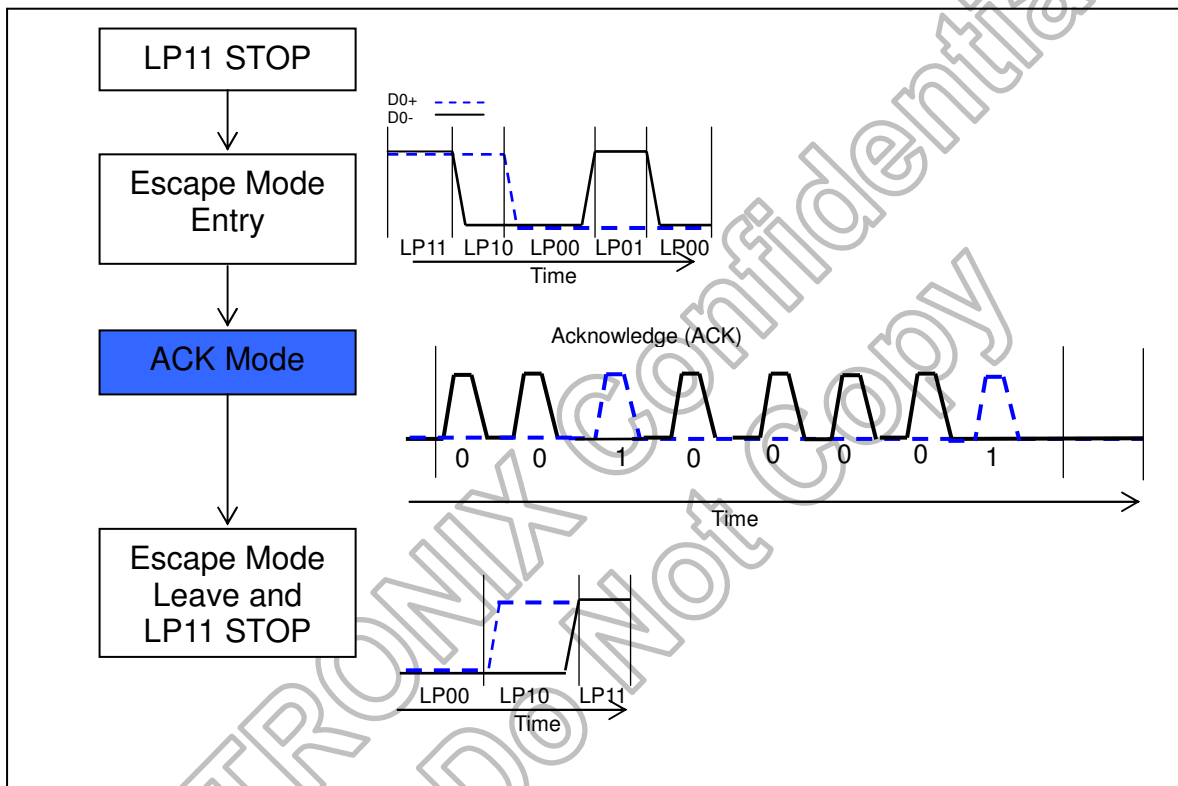
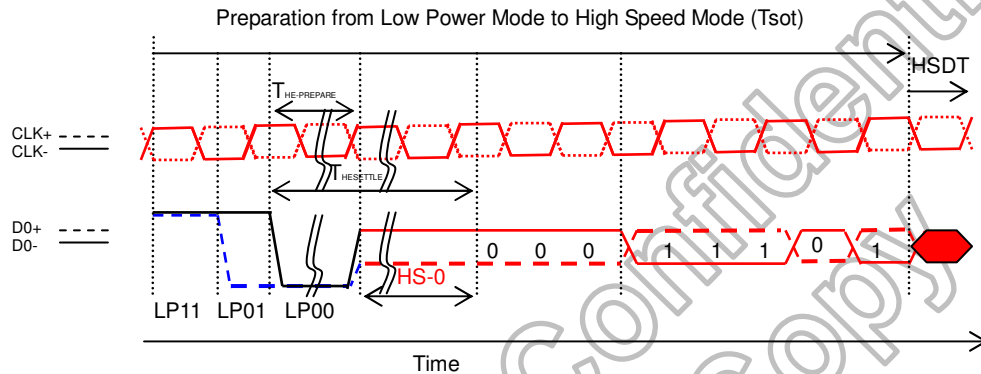


Figure 5.18: DSI Data Lane D0 ACK sequence

### High Speed Data Transmission

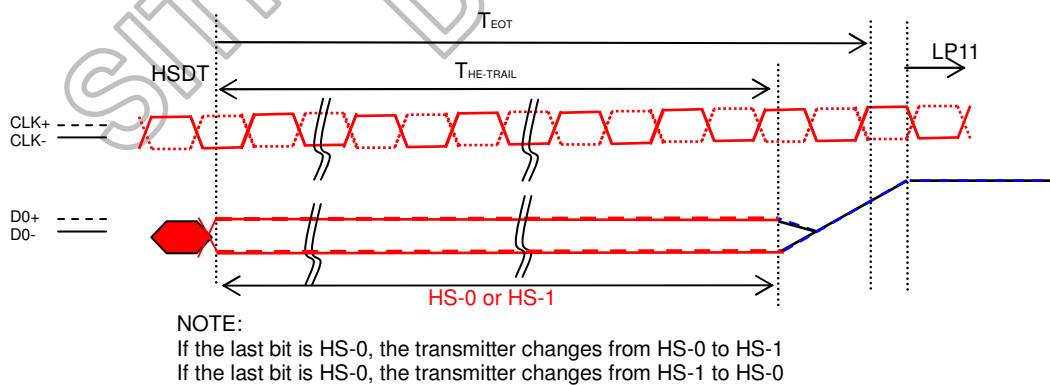
The driver IC is entering High-Speed Data Transmission when Clock lanes DSI-CLK+/- have already been entered in the High-Speed Clock Mode by the HOST. Data lanes of the driver IC are entering (TSOT) in the High-Speed Data Transmission (HSDT) as below figure.



**Figure 5.19: DSI Data Lane Entering High Speed Data Transmission**

### Leaving High Speed Data Transmission

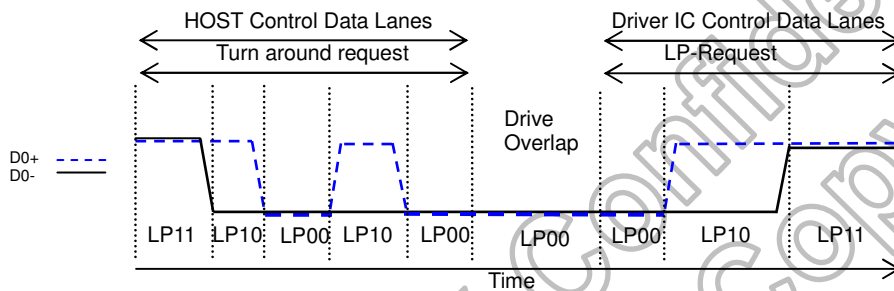
The driver IC is leaving the High-Speed Data Transmission (TEOT of HSDT) when Clock lanes DSI-CLK+/- are in the High-Speed Clock Mode by the HOST and it is kept until data lanes are in LP-11 mode. Data lanes of the driver IC are leaving from the High-Speed Data Transmission (TEOT of HSDT) as follows



**Figure 5.20: DSI Data Lane Entering High Speed Data Transmission**

### Bus Turnaround (BTA)

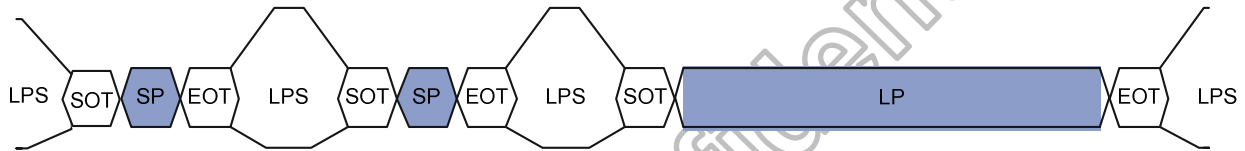
The HOST or driver IC, which is controlling DSI-D0+/- Data Lanes, can start a bus turnaround procedure when it wants information from a receiver, which can be the HOST or driver IC. The HOST or driver IC is using the same sequence when this bus turnaround procedure is used. This sequence is described for reference purposes, when the HOST wants to do the bus turnaround procedure to the driver IC, as below.



**Figure 5.21: DSI Bus Turn Around Procedure**

### 5.2.2 DSI Packet Level Communication

The DSI protocol permits multiple packets which is useful for events such as peripheral initialization, where many registers may be loaded separate write commands at system startup. Below figure illustrates multiple HS Transmission packets.



LPS : Low power state  
SOT : Start of Transmission  
SP : Short Packet  
LP : Long Packet  
EOT : End of Transmission

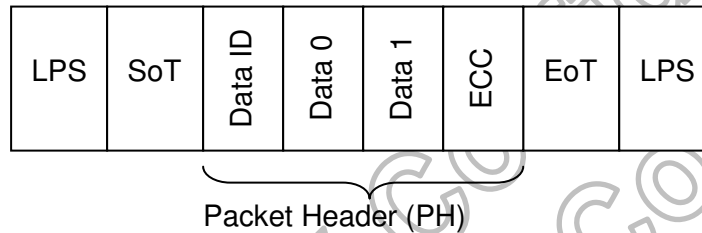
**Figure 5.22: DSI multiple HS transmission packets**

The packet includes two types which are Long packet and short packet. The first byte of the packet, the Data Identifier (DI), includes information specifying the length of the packet. Command Mode systems send commands and an associated set of parameters, with the number of parameters depending on the command type.

### 5.2.2.1 General Packet Structure

#### Short packets

Specify the payload length using the Data Type field and are from two to nine bytes in length. Short packet is used for most Command Mode commands and associated parameters. Where short packets format include an 8-bit Data ID followed by zero to seven bytes and an 8-bit ECC. Below figure shows the structure of the Short packet.



**SOT:** Start of Transmission

**DI(Data ID):** 8-bit Contain Virtual Channel Identifier and Data Type.

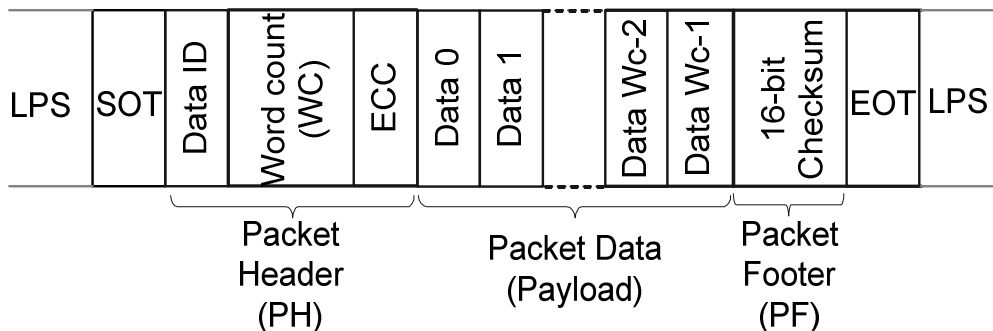
**Data 0 and Data 1:** Packet Data (8+8bit)

**ECC(Error Correction Code):** The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header.

**Figure 5.23: Structure of the short packet**

#### Long packets

Specify the payload length using a two-byte Word Count field and then the payload maybe from 0 to 65,541 bytes in length. Long packets permit transmission of large blocks of pixel or other data. Below figure shows the structure of the Long packet. Long Packet Header composed of three elements: an 8-bit Data Identifier, a 16-bit Word Count, and 8-bit ECC. The Packet Footer has one element, a 16-bit checksum. Long packets can be from 6 to 65,541 bytes in length. Where  $65,541 \text{ bytes} = (216-1) + 4 \text{ bytes PH} + 2 \text{ bytes PF}$



**DI (Data ID) :** Contain Virtual Channel Identifier and Data Type.

**WC (Word Count) :** 8+8 bits The receiver use WC to define packet end.

**ECC (Error Correction Code) :** The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header.

**PF(Packet Footer) :** Mean 16-bit Checksum.

**Figure 5.24: Structure of the long packet**



According to packet form, basic elements include DI and ECC. Figure 4.44 the shows format of Data ID.

| DI7                  | DI6 | DI5            | DI4 | DI3 | DI2 | DI1 | DI0 |
|----------------------|-----|----------------|-----|-----|-----|-----|-----|
| VC (Virtual Channel) |     | DT (Data Type) |     |     |     |     |     |

DI[7:6] → These two bits identify the data as directed to one of four virtual channels.

DI[5:0]: These six bits specify the Data Type, which specifies the size, format and, in some cases, the interpretation of the packet contents.

**Figure 5.25: Format of data ID**

### Virtual Channel (VC)

Virtual Channel (VC) is a part of Data Identification (DI[7...6]) structure and it is used to address where a packet is wanted to send from the HOST. The ST7703 supports Virtual Channel only when VC = 00.

### Data Type (DT)

Data Type (DT) is a part of Data Identification (DI[5...0]) structure and it is used to define a type of the used data on a packet.

| Data type, hex           | Data type, binary  | Description packet                                    | Size  |
|--------------------------|--------------------|-------------------------------------------------------|-------|
| 05h                      | 00 0101            | DCS Write, no parameters                              | Short |
| 15h                      | 01 0101            | DCS Write, 1 parameter                                | Short |
| 01h                      | 00 0001            | Sync Event, V Sync Start                              | Short |
| 11h                      | 01 0001            | Sync Event, V Sync End                                | Short |
| 21h                      | 10 0001            | Sync Event, H Sync Start                              | Short |
| 31h                      | 11 0001            | Sync Event, H Sync End                                | Short |
| 02h                      | 00 0010            | Color Mode (CM) Off Command                           | Short |
| 12h                      | 01 0010            | Color Mode (CM) On Command                            | Short |
| 22h                      | 10 0010            | Shut Down Peripheral Command                          | Short |
| 32h                      | 11 0010            | Turn On Peripheral Command                            | Short |
| 13h                      | 01 0011            | Generic Short Write, 1 parameter                      | Short |
| 23h                      | 10 0011            | Generic Short Write, 2 parameter                      | Short |
| 14h                      | 01 0100            | Generic Read, 1 parameter                             | Short |
| 29h                      | 10 1001            | Generic Long Write                                    | Long  |
| 06h                      | 00 0110            | DCS READ, no parameters                               | Short |
| 37h                      | 11 0111            | Set Maximum Return Packet Size                        | Short |
| 09h                      | 00 1001            | Null Packet, no data                                  | Long  |
| 19h                      | 01 1001            | Blanking Packet, no data                              | Long  |
| 39h                      | 11 1001            | DCS Long Write/write_LUT Command Packet               | Long  |
| 0Eh                      | 00 1110            | Packed Pixel Stream, 16-bit RGB, 5-6-5 Format         | Long  |
| 1Eh                      | 01 1110            | Packed Pixel Stream, 18-bit RGB, 6-6-6 Format         | Long  |
| 2Eh                      | 10 1110            | Loosely Packed Pixel Stream, 18-bit RGB, 6-6-6 Format | Long  |
| 3Eh                      | 11 1110            | Packed Pixel Stream, 24-bit RGB, 8-8-8 Format         | Long  |
| X0h and XFh, unspecified | xx 0000<br>xx 1111 | DO NOT USE<br>All unspecified codes are reserved      | -     |

**Table 5.6: Data Types from Host to the Driver IC**

| Data type, hex | Data type, binary | Description packet                           | Size  |
|----------------|-------------------|----------------------------------------------|-------|
| 02h            | 00 0010           | Acknowledge with Error Report                | Short |
| 21h            | 01 0001           | Generic short READ Response, 1 byte returned | Short |
| 22h            | 01 0010           | Generic short READ Response, 2 byte returned | Short |
| 1Ah            | 01 1010           | Generic Read Long Response                   | Long  |
| 1Ch            | 01 1100           | DCS Long READ Response                       | Long  |

**Table 5.7: Data Types from the Driver IC to Host**

**Error Correction Code (ECC)**

Error Correction Code (ECC) is a part of Packet Header (PH) and its purpose is to identify an error or errors on the Packet Header (PH).

Bits (P[7...0]) of the Error Correction Code (ECC) are defined, where the symbol '^' is presenting XOR function, as follows.

- P7 = 0
- P6 = 0
- P5 = D10^D11^D12^D13^D14^D15^D16^D17^D18^D19^D21^D22^D23
- P4 = D4^D5^D6^D7^D8^D9^D16^D17^D18^D19^D20^D22^D23
- P3 = D1^D2^D3^D7^D8^D9^D13^D14^D15^D19^D20^D21^D23
- P2 = D0^D2^D3^D5^D6^D9^D11^D12^D15^D18^D20^D21^D22
- P1 = D0^D1^D3^D4^D6^D8^D10^D12^D14^D17^D20^D21^D22^D23
- P0 = D0^D1^D2^D4^D5^D7^D10^D11^D13^D16^D20^D21^D22^D23

**PACKET DATA (PD) ON THE LONG PACKET**

Packet Data (PD) of the Long Packet is defined after Packet Header (PH) of the Long Packet. The number of the data bytes is defined on chapter "Word Count (WC) on the Long Packet".

**PACKET FOOTER (PF) ON THE LONG PACKET**

Packet Footer (PF) of the Long Packet is defined after the Packet Data (PD) of the Long Packet. The Packet Footer (PF) is a checksum value what is calculated from the Packet Data of the Long Packet. The checksum is using a 16-bit Cyclic Redundancy Check (CRC) value which is generated with a polynomial  $X^{16}+X^{12}+X^5+X^0$ .

### 5.2.2.2 Detail Format Description

**Generic Write, 1 Parameter, Data Type = 01 0011 (13h)**

Generic Write, 1 Parameter is always using a Short Packet from the HOST to the driver IC. The content of 2 payload bytes is “command” and “00h”.

**Generic Write, 2 Parameter, Data Type = 10 0011 (23h)**

Generic Write, 2 Parameter is always using a Short Packet from the HOST to the driver IC. The content of 2 payload bytes is “command” and “parameter”.

**Generic Long Write, Data Type = 10 1001 (29h)**

Generic Long Write is always using a Long Packet from the HOST to the driver IC. The content can include Command (No Parameters) or Command with 1 or more parameters.

**Generic Read, 1 Parameter, Data Type = 01 0100 (14h)**

Generic Read, 1 Parameter is always using a Short Packet from the HOST to the driver IC. The HOST has to define to the driver IC, what is the maximum size of the return packet. A command, what is used for this purpose, is “Set Maximum Return Packet Size”, which Data Type is 11 0111 and which is using Short Packet before the HOST can send “Generic Read, 1 Parameter” to the driver IC.

**Display Command Set Write, No Parameter, Data Type = 00 0101 (05h)**

Display Command Set (DCS) Write, No Parameter is always using a Short Packet from the HOST to the driver IC. The content of 2 payload bytes is “command” and “00h”.

**Display Command Set Write, 1 Parameter, Data Type = 01 0101 (15h)**

Display Command Set (DCS) Write, 1 Parameter is always using a Short Packet from the HOST to the driver IC. The content of 2 payload bytes is “command” and “parameter”.

**Display Command Set Long Write, Data Type = 11 1001 (39h)**

Display Command Set (DCS) Long Write is always using a Long Packet from the HOST to the driver IC. The content can include Command (No Parameters) or Command with 1 or more parameters.

**Display Command Set (DCS) Read, No Parameter, Data Type = 00 0110 (06h)**

Display Command Set (DCS) Read, No Parameter is always using a Short Packet, from the HOST to the driver IC. The HOST has to define to the driver IC, what is the maximum size of the return packet. A command, what is used for this purpose, is “Set Maximum Return Packet Size”, which Data Type is 11 0111 and which is using Short Packet before the HOST can send “Display Command Set (DCS) Read, No Parameter” to the driver IC.

**Null Packet, No Data , Data Type = 00 1001 (09h)**

Null Packet, No Data is always using a Long Packet, what is defined on Data Type from the HOST to the driver IC. The purpose of this command is keeping data lanes in the high speed mode, if it is needed. The driver IC is ignored Packet Data what the HOST is sending.

**Set Maximum Return Packet Size, Data Type = 11 0011 (37h)**

Set Maximum Return Packet Size is always using a Long Packet, what is defined on Data Type from the HOST to the driver IC. The purpose of this command is specifies the maximum size of the payload in a Long packet transmitted from peripheral back to the host processor.

**Sync Event (H Start, H End, V Start, V End), Data Type = XX 0001 (0xX1)**

Sync Events are Short packets and, therefore, can time-accurately represent events like the start and end of sync pulses. As “start” and “end” are separate and distinct events, the length of sync pulses, as well as position relative to active pixel data, The Sync Events are defined as follows:

- Data Type = 00 0001 (01h) V Sync Start
- Data Type = 01 0001 (11h) V Sync End
- Data Type = 10 0001 (21h) H Sync Start
- Data Type = 11 0001 (31h) H Sync End

**Color Mode On Command, and, Data Type = 01 0010 (12h)**

Color Mode On is a Short packet command that switches a Video Mode driver IC to 8-colors mode for power saving.

**Color Mode Off Command, Data Type = 00 0010 (02h)**

Color Mode Off is a Short packet command that returns a Video Mode driver IC from 8-colors mode to normal display operation.

**Shutdown Peripheral Command, Data Type = 10 0010 (22h)**

Shutdown Peripheral command is a Short packet command that turns off the display in a Video Mode driver IC for power saving. Note the interface shall remain powered in order to receive the turn-on, or wake-up command.

**Turn On Peripheral Command, Data Type = 11 0010 (32h)**

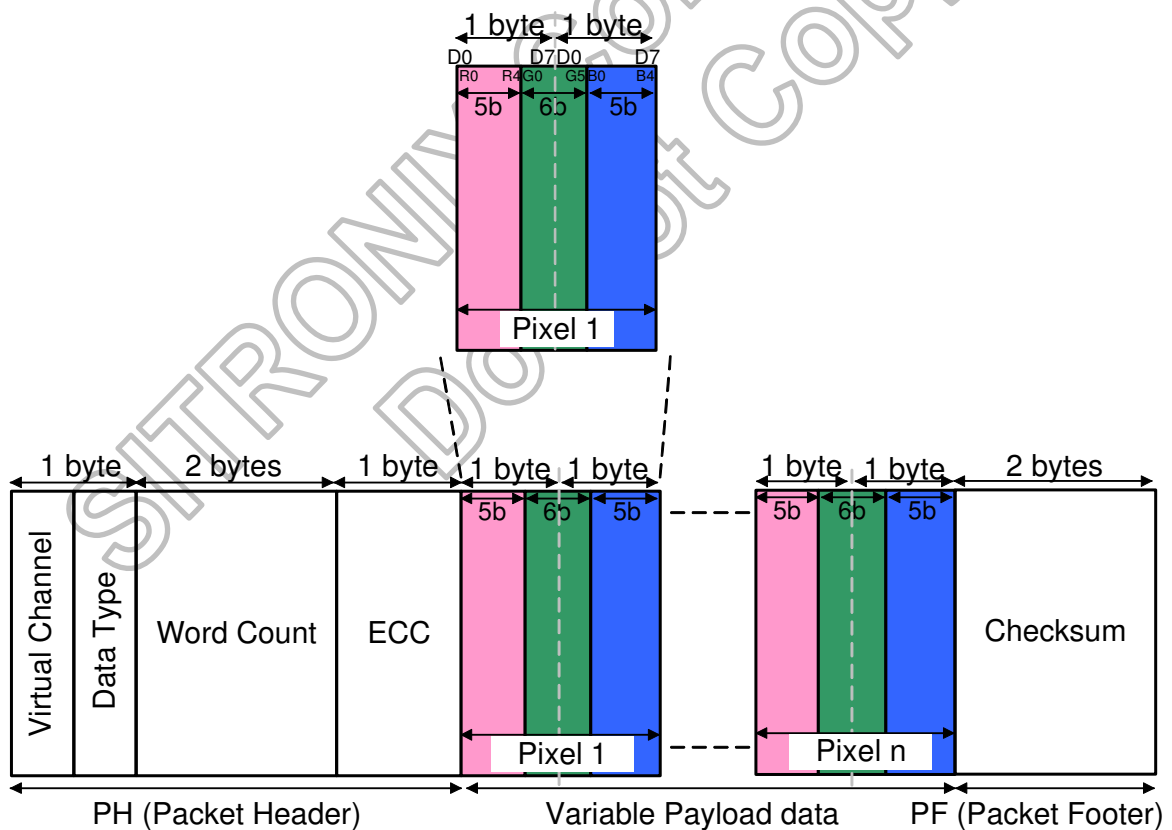
Turn On Peripheral command is Short packet command that turns on the display in a Video Mode driver IC for normal display operation.

### Blanking Packet (Long), Data Type = 01 1001 (19h)

A Blanking packet is used to convey blanking timing information in a Long packet. Normally, the packet represents a period between active scan lines of a Video Mode display, where traditional display timing is provided from the host processor to the driver IC.

### Packed Pixel Stream, 16-bit Format, Long packet, Data Type = 00 1110 (0Eh)

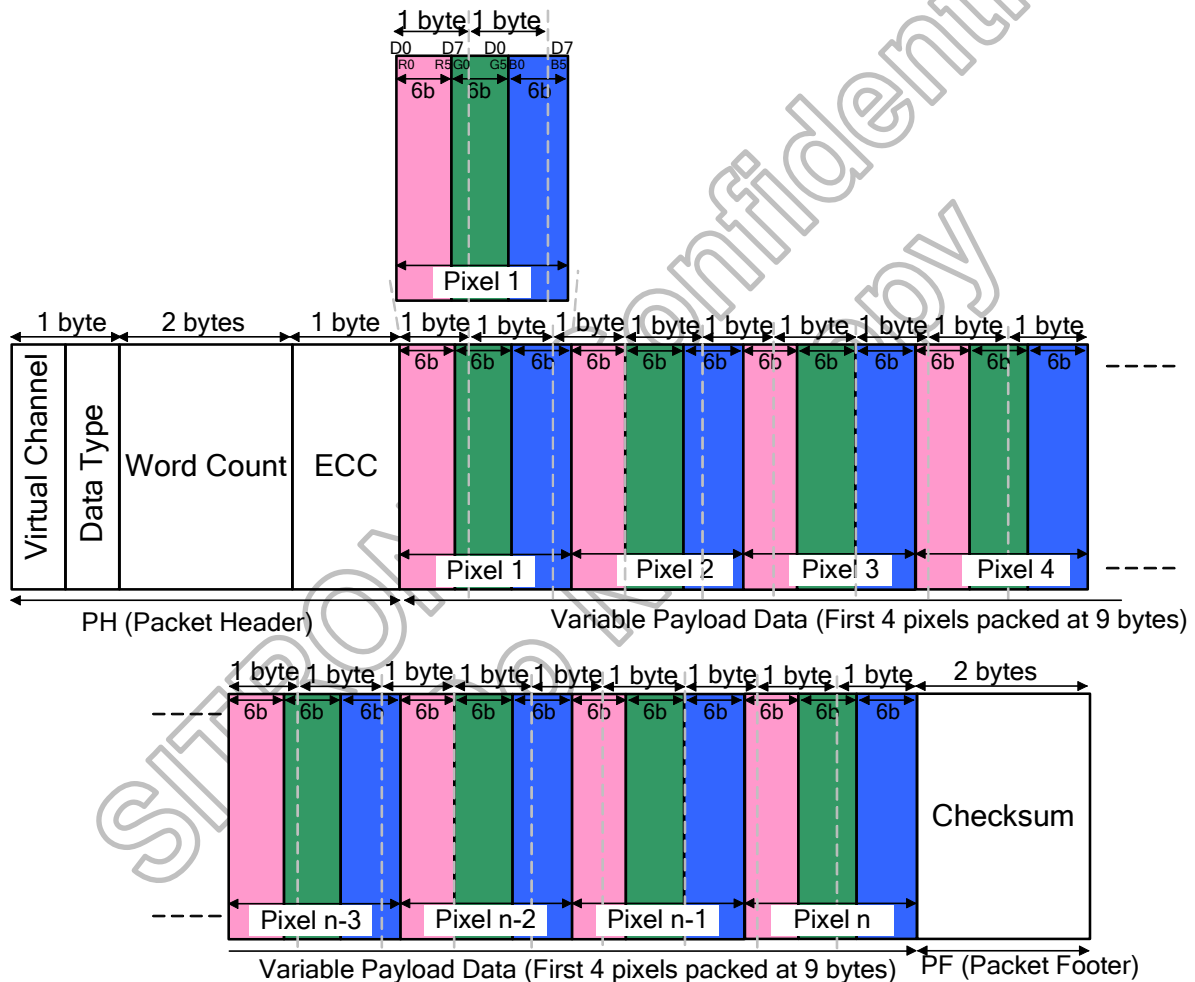
Packed Pixel Stream 16-Bit Format is a Long packet used to transmit image data formatted as 16-bit pixels to a Video Mode driver IC. Pixel format is five bits red, six bits green, five bits blue, in that order. Note that the “Green” component is split across two bytes. Within a color component, the LSB is sent first, the MSB last.



**Figure 5.26: 16-bit RGB Color Format, Long packet**

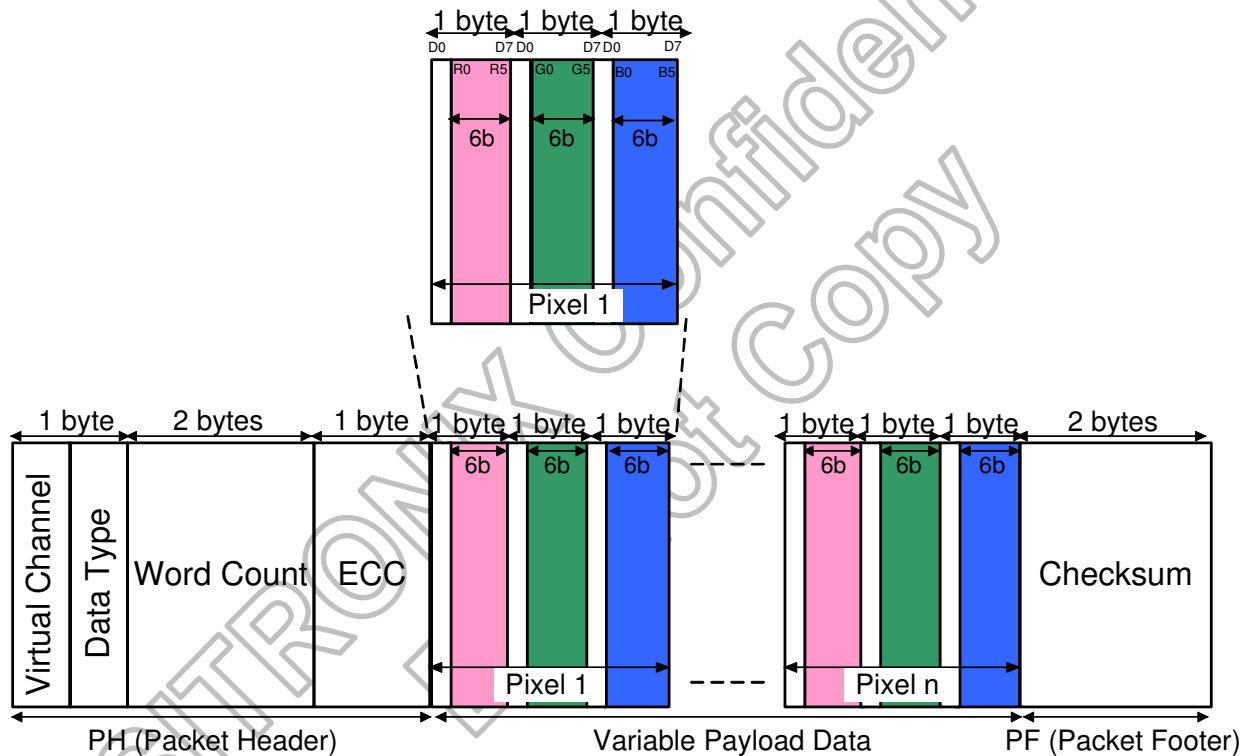
**Packed Pixel Stream, 18-bit Format, Long packet, Data type = 01 1110 (1Eh)**

Packed Pixel Stream 18-Bit Format is a Long packet used to transmit image data formatted as 18-bit pixels to a Video Mode driver IC. Pixel format is six bits red, six bits green, six bits blue, in that order.



**Pixel Stream, 18-bit Format in Three Bytes, Long packet, Data Type = 101110 (2Eh)**

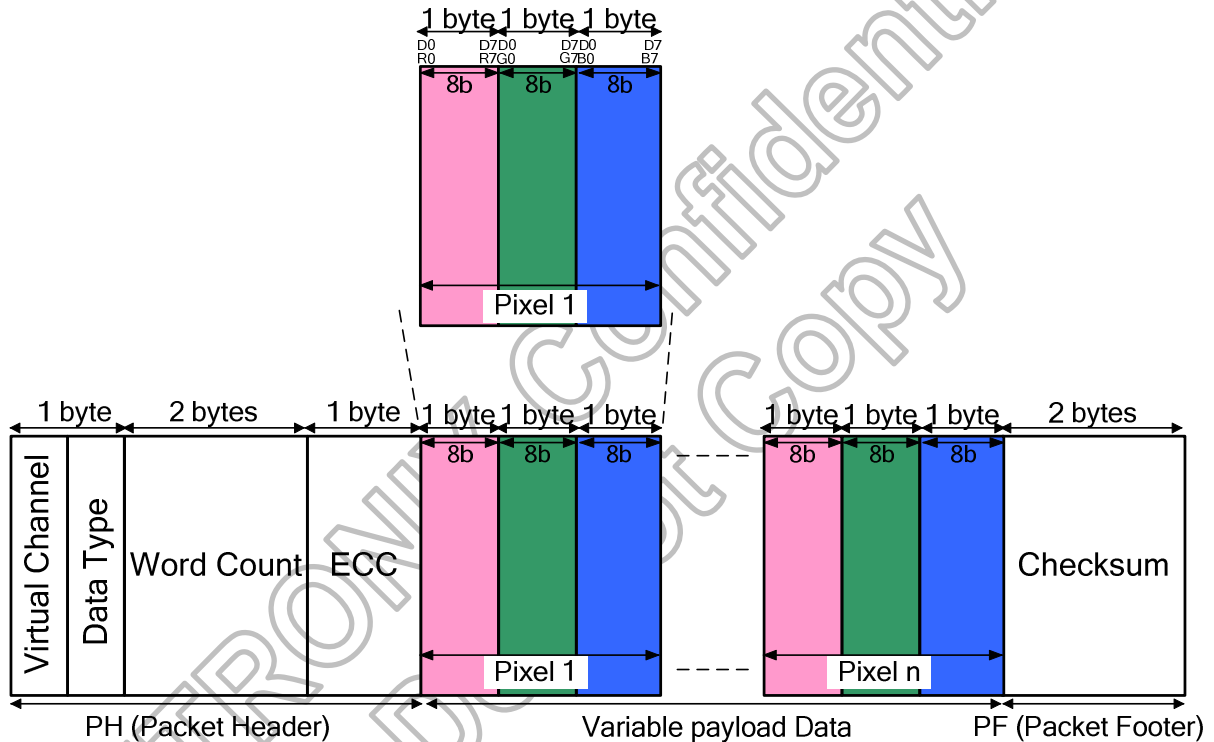
Packed Pixel Stream 18-Bit Format is a Long packet used to transmit image data formatted as 18-bit pixels to a Video Mode driver IC. Each R, G, or B color component is six bits but is shifted to the upper bits of the byte, such that the valid pixel bits occupy bits [7:2] of each byte. Bits [1:0] of each payload byte representing active pixels are ignored.





**Packed Pixel Stream, 24-bit Format, Long packet, Data Type = 11 1110 (3Eh)**

Packed Pixel Stream 24-Bit Format is a Long packet used to transmit image data formatted as 18-bit pixels to a Video Mode driver IC. Pixel format is eight bits red, eight bits green, eight bits blue, in that order.



### Acknowledge with Error Report, Data Type = 00 0010(02h)

“Acknowledge with Error Report” is always using a Short Packet, from the driver IC to the HOST. The Packet Data can include bits, which are defining the current error, when a corresponding bit is set to ‘1’, as they are defined on the following table.

| Bit | Description                                    |
|-----|------------------------------------------------|
| 0   | SoT Error                                      |
| 1   | SoT Sync Error                                 |
| 2   | EoT Sync Error                                 |
| 3   | Escape Mode Entry Command Error                |
| 4   | Low-Power Transmit Sync Error                  |
| 5   | HS Receive Timeout Error                       |
| 6   | Reserved                                       |
| 7   | Reserved                                       |
| 8   | ECC Error, single-bit (detected and corrected) |
| 9   | ECC Error, multi-bit (detected, not corrected) |
| 10  | Checksum Error (long packet only)              |
| 11  | DSI Data Type Not Recognized                   |
| 12  | DSI VC ID Invalid                              |
| 13  | Reserved                                       |
| 14  | Reserved                                       |
| 15  | Reserved                                       |

These errors of the previous packets can check “Read Display Signal Mode (0Eh)” and “Read Number of the Errors on DSI (05h)” commands. The bit D0 of the “Read Display Signal Mode (0Eh)” command has been set to ‘1’ if a received packet includes an error.

The number of the packets, which are including an ECC or CRC error, are calculated on the RDNUMED register, which can read “Read Number of the Errors on DSI (05h)” command.

### DCS Read Long Response, Data Type = 01 1100(1Ch)

DCS Read Long Response is always using a Long Packet, from the driver IC to the Host. “DCS Read Long Response” is used when the driver IC wants to response a DCS Read command, which the Host has sent to the driver IC.

### DCS Read Short Response, 1 Byte Returned, Data Type = 10 0001(21h)

DCS Read Short Response, 1 Byte Returned is always using a Short Packet from the driver IC to the Host. “DCS Read Short Response, 1 Byte Returned” is used when the driver IC wants to response a DCS Read command, which the Host has sent to the driver IC.

### DCS Read Short Response, 2 Bytes Returned, Data Type = 10 0010(22h)

DCS Read Short Response, 2 Bytes Returned is always using a Short Packet from the driver IC to the Host. “DCS Read Short Response, 2 Bytes Returned” is used when the driver IC wants to response a DCS Read command, which the Host has sent to the driver IC.

**Generic Read Long Response, Data Type = 01 1010(1Ah)**

Generic Read Long Response is always using a Long Packet from the driver IC to the HOST. "Generic Read Long Response" is used when the driver IC wants to response a Generic Read command.

**Generic Read Short Response, 1 Byte Returned, Data Type = 01 0001(11h)**

Generic Read Short Response, 1 Byte Returned is always using a Short Packet from the driver IC to the HOST. "Generic Read Short Response, 1 Byte Returned" is used when the driver IC wants to response a Generic Read command, which the HOST has sent to the driver IC.

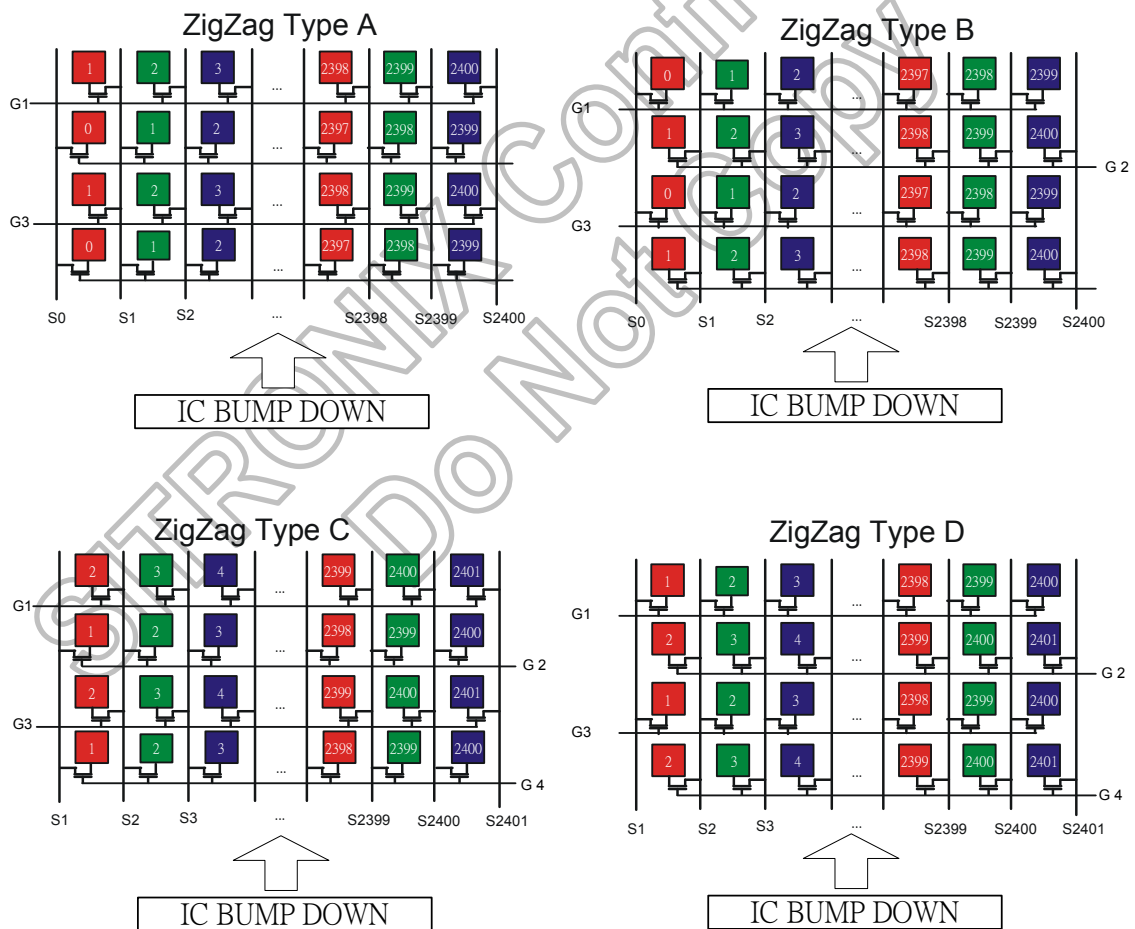
**Generic Read Short Response, 2 Bytes Returned, Data Type = 01 0010(12h)**

Generic Read Short Response, 2 Bytes Returned is always using a Short Packet from the driver IC to the HOST. "Generic Read Short Response, 2 Bytes Returned" is used when the driver IC wants to response a Generic Read command, which the HOST has sent to the driver IC.

### 5.3 Inversion

The ST7703 can support the column, 1-dot, 1+2-dot, 2-dot, 3-dot, 4-dot and 8-dot inversion of liquid crystal. These inversions can provide a solution for improving display quality. In determining the inversion drive for the inversion cycle, check the quality of display on the liquid crystal panel.

The ST7703 also can support the ZigZag inversion. Eight kinds of ZigZag inversion can be selected by software setting. These kinds are as below figure:



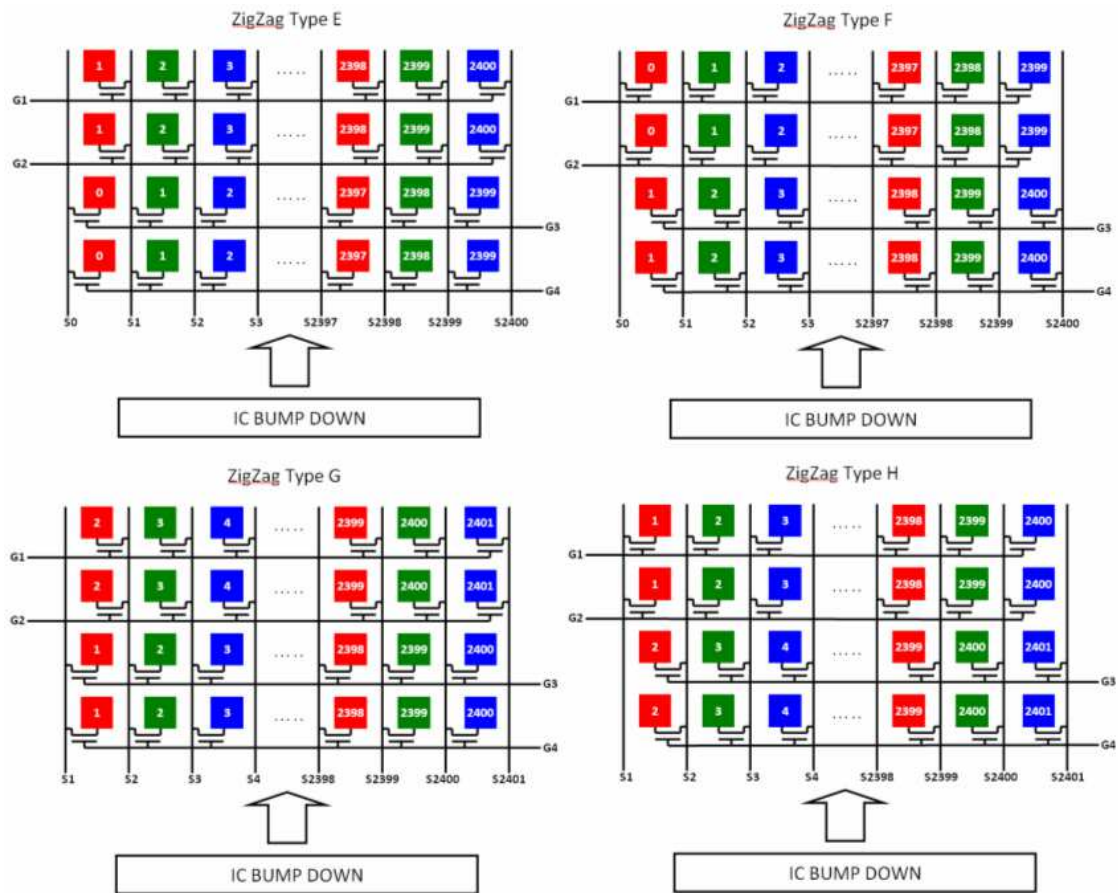
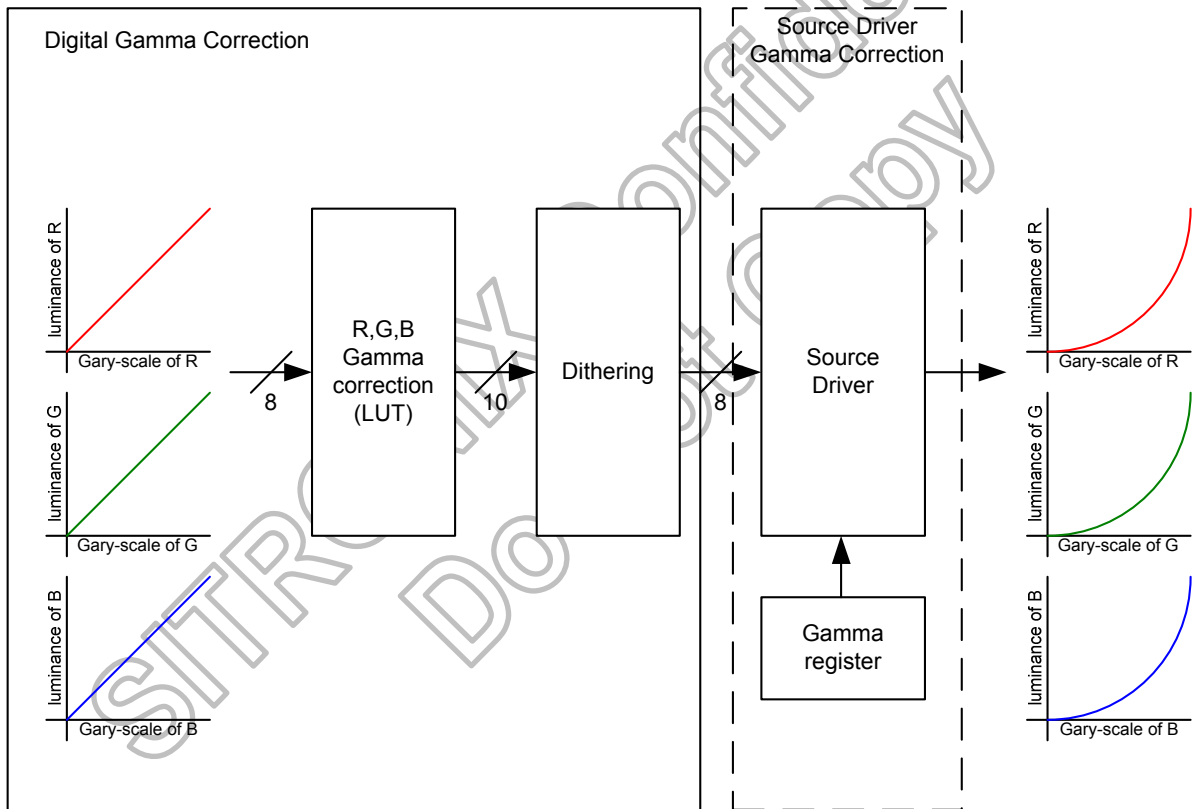


Figure 5.27: Source channels of ZigZag inversion mode

## 5.4 Gamma Function

The ST7703 offers two kinds of Gamma adjustment ways to come to accord with LC characteristic, one kind is through Source Driver directly, another one is adjusted by the digital gamma correction. The adjustment of digital gamma is selected by internal register DGC\_EN bit.

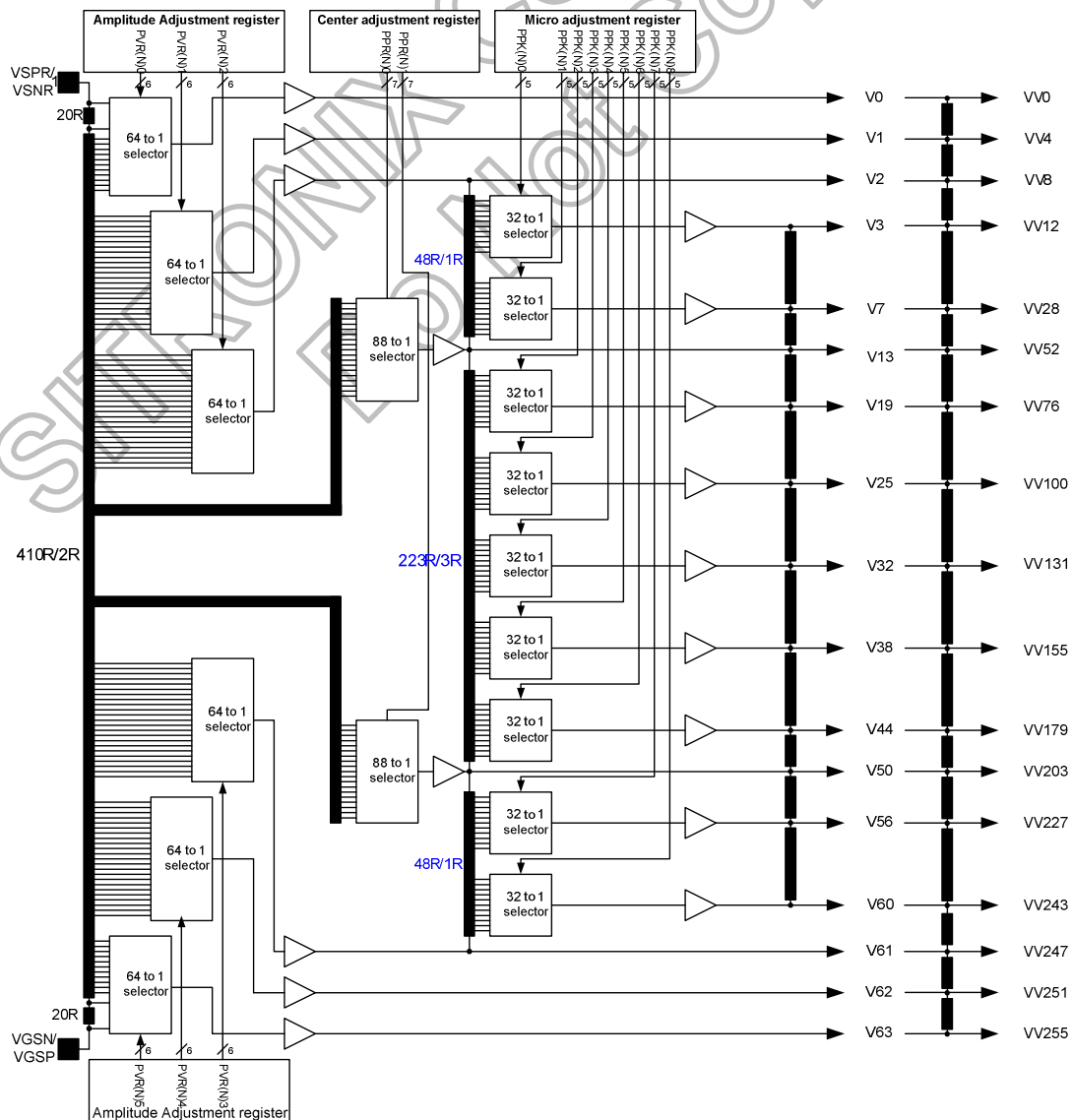


**Figure 5.28: Gamma adjustments different of source driver with digital gamma correction**

### 5.4.1 Gamma Characteristic Correction Function

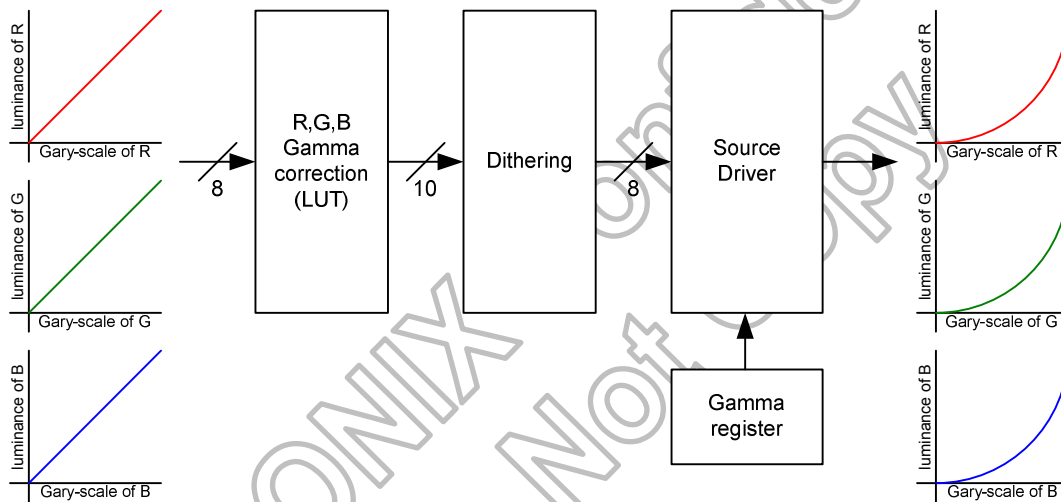
The ST7703 incorporates gamma adjustment function for the 16.7m-color display. Gamma adjustment operation is implemented by deciding the 17 grayscale levels firstly in gamma adjustment control registers to match the LCD panel. These registers are available both for positive polarities and negative polarities.

The block consists of two gamma resistor streams one is for positive polarity and the other is for negative polarity, each one including 17 gamma reference voltages. VgP/N (0, 4, 8, 12, 28, 52, 76, 100, 131, 155, 179, 203, 227, 273, 247, 251, 255).



#### 5.4.2 Gray Voltage Generator for Digital Gamma Correction

The ST7703 digital gamma correction can reach the independent GAMMA curve of RGB. The ST7703 utilizes DGC\_LUT (Digital Gamma Correction Look Up Table) to change input data from 8-bit into 10-bit and sends 10-bit data to Dithering circuit, and then drive Source Driver via Dithering circuit. The following of the block diagram of the function.



**Figure 5.29: Block diagram of digital gamma correction**



There are 99 bytes DGC LUT to set R, G, B gamma independently. When DGC\_EN=1, R, G, B gamma will mapping V0, V8, V16, ....., V240, V248, V255 voltage to the LUT register setting gray level voltage.

| CDh       | DGC_R           |                 |                 |                 |                 |                 |                 |                 |         |
|-----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|---------|
| Parameter | D7              | D6              | D5              | D4              | D3              | D2              | D1              | D0              | Default |
| 1st       | x               | x               | x               | x               | x               | x               | x               | DGC_EN          | 00      |
| 2nd       | DGC_LU<br>T_R00 | DGC_LU<br>T_R00 | DGC_LU<br>T_R00 | DGC_LU<br>T_R00 | DGC_LU<br>T_R00 | DGC_LU<br>T_R00 | DGC_LU<br>T_R00 | DGC_LU<br>T_R00 | 80      |
| 3rd       | DGC_LU<br>T_R01 | DGC_LU<br>T_R01 | DGC_LU<br>T_R01 | DGC_LU<br>T_R01 | DGC_LU<br>T_R01 | DGC_LU<br>T_R01 | DGC_LU<br>T_R01 | DGC_LU<br>T_R01 | 80      |
| ⋮         | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮       |
| 33rd      | DGC_LU<br>T_R31 | DGC_LU<br>T_R31 | DGC_LU<br>T_R31 | DGC_LU<br>T_R31 | DGC_LU<br>T_R31 | DGC_LU<br>T_R31 | DGC_LU<br>T_R31 | DGC_LU<br>T_R31 | 80      |
| 34th      | DGC_LU<br>T_R32 | DGC_LU<br>T_R32 | DGC_LU<br>T_R32 | DGC_LU<br>T_R32 | DGC_LU<br>T_R32 | DGC_LU<br>T_R32 | DGC_LU<br>T_R32 | DGC_LU<br>T_R32 | 80      |

| CEh       | DGC_G           |                 |                 |                 |                 |                 |                 |                 |         |
|-----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|---------|
| Parameter | D7              | D6              | D5              | D4              | D3              | D2              | D1              | D0              | Default |
| 1st       | DGC_LU<br>T_G00 | DGC_LU<br>T_G00 | DGC_LU<br>T_G00 | DGC_LU<br>T_G00 | DGC_LU<br>T_G00 | DGC_LU<br>T_G00 | DGC_LU<br>T_G00 | DGC_LU<br>T_G00 | 80      |
| 2nd       | DGC_LU<br>T_G01 | DGC_LU<br>T_G01 | DGC_LU<br>T_G01 | DGC_LU<br>T_G01 | DGC_LU<br>T_G01 | DGC_LU<br>T_G01 | DGC_LU<br>T_G01 | DGC_LU<br>T_G01 | 80      |
| ⋮         | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮       |
| 32rd      | DGC_LU<br>T_G31 | DGC_LU<br>T_G31 | DGC_LU<br>T_G31 | DGC_LU<br>T_G31 | DGC_LU<br>T_G31 | DGC_LU<br>T_G31 | DGC_LU<br>T_G31 | DGC_LU<br>T_G31 | 80      |
| 33th      | DGC_LU<br>T_G32 | DGC_LU<br>T_G32 | DGC_LU<br>T_G32 | DGC_LU<br>T_G32 | DGC_LU<br>T_G32 | DGC_LU<br>T_G32 | DGC_LU<br>T_G32 | DGC_LU<br>T_G32 | 80      |

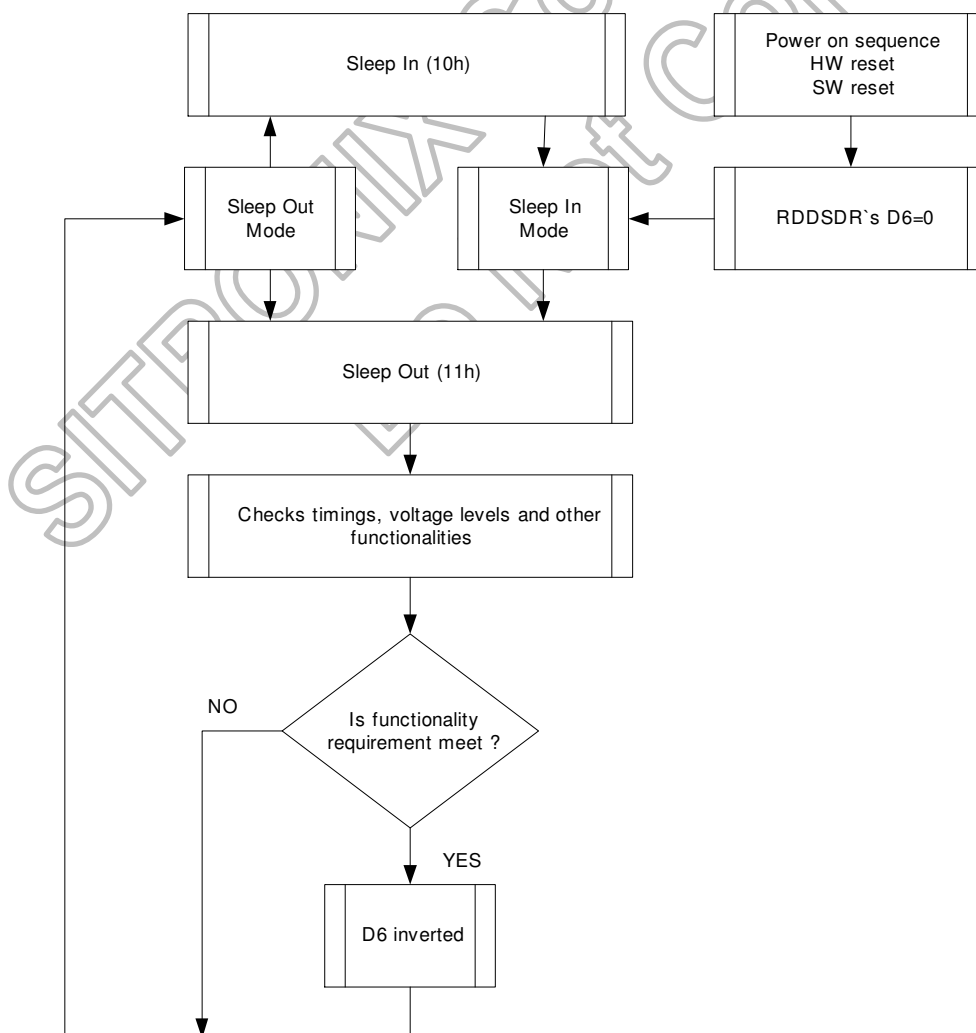
| CFh       | DGC_B           |                 |                 |                 |                 |                 |                 |                 |         |
|-----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|---------|
| Parameter | D7              | D6              | D5              | D4              | D3              | D2              | D1              | D0              | Default |
| 1st       | DGC_LU<br>T_B00 | DGC_LU<br>T_B00 | DGC_LU<br>T_B00 | DGC_LU<br>T_B00 | DGC_LU<br>T_B00 | DGC_LU<br>T_B00 | DGC_LU<br>T_B00 | DGC_LU<br>T_B00 | 80      |
| 2nd       | DGC_LU<br>T_B01 | DGC_LU<br>T_B01 | DGC_LU<br>T_B01 | DGC_LU<br>T_B01 | DGC_LU<br>T_B01 | DGC_LU<br>T_B01 | DGC_LU<br>T_B01 | DGC_LU<br>T_B01 | 80      |
| ⋮         | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮               | ⋮       |
| 32rd      | DGC_LU<br>T_B31 | DGC_LU<br>T_B31 | DGC_LU<br>T_B31 | DGC_LU<br>T_B31 | DGC_LU<br>T_B31 | DGC_LU<br>T_B31 | DGC_LU<br>T_B31 | DGC_LU<br>T_B31 | 80      |
| 33th      | DGC_LU<br>T_B32 | DGC_LU<br>T_B32 | DGC_LU<br>T_B32 | DGC_LU<br>T_B32 | DGC_LU<br>T_B32 | DGC_LU<br>T_B32 | DGC_LU<br>T_B32 | DGC_LU<br>T_B32 | 80      |



## 5.5.2 Functionality Detection

Sleep Out-command (11h) is a trigger for an internal function of the display module, which indicates, if the display module is still running and meets functionality requirements.

The internal function is comparing, if the display module still meets functionality requirements (e.g. booster voltage levels, timings, etc.). If functionality requirement is met, 1 bit will be inverted (=increased by 1), which is defined in command “Read Display Self- Diagnostic Result (0Fh)” (=RDDSDR) (The used bit of this command is D6). If functionality requirement is not the same, this bit (D6) is not inverted (=increased by 1). The flow chart for this internal function is shown as below.



**Figure 5.31: Sleep out flow chart internal function detection**

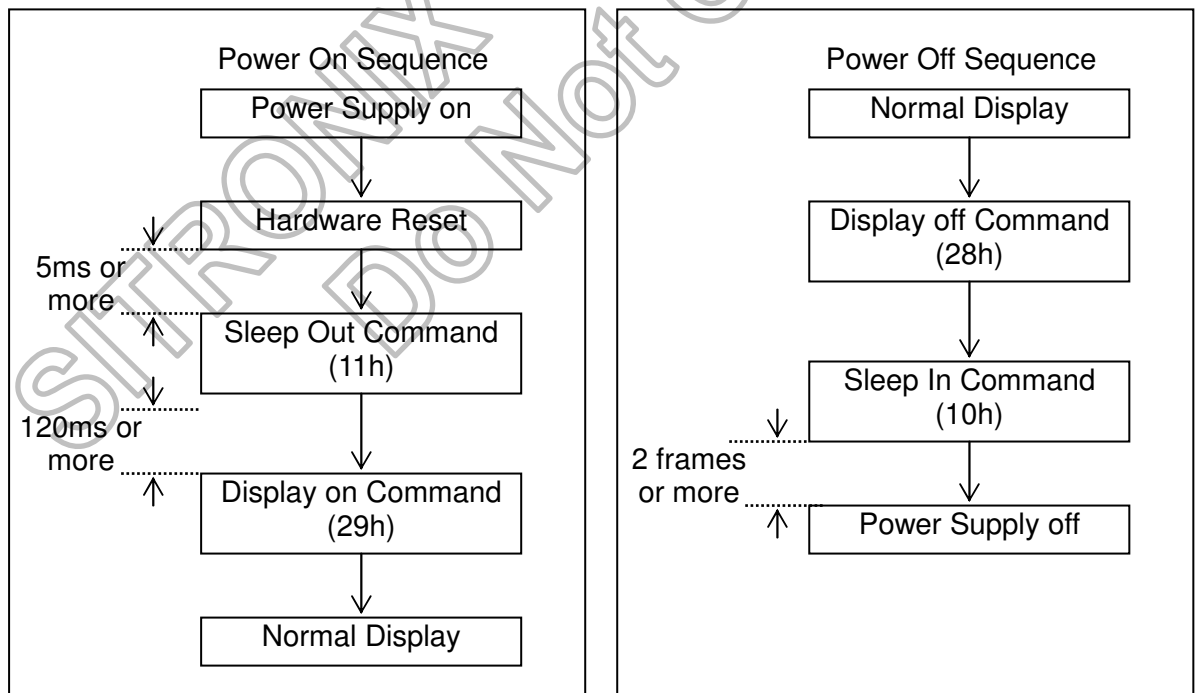
## 5.6 Power on/off Sequence

Power source IOVCC, VCI can be applied and powered down in any order. IOVCC, VCI can be powered down in any order.

During power off, if LCD is in the Sleep Out mode, IOVCC, VCI must be powered down minimum 120msec after NRESET has been released.

During power off, if LCD is in the Sleep In mode, IOVCC, VCI can be powered down minimum 0msec after NRESET has been released.

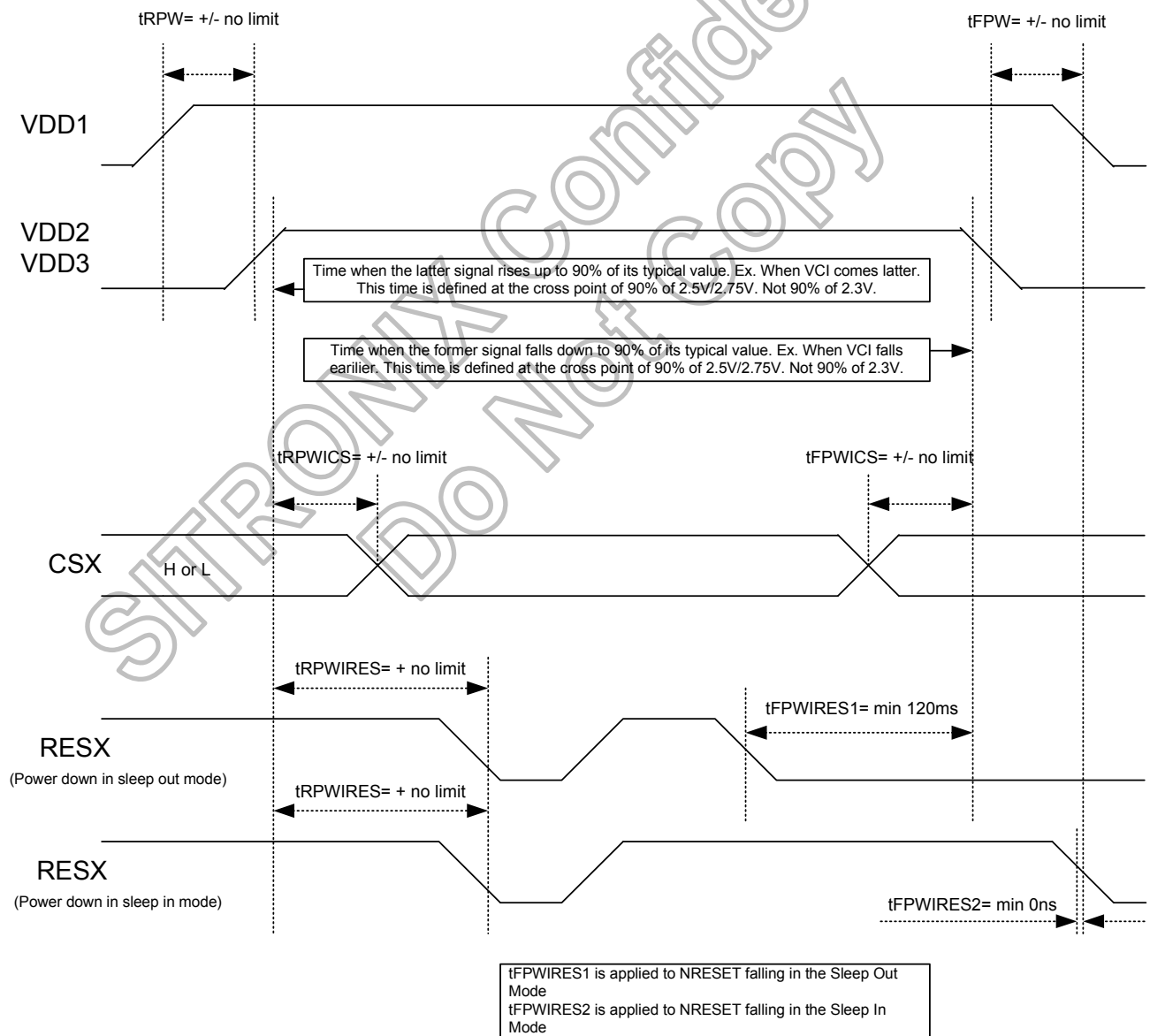
NCS can be applied at any timing or can be permanently grounded. NRESET has priority over NCS.



**Figure 5.32: The power supply ON/OFF setting for Display ON/OFF and Sleep In/out**

### 5.6.1 Case 1: RESX line is held high or unstable by host at power on

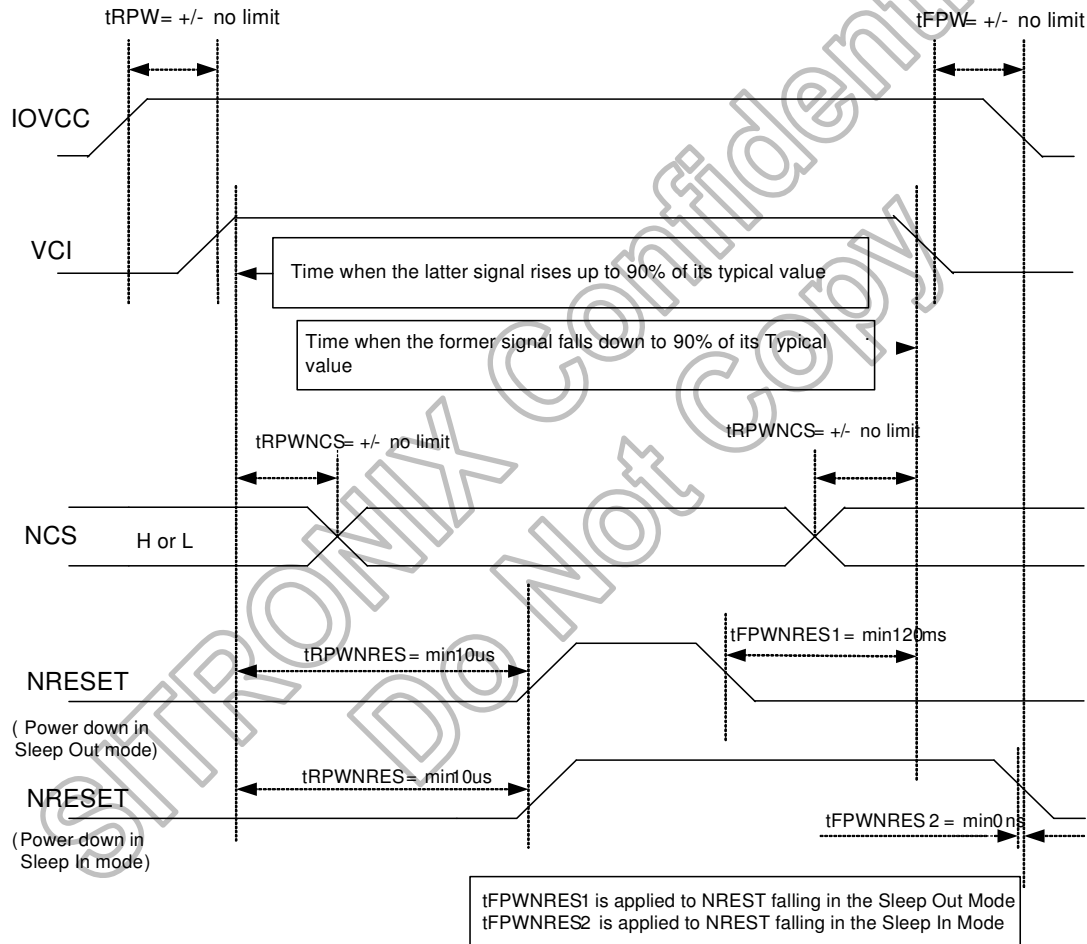
If RESX line is held high or unstable by the host during power on, then a Hardware Reset must be applied after both VDD1, VDD2 and VDD3 have been applied- otherwise correct functionality is not guaranteed. There is no timing restriction upon this hardware reset.



**Figure 5.33: Case 1: RESX line is held high or unstable by host at power on**

### 5.6.2 Case 2: RESX line is held low by host at power on

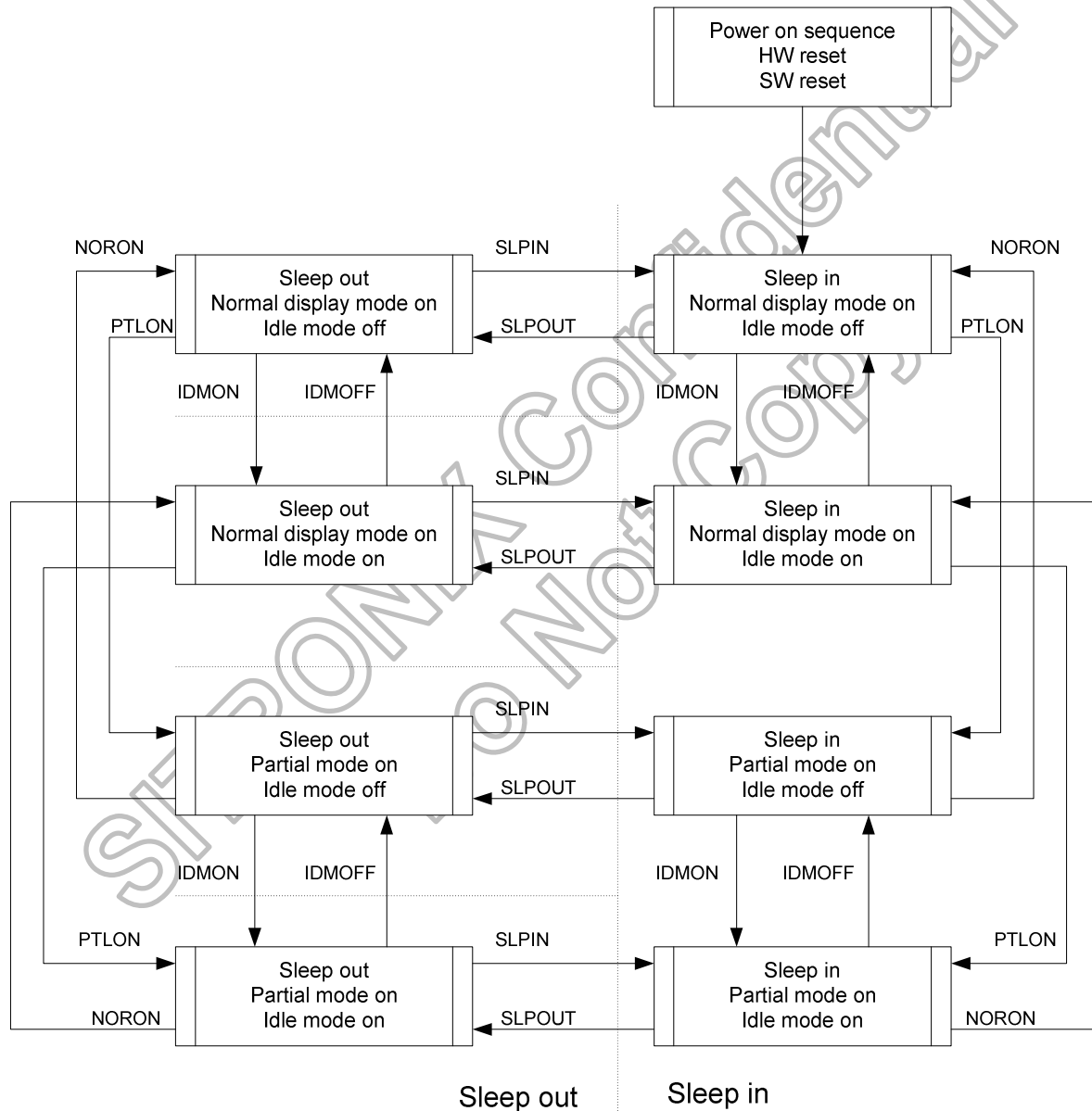
If RESX line is held low (and stable) by the host during power on, then the RESX must be held low for minimum 10 $\mu$ sec after both VDD1, VDD2 and VDD3 have been applied.



Note: Unless otherwise specified timings herein show cross point at 50% of signal/power level

**Figure 5.34: Case 2: RESX line is held low by host at power on**

## 5.7 Power Levels Definition



**Figure 5.35: Power flow chart for different power modes**

## 5.8 Reset function

### 5.8.1 Standard Command Default Value

| Item                                   | Register | After Power On | After Hardware Reset | After Software Reset |
|----------------------------------------|----------|----------------|----------------------|----------------------|
| RDDPM                                  | 0Ah      | 08h            | 08h                  | 08h                  |
| RDDMADCTR                              | 0Bh      | 00h            | 00h                  | No Change            |
| RDDCOLMOD                              | 0Ch      | 70h            | 70h                  | No Change            |
| RDDIM                                  | 0Dh      | 00h            | 00h                  | 00h                  |
| RDDSM                                  | 0Eh      | 00h            | 00h                  | 00h                  |
| RDDSDR                                 | 0Fh      | 00h            | 00h                  | 00h                  |
| Sleep In/Out                           | 10h/ 11h | In             | In                   | In                   |
| Display On/Off                         | 29h/ 28h | Off            | Off                  | Off                  |
| Display mode (normal/partial)          | 13h/ 12h | Normal         | Normal               | Normal               |
| Display Inversion On/Off               | 21h/ 20h | Off            | Off                  | Off                  |
| All pixel On/Off                       | 23/ 22h  | Off            | Off                  | Off                  |
| Display Idle Mode On/Off               | 39/ 38h  | Off            | Off                  | Off                  |
| Gamma setting                          | 26H      | GC0            | GC0                  | GC0                  |
| Memory Data Access Control (MY/MX/RGB) | 36H      | 0/0/0/0/0      | 0/0/0/0/0            | No Change            |
| Interface Pixel Color Format           | 3AH      | 70h            | 70h                  | No Change            |
| Display Brightness                     | 51h/ 52h | 00h            | 00h                  | 00h                  |
| CTRL Display                           | 53h/ 54h | 00h            | 00h                  | 00h                  |
| CABC Control                           | 55h/ 56h | 00h            | 00h                  | 00h                  |
| ID1                                    | DAH      | NO OTP         | 38h                  | 38h                  |
|                                        |          | OTPed          | OTP Value            | OTP Value            |
| ID2                                    | DBH      | NO OTP         | 21h                  | 21h                  |
|                                        |          | OTPed          | OTP Value            | OTP Value            |
| ID3                                    | DCH      | NO OTP         | 1Fh                  | 1Fh                  |
|                                        |          | OTPed          | OTP Value            | OTP Value            |

**Table 5.8: Standard Command Default Value**



### 5.8.2 Input Pins

| Input pins               | After Power On | After Hardware Reset | After Software Reset |
|--------------------------|----------------|----------------------|----------------------|
| RESX                     | Input valid    | Input valid          | Input valid          |
| HS_DSI_D0P<br>HS_DSI_D0P | Input valid    | Input valid          | Input valid          |
| HS_DSI_D1P<br>HS_DSI_D1P | Input valid    | Input valid          | Input valid          |
| HS_DSI_D2P<br>HS_DSI_D2P | Input valid    | Input valid          | Input valid          |
| HS_DSI_D3P<br>HS_DSI_D3N | Input valid    | Input valid          | Input valid          |
| HS_DSI_CP<br>HS_DSI_CN   | Input valid    | Input valid          | Input valid          |

Table 5.9: Characteristics of Input Pins

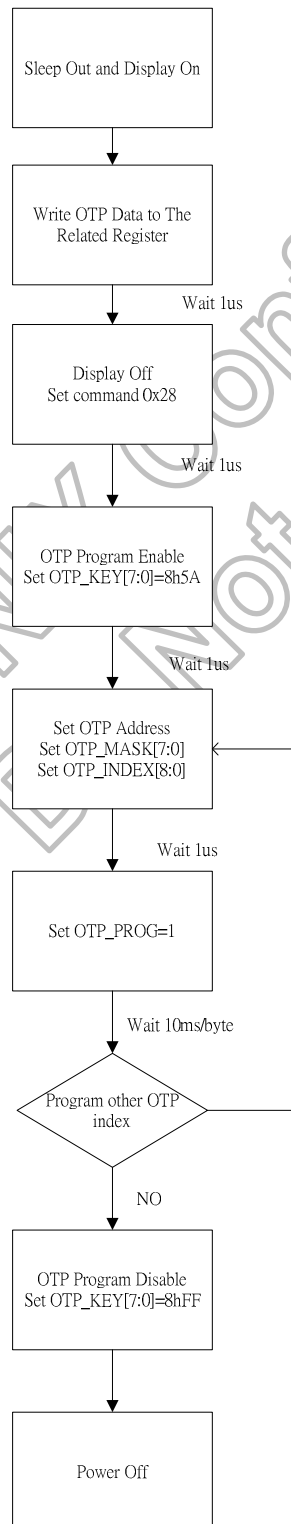
### 5.8.3 Output or Bi-directional Pins

| Output or Bi-directional pins | After Power On    | After Hardware Reset | After Software Reset |
|-------------------------------|-------------------|----------------------|----------------------|
| Source Output                 | VSSA              | VSSA                 | VSSA                 |
| DSI_D0P<br>DSI_D0P            | High-Z (Inactive) | High-Z (Inactive)    | High-Z (Inactive)    |
| CGOUTL1~22                    | VSSA              | VSSA                 | VSSA                 |
| CGOUTR1~22                    | VSSA              | VSSA                 | VSSA                 |

Table 5.10: Characteristics of Output or Bi-direction Pins

## 5.9 OTP Sequence

### 5.9.1 OTP Sequence



## 5.9.2 OTP Standard Programming Table

| Programming Index<br>(Hex) | D7           | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Storage Index<br>(Hex) |
|----------------------------|--------------|----|----|----|----|----|----|----|------------------------|
| 01                         | ID1_1[7:0]   |    |    |    |    |    |    |    | 01                     |
|                            | ID2_1[7:0]   |    |    |    |    |    |    |    | 02                     |
|                            | ID3_1[7:0]   |    |    |    |    |    |    |    | 03                     |
|                            | ID1_2[7:0]   |    |    |    |    |    |    |    | 04                     |
|                            | ID2_2[7:0]   |    |    |    |    |    |    |    | 05                     |
|                            | ID3_2[7:0]   |    |    |    |    |    |    |    | 06                     |
|                            | ID1_3[7:0]   |    |    |    |    |    |    |    | 07                     |
|                            | ID2_3[7:0]   |    |    |    |    |    |    |    | 08                     |
|                            | ID3_3[7:0]   |    |    |    |    |    |    |    | 09                     |
|                            | ID1_4[7:0]   |    |    |    |    |    |    |    | 0A                     |
|                            | ID2_4[7:0]   |    |    |    |    |    |    |    | 0B                     |
|                            | ID3_4[7:0]   |    |    |    |    |    |    |    | 0C                     |
| 0D                         | VCOM_F1[7:0] |    |    |    |    |    |    |    | 0D                     |
|                            | VCOM_B1[7:0] |    |    |    |    |    |    |    | 0E                     |
|                            | VCOM_F2[7:0] |    |    |    |    |    |    |    | 0F                     |
|                            | VCOM_B2[7:0] |    |    |    |    |    |    |    | 10                     |
|                            | VCOM_F3[7:0] |    |    |    |    |    |    |    | 11                     |
|                            | VCOM_B3[7:0] |    |    |    |    |    |    |    | 12                     |
|                            | VCOM_F4[7:0] |    |    |    |    |    |    |    | 13                     |
|                            | VCOM_B4[7:0] |    |    |    |    |    |    |    | 14                     |

## 5.10 CABC

This driver IC provides a dynamic backlight control function as CABC (Content adaptive brightness control) to reduce the power consumption of the luminance source. There are two module architectures for CABC operation as below figure.

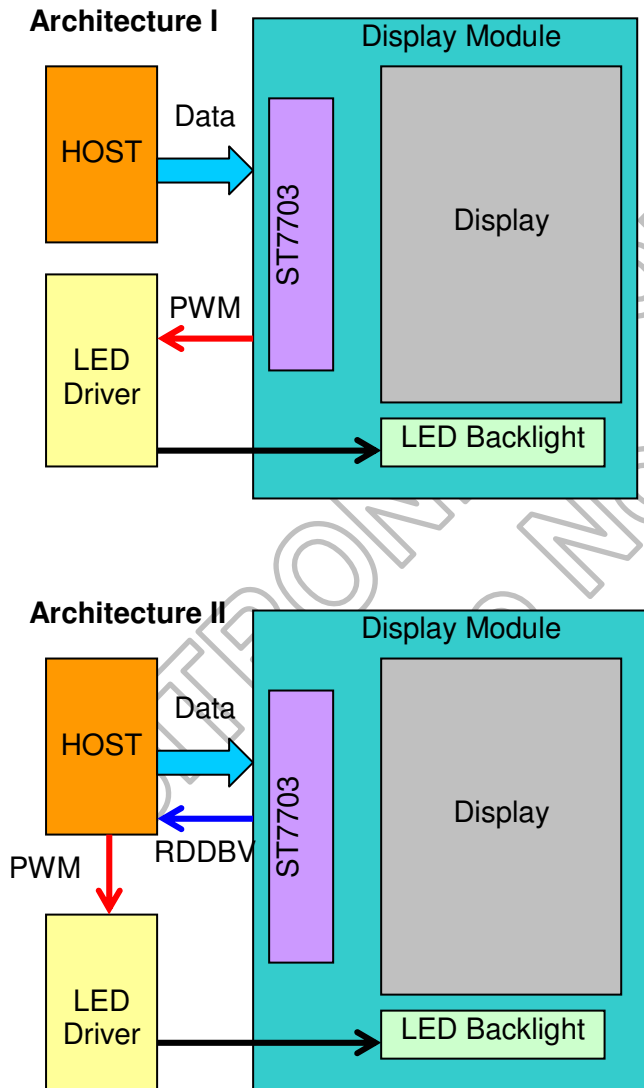
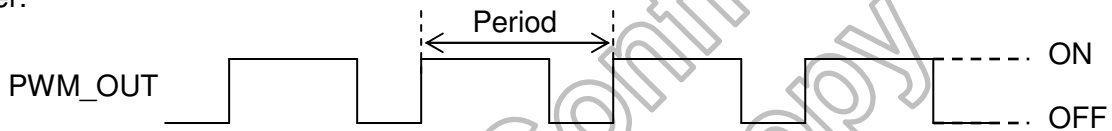


Figure 5.36: Two kinds of CABC Architecture

The driver IC will refer the gray scale content of display image to output a PWM waveform to LED driver for backlight brightness control. The PWM frequency can be adjusted by PWM\_DIV parameters and the calculating equation as below:

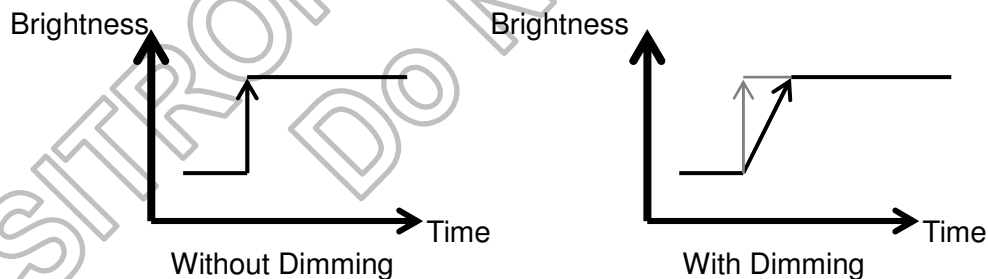
$$F_{pwm\_out} = \frac{36MHz / (2^{PWM\_DIV[2:0]})}{(PWM\_PERIOD[7:0] + 1) \times 255}$$

The figure in the following is the basic timing diagram which is applied ST7703 to control LED driver.



### Display Backlight Dimming Control

A dimming function is used when changing from brightness level to another. This dimming function curve is the same in increment and decrement directions. Dimming function can be enabled and disabled by software register. The basic idea is described below.



## 6. Command Description

### 6.1 Command Table

#### Standard Command

| (Hex) | Operation code | R/W | D7       | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Function                            | Default (Hex) |
|-------|----------------|-----|----------|----|----|----|----|----|----|----|-------------------------------------|---------------|
| 00h   | NOP            | W   | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | No Operation                        | -             |
| 01h   | SWRESET        | W   | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 1  | Software Reset                      | -             |
|       |                |     | 0        | 0  | 0  | 0  | 0  | 1  | 0  | 0  | Read Display ID                     | -             |
| 04h   | RDDIDIF        | R   | ID1[7:0] |    |    |    |    |    |    |    | ID1                                 | 38h           |
|       |                |     | ID2[7:0] |    |    |    |    |    |    |    | ID2                                 | 21h           |
|       |                |     | ID3[7:0] |    |    |    |    |    |    |    | ID3                                 | 1Fh           |
| 0Ah   | RDDPM          | R   | 0        | 0  | 0  | 0  | 1  | 0  | 1  | 0  | Read display power mode             | -             |
|       |                |     | D7       | D6 | D5 | D4 | D3 | D2 | 0  | 0  | -                                   | 08h           |
| 0Bh   | RDDMADCTL      | R   | 0        | 0  | 0  | 0  | 1  | 0  | 1  | 1  | Read display MADCTL                 | -             |
|       |                |     | D7       | D6 | D5 | D4 | D3 | D2 | 0  | 0  | -                                   | 00h           |
| 0Ch   | RDDCOLMOD      | R   | 0        | 0  | 0  | 0  | 1  | 1  | 0  | 0  | Read display pixel format           | -             |
|       |                |     | -        | D6 | D5 | D4 | -  | -  | -  | -  | -                                   | 70h           |
| 0Dh   | RDDIM          | R   | 0        | 0  | 0  | 0  | 1  | 1  | 0  | 1  | Read display image mode             | -             |
|       |                |     | D7       | D6 | D5 | 0  | 0  | D2 | D1 | D0 | -                                   | 00h           |
| 0Eh   | RDDSM          | R   | 0        | 0  | 0  | 0  | 1  | 1  | 1  | 0  | Read display signal mode            | -             |
|       |                |     | D7       | D6 | 0  | 0  | 0  | 0  | 0  | D0 | -                                   | 00h           |
| 0Fh   | RDDSDR         | R   | 0        | 0  | 0  | 0  | 1  | 1  | 1  | 1  | Read display self-diagnostic result | -             |
|       |                |     | D7       | D6 | 0  | 0  | 0  | 0  | 0  | D0 | -                                   | 00h           |
| 10h   | SLPIN          | W   | 0        | 0  | 0  | 1  | 0  | 0  | 0  | 0  | Sleep In                            | -             |
| 11h   | SLPOUT         | W   | 0        | 0  | 0  | 1  | 0  | 0  | 0  | 1  | Sleep Out                           | -             |

| (Hex) | Operation code | R/W | D7   | D6   | D5    | D4   | D3   | D2   | D1   | D0   | Function                        | Default (Hex) |
|-------|----------------|-----|------|------|-------|------|------|------|------|------|---------------------------------|---------------|
| 13h   | NORON          | W   | 0    | 0    | 0     | 1    | 0    | 0    | 1    | 1    | Normal Display mode             | -             |
| 20h   | INVOFF         | W   | 0    | 0    | 1     | 0    | 0    | 0    | 0    | 0    | Display inversion off           | -             |
| 21h   | INVON          | W   | 0    | 0    | 1     | 0    | 0    | 0    | 0    | 1    | Display inversion on            | -             |
| 22h   | ALLPOFF        | W   | 0    | 0    | 1     | 0    | 0    | 0    | 1    | 0    | All pixel off                   | -             |
| 23h   | ALLPON         | W   | 0    | 0    | 1     | 0    | 0    | 0    | 1    | 1    | All pixel on                    | -             |
| 28h   | DISPOFF        | W   | 0    | 0    | 1     | 0    | 1    | 0    | 0    | 0    | Display off                     | -             |
| 29h   | DISPON         | W   | 0    | 0    | 1     | 0    | 1    | 0    | 0    | 1    | Display on                      | -             |
| 36h   | MADCTL         | W   | 0    | 0    | 1     | 1    | 0    | 1    | 1    | 0    | Memory Access Control           | -             |
|       |                |     | MY   | MX   | x     | x    | RGB  | x    | x    | x    |                                 | 00h           |
| 38h   | IDMOFF         | W   | 0    | 0    | 1     | 1    | 1    | 0    | 0    | 0    | Idle mode off                   | -             |
| 39h   | IDMON          | W   | 0    | 0    | 1     | 1    | 1    | 0    | 0    | 1    | Idle mode on                    | -             |
| 51h   | WRDISBV        | W   | 0    | 1    | 0     | 1    | 0    | 0    | 0    | 1    | Write display brightness value  | -             |
|       |                |     | DBV7 | DBV6 | DBV5  | DBV4 | DBV3 | DBV2 | DBV1 | DBV0 |                                 | 00h           |
| 52h   | RDRDISBV       | R   | 0    | 1    | 0     | 1    | 0    | 0    | 1    | 0    | Read display brightness value   | -             |
|       |                |     | DBV7 | DBV6 | DBV5  | DBV4 | DBV3 | DBV2 | DBV1 | DBV0 |                                 | 00h           |
| 53h   | WRCTRLD        | W   | 0    | 1    | 0     | 1    | 0    | 0    | 1    | 1    | Write CTRL display              | -             |
|       |                |     | x    | x    | BCTRL | x    | x    | DD   | BL   | x    |                                 | 00h           |
| 54h   | RDCTRLD        | R   | 0    | 1    | 0     | 1    | 0    | 1    | 0    | 0    | Read CTRL display               | -             |
|       |                |     | x    | x    | BCTRL | x    | x    | DD   | BL   | x    |                                 | 00h           |
| 55h   | WRCABC         | W   | 0    | 1    | 0     | 1    | 0    | 1    | 0    | 1    | Write CABC mode                 | -             |
|       |                |     | x    | x    | x     | x    | x    | x    | C1   | C0   |                                 | 00h           |
| 56h   | RDCABC         | R   | 0    | 1    | 0     | 1    | 0    | 1    | 1    | 0    | Read CABC mode                  | -             |
|       |                |     | x    | x    | x     | x    | x    | x    | C1   | C0   |                                 | 00h           |
| 68h   | RDABCSDR       | R   | 0    | 1    | 0     | 1    | 0    | 1    | 1    | 0    | Read ABC Self-diagnostic result | -             |
|       |                |     | D7   | D6   | x     | x    | x    | x    | x    | x    |                                 | 00h           |
| 70    | RDBWLB         | R   | 0    | 1    | 1     | 1    | 0    | 0    | 0    | 0    | Read Black/White Low Bits       | -             |
|       |                |     | Bkx1 | Bkx0 | Bky1  | Bky0 | Wx1  | Wx0  | Wy1  | Wy0  |                                 | -             |
| 71    | RDBkx          | R   | 0    | 1    | 1     | 1    | 0    | 0    | 0    | 1    | Read Bkx                        | -             |
|       |                |     | Bkx9 | Bkx8 | Bkx7  | Bkx6 | Bkx5 | Bkx4 | Bkx3 | Bkx2 |                                 | -             |
| 72    | RDBky          | R   | 0    | 1    | 1     | 1    | 0    | 0    | 1    | 0    | Read Bky                        | -             |
|       |                |     | Bky9 | Bky8 | Bky7  | Bky6 | Bky5 | Bky4 | Bky3 | Bky2 |                                 | -             |
| 73    | RDWx           | R   | 0    | 1    | 1     | 1    | 0    | 0    | 1    | 1    | Read Wx                         | -             |
|       |                |     | Wx9  | Wx8  | Wx7   | Wx6  | Wx5  | Wx4  | Wx3  | Wx2  |                                 | -             |
| 74    | RDWy           | R   | 0    | 1    | 1     | 1    | 0    | 1    | 0    | 0    | Read Wy                         | -             |
|       |                |     | Wy9  | Wy8  | Wy7   | Wy6  | Wy5  | Wy4  | Wy3  | Wy2  |                                 | -             |
| 75    | RDRGLB         | R   | 0    | 1    | 1     | 1    | 0    | 1    | 0    | 1    | Read Red/Green Low Bits         | -             |
|       |                |     | Rx1  | Rx0  | Ry1   | Ry0  | Gx1  | Gx0  | Gy1  | Gy0  |                                 | -             |
| 76    | RDRx           | R   | 0    | 1    | 1     | 1    | 0    | 1    | 1    | 0    | Read Rx                         | -             |
|       |                |     | Rx9  | Rx8  | Rx7   | Rx6  | Rx5  | Rx4  | Rx3  | Rx2  |                                 | -             |

| (Hex) | Operation code    | R/W | D7                              | D6  | D5  | D4  | D3  | D2  | D1  | D0  | Function                                              | Default (Hex) |
|-------|-------------------|-----|---------------------------------|-----|-----|-----|-----|-----|-----|-----|-------------------------------------------------------|---------------|
| 77    | RDRy              | R   | 0                               | 1   | 1   | 1   | 0   | 1   | 1   | 1   | Read Ry                                               | -             |
|       |                   |     | Ry9                             | Ry8 | Ry7 | Ry6 | Ry5 | Ry4 | Ry3 | Ry2 |                                                       | -             |
| 78    | RDGx              | R   | 0                               | 1   | 1   | 1   | 1   | 0   | 0   | 0   | Read Gx                                               | -             |
|       |                   |     | Gx9                             | Gx8 | Gx7 | Gx6 | Gx5 | Gx4 | Gx3 | Gx2 |                                                       | -             |
| 79    | RDRy              | R   | 0                               | 1   | 1   | 1   | 1   | 0   | 0   | 1   | Read Gy                                               | -             |
|       |                   |     | Gy9                             | Gy8 | Gy7 | Gy6 | Gy5 | Gy4 | Gy3 | Gy2 |                                                       | -             |
| 7A    | RDBALB            | R   | 0                               | 1   | 1   | 1   | 1   | 0   | 1   | 0   | Read Biue/AColour Low Bits                            | -             |
|       |                   |     | Bx1                             | Bx0 | By1 | Bx0 | Ax1 | Ax0 | Ay1 | Ay0 |                                                       | -             |
| 7B    | RDBx              | R   | 0                               | 1   | 1   | 1   | 1   | 0   | 1   | 1   | Read Bx                                               | -             |
|       |                   |     | Bx9                             | Bx8 | Bx7 | Bx6 | Bx5 | Bx4 | Bx3 | Bx2 |                                                       | -             |
| 7C    | RDBy              | R   | 0                               | 1   | 1   | 1   | 1   | 1   | 0   | 0   | Read By                                               | -             |
|       |                   |     | By9                             | By8 | By7 | By6 | By5 | By4 | By3 | By2 |                                                       | -             |
| 7D    | RDAx              | R   | 0                               | 1   | 1   | 1   | 1   | 1   | 0   | 1   | Read Ax                                               | -             |
|       |                   |     | Ax9                             | Ax8 | Ax7 | Ax6 | Ax5 | Ax4 | Ax3 | Ax2 |                                                       | -             |
| 7E    | RDAy              | R   | 0                               | 1   | 1   | 1   | 1   | 1   | 1   | 0   | Read Ay                                               | -             |
|       |                   |     | Ay9                             | Ay8 | Ay7 | Ay6 | Ay5 | Ay4 | Ay3 | Ay2 |                                                       | -             |
| A1    | Read_DDB_start    | R   | 1                               | 0   | 1   | 0   | 0   | 0   | 0   | 1   | Read the DDB from the provided location.              | -             |
|       |                   |     | x                               | x   | x   | x   | x   | x   | x   | x   |                                                       | -             |
|       |                   |     | x                               | x   | x   | x   | x   | x   | x   | x   |                                                       | 00h           |
|       |                   |     | x                               | x   | x   | x   | x   | x   | x   | x   |                                                       | 00h           |
|       |                   |     | x                               | x   | x   | x   | x   | x   | x   | x   |                                                       | 00h           |
|       |                   |     | x                               | x   | x   | x   | x   | x   | x   | x   |                                                       | 00h           |
|       |                   |     | 1                               | 1   | 1   | 1   | 1   | 1   | 1   | 1   |                                                       | FFh           |
| A8    | Read_DDB_continue | R   | 1                               | 0   | 1   | 0   | 1   | 0   | 0   | 0   | Continue reading the DDB from the last read location. | -             |
|       |                   |     | x                               | x   | x   | x   | x   | x   | x   | x   |                                                       | 00h           |
|       |                   |     | x                               | x   | x   | x   | x   | x   | x   | x   |                                                       | 00h           |
|       |                   |     | x                               | x   | x   | x   | x   | x   | x   | x   |                                                       | 00h           |
|       |                   |     | x                               | x   | x   | x   | x   | x   | x   | x   |                                                       | 00h           |
| DA    | RDID1             | R   | 1                               | 1   | 0   | 1   | 1   | 0   | 1   | 0   | Read ID1                                              | -             |
|       |                   |     | module's manufacturer[7:0]      |     |     |     |     |     |     |     |                                                       | 38h           |
| DB    | RDID2             | R   | 1                               | 1   | 0   | 1   | 1   | 0   | 1   | 1   | Read ID2                                              | -             |
|       |                   |     | LCD module/driver version [7:0] |     |     |     |     |     |     |     |                                                       | 21h           |
| DC    | RDID3             | R   | 1                               | 1   | 0   | 1   | 1   | 1   | 0   | 0   | Read ID3                                              | -             |
|       |                   |     | LCD module/driver ID[7:0]       |     |     |     |     |     |     |     |                                                       | 1Fh           |



**User Command**

| (Hex) | Operation code | Para. | D7               | D6                   | D5                  | D4                   | D3               | D2               | D1               | D0               | Default (Hex) |
|-------|----------------|-------|------------------|----------------------|---------------------|----------------------|------------------|------------------|------------------|------------------|---------------|
| B9h   | SETEXTC        |       | 1                | 0                    | 1                   | 1                    | 1                | 0                | 0                | 1                | -             |
|       |                | 1     | EXTC1[7]         | EXTC1[6]             | EXTC1[5]            | EXTC1[4]             | EXTC1[3]         | EXTC1[2]         | EXTC1[1]         | EXTC1[0]         | 00            |
|       |                | 2     | EXTC2[7]         | EXTC2[6]             | EXTC2[5]            | EXTC2[4]             | EXTC2[3]         | EXTC2[2]         | EXTC2[1]         | EXTC2[0]         | 00            |
|       |                | 3     | EXTC3[7]         | EXTC3[6]             | EXTC3[5]            | EXTC3[4]             | EXTC3[3]         | EXTC3[2]         | EXTC3[1]         | EXTC3[0]         | 00            |
| B2h   | SETDISP        |       | 1                | 0                    | 1                   | 1                    | 0                | 0                | 1                | 0                | -             |
|       |                | 1     | NL[7]            | NL[6]                | NL[5]               | NL[4]                | NL[3]            | NL[2]            | NL[1]            | NL[0]            | C8            |
|       |                | 2     | RES_V_LSB[1]     | RES_V_L_SB[0]        | BLK_CON[1]          | BLK_CON[0]           | x                | RESO_SE_L[2]     | RESO_SE_L[1]     | RESO_SE_L[0]     | 12            |
|       |                | 3     | WHITE_GND_EN     | WHITE_F_RAME_S_EL[2] | WHITE_F_RAME_S_L[1] | WHITE_F_RAME_S_EL[0] | x                | ISC[2]           | ISC[1]           | ISC[0]           | A0            |
| B3h   | SETRGBIF       |       | 1                | 0                    | 1                   | 1                    | 0                | 0                | 1                | 1                | -             |
|       |                | 1     | VBP_RGB_GEN[7]   | VBP_RGB_GEN[6]       | VBP_RGB_GEN[5]      | VBP_RGB_GEN[4]       | VBP_RGB_GEN[3]   | VBP_RGB_GEN[2]   | VBP_RGB_GEN[1]   | VBP_RGB_GEN[0]   | 28            |
|       |                | 2     | VFP_RGB_GEN[7]   | VFP_RGB_GEN[6]       | VFP_RGB_GEN[5]      | VFP_RGB_GEN[4]       | VFP_RGB_GEN[3]   | VFP_RGB_GEN[2]   | VFP_RGB_GEN[1]   | VFP_RGB_GEN[0]   | 28            |
|       |                | 3     | DE_BP_RGB_GEN[7] | DE_BP_RGB_GEN[6]     | DE_BP_RGB_GEN[5]    | DE_BP_RGB_GEN[4]     | DE_BP_RGB_GEN[3] | DE_BP_RGB_GEN[2] | DE_BP_RGB_GEN[1] | DE_BP_RGB_GEN[0] | 28            |
|       |                | 4     | DE_FP_RGB_GEN[7] | DE_FP_RGB_GEN[6]     | DE_FP_RGB_GEN[5]    | DE_FP_RGB_GEN[4]     | DE_FP_RGB_GEN[3] | DE_FP_RGB_GEN[2] | DE_FP_RGB_GEN[1] | DE_FP_RGB_GEN[0] | 28            |
| B4h   | SETCYC         |       | 1                | 0                    | 1                   | 1                    | 0                | 1                | 0                | 0                | -             |
|       |                | 1     | ZINV_S2_401_EN   | ZINV_G_EVEN_EN       | ZINV_EN             | ZINV2_EN             | x                | N_NW[2]          | N_NW[1]          | N_NW[0]          | 80            |
|       |                | 2     | x                | x                    | x                   | x                    | x                | I_NW[2]          | I_NW[1]          | I_NW[0]          | 00            |

| (Hex) | Operation code | Para. | D7             | D6             | D5             | D4            | D3             | D2                | D1                | D0                | Default (Hex) |
|-------|----------------|-------|----------------|----------------|----------------|---------------|----------------|-------------------|-------------------|-------------------|---------------|
| B5h   | SETBGP         |       | 1              | 0              | 1              | 1             | 0              | 1                 | 0                 | 1                 |               |
|       |                | 1     | x              | x              | x              | VREF_SE L[4]  | VREF_SE L[3]   | VREF_SE L[2]      | VREF_SE L[1]      | VREF_SE L[0]      | 09            |
| B6h   | SETCOM         | 2     | x              | x              | x              | NVREF_S EL[4] | NVREF_S EL[3]  | NVREF_S EL[2]     | NVREF_S EL[1]     | NVREF_S EL[0]     | 09            |
|       |                | 1     | VCOMD C_F[7]   | VCOMDC F[6]    | VCOMDC F[5]    | VCOMDC F[4]   | VCOMDC F[3]    | VCOMDC F[2]       | VCOMDC F[1]       | VCOMDC F[0]       | 4D            |
|       |                | 2     | VCOMD C_B[7]   | VCOMDC B[6]    | VCOMDC B[5]    | VCOMDC B[4]   | VCOMDC B[3]    | VCOMDC B[2]       | VCOMDC B[1]       | VCOMDC B[0]       | 4D            |
| B7h   | SETOTP         | 3     | x              | x              | x              | x             | x              | VCOM_O TP_TIME[2] | VCOM_O TP_TIME[1] | VCOM_O TP_TIME[0] | 00            |
|       |                |       | 1              | 0              | 1              | 1             | 0              | 1                 | 1                 | 1                 | -             |
|       |                | 1     | OTP_KEY Y[7]   | OTP_KEY [6]    | OTP_KEY [5]    | OTP_KEY [4]   | OTP_KEY [3]    | OTP_KEY [2]       | OTP_KEY [1]       | OTP_KEY [0]       | FF            |
|       |                | 2     | OTP_MA SK[7]   | OTP_MA SK[6]   | OTP_MA SK[5]   | OTP_MA SK[4]  | OTP_MA SK[3]   | OTP_MA SK[2]      | OTP_MA SK[1]      | OTP_MA SK[0]      | 00            |
|       |                | 3     | x              | VPP_SEL[2]     | VPP_SEL[1]     | VPP_SEL[0]    | x              | x                 | x                 | OTP_IND EX[8]     | 30            |
|       |                | 4     | OTP_IND EX[7]  | OTP_IND EX[6]  | OTP_IND EX[5]  | OTP_IND EX[4] | OTP_IND EX[3]  | OTP_IND EX[2]     | OTP_IND EX[1]     | OTP_IND EX[0]     | 00            |
| B8h   | SET POWER_EXT  | 5     | Load_DI S      | VPP_EN         | OTP_SEL        | OTP_PW E      | OTP_PTM [1]    | OTP_PTM [0]       | OTP_PO R          | OTP_PO RG         | 00            |
|       |                | 6     | OTP_DA TA[7]   | OTP_DAT A[6]   | OTP_DAT A[5]   | OTP_DAT A[4]  | OTP_DAT A[3]   | OTP_DAT A[2]      | OTP_DAT A[1]      | OTP_DAT A[0]      | 00            |
|       |                | 1     | x              | PCCS[2]        | PCCS[1]        | PCCS[0]       | ECP_DC_DIV[3]  | ECP_DC_DIV[2]     | ECP_DC_DIV[1]     | ECP_DC_DIV[0]     | 34            |
|       |                | 2     | x              | SINGLE_PUMP    | DT[1]          | DT[0]         | SYNC_ST B      | XDK_ECP [1]       | XDK_ECP [0]       | x                 | 22            |
|       |                | 3     | x              | x              | x              | x             | x              | PFM_DC_DIV[2]     | PFM_DC_DIV[1]     | PFM_DC_DIV[0]     | 00            |
|       |                | 4     | x              | x              | x              | x             | x              | x                 | ECP_SYN C_EN      | VGX_SYN C_EN      | 03            |
| BAh   | SETMIPI        | 5     | A_DC[1]        | A_DC[0]        | A_DTP[2]       | A_DTP[1]      | A_DTP[0]       | A_DTN[2]          | A_DTN[1]          | A_DTN[0]          | ED            |
|       |                | 6     | DCS[1]         | DCS[0]         | DTPS[2]        | DTPS[1]       | DTPS[0]        | DTNS[2]           | DTNS[1]           | DTNS[0]           | 00            |
|       |                | 1     | x              | x              | x              | x             | VC_Main[1]     | VC_Main[0]        | Lane_Nu mber[1]   | Lane_Nu mber[0]   | 33            |
|       |                | 2     | DSI_LDO_SEL[2] | DSI_LDO_SEL[1] | DSI_LDO_SEL[0] | x             | x              | x                 | RTERM[1]          | RTERM[0]          | 61            |
|       |                | 3     | x              | x              | x              | x             | IHSRX[3]       | IHSRX[2]          | IHSRX[1]          | IHSRX[0]          | 06            |
|       |                | 4     | x              | x              | x              | x             | Tx_clk_sel [1] | Tx_clk_sel [0]    | x                 | x                 | F9            |
|       |                | 5     | HFP_OS C[7]    | HFP_OSC [6]    | HFP_OSC [5]    | HFP_OSC [4]   | HFP_OSC [3]    | HFP_OSC [2]       | HFP_OSC [1]       | HFP_OSC [0]       | FF            |
|       |                | 6     | HBP_OS C[7]    | HBP_OSC [6]    | HBP_OSC [5]    | HBP_OSC [4]   | HBP_OSC [3]    | HBP_OSC [2]       | HBP_OSC [1]       | HBP_OSC [0]       | 0A            |

| (Hex) | Operation code | Para. | D7             | D6             | D5             | D4             | D3             | D2             | D1             | D0             | Default (Hex) |
|-------|----------------|-------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|
| BCh   | SET VDC        |       | 1              | 0              | 1              | 1              | 1              | 1              | 0              | 0              |               |
|       |                | 1     | x              | NVDDD_SEL[2]   | NVDDD_SEL[1]   | NVDDD_SEL[0]   | x              | VDDD_SE L[2]   | VDDD_SE L[1]   | VDDD_SE L[0]   | 47            |
| C0h   | SETSCR         |       | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | -             |
|       |                | 1     | N_POPO N[7]    | N_POPO N[6]    | N_POPO N[5]    | N_POPO N[4]    | N_POPO N[3]    | N_POPO N[2]    | N_POPO N[1]    | N_POPO N[0]    | 73            |
|       |                | 2     | N_NOPO N[7]    | N_NOPO N[6]    | N_NOPO N[5]    | N_NOPO N[4]    | N_NOPO N[3]    | N_NOPO N[2]    | N_NOPO N[1]    | N_NOPO N[0]    | 73            |
|       |                | 3     | I_POPON[7]     | I_POPON[6]     | I_POPON[5]     | I_POPON[4]     | I_POPON[3]     | I_POPON[2]     | I_POPON[1]     | I_POPON[0]     | 50            |
|       |                | 4     | I_NOPON[7]     | I_NOPON[6]     | I_NOPON[5]     | I_NOPON[4]     | I_NOPON[3]     | I_NOPON[2]     | I_NOPON[1]     | I_NOPON[0]     | 50            |
|       |                | 5     | SCR[31]        | SCR[30]        | SCR[29]        | SCR[28]        | SCR[27]        | SCR[26]        | SCR[25]        | SCR[24]        | C0            |
|       |                | 6     | SCR[23]        | SCR[22]        | SCR[21]        | SCR[20]        | SCR[19]        | SCR[18]        | SCR[17]        | SCR[16]        | 00            |
|       |                | 7     | SCR[15]        | SCR[14]        | SCR[13]        | SCR[12]        | SCR[11]        | SCR[10]        | SCR[9]         | SCR[8]         | 08            |
|       |                | 8     | SCR[7]         | SCR[6]         | SCR[5]         | SCR[4]         | SCR[3]         | SCR[2]         | SCR[1]         | SCR[0]         | 70            |
| C1h   | SETPower       |       | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 1              | -             |
|       |                | 1     | VBTHS[3]       | VBTHS[2]       | VBTHS[1]       | VBTHS[0]       | VBTLs[3]       | VBTLs[2]       | VBTLs[1]       | VBTLs[0]       | 53            |
|       |                | 2     | FBOFF_V GH     | FBOFF_V GL     | x              | x              | x              | x              | x              | x              | 53            |
|       |                | 3     | x              | x              | VRP[5]         | VRP[4]         | VRP[3]         | VRP[2]         | VRP[1]         | VRP[0]         | C0            |
|       |                | 4     | x              | x              | VRN[5]         | VRN[4]         | VRN[3]         | VRN[2]         | VRN[1]         | VRN[0]         | 16            |
|       |                | 5     | x              | x              | x              | x              | x              | x              | x              | x              | 00            |
|       |                | 6     | VGL_DET EN     | VGH_DET EN     | VGL_TUR BO     | VGH_TU RBO     | x              | APS[2]         | APS[1]         | APS[0]         | 77            |
|       |                | 7     | VGH1_L_ DIV[3] | VGH1_L_ DIV[2] | VGH1_L_ DIV[1] | VGH1_L_ DIV[0] | VGL1_L_ DIV[3] | VGL1_L_ DIV[2] | VGL1_L_ DIV[1] | VGL1_L_ DIV[0] | D1            |
|       |                | 8     | VGH1_R_ DIV[3] | VGH1_R_ DIV[2] | VGH1_R_ DIV[1] | VGH1_R_ DIV[0] | VGL1_R_ DIV[3] | VGL1_R_ DIV[2] | VGL1_R_ DIV[1] | VGL1_R_ DIV[0] | CC            |
|       |                | 9     | VGH2_L_ DIV[3] | VGH2_L_ DIV[2] | VGH2_L_ DIV[1] | VGH2_L_ DIV[0] | VGL2_L_ DIV[3] | VGL2_L_ DIV[2] | VGL2_L_ DIV[1] | VGL2_L_ DIV[0] | DD            |
|       |                | 10    | VGH2_R_ DIV[3] | VGH2_R_ DIV[2] | VGH2_R_ DIV[1] | VGH2_R_ DIV[0] | VGL2_R_ DIV[3] | VGL2_R_ DIV[2] | VGL2_R_ DIV[1] | VGL2_R_ DIV[0] | 67            |
|       |                | 11    | VGH3_L_ DIV[3] | VGH3_L_ DIV[2] | VGH3_L_ DIV[1] | VGH3_L_ DIV[0] | VGL3_L_ DIV[3] | VGL3_L_ DIV[2] | VGL3_L_ DIV[1] | VGL3_L_ DIV[0] | 77            |
|       |                | 12    | VGH3_R_ DIV[3] | VGH3_R_ DIV[2] | VGH3_R_ DIV[1] | VGH3_R_ DIV[0] | VGL3_R_ DIV[3] | VGL3_R_ DIV[2] | VGL3_R_ DIV[1] | VGL3_R_ DIV[0] | 33            |
|       |                | 13    | VGH4_L_ DIV[3] | VGH4_L_ DIV[2] | VGH4_L_ DIV[1] | VGH4_L_ DIV[0] | VGL4_L_ DIV[3] | VGL4_L_ DIV[2] | VGL4_L_ DIV[1] | VGL4_L_ DIV[0] | 33            |
|       |                | 14    | VGH4_R_ DIV[3] | VGH4_R_ DIV[2] | VGH4_R_ DIV[1] | VGH4_R_ DIV[0] | VGL4_R_ DIV[3] | VGL4_R_ DIV[2] | VGL4_R_ DIV[1] | VGL4_R_ DIV[0] | 11            |
|       |                | 15    | VGH5_L_ DIV[3] | VGH5_L_ DIV[2] | VGH5_L_ DIV[1] | VGH5_L_ DIV[0] | VGL5_L_ DIV[3] | VGL5_L_ DIV[2] | VGL5_L_ DIV[1] | VGL5_L_ DIV[0] | 11            |
|       |                | 16    | VGH5_R_ DIV[3] | VGH5_R_ DIV[2] | VGH5_R_ DIV[1] | VGH5_R_ DIV[0] | VGL5_R_ DIV[3] | VGL5_R_ DIV[2] | VGL5_R_ DIV[1] | VGL5_R_ DIV[0] | 00            |
| C3h   | SETID          |       | 1              | 1              | 0              | 0              | 0              | 0              | 1              | 1              | -             |
|       |                | 1     | ID1[7]         | ID1[6]         | ID1[5]         | ID1[4]         | ID1[3]         | ID1[2]         | ID1[1]         | ID1[0]         | 38h           |
|       |                | 2     | ID2[7]         | ID2[6]         | ID2[5]         | ID2[4]         | ID2[3]         | ID2[2]         | ID2[1]         | ID2[0]         | 21h           |
|       |                | 3     | ID3[7]         | ID3[6]         | ID3[5]         | ID3[4]         | ID3[3]         | ID3[2]         | ID3[1]         | ID3[0]         | 1Fh           |
|       |                | 4     |                |                |                |                |                | ID_TIMES 2     | ID_TIMES 1     | ID_TIMES 0     |               |
| C4h   | SETDDB         |       | 1              | 1              | 0              | 0              | 0              | 1              | 0              | 0              | -             |
|       |                | 1     | DDB1[7]        | DDB1[6]        | DDB1[5]        | DDB1[4]        | DDB1[3]        | DDB1[2]        | DDB1[1]        | DDB1[0]        | 00h           |
|       |                | 2     | DDB2[7]        | DDB2[6]        | DDB2[5]        | DDB2[4]        | DDB2[3]        | DDB2[2]        | DDB2[1]        | DDB2[0]        | 00h           |
|       |                | 3     | DDB3[7]        | DDB3[6]        | DDB3[5]        | DDB3[4]        | DDB3[3]        | DDB3[2]        | DDB3[1]        | DDB3[0]        | 00h           |
|       |                | 4     | DDB4[7]        | DDB4[6]        | DDB4[5]        | DDB4[4]        | DDB4[3]        | DDB4[2]        | DDB4[1]        | DDB4[0]        | 00h           |

| (Hex) | Operation code | Para. | D7             | D6             | D5             | D4             | D3             | D2             | D1             | D0             | Default (Hex) |
|-------|----------------|-------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|
| C7h   | SETIO          |       | 1              | 1              | 0              | 0              | 0              | 1              | 1              | 1              | -             |
|       |                | 1     | PWM_OE         | x              | INVPWM         | VOUT_OE        | HOUT_OE        | x              | x              | x              | 00            |
|       |                | 2     | x              | V_DELAY[2]     | V_DELAY[1]     | V_DELAY[0]     | x              | H_DELAY[2]     | H_DELAY[1]     | H_DELAY[0]     | 00            |
| C8h   | SETCABC        |       | 1              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | -             |
|       |                | 1     | x              | PWMDIV[2]      | PWMDIV[1]      | PWMDIV[0]      | x              | x              | x              | x              | 10            |
|       |                | 2     | PWM_PE RIOD[7] | PWM_PE RIOD[6] | PWM_PE RIOD[5] | PWM_PE RIOD[4] | PWM_PE RIOD[3] | PWM_PE RIOD[2] | PWM_PE RIOD[1] | PWM_PE RIOD[0] | 40            |
| CCh   | SETPANEL       |       | 1              | 1              | 0              | 0              | 1              | 1              | 0              | 0              | -             |
|       |                | 1     | x              | x              | x              | x              | SS_PANELL      | GS_PANELL      | REV_PANELL     | BGR_PANELL     | 00h           |
| CDh   | DGC_R          |       | 1              | 1              | 0              | 0              | 1              | 1              | 0              | 1              | -             |
|       |                | 1     | x              | x              | x              | x              | x              | DITH_EN        | DITH_OP        | DGC_EN         | 00            |
|       |                | 2     | DGC_LUT_R00[7] | DGC_LUT_R00[6] | DGC_LUT_R00[5] | DGC_LUT_R00[4] | DGC_LUT_R00[3] | DGC_LUT_R00[2] | DGC_LUT_R00[1] | DGC_LUT_R00[0] | 80            |
|       |                | 3     | DGC_LUT_R01[7] | DGC_LUT_R01[6] | DGC_LUT_R01[5] | DGC_LUT_R01[4] | DGC_LUT_R01[3] | DGC_LUT_R01[2] | DGC_LUT_R01[1] | DGC_LUT_R01[0] | 80            |
|       |                | 4     | DGC_LUT_R02[7] | DGC_LUT_R02[6] | DGC_LUT_R02[5] | DGC_LUT_R02[4] | DGC_LUT_R02[3] | DGC_LUT_R02[2] | DGC_LUT_R02[1] | DGC_LUT_R02[0] | 80            |
|       |                | :     | :              | :              | :              | :              | :              | :              | :              | :              |               |
|       |                | 10    | DGC_LUT_R08[7] | DGC_LUT_R08[6] | DGC_LUT_R08[5] | DGC_LUT_R08[4] | DGC_LUT_R08[3] | DGC_LUT_R08[2] | DGC_LUT_R08[1] | DGC_LUT_R08[0] | 80            |
|       |                | 11    | DGC_LUT_R09[7] | DGC_LUT_R09[6] | DGC_LUT_R09[5] | DGC_LUT_R09[4] | DGC_LUT_R09[3] | DGC_LUT_R09[2] | DGC_LUT_R09[1] | DGC_LUT_R09[0] | 80            |
|       |                | 12    | DGC_LUT_R10[7] | DGC_LUT_R10[6] | DGC_LUT_R10[5] | DGC_LUT_R10[4] | DGC_LUT_R10[3] | DGC_LUT_R10[2] | DGC_LUT_R10[1] | DGC_LUT_R10[0] | 80            |
|       |                | 13    | DGC_LUT_R11[7] | DGC_LUT_R11[6] | DGC_LUT_R11[5] | DGC_LUT_R11[4] | DGC_LUT_R11[3] | DGC_LUT_R11[2] | DGC_LUT_R11[1] | DGC_LUT_R11[0] | 80            |
|       |                | :     | :              | :              | :              | :              | :              | :              | :              | :              |               |
|       |                | 20    | DGC_LUT_R18[7] | DGC_LUT_R18[6] | DGC_LUT_R18[5] | DGC_LUT_R18[4] | DGC_LUT_R18[3] | DGC_LUT_R18[2] | DGC_LUT_R18[1] | DGC_LUT_R18[0] | 80            |
|       |                | 21    | DGC_LUT_R19[7] | DGC_LUT_R19[6] | DGC_LUT_R19[5] | DGC_LUT_R19[4] | DGC_LUT_R19[3] | DGC_LUT_R19[2] | DGC_LUT_R19[1] | DGC_LUT_R19[0] | 80            |
|       |                | 22    | DGC_LUT_R20[7] | DGC_LUT_R20[6] | DGC_LUT_R20[5] | DGC_LUT_R20[4] | DGC_LUT_R20[3] | DGC_LUT_R20[2] | DGC_LUT_R20[1] | DGC_LUT_R20[0] | 80            |
|       |                | 23    | DGC_LUT_R21[7] | DGC_LUT_R21[6] | DGC_LUT_R21[5] | DGC_LUT_R21[4] | DGC_LUT_R21[3] | DGC_LUT_R21[2] | DGC_LUT_R21[1] | DGC_LUT_R21[0] | 80            |
|       |                | :     | :              | :              | :              | :              | :              | :              | :              | :              |               |
|       |                | 31    | DGC_LUT_R29[7] | DGC_LUT_R29[6] | DGC_LUT_R29[5] | DGC_LUT_R29[4] | DGC_LUT_R29[3] | DGC_LUT_R29[2] | DGC_LUT_R29[1] | DGC_LUT_R29[0] | 80            |
|       |                | 32    | DGC_LUT_R30[7] | DGC_LUT_R30[6] | DGC_LUT_R30[5] | DGC_LUT_R30[4] | DGC_LUT_R30[3] | DGC_LUT_R30[2] | DGC_LUT_R30[1] | DGC_LUT_R30[0] | 80            |
|       |                | 33    | DGC_LUT_R31[7] | DGC_LUT_R31[6] | DGC_LUT_R31[5] | DGC_LUT_R31[4] | DGC_LUT_R31[3] | DGC_LUT_R31[2] | DGC_LUT_R31[1] | DGC_LUT_R31[0] | 80            |
|       |                | 34    | DGC_LUT_R32[7] | DGC_LUT_R32[6] | DGC_LUT_R32[5] | DGC_LUT_R32[4] | DGC_LUT_R32[3] | DGC_LUT_R32[2] | DGC_LUT_R32[1] | DGC_LUT_R32[0] | 80            |

| (Hex) | Operation code | Para. | D7             | D6             | D5             | D4             | D3             | D2             | D1             | D0             | Default (Hex) |
|-------|----------------|-------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|
| CEh   | DGC_G          |       | 1              | 1              | 0              | 0              | 1              | 1              | 1              | 0              |               |
|       |                | 1     | DGC_LUT_G00[7] | DGC_LUT_G00[6] | DGC_LUT_G00[5] | DGC_LUT_G00[4] | DGC_LUT_G00[3] | DGC_LUT_G00[2] | DGC_LUT_G00[1] | DGC_LUT_G00[0] | 80            |
|       |                | 2     | DGC_LUT_G01[7] | DGC_LUT_G01[6] | DGC_LUT_G01[5] | DGC_LUT_G01[4] | DGC_LUT_G01[3] | DGC_LUT_G01[2] | DGC_LUT_G01[1] | DGC_LUT_G01[0] | 80            |
|       |                | 3     | DGC_LUT_G02[7] | DGC_LUT_G02[6] | DGC_LUT_G02[5] | DGC_LUT_G02[4] | DGC_LUT_G02[3] | DGC_LUT_G02[2] | DGC_LUT_G02[1] | DGC_LUT_G02[0] | 80            |
|       |                | 4     | DGC_LUT_G03[7] | DGC_LUT_G03[6] | DGC_LUT_G03[5] | DGC_LUT_G03[4] | DGC_LUT_G03[3] | DGC_LUT_G03[2] | DGC_LUT_G03[1] | DGC_LUT_G03[0] | 80            |
|       |                | :     | :              | :              | :              | :              | :              | :              | :              | :              |               |
|       |                | 10    | DGC_LUT_G09[7] | DGC_LUT_G09[6] | DGC_LUT_G09[5] | DGC_LUT_G09[4] | DGC_LUT_G09[3] | DGC_LUT_G09[2] | DGC_LUT_G09[1] | DGC_LUT_G09[0] | 80            |
|       |                | 11    | DGC_LUT_G10[7] | DGC_LUT_G10[6] | DGC_LUT_G10[5] | DGC_LUT_G10[4] | DGC_LUT_G10[3] | DGC_LUT_G10[2] | DGC_LUT_G10[1] | DGC_LUT_G10[0] | 80            |
|       |                | 12    | DGC_LUT_G11[7] | DGC_LUT_G11[6] | DGC_LUT_G11[5] | DGC_LUT_G11[4] | DGC_LUT_G11[3] | DGC_LUT_G11[2] | DGC_LUT_G11[1] | DGC_LUT_G11[0] | 80            |
|       |                | 13    | DGC_LUT_G12[7] | DGC_LUT_G12[6] | DGC_LUT_G12[5] | DGC_LUT_G12[4] | DGC_LUT_G12[3] | DGC_LUT_G12[2] | DGC_LUT_G12[1] | DGC_LUT_G12[0] | 80            |
|       |                | :     | :              | :              | :              | :              | :              | :              | :              | :              |               |
|       |                | 19    | DGC_LUT_G18[7] | DGC_LUT_G18[6] | DGC_LUT_G18[5] | DGC_LUT_G18[4] | DGC_LUT_G18[3] | DGC_LUT_G18[2] | DGC_LUT_G18[1] | DGC_LUT_G18[0] | 80            |
|       |                | 20    | DGC_LUT_G19[7] | DGC_LUT_G19[6] | DGC_LUT_G19[5] | DGC_LUT_G19[4] | DGC_LUT_G19[3] | DGC_LUT_G19[2] | DGC_LUT_G19[1] | DGC_LUT_G19[0] | 80            |
|       |                | 21    | DGC_LUT_G20[7] | DGC_LUT_G20[6] | DGC_LUT_G20[5] | DGC_LUT_G20[4] | DGC_LUT_G20[3] | DGC_LUT_G20[2] | DGC_LUT_G20[1] | DGC_LUT_G20[0] | 80            |
|       |                | 22    | DGC_LUT_G21[7] | DGC_LUT_G21[6] | DGC_LUT_G21[5] | DGC_LUT_G21[4] | DGC_LUT_G21[3] | DGC_LUT_G21[2] | DGC_LUT_G21[1] | DGC_LUT_G21[0] | 80            |
|       |                | :     | :              | :              | :              | :              | :              | :              | :              | :              |               |
|       |                | 30    | DGC_LUT_G29[7] | DGC_LUT_G29[6] | DGC_LUT_G29[5] | DGC_LUT_G29[4] | DGC_LUT_G29[3] | DGC_LUT_G29[2] | DGC_LUT_G29[1] | DGC_LUT_G29[0] | 80            |
|       |                | 31    | DGC_LUT_G30[7] | DGC_LUT_G30[6] | DGC_LUT_G30[5] | DGC_LUT_G30[4] | DGC_LUT_G30[3] | DGC_LUT_G30[2] | DGC_LUT_G30[1] | DGC_LUT_G30[0] | 80            |
|       |                | 32    | DGC_LUT_G31[7] | DGC_LUT_G31[6] | DGC_LUT_G31[5] | DGC_LUT_G31[4] | DGC_LUT_G31[3] | DGC_LUT_G31[2] | DGC_LUT_G31[1] | DGC_LUT_G31[0] | 80            |
|       |                | 33    | DGC_LUT_G32[7] | DGC_LUT_G32[6] | DGC_LUT_G32[5] | DGC_LUT_G32[4] | DGC_LUT_G32[3] | DGC_LUT_G32[2] | DGC_LUT_G32[1] | DGC_LUT_G32[0] | 80            |
| CFh   | DGC_B          |       | 1              | 1              | 0              | 0              | 1              | 1              | 1              | 1              |               |
|       |                | 1     | DGC_LUT_B00[7] | DGC_LUT_B00[6] | DGC_LUT_B00[5] | DGC_LUT_B00[4] | DGC_LUT_B00[3] | DGC_LUT_B00[2] | DGC_LUT_B00[1] | DGC_LUT_B00[0] | 80            |
|       |                | 2     | DGC_LUT_B01[7] | DGC_LUT_B01[6] | DGC_LUT_B01[5] | DGC_LUT_B01[4] | DGC_LUT_B01[3] | DGC_LUT_B01[2] | DGC_LUT_B01[1] | DGC_LUT_B01[0] | 80            |
|       |                | 3     | DGC_LUT_B02[7] | DGC_LUT_B02[6] | DGC_LUT_B02[5] | DGC_LUT_B02[4] | DGC_LUT_B02[3] | DGC_LUT_B02[2] | DGC_LUT_B02[1] | DGC_LUT_B02[0] | 80            |
|       |                | 4     | DGC_LUT_B03[7] | DGC_LUT_B03[6] | DGC_LUT_B03[5] | DGC_LUT_B03[4] | DGC_LUT_B03[3] | DGC_LUT_B03[2] | DGC_LUT_B03[1] | DGC_LUT_B03[0] | 80            |
|       |                | :     | :              | :              | :              | :              | :              | :              | :              | :              |               |
|       |                | 10    | DGC_LUT_B09[7] | DGC_LUT_B09[6] | DGC_LUT_B09[5] | DGC_LUT_B09[4] | DGC_LUT_B09[3] | DGC_LUT_B09[2] | DGC_LUT_B09[1] | DGC_LUT_B09[0] | 80            |
|       |                | 11    | DGC_LUT_B10[7] | DGC_LUT_B10[6] | DGC_LUT_B10[5] | DGC_LUT_B10[4] | DGC_LUT_B10[3] | DGC_LUT_B10[2] | DGC_LUT_B10[1] | DGC_LUT_B10[0] | 80            |
|       |                | 12    | DGC_LUT_B11[7] | DGC_LUT_B11[6] | DGC_LUT_B11[5] | DGC_LUT_B11[4] | DGC_LUT_B11[3] | DGC_LUT_B11[2] | DGC_LUT_B11[1] | DGC_LUT_B11[0] | 80            |
|       |                | 13    | DGC_LUT_B12[7] | DGC_LUT_B12[6] | DGC_LUT_B12[5] | DGC_LUT_B12[4] | DGC_LUT_B12[3] | DGC_LUT_B12[2] | DGC_LUT_B12[1] | DGC_LUT_B12[0] | 80            |
|       |                | :     | :              | :              | :              | :              | :              | :              | :              | :              |               |
|       |                | 19    | DGC_LUT_B18[7] | DGC_LUT_B18[6] | DGC_LUT_B18[5] | DGC_LUT_B18[4] | DGC_LUT_B18[3] | DGC_LUT_B18[2] | DGC_LUT_B18[1] | DGC_LUT_B18[0] | 80            |
|       |                | 20    | DGC_LUT_B19[7] | DGC_LUT_B19[6] | DGC_LUT_B19[5] | DGC_LUT_B19[4] | DGC_LUT_B19[3] | DGC_LUT_B19[2] | DGC_LUT_B19[1] | DGC_LUT_B19[0] | 80            |
|       |                | 21    | DGC_LUT_B20[7] | DGC_LUT_B20[6] | DGC_LUT_B20[5] | DGC_LUT_B20[4] | DGC_LUT_B20[3] | DGC_LUT_B20[2] | DGC_LUT_B20[1] | DGC_LUT_B20[0] | 80            |
|       |                | :     | :              | :              | :              | :              | :              | :              | :              | :              |               |
|       |                | 30    | DGC_LUT_B29[7] | DGC_LUT_B29[6] | DGC_LUT_B29[5] | DGC_LUT_B29[4] | DGC_LUT_B29[3] | DGC_LUT_B29[2] | DGC_LUT_B29[1] | DGC_LUT_B29[0] | 80            |
|       |                | 31    | DGC_LUT_B30[7] | DGC_LUT_B30[6] | DGC_LUT_B30[5] | DGC_LUT_B30[4] | DGC_LUT_B30[3] | DGC_LUT_B30[2] | DGC_LUT_B30[1] | DGC_LUT_B30[0] | 80            |

|  |  |    |                    |                    |                    |                    |                    |                    |                    |                    |    |
|--|--|----|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|----|
|  |  | 32 | DGC_LUT<br>_B31[7] | DGC_LUT<br>_B31[6] | DGC_LUT<br>_B31[5] | DGC_LUT<br>_B31[4] | DGC_LUT<br>_B31[3] | DGC_LUT<br>_B31[2] | DGC_LUT<br>_B31[1] | DGC_LUT<br>_B31[0] | 80 |
|  |  | 33 | DGC_LUT<br>_B32[7] | DGC_LUT<br>_B32[6] | DGC_LUT<br>_B32[5] | DGC_LUT<br>_B32[4] | DGC_LUT<br>_B32[3] | DGC_LUT<br>_B32[2] | DGC_LUT<br>_B32[1] | DGC_LUT<br>_B32[0] | 80 |

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| (Hex) | Operation code | Para. | D7 | D6      | D5      | D4      | D3      | D2      | D1      | D0      | Default (Hex) |
|-------|----------------|-------|----|---------|---------|---------|---------|---------|---------|---------|---------------|
| E0h   | SETGAMMA       |       | 1  | 1       | 0       | 0       | 0       | 0       | 0       | 1       |               |
|       |                | 1     | x  | x       | PVR0[5] | PVR0[4] | PVR0[3] | PVR0[2] | PVR0[1] | PVR0[0] | 0B            |
|       |                | 2     | x  | x       | PVR1[5] | PVR1[4] | PVR1[3] | PVR1[2] | PVR1[1] | PVR1[0] | 0C            |
|       |                | 3     | x  | x       | PVR2[5] | PVR2[4] | PVR2[3] | PVR2[2] | PVR2[1] | PVR2[0] | 11            |
|       |                | 4     | x  | x       | PVR3[5] | PVR3[4] | PVR3[3] | PVR3[2] | PVR3[1] | PVR3[0] | 1D            |
|       |                | 5     | x  | x       | PVR4[5] | PVR4[4] | PVR4[3] | PVR4[2] | PVR4[1] | PVR4[0] | 25            |
|       |                | 6     | x  | x       | PVR5[5] | PVR5[4] | PVR5[3] | PVR5[2] | PVR5[1] | PVR5[0] | 37            |
|       |                | 7     | x  | PPR0[6] | PPR0[5] | PPR0[4] | PPR0[3] | PPR0[2] | PPR0[1] | PPR0[0] | 43            |
|       |                | 8     | x  | PPR1[6] | PPR1[5] | PPR1[4] | PPR1[3] | PPR1[2] | PPR1[1] | PPR1[0] | 4B            |
|       |                | 9     | x  | x       | x       | PPK0[4] | PPK0[3] | PPK0[2] | PPK0[1] | PPK0[0] | 4E            |
|       |                | 10    | x  | x       | x       | PPK1[4] | PPK1[3] | PPK1[2] | PPK1[1] | PPK1[0] | 47            |
|       |                | 11    | x  | x       | x       | PPK2[4] | PPK2[3] | PPK2[2] | PPK2[1] | PPK2[0] | 41            |
|       |                | 12    | x  | x       | x       | PPK3[4] | PPK3[3] | PPK3[2] | PPK3[1] | PPK3[0] | 39            |
|       |                | 13    | x  | x       | x       | PPK4[4] | PPK4[3] | PPK4[2] | PPK4[1] | PPK4[0] | 35            |
|       |                | 14    | x  | x       | x       | PPK5[4] | PPK5[3] | PPK5[2] | PPK5[1] | PPK5[0] | 31            |
|       |                | 15    | x  | x       | x       | PPK6[4] | PPK6[3] | PPK6[2] | PPK6[1] | PPK6[0] | 2E            |
|       |                | 16    | x  | x       | x       | PPK7[4] | PPK7[3] | PPK7[2] | PPK7[1] | PPK7[0] | 21            |
|       |                | 17    | x  | x       | x       | PPK8[4] | PPK8[3] | PPK8[2] | PPK8[1] | PPK8[0] | 1C            |
|       |                | 18    | x  | x       | NVR0[5] | NVR0[4] | NVR0[3] | NVR0[2] | NVR0[1] | NVR0[0] | 1D            |
|       |                | 19    | x  | x       | NVR1[5] | NVR1[4] | NVR1[3] | NVR1[2] | NVR1[1] | NVR1[0] | 1D            |
|       |                | 20    | x  | x       | NVR2[5] | NVR2[4] | NVR2[3] | NVR2[2] | NVR2[1] | NVR2[0] | 26            |
|       |                | 21    | x  | x       | NVR3[5] | NVR3[4] | NVR3[3] | NVR3[2] | NVR3[1] | NVR3[0] | 31            |
|       |                | 22    | x  | x       | NVR4[5] | NVR4[4] | NVR4[3] | NVR4[2] | NVR4[1] | NVR4[0] | 44            |
|       |                | 23    | x  | x       | NVR5[5] | NVR5[4] | NVR5[3] | NVR5[2] | NVR5[1] | NVR5[0] | 4E            |
|       |                | 24    | x  | NPR0[6] | NPR0[5] | NPR0[4] | NPR0[3] | NPR0[2] | NPR0[1] | NPR0[0] | 56            |
|       |                | 25    | x  | NPR1[6] | NPR1[5] | NPR1[4] | NPR1[3] | NPR1[2] | NPR1[1] | NPR1[0] | 44            |
|       |                | 26    | x  | x       | x       | NPK0[4] | NPK0[3] | NPK0[2] | NPK0[1] | NPK0[0] | 3F            |
|       |                | 27    | x  | x       | x       | NPK1[4] | NPK1[3] | NPK1[2] | NPK1[1] | NPK1[0] | 39            |
|       |                | 28    | x  | x       | x       | NPK2[4] | NPK2[3] | NPK2[2] | NPK2[1] | NPK2[0] | 33            |
|       |                | 29    | x  | x       | x       | NPK3[4] | NPK3[3] | NPK3[2] | NPK3[1] | NPK3[0] | 31            |
|       |                | 30    | x  | x       | x       | NPK4[4] | NPK4[3] | NPK4[2] | NPK4[1] | NPK4[0] | 2E            |
|       |                | 31    | x  | x       | x       | NPK5[4] | NPK5[3] | NPK5[2] | NPK5[1] | NPK5[0] | 28            |
|       |                | 32    | x  | x       | x       | NPK6[4] | NPK6[3] | NPK6[2] | NPK6[1] | NPK6[0] | 1D            |
|       |                | 33    | x  | x       | x       | NPK7[4] | NPK7[3] | NPK7[2] | NPK7[1] | NPK7[0] | E0            |
|       |                | 34    | x  | x       | x       | NPK8[4] | NPK8[3] | NPK8[2] | NPK8[1] | NPK8[0] | 00            |

| (Hex) | Operation code | Para. | D7                 | D6           | D5           | D4           | D3                | D2               | D1                  | D0                  | Default (Hex) |
|-------|----------------|-------|--------------------|--------------|--------------|--------------|-------------------|------------------|---------------------|---------------------|---------------|
| E3h   | SETEQ          |       | 1                  | 1            | 1            | 0            | 0                 | 0                | 1                   | 1                   | -             |
|       |                | 1     | PNOEQ[7]           | PNOEQ[6]     | PNOEQ[5]     | PNOEQ[4]     | PNOEQ[3]          | PNOEQ[2]         | PNOEQ[1]            | PNOEQ[0]            | 00            |
|       |                | 2     | NNOEQ[7]           | NNOEQ[6]     | PEOEQ[5]     | NNOEQ[4]     | NNOEQ[3]          | NNOEQ[2]         | NNOEQ[1]            | NNOEQ[0]            | 00            |
|       |                | 3     | PEQGND[7]          | PEQGND[6]    | PEQGND[5]    | PEQGND[4]    | PEQGND[3]         | PEQGND[2]        | PEQGND[1]           | PEQGND[0]           | 03            |
|       |                | 4     | NEQGND[7]          | NEQGND[6]    | NEQGND[5]    | NEQGND[4]    | NEQGND[3]         | NEQGND[2]        | NEQGND[1]           | NEQGND[0]           | 03            |
|       |                | 5     | PEQVCI[7]          | PEQVCI[6]    | PEQVCI[5]    | PEQVCI[4]    | PEQVCI[3]         | PEQVCI[2]        | PEQVCI[1]           | PEQVCI[0]           | 03            |
|       |                | 6     | NEQVCI[7]          | NEQVCI[6]    | NEQVCI[5]    | NEQVCI[4]    | NEQVCI[3]         | NEQVCI[2]        | NEQVCI[1]           | NEQVCI[0]           | 03            |
|       |                | 7     | PEQVCI1[7]         | PEQVCI1[6]   | PEQVCI1[5]   | PEQVCI1[4]   | PEQVCI1[3]        | PEQVCI1[2]       | PEQVCI1[1]          | PEQVCI1[0]          | 03            |
|       |                | 8     | NEQVCI1[7]         | NEQVCI1[6]   | NEQVCI1[5]   | NEQVCI1[4]   | NEQVCI1[3]        | NEQVCI1[2]       | NEQVCI1[1]          | NEQVCI1[0]          | 03            |
|       |                | 9     | x                  | x            | x            | x            | x                 | x                | x                   | x                   | 00            |
|       |                | 10    | x                  | x            | x            | x            | x                 | x                | x                   | x                   | 00            |
|       |                | 11    | x                  | x            | x            | x            | x                 | x                | x                   | x                   | 00            |
|       |                | 12    | x                  | x            | x            | x            | x                 | x                | x                   | x                   | 00            |
|       |                | 13    | ESD_DET_DATA_WHITE | ESD_WHITE_EN | ESD_OPT_P[1] | ESD_OPT_P[0] | ESD_OPT_G[1]      | ESD_OPT_G[0]     | ESD_OPT_S[1]        | ESD_OPT_S[0]        | C0            |
|       |                | 14    | x                  | x            | x            | SLPIN_OPTION | VEDIO_NO_CHECK_EN | ESD_WHITE_GND_EN | ESD_DET_TIME_SEL[1] | ESD_DET_TIME_SEL[0] | 14            |
| E4h   | SET_ROI        |       | 1                  | 1            | 1            | 0            | 0                 | 1                | 0                   | 0                   | -             |
|       |                | 1     | x                  | x            | x            | x            | x                 | x                | x                   | CE_MASK             | 01            |
| E5h   | SETCOLOR_EN    |       | 1                  | 1            | 1            | 0            | 0                 | 1                | 0                   | 1                   | -             |
|       |                | 1     | SP_LEVEL[1]        | SP_LEVEL[0]  | CE_EN        | x            | x                 | x                | x                   | x                   | 00            |
|       |                | 2     | x                  | x            | x            | x            | x                 | x                | x                   | x                   | 00            |
|       |                | 3     | x                  | x            | x            | Blue_3[12]   | Blue_3[11]        | Blue_3[10]       | Blue_3[9]           | Blue_3[8]           | 08            |
|       |                | 4     | Blue_3[7]          | Blue_3[6]    | Blue_3[5]    | Blue_3[4]    | Blue_3[3]         | Blue_3[2]        | Blue_3[1]           | Blue_3[0]           | 32            |
|       |                | 5     | x                  | x            | x            | Blue_2[12]   | Blue_2[11]        | Blue_2[10]       | Blue_2[9]           | Blue_2[8]           | 1c            |
|       |                | 6     | Blue_2[7]          | Blue_2[6]    | Blue_2[5]    | Blue_2[4]    | Blue_2[3]         | Blue_2[2]        | Blue_2[1]           | Blue_2[0]           | 71            |
|       |                | 7     | x                  | x            | x            | Blue_1[12]   | Blue_1[11]        | Blue_1[10]       | Blue_1[9]           | Blue_1[8]           | 1F            |
|       |                | 8     | Blue_1[7]          | Blue_1[6]    | Blue_1[5]    | Blue_1[4]    | Blue_1[3]         | Blue_1[2]        | Blue_1[1]           | Blue_1[0]           | 90            |
|       |                | 9     | x                  | x            | x            | Green_3[12]  | Green_3[11]       | Green_3[10]      | Green_3[9]          | Green_3[8]          | 1E            |
|       |                | 10    | Green_3[7]         | Green_3[6]   | Green_3[5]   | Green_3[4]   | Green_3[3]        | Green_3[2]       | Green_3[1]          | Green_3[0]          | CD            |
|       |                | 11    | x                  | x            | x            | Green_2[12]  | Green_2[11]       | Green_2[10]      | Green_2[9]          | Green_2[8]          | 07            |
|       |                | 12    | Green_2[7]         | Green_2[6]   | Green_2[5]   | Green_2[4]   | Green_2[3]        | Green_2[2]       | Green_2[1]          | Green_2[0]          | 4C            |
|       |                | 13    | x                  | x            | x            | Green_1[12]  | Green_1[11]       | Green_1[10]      | Green_1[9]          | Green_1[8]          | 1E            |
|       |                | 14    | Green_1[7]         | Green_1[6]   | Green_1[5]   | Green_1[4]   | Green_1[3]        | Green_1[2]       | Green_1[1]          | Green_1[0]          | 06            |
|       |                | 15    | x                  | x            | x            | Red_3[12]    | Red_3[11]         | Red_3[10]        | Red_3[9]            | Red_3[8]            | 1F            |
|       |                | 16    | Red_3[7]           | Red_3[6]     | Red_3[5]     | Red_3[4]     | Red_3[3]          | Red_3[2]         | Red_3[1]            | Red_3[0]            | 66            |
|       |                | 17    | x                  | x            | x            | Red_2[12]    | Red_2[11]         | Red_2[10]        | Red_2[9]            | Red_2[8]            | 1E            |
|       |                | 18    | Red_2[7]           | Red_2[6]     | Red_2[5]     | Red_2[4]     | Red_2[3]          | Red_2[2]         | Red_2[1]            | Red_2[0]            | 2A            |
|       |                | 19    | x                  | x            | x            | Red_1[12]    | Red_1[11]         | Red_1[10]        | Red_1[9]            | Red_1[8]            | 06            |



|  |  |    |           |           |           |           |           |           |           |           |    |
|--|--|----|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----|
|  |  | 20 | Red _1[7] | Red _1[6] | Red _1[5] | Red _1[4] | Red _1[3] | Red _1[2] | Red _1[1] | Red _1[0] | 70 |
|--|--|----|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----|

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| (Hex) | Operation code | Para. | D7                   | D6                   | D5                   | D4                   | D3                   | D2                   | D1                   | D0                   | Default (Hex) |
|-------|----------------|-------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|---------------|
| E9h   | SETGIP1        |       | 1                    | 1                    | 1                    | 0                    | 1                    | 0                    | 0                    | 1                    | -             |
|       |                | 1     | REF_EN               | x                    | x                    | x                    | PANEL_S<br>EL[3]     | PANEL_S<br>EL[2]     | PANEL_S<br>EL[1]     | PANEL_S<br>EL[0]     | 24            |
|       |                | 2     | x                    | x                    | x                    | SHR0[12]             | SHR0[11]             | SHR0[10]             | SHR0[9]              | SHR0[8]              | 00            |
|       |                | 3     | SHR0[7]              | SHR0[6]              | SHR0[5]              | SHR0[4]              | SHR0[3]              | SHR0[2]              | SHR0[1]              | SHR0[0]              | 00            |
|       |                | 4     | x                    | x                    | x                    | SHR0[12]             | SHR1[11]             | SHR1[10]             | SHR1[9]              | SHR1[8]              | 00            |
|       |                | 5     | SHR1[7]              | SHR1[6]              | SHR1[5]              | SHR1[4]              | SHR1[3]              | SHR1[2]              | SHR1[1]              | SHR1[0]              | 00            |
|       |                | 6     | SPON[7]              | SPON[6]              | SPON[5]              | SPON[4]              | SPON[3]              | SPON[2]              | SPON[1]              | SPON[0]              | 00            |
|       |                | 7     | SPOFF[7]             | SPOFF[6]             | SPOFF[5]             | SPOFF[4]             | SPOFF[3]             | SPOFF[2]             | SPOFF[1]             | SPOFF[0]             | 00            |
|       |                | 8     | SHR0_1[3]            | SHR0_1[2]            | SHR0_1[1]            | SHR0_1[0]            | SHR0_2[3]            | SHR0_2[2]            | SHR0_2[1]            | SHR0_2[0]            | 00            |
|       |                | 9     | SHR0_3[3]            | SHR0_3[2]            | SHR0_3[1]            | SHR0_3[0]            | SHR1_1[3]            | SHR1_1[2]            | SHR1_1[1]            | SHR1_1[0]            | 00            |
|       |                | 10    | SHR1_2[3]            | SHR1_2[2]            | SHR1_2[1]            | SHR1_2[0]            | SHR1_3[3]            | SHR1_3[2]            | SHR1_3[1]            | SHR1_3[0]            | 00            |
|       |                | 11    | SHP[3]               | SHP[2]               | SHP[1]               | SHP[0]               | SCP[3]               | SCP[2]               | SCP[1]               | SCP[0]               | 00            |
|       |                | 12    | CHR [7]              | CHR [6]              | CHR [5]              | CHR [4]              | CHR [3]              | CHR [2]              | CHR [1]              | CHR [0]              | 02            |
|       |                | 13    | CON [7]              | CON [6]              | CON [5]              | CON [4]              | CON [3]              | CON [2]              | CON [1]              | CON [0]              | 02            |
|       |                | 14    | COF [7]              | COF [6]              | COF [5]              | COF [4]              | COF [3]              | COF [2]              | COF [1]              | COF [0]              | 00            |
|       |                | 15    | CHP[3]               | CHP[2]               | CHP[1]               | CHP[0]               | CCP[3]               | CCP[2]               | CCP[1]               | CCP[0]               | 00            |
|       |                | 16    | USER_GI<br>P_GATE[7] | USER_GI<br>P_GATE[6] | USER_GI<br>P_GATE[5] | USER_GI<br>P_GATE[4] | USER_GI<br>P_GATE[3] | USER_GI<br>P_GATE[2] | USER_GI<br>P_GATE[1] | USER_GI<br>P_GATE[0] | 00            |
|       |                | 17    | x                    | x                    | CGTS_L[2]<br>1]      | CGTS_L[2]<br>0]      | CGTS_L[1]<br>9]      | CGTS_L[1]<br>8]      | CGTS_L[1]<br>7]      | CGTS_L[1]<br>6]      | 00            |
|       |                | 18    | CGTS_L[1]<br>5]      | CGTS_L[1]<br>4]      | CGTS_L[1]<br>3]      | CGTS_L[1]<br>2]      | CGTS_L[1]<br>1]      | CGTS_L[1]<br>0]      | CGTS_L[9]            | CGTS_L[8]            | 00            |
|       |                | 19    | CGTS_L[7]<br>]       | CGTS_L[6]<br>]       | CGTS_L[5]<br>]       | CGTS_L[4]<br>]       | CGTS_L[3]<br>]       | CGTS_L[2]<br>]       | CGTS_L[1]<br>]       | CGTS_L[0]<br>]       | 00            |
|       |                | 20    | x                    | x                    | CGTS_IN<br>V_L[21]   | CGTS_IN<br>V_L[20]   | CGTS_IN<br>V_L[19]   | CGTS_IN<br>V_L[18]   | CGTS_IN<br>V_L[17]   | CGTS_IN<br>V_L[16]   | 00            |
|       |                | 21    | CGTS_IN<br>V_L[15]   | CGTS_IN<br>V_L[14]   | CGTS_IN<br>V_L[13]   | CGTS_IN<br>V_L[12]   | CGTS_IN<br>V_L[11]   | CGTS_IN<br>V_L[10]   | CGTS_IN<br>V_L[9]    | CGTS_IN<br>V_L[8]    | 00            |
|       |                | 22    | CGTS_IN<br>V_L[7]    | CGTS_IN<br>V_L[6]    | CGTS_IN<br>V_L[5]    | CGTS_IN<br>V_L[4]    | CGTS_IN<br>V_L[3]    | CGTS_IN<br>V_L[2]    | CGTS_IN<br>V_L[1]    | CGTS_IN<br>V_L[0]    | 00            |
|       |                | 23    | x                    | x                    | CGTS_R[<br>21]       | CGTS_R[<br>20]       | CGTS_R[<br>19]       | CGTS_R[<br>18]       | CGTS_R[<br>17]       | CGTS_R[<br>16]       | 00            |
|       |                | 24    | CGTS_R[<br>15]       | CGTS_R[<br>14]       | CGTS_R[<br>13]       | CGTS_R[<br>12]       | CGTS_R[<br>11]       | CGTS_R[<br>10]       | CGTS_R[<br>9]        | CGTS_R[<br>8]        | 00            |
|       |                | 25    | CGTS_R[<br>7]        | CGTS_R[<br>6]        | CGTS_R[<br>5]        | CGTS_R[<br>4]        | CGTS_R[<br>3]        | CGTS_R[<br>2]        | CGTS_R[<br>1]        | CGTS_R[<br>0]        | 00            |
|       |                | 26    | x                    | x                    | CGTS_IN<br>V_R[21]   | CGTS_IN<br>V_R[20]   | CGTS_IN<br>V_R[19]   | CGTS_IN<br>V_R[18]   | CGTS_IN<br>V_R[17]   | CGTS_IN<br>V_R[16]   | 00            |
|       |                | 27    | CGTS_IN<br>V_R[15]   | CGTS_IN<br>V_R[14]   | CGTS_IN<br>V_R[13]   | CGTS_IN<br>V_R[12]   | CGTS_IN<br>V_R[11]   | CGTS_IN<br>V_R[10]   | CGTS_IN<br>V_R[9]    | CGTS_IN<br>V_R[8]    | 00            |
|       |                | 28    | CGTS_IN<br>V_R[7]    | CGTS_IN<br>V_R[6]    | CGTS_IN<br>V_R[5]    | CGTS_IN<br>V_R[4]    | CGTS_IN<br>V_R[3]    | CGTS_IN<br>V_R[2]    | CGTS_IN<br>V_R[1]    | CGTS_IN<br>V_R[0]    | 00            |
|       |                | 29    | COS1_L[3]<br>]       | COS1_L[2]<br>]       | COS1_L[1]<br>]       | COS1_L[0]<br>]       | COS2_L[3]<br>]       | COS2_L[2]<br>]       | COS2_L[1]<br>]       | COS2_L[0]<br>]       | 00            |
|       |                | 30    | COS3_L[3]<br>]       | COS3_L[2]<br>]       | COS3_L[1]<br>]       | COS3_L[0]<br>]       | COS4_L[3]<br>]       | COS4_L[2]<br>]       | COS4_L[1]<br>]       | COS4_L[0]<br>]       | 00            |
|       |                | 31    | COS5_L[3]<br>]       | COS5_L[2]<br>]       | COS5_L[1]<br>]       | COS5_L<br>[0]        | COS6_L[3]<br>]       | COS6_L<br>[2]        | COS6_L[1]<br>]       | COS6_L[0]<br>]       | 00            |
|       |                | :     | :                    | :                    | :                    | :                    | :                    | :                    | :                    | :                    |               |
|       |                | 37    | COS17_L[<br>3]       | COS17_L[<br>2]       | COS17_L[<br>1]       | COS17_L[<br>0]       | COS18_L[<br>3]       | COS18_L[<br>2]       | COS18_L[<br>1]       | COS18_L[<br>0]       | 00            |
|       |                | 38    | COS19_L[<br>3]       | COS19_L[<br>2]       | COS19_L[<br>1]       | COS19_L[<br>0]       | COS20_L[<br>3]       | COS20_L[<br>2]       | COS20_L[<br>1]       | COS20_L[<br>0]       | 00            |
|       |                | 39    | COS21_L[<br>3]       | COS21_L[<br>2]       | COS21_L[<br>1]       | COS21_L[<br>0]       | COS22_L[<br>3]       | COS22_L[<br>2]       | COS22_L[<br>1]       | COS22_L[<br>0]       | 00            |
|       |                | 40    | COS1_R[<br>3]        | COS1_R[<br>2]        | COS1_R[<br>1]        | COS1_R[<br>0]        | COS2_R[<br>3]        | COS2_R[<br>2]        | COS2_R[<br>1]        | COS2_R[<br>0]        | 00            |

|     |         |     |              |              |              |              |              |              |              |              |    |
|-----|---------|-----|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|----|
| E9h | SETGIP1 | 41  | COS3_R[3]    | COS3_R[2]    | COS3_R[1]    | COS3_R[0]    | COS4_R[3]    | COS4_R[2]    | COS4_R[1]    | COS4_R[0]    | 00 |
|     |         | 42  | COS5_R[3]    | COS5_R[2]    | COS5_R[1]    | COS5_R[0]    | COS6_R[3]    | COS6_R[2]    | COS6_R[1]    | COS6_R[0]    | 00 |
|     |         | :   | :            | :            | :            | :            | :            | :            | :            | :            |    |
|     |         | 48  | COS17_R[3]   | COS17_R[2]   | COS17_R[1]   | COS17_R[0]   | COS18_R[3]   | COS18_R[2]   | COS18_R[1]   | COS18_R[0]   | 00 |
|     |         | 49h | COS19_R[3]   | COS19_R[2]   | COS19_R[1]   | COS19_R[0]   | COS20_R[3]   | COS20_R[2]   | COS20_R[1]   | COS20_R[0]   | 00 |
|     |         | 50  | COS21_R[3]   | COS21_R[2]   | COS21_R[1]   | COS21_R[0]   | COS22_R[3]   | COS22_R[2]   | COS22_R[1]   | COS22_R[0]   | 00 |
|     |         | 51  | TCON_O PT[7] | TCON_O PT[6] | TCON_O PT[5] | TCON_O PT[4] | TCON_O PT[3] | TCON_O PT[2] | TCON_O PT[1] | TCON_O PT[0] | 00 |
|     |         | 52  | x            | GIP_OPT[22]  | GIP_OPT[21]  | GIP_OPT[20]  | GIP_OPT[19]  | GIP_OPT[18]  | GIP_OPT[17]  | GIP_OPT[16]  | 00 |
|     |         | 53  | GIP_OPT[15]  | GIP_OPT[14]  | GIP_OPT[13]  | GIP_OPT[12]  | GIP_OPT[11]  | GIP_OPT[10]  | GIP_OPT[9]   | GIP_OPT[8]   | 00 |
|     |         | 54  | GIP_OPT[7]   | GIP_OPT[6]   | GIP_OPT[5]   | GIP_OPT[4]   | GIP_OPT[3]   | GIP_OPT[2]   | GIP_OPT[1]   | GIP_OPT[0]   | 00 |
|     |         | 55  | CHR2[7]      | CHR2[6]      | CHR2[5]      | CHR2[4]      | CHR2[3]      | CHR2[2]      | CHR2[1]      | CHR2[0]      | 00 |
|     |         | 56  | CON2[7]      | CON2[6]      | CON2[5]      | CON2[4]      | CON2[3]      | CON2[2]      | CON2[1]      | CON2[0]      | 02 |
|     |         | 57  | COFF2[7]     | COFF2[6]     | COFF2[5]     | COFF2[4]     | COFF2[3]     | COFF2[2]     | COFF2[1]     | COFF2[0]     | 02 |
|     |         | 58  | CHP2[3]      | CHP2[2]      | CHP2[1]      | CHP2[0]      | CCP2[3]      | CCP2[2]      | CCP2[1]      | CCP2[0]      | 00 |
|     |         | 59  | x            | x            | CKS[21]      | CKS[20]      | CKS[19]      | CKS[18]      | CKS[17]      | CKS[16]      | 00 |
|     |         | 60  | CKS[15]      | CKS[14]      | CKS[13]      | CKS[12]      | CKS[11]      | CKS[10]      | CKS[9]       | CKS[8]       | 00 |
|     |         | 61  | CKS[7]       | CKS[6]       | CKS[5]       | CKS[4]       | CKS[3]       | CKS[2]       | CKS[1]       | CKS[0]       | 00 |
|     |         | 62  | COFF[9]      | COFF[8]      | CON[9]       | CON[8]       | SPOFF[9]     | SPOFF[8]     | SPON[9]      | SPON[8]      | 00 |
|     |         | 63  | COFF2[9]     | COFF[8]      | CON2[9]      | CON2[8]      | x            | x            | x            | x            | 00 |

| (Hex) | Operation code | Para. | D7                  | D6                  | D5                     | D4                     | D3                     | D2                     | D1                     | D0                     | Default (Hex) |
|-------|----------------|-------|---------------------|---------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|---------------|
| EAh   | SETGIP2        |       | 1                   | 1                   | 1                      | 0                      | 1                      | 0                      | 1                      | 0                      | -             |
|       |                | 1     | YS2_SEL[1]          | YS2_SEL[0]          | YS1_SEL[1]             | YS1_SEL[0]             | YS2_XOR                | YS1_XOR                | YS_FLAG_EN             | ALL_ON_EN              | 94            |
|       |                | 2     | GATE[7]             | GATE[6]             | GATE[5]                | GATE[4]                | GATE[3]                | GATE[2]                | GATE[1]                | GATE[0]                | 00            |
|       |                | 3     | CK_ALL_ON_EN        | STV_ALL_ON_EN       | CK_ALL_ON_WIDT_H1[5]   | CK_ALL_ON_WIDT_H1[4]   | CK_ALL_ON_WIDT_H1[3]   | CK_ALL_ON_WIDT_H1[2]   | CK_ALL_ON_WIDT_H1[1]   | CK_ALL_ON_WIDT_H1[0]   | 00            |
|       |                | 4     | x                   | x                   | CK_ALL_ON_WIDT_H2[5]   | CK_ALL_ON_WIDT_H2[4]   | CK_ALL_ON_WIDT_H2[3]   | CK_ALL_ON_WIDT_H2[2]   | CK_ALL_ON_WIDT_H2[1]   | CK_ALL_ON_WIDT_H2[0]   | 00            |
|       |                | 5     | x                   | x                   | CK_ALL_ON_WIDT_H3[5]   | CK_ALL_ON_WIDT_H3[4]   | CK_ALL_ON_WIDT_H3[3]   | CK_ALL_ON_WIDT_H3[2]   | CK_ALL_ON_WIDT_H3[1]   | CK_ALL_ON_WIDT_H3[0]   | 00            |
|       |                | 6     | YS_FLAG_PERIOD[7]   | YS_FLAG_PERIOD[6]   | YS_FLAG_PERIOD[5]      | YS_FLAG_PERIOD[4]      | YS_FLAG_PERIOD[3]      | YS_FLAG_PERIOD[2]      | YS_FLAG_PERIOD[1]      | YS_FLAG_PERIOD[0]      | 00            |
|       |                | 7     | YS2_SEL_2[1]        | YS2_SEL_2[0]        | YS1_SEL_2[1]           | YS1_SEL_2[0]           | YS2_XOR_2              | YS1_XOR_2              | YS_FLAG_EN_2           | ALL_ON_EN_2            | 94            |
|       |                | 8     | USER_GIP_GATE1_2[7] | USER_GIP_GATE1_2[6] | USER_GIP_GATE1_2[5]    | USER_GIP_GATE1_2[4]    | USER_GIP_GATE1_2[3]    | USER_GIP_GATE1_2[2]    | USER_GIP_GATE1_2[1]    | USER_GIP_GATE1_2[0]    | 00            |
|       |                | 9     | CK_ALL_ON_EN_2      | STV_ALL_ON_EN_2     | CK_ALL_ON_WIDT_H1_2[5] | CK_ALL_ON_WIDT_H1_2[4] | CK_ALL_ON_WIDT_H1_2[3] | CK_ALL_ON_WIDT_H1_2[2] | CK_ALL_ON_WIDT_H1_2[1] | CK_ALL_ON_WIDT_H1_2[0] | 00            |
|       |                | 10    | x                   | x                   | CK_ALL_ON_WIDT_H2_2[5] | CK_ALL_ON_WIDT_H2_2[4] | CK_ALL_ON_WIDT_H2_2[3] | CK_ALL_ON_WIDT_H2_2[2] | CK_ALL_ON_WIDT_H2_2[1] | CK_ALL_ON_WIDT_H2_2[0] | 00            |
|       |                | 11    | x                   | x                   | CK_ALL_ON_WIDT_H3_2[5] | CK_ALL_ON_WIDT_H3_2[4] | CK_ALL_ON_WIDT_H3_2[3] | CK_ALL_ON_WIDT_H3_2[2] | CK_ALL_ON_WIDT_H3_2[1] | CK_ALL_ON_WIDT_H3_2[0] | 00            |
|       |                | 12    | YS_FLAG_PERIOD_2[7] | YS_FLAG_PERIOD_2[6] | YS_FLAG_PERIOD_2[5]    | YS_FLAG_PERIOD_2[4]    | YS_FLAG_PERIOD_2[3]    | YS_FLAG_PERIOD_2[2]    | YS_FLAG_PERIOD_2[1]    | YS_FLAG_PERIOD_2[0]    | 00            |
|       |                | 13    | COS1_L_GS[3]        | COS1_L_GS[2]        | COS1_L_GS[1]           | COS1_L_GS[0]           | COS2_L_GS[3]           | COS2_L_GS[2]           | COS2_L_GS[1]           | COS2_L_GS[0]           | 00            |
|       |                | 14    | COS3_L_GS[3]        | COS3_L_GS[2]        | COS3_L_GS[1]           | COS3_L_GS[0]           | COS4_L_GS[3]           | COS4_L_GS[2]           | COS4_L_GS[1]           | COS4_L_GS[0]           | 00            |
|       |                | 15    | COS5_L_GS[3]        | COS5_L_GS[2]        | COS5_L_GS[1]           | COS5_L_GS[0]           | COS6_L_GS[3]           | COS6_L_GS[2]           | COS6_L_GS[1]           | COS6_L_GS[0]           | 00            |
|       |                | :     | :                   | :                   | :                      | :                      | :                      | :                      | :                      | :                      | :             |
|       |                | 21    | COS17_L_GS[3]       | COS17_L_GS[2]       | COS17_L_GS[1]          | COS17_L_GS[0]          | COS18_L_GS[3]          | COS18_L_GS[2]          | COS18_L_GS[1]          | COS18_L_GS[0]          | 00            |
|       |                | 22    | COS19_L_GS[3]       | COS19_L_GS[2]       | COS19_L_GS[1]          | COS19_L_GS[0]          | COS20_L_GS[3]          | COS20_L_GS[2]          | COS20_L_GS[1]          | COS20_L_GS[0]          | 00            |
|       |                | 23    | COS21_L_GS[3]       | COS21_L_GS[2]       | COS21_L_GS[1]          | COS21_L_GS[0]          | COS22_L_GS[3]          | COS22_L_GS[2]          | COS22_L_GS[1]          | COS22_L_GS[0]          | 00            |
|       |                | 24    | COS1_R_GS[3]        | COS1_R_GS[2]        | COS1_R_GS[1]           | COS1_R_GS[0]           | COS2_R_GS[3]           | COS2_R_GS[2]           | COS2_R_GS[1]           | COS2_R_GS[0]           | 00            |
|       |                | 25    | COS3_R_GS[3]        | COS3_R_GS[2]        | COS3_R_GS[1]           | COS3_R_GS[0]           | COS4_R_GS[3]           | COS4_R_GS[2]           | COS4_R_GS[1]           | COS4_R_GS[0]           | 00            |
|       |                | 26    | COS5_R_GS[3]        | COS5_R_GS[2]        | COS5_R_GS[1]           | COS5_R_GS[0]           | COS6_R_GS[3]           | COS6_R_GS[2]           | COS6_R_GS[1]           | COS6_R_GS[0]           | 00            |
|       |                | :     | :                   | :                   | :                      | :                      | :                      | :                      | :                      | :                      | :             |
|       |                | 32    | COS17_R_GS[3]       | COS17_R_GS[2]       | COS17_R_GS[1]          | COS17_R_GS[0]          | COS18_R_GS[3]          | COS18_R_GS[2]          | COS18_R_GS[1]          | COS18_R_GS[0]          | 00            |
|       |                | 33    | COS19_R_GS[3]       | COS19_R_GS[2]       | COS19_R_GS[1]          | COS19_R_GS[0]          | COS20_R_GS[3]          | COS20_R_GS[2]          | COS20_R_GS[1]          | COS20_R_GS[0]          | 00            |
|       |                | 34    | COS21_R_GS[3]       | COS21_R_GS[2]       | COS21_R_GS[1]          | COS21_R_GS[0]          | COS22_R_GS[3]          | COS22_R_GS[2]          | COS22_R_GS[1]          | COS22_R_GS[0]          | 00            |
|       |                | 35    | x                   | x                   | EQOPT[1]               | EQOPT[0]               | x                      | x                      | EQ_SEL[1]              | EQ_SEL[0]              | 30            |
|       |                | 36    | EQ_DELA_Y[7]        | EQ_DELA_Y[6]        | EQ_DELA_Y[5]           | EQ_DELA_Y[4]           | EQ_DELA_Y[3]           | EQ_DELA_Y[2]           | EQ_DELA_Y[1]           | EQ_DELA_Y[0]           | 00            |
|       |                | 37    | x                   | x                   | x                      | x                      | EQ_DELA_Y_HSYNC[3]     | EQ_DELA_Y_HSYNC[2]     | EQ_DELA_Y_HSYNC[1]     | EQ_DELA_Y_HSYNC[0]     | 00            |

|     |                 |    |                               |                               |                               |                               |                               |                               |                               |                               |    |
|-----|-----------------|----|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|----|
|     |                 | 38 | x                             | x                             | x                             | x                             | x                             | x                             | HSYNC_T<br>O_CL1_C<br>NT10[9] | HSYNC_T<br>O_CL1_C<br>NT10[8] | 00 |
|     |                 | 39 | HSYNC_T<br>O_CL1_C<br>NT10[7] | HSYNC_T<br>O_CL1_C<br>NT10[6] | HSYNC_T<br>O_CL1_C<br>NT10[5] | HSYNC_T<br>O_CL1_C<br>NT10[4] | HSYNC_T<br>O_CL1_C<br>NT10[3] | HSYNC_T<br>O_CL1_C<br>NT10[2] | HSYNC_T<br>O_CL1_C<br>NT10[1] | HSYNC_T<br>O_CL1_C<br>NT10[0] | 02 |
| EBh | SETCOLOR        |    | 1                             | 1                             | 1                             | 0                             | 1                             | 0                             | 1                             | 1                             | -  |
|     |                 | 1  | Bkx[1]                        | Bkx[0]                        | Bky[1]                        | Bky[0]                        | Wx[1]                         | Wx[0]                         | Wy[1]                         | Wy[0]                         | 00 |
|     |                 | 2  | BKx[9]                        | BKx[8]                        | BKx[7]                        | BKx[6]                        | BKx[5]                        | BKx[4]                        | BKx[3]                        | BKx[2]                        | 00 |
|     |                 | 3  | BKy[9]                        | BKy[8]                        | BKy[7]                        | BKy[6]                        | BKy[5]                        | BKy[4]                        | BKy[3]                        | BKy[2]                        | 00 |
|     |                 | 4  | Wx[9]                         | Wx[8]                         | Wx[7]                         | Wx[6]                         | Wx[5]                         | Wx[4]                         | Wx[3]                         | Wx[2]                         | 00 |
|     |                 | 5  | Wy[9]                         | Wy[8]                         | Wy[7]                         | Wy[6]                         | Wy[5]                         | Wy[4]                         | Wy[3]                         | Wy[2]                         | 00 |
|     |                 | 6  | Rx[1]                         | Rx[0]                         | Ry[1]                         | Ry[0]                         | Gx[1]                         | Gx[0]                         | Gy[1]                         | Gy[0]                         | 00 |
|     |                 | 7  | Rx[9]                         | Rx[8]                         | Rx[7]                         | Rx[6]                         | Rx[5]                         | Rx[4]                         | Rx[3]                         | Rx[2]                         | 00 |
|     |                 | 8  | Ry[9]                         | Ry[8]                         | Ry[7]                         | Ry[6]                         | Ry[5]                         | Ry[4]                         | Ry[3]                         | Ry[2]                         | 00 |
|     |                 | 9  | Gx[9]                         | Gx[8]                         | Gx[7]                         | Gx[6]                         | Gx[5]                         | Gx[4]                         | Gx[3]                         | Gx[2]                         | 00 |
|     |                 | 10 | Gy[9]                         | Gy[8]                         | Gy[7]                         | Gy[6]                         | Gy[5]                         | Gy[4]                         | Gy[3]                         | Gy[2]                         | 00 |
|     |                 | 11 | Bx[1]                         | Bx[0]                         | By[1]                         | By[0]                         | Ax[1]                         | Ax[0]                         | Ay[1]                         | Ay[0]                         | 00 |
|     |                 | 12 | Bx[9]                         | Bx[8]                         | Bx[7]                         | Bx[6]                         | Bx[5]                         | Bx[4]                         | Bx[3]                         | Bx[2]                         | 00 |
|     |                 | 13 | By[9]                         | By[8]                         | By[7]                         | By[6]                         | By[5]                         | By[4]                         | By[3]                         | By[2]                         | 00 |
|     |                 | 14 | Ax[9]                         | Ax[8]                         | Ax[7]                         | Ax[6]                         | Ax[5]                         | Ax[4]                         | Ax[3]                         | Ax[2]                         | 00 |
|     |                 | 15 | Ay[9]                         | Ay[8]                         | Ay[7]                         | Ay[6]                         | Ay[5]                         | Ay[4]                         | Ay[3]                         | Ay[2]                         | 00 |
| F1h | TEMP_SEN<br>SOR |    | 1                             | 1                             | 1                             | 1                             | 0                             | 0                             | 0                             | 1                             | -  |
|     |                 | 1  | PASSWO<br>RD[7]               | PASSWO<br>RD[6]               | PASSWO<br>RD[5]               | PASSWO<br>RD[4]               | PASSWO<br>RD[3]               | PASSWO<br>RD[2]               | PASSWO<br>RD[1]               | PASSWO<br>RD[0]               | 00 |
|     |                 | 2  | TEMP_ST<br>[3]                | TEMP_ST<br>[2]                | TEMP_ST<br>[1]                | TEMP_ST<br>[0]                | TEMP_FR<br>A [3]              | TEMP_FR<br>A [2]              | TEMP_FR<br>A [1]              | TEMP_FR<br>A [0]              | 3F |
|     |                 | 3  | x                             | x                             | L_TH[5]                       | L_TH[4]                       | L_TH[3]                       | L_TH[2]                       | L_TH[1]                       | L_TH[0]                       | 14 |
|     |                 | 4  | x                             | x                             | H_TH[5]                       | H_TH[4]                       | H_TH[3]                       | H_TH[2]                       | H_TH[1]                       | H_TH[0]                       | 30 |
|     |                 | 5  | x                             | L_PLUS[2]                     | L_PLUS[1]                     | L_PLUS[0]                     | x                             | L_MINUS[2]                    | L_MINUS[1]                    | L_MINUS[0]                    | 55 |
|     |                 | 6  | x                             | H_PLUS[2]                     | H_PLUS[1]                     | H_PLUS[0]                     | x                             | H_MINUS[2]                    | H_MINUS[1]                    | H_MINUS[0]                    | 55 |
|     |                 | 7  | TEMP_FI<br>LTER_EN            | x                             | x                             | x                             | SAME_N<br>UM[3]               | SAME_N<br>UM[2]               | SAME_N<br>UM[1]               | SAME_N<br>UM[0]               | 0F |

| (Hex) | Operation code   | Para. | D7                  | D6                  | D5                  | D4                  | D3                | D2                | D1                | D0                     | Default (Hex) |
|-------|------------------|-------|---------------------|---------------------|---------------------|---------------------|-------------------|-------------------|-------------------|------------------------|---------------|
| F2h   | TEMP_VOLT<br>AGE |       | 1                   | 1                   | 1                   | 1                   | 0                 | 0                 | 1                 | 0                      | -             |
|       |                  | 1     | VBTHS1[3]<br>]      | VBTHS1[2]<br>]      | VBTHS1[1]<br>]      | VBTHS1[0]<br>]      | VBTLS1[3]<br>]    | VBTLS1[2]<br>]    | VBTLS1[1]<br>]    | VBTLS1[0]<br>]         | 53            |
|       |                  | 2     | ECP_DC_DIV1[3]<br>] | ECP_DC_DIV1[2]<br>] | ECP_DC_DIV1[1]<br>] | ECP_DC_DIV1[0]<br>] | VGLREG1[3]<br>]   | VGLREG1[2]<br>]   | VGLREG1[1]<br>]   | VGLREG1[0]<br>]        | 43            |
|       |                  | 3     | VCOMDC_F1[7]<br>]   | VCOMDC_F1[6]<br>]   | VCOMDC_F1[5]<br>]   | VCOMDC_F1[4]<br>]   | VCOMDC_F1[3]<br>] | VCOMDC_F1[2]<br>] | VCOMDC_F1[1]<br>] | VCOMDC_F1[0]<br>]      | 4D            |
|       |                  | 4     | VCOMDC_B1[7]<br>]   | VCOMDC_B1[6]<br>]   | VCOMDC_B1[5]<br>]   | VCOMDC_B1[4]<br>]   | VCOMDC_B1[3]<br>] | VCOMDC_B1[2]<br>] | VCOMDC_B1[1]<br>] | VCOMDC_B1[0]<br>]      | 4D            |
|       |                  | 5     | x                   | x                   | PVR01[5]<br>]       | PVR01[4]<br>]       | PVR01[3]<br>]     | PVR01[2]<br>]     | PVR01[1]<br>]     | PVR01[0]<br>]          | 00            |
|       |                  | 6     | x                   | x                   | PVR51[5]<br>]       | PVR51[4]<br>]       | PVR51[3]<br>]     | PVR51[2]<br>]     | PVR51[1]<br>]     | PVR51[0]<br>]          | 00            |
|       |                  | 7     | x                   | x                   | NVR01[5]<br>]       | NVR01[4]<br>]       | NVR01[3]<br>]     | NVR01[2]<br>]     | NVR01[1]<br>]     | NVR01[0]<br>]          | 00            |
|       |                  | 8     | x                   | x                   | NVR51[5]<br>]       | NVR51[4]<br>]       | NVR51[3]<br>]     | NVR51[2]<br>]     | NVR51[1]<br>]     | NVR51[0]<br>]          | 00            |
|       |                  | 9     | x                   | x                   | x                   | x                   | x                 | N_NW_H_T[2]<br>]  | N_NW_H_T[1]<br>]  | N_NW_H_T[0]<br>]       | 00            |
|       |                  | 10    | VBTHS4[3]<br>]      | VBTHS4[2]<br>]      | VBTHS4[1]<br>]      | VBTHS4[0]<br>]      | VBTLS4[3]<br>]    | VBTLS4[2]<br>]    | VBTLS4[1]<br>]    | VBTLS4[0]<br>]         | 53            |
|       |                  | 11    | ECP_DC_DIV4[3]<br>] | ECP_DC_DIV4[2]<br>] | ECP_DC_DIV4[1]<br>] | ECP_DC_DIV4[0]<br>] | VGLREG4[3]<br>]   | VGLREG4[2]<br>]   | VGLREG4[1]<br>]   | VGLREG4[0]<br>]        | 43            |
|       |                  | 12    | VCOMDC_F4[7]<br>]   | VCOMDC_F4[6]<br>]   | VCOMDC_F4[5]<br>]   | VCOMDC_F4[4]<br>]   | VCOMDC_F4[3]<br>] | VCOMDC_F4[2]<br>] | VCOMDC_F4[1]<br>] | VCOMDC_F4[0]<br>]      | 4D            |
|       |                  | 13    | VCOMDC_B4[7]<br>]   | VCOMDC_B4[6]<br>]   | VCOMDC_B4[5]<br>]   | VCOMDC_B4[4]<br>]   | VCOMDC_B4[3]<br>] | VCOMDC_B4[2]<br>] | VCOMDC_B4[1]<br>] | VCOMDC_B4[0]<br>]      | 4D            |
|       |                  | 14    | x                   | x                   | PVR04[5]<br>]       | PVR04[4]<br>]       | PVR04[3]<br>]     | PVR04[2]<br>]     | PVR04[1]<br>]     | PVR04[0]<br>]          | 00            |
|       |                  | 15    | x                   | x                   | PVR54[5]<br>]       | PVR54[4]<br>]       | PVR54[3]<br>]     | PVR54[2]<br>]     | PVR54[1]<br>]     | PVR54[0]<br>]          | 00            |
|       |                  | 16    | x                   | x                   | NVR04[5]<br>]       | NVR04[4]<br>]       | NVR04[3]<br>]     | NVR04[2]<br>]     | NVR04[1]<br>]     | NVR04[0]<br>]          | 00            |
|       |                  | 17    | x                   | x                   | NVR54[5]<br>]       | NVR54[4]<br>]       | NVR54[3]<br>]     | NVR54[2]<br>]     | NVR54[1]<br>]     | NVR54[0]<br>]          | 00            |
|       |                  | 18    | x                   | x                   | x                   | x                   | x                 | N_NW_LT[2]<br>]   | N_NW_LT[1]<br>]   | N_NW_LT[0]<br>]        | 00            |
|       |                  | 19    | x                   | x                   | x                   | x                   | x                 | x                 | x                 | TEMP_SE<br>NSOR_O<br>N | 00            |

## 6.2 Command description

### 6.2.1 NOP (00h)

| 00 H                  |                                                                                                                                                                                                                                                                                                                                                                                                           | NOP (No Operation) |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----|----|----|----|----|----|----|-----|-------------------------------------------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|-------------------------|-----|
|                       | R/W                                                                                                                                                                                                                                                                                                                                                                                                       | D7                 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Command               | Write                                                                                                                                                                                                                                                                                                                                                                                                     | 0                  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 00  |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Parameter             | NO PARAMETER                                                                                                                                                                                                                                                                                                                                                                                              |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Description           | This command is an empty command; it does not have any effect on the display module.                                                                                                                                                                                                                                                                                                                      |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Restriction           | -                                                                                                                                                                                                                                                                                                                                                                                                         |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table> |                    |    |    |    |    |    |    |    |     | Status                                    | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | Status                                    | Availability  |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | Normal Mode On, Idle Mode Off, Sleep Out  | Yes           |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | Normal Mode On, Idle Mode On, Sleep Out   | Yes           |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | Partial Mode On, Idle Mode Off, Sleep Out | Yes           |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | Partial Mode On, Idle Mode On, Sleep Out  | Yes           |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | Sleep In or Booster Off                   | Yes           |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Default               | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>S/W Reset</td><td>N/A</td></tr><tr><td>H/W Reset</td><td>N/A</td></tr></table>                                                                                                                                                                                                                |                    |    |    |    |    |    |    |    |     | Status                                    | Default Value | Power On Sequence                        | N/A | S/W Reset                               | N/A | H/W Reset                                 | N/A |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | Status                                    | Default Value |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | Power On Sequence                         | N/A           |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | S/W Reset                                 | N/A           |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     | H/W Reset                                 | N/A           |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                           |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Flow Chart            | -                                                                                                                                                                                                                                                                                                                                                                                                         |                    |    |    |    |    |    |    |    |     |                                           |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |

## 6.2.2 Software reset (01h)

| 01 H                                      | SWRESET (Software Reset)                                                                                                                                                                                                                                                                                                                                                                                  |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                     | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 01  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Parameter                                 | NO PARAMETER                                                                                                                                                                                                                                                                                                                                                                                              |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Description                               | When the Software Reset command is written, it causes a software reset. It resets the commands and parameters to their S/W Reset default values. (See default tables in each command description.)<br>The display is blank immediately.                                                                                                                                                                   |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Restriction                               | It will be necessary to wait 5msec before sending new command following software reset. The display module loads all display suppliers' factory default values to the registers during this 5m sec.<br>If SW Reset is applied during Sleep Out mode, it will be necessary to wait 120m sec before sending Sleep Out command.<br>SW Reset command cannot be sent during Sleep Out sequence.                |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table> |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                              |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>S/W Reset</td><td>N/A</td></tr><tr><td>H/W Reset</td><td>N/A</td></tr></table>                                                                                                                                                                                                                |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | N/A | S/W Reset                               | N/A | H/W Reset                                 | N/A |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Power On Sequence                         | N/A                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| S/W Reset                                 | N/A                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| H/W Reset                                 | N/A                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Flow Chart                                | <div><div><div>SWRESET (01H)</div><div>Display whole blank screen</div><div>Set Commands to S/W Default Value</div><div>Sleep In Mode</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |



### 6.2.3 Read Display ID (04h)

| 04 H                      |                                                                                                                                                                                                                                                                                                          | RDDID (Read Display ID) |      |      |      |      |               |      |      |     |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------|------|------|------|---------------|------|------|-----|
|                           | R/W                                                                                                                                                                                                                                                                                                      | D7                      | D6   | D5   | D4   | D3   | D2            | D1   | D0   | HEX |
| Command                   | Read                                                                                                                                                                                                                                                                                                     | 0                       | 0    | 0    | 0    | 0    | 1             | 0    | 0    | 04  |
| 1 <sup>st</sup> parameter | -                                                                                                                                                                                                                                                                                                        | ID17                    | ID16 | ID15 | ID14 | ID13 | ID12          | ID11 | ID10 | 38  |
| 2 <sup>nd</sup> parameter | -                                                                                                                                                                                                                                                                                                        | ID27                    | ID26 | ID25 | ID24 | ID23 | ID22          | ID21 | ID20 | 21  |
| 3 <sup>rd</sup> parameter | -                                                                                                                                                                                                                                                                                                        | ID37                    | ID36 | ID35 | ID34 | ID33 | ID32          | ID31 | ID30 | 1F  |
| Description               | This read byte returns 24-bit display identification information.                                                                                                                                                                                                                                        |                         |      |      |      |      |               |      |      |     |
| Restriction               | -                                                                                                                                                                                                                                                                                                        |                         |      |      |      |      |               |      |      |     |
| Register Availability     | Status                                                                                                                                                                                                                                                                                                   |                         |      |      |      |      | Availability  |      |      |     |
|                           | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                 |                         |      |      |      |      | Yes           |      |      |     |
|                           | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                  |                         |      |      |      |      | Yes           |      |      |     |
|                           | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                |                         |      |      |      |      | Yes           |      |      |     |
|                           | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                 |                         |      |      |      |      | Yes           |      |      |     |
|                           | Sleep In or Booster Off                                                                                                                                                                                                                                                                                  |                         |      |      |      |      | Yes           |      |      |     |
| Default                   | Status                                                                                                                                                                                                                                                                                                   |                         |      |      |      |      | Default Value |      |      |     |
|                           | Power On Sequence                                                                                                                                                                                                                                                                                        |                         |      |      |      |      | 38            |      |      |     |
|                           | S/W Reset                                                                                                                                                                                                                                                                                                |                         |      |      |      |      | 21            |      |      |     |
|                           | H/W Reset                                                                                                                                                                                                                                                                                                |                         |      |      |      |      | 1F            |      |      |     |
| Flow Chart                | <div><div><div>RDDID (04H)</div><div>Send ID1[7:0]</div><div>Send ID2[7:0]</div><div>Send ID3[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                         |      |      |      |      |               |      |      |     |

## 6.2.4 Read Display Power Mode (0Ah)

| 0A H                      |                                                                                                                                                               | RDDPM (Read Display Power Mode) |    |    |    |               |                                               |    |    |     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----|----|----|---------------|-----------------------------------------------|----|----|-----|
|                           | R/W                                                                                                                                                           | D7                              | D6 | D5 | D4 | D3            | D2                                            | D1 | D0 | HEX |
| Command                   | Read                                                                                                                                                          | 0                               | 0  | 0  | 0  | 1             | 0                                             | 1  | 0  | 0A  |
| 1 <sup>st</sup> parameter | -                                                                                                                                                             | D7                              | D6 | D5 | D4 | D3            | D2                                            | 0  | 0  | -   |
| Description               | This command indicates the current status of the display as described in the table below:                                                                     |                                 |    |    |    |               |                                               |    |    |     |
|                           | Bit                                                                                                                                                           | Description                     |    |    |    |               | Value                                         |    |    |     |
|                           | D7                                                                                                                                                            | Booster Voltage Status          |    |    |    |               | '1'=Booster on, '0'=Booster off               |    |    |     |
|                           | D6                                                                                                                                                            | Idle Mode On/Off                |    |    |    |               | '1'=Idle Mode on, '0'=Idle Mode off           |    |    |     |
|                           | D5                                                                                                                                                            | Partial Mode On/Off             |    |    |    |               | '1'=Partial Mode on, '0'= Partial Mode off    |    |    |     |
|                           | D4                                                                                                                                                            | Sleep In/Out                    |    |    |    |               | '1'=Sleep out , '0'=Sleep in                  |    |    |     |
|                           | D3                                                                                                                                                            | Display Normal Mode On/Off      |    |    |    |               | '1'=Display Normal on, '0'=Display Normal off |    |    |     |
|                           | D2                                                                                                                                                            | Display On/Off                  |    |    |    |               | '1'=Display on, '0'=Display off               |    |    |     |
|                           | D1                                                                                                                                                            | Not Defined                     |    |    |    |               | Set to '0'                                    |    |    |     |
|                           | D0                                                                                                                                                            | Not Defined                     |    |    |    |               | Set to '0'                                    |    |    |     |
| Restriction               | -                                                                                                                                                             |                                 |    |    |    |               |                                               |    |    |     |
| Register Availability     | Status                                                                                                                                                        |                                 |    |    |    | Availability  |                                               |    |    |     |
|                           | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                      |                                 |    |    |    | Yes           |                                               |    |    |     |
|                           | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                       |                                 |    |    |    | Yes           |                                               |    |    |     |
|                           | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                     |                                 |    |    |    | Yes           |                                               |    |    |     |
|                           | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                      |                                 |    |    |    | Yes           |                                               |    |    |     |
|                           | Sleep In or Booster Off                                                                                                                                       |                                 |    |    |    | Yes           |                                               |    |    |     |
| Default                   | Status                                                                                                                                                        |                                 |    |    |    | Default Value |                                               |    |    |     |
|                           | Power On Sequence                                                                                                                                             |                                 |    |    |    | 08h           |                                               |    |    |     |
|                           | S/W Reset                                                                                                                                                     |                                 |    |    |    | 08h           |                                               |    |    |     |
|                           | H/W Reset                                                                                                                                                     |                                 |    |    |    | 08h           |                                               |    |    |     |
| Flow Chart                | <div><div><div>RDDPM (0AH)</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div>                                                          |                                 |    |    |    |               |                                               |    |    |     |
|                           | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                 |    |    |    |               |                                               |    |    |     |

## 6.2.5 Read Display MADCTR (0Bh)

| 0B H                      |                                                                                                                                                               | RDDMADCTR (Read Display MADCTR) |    |    |    |               |                                |    |    |     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----|----|----|---------------|--------------------------------|----|----|-----|
|                           | R/W                                                                                                                                                           | D7                              | D6 | D5 | D4 | D3            | D2                             | D1 | D0 | HEX |
| Command                   | Read                                                                                                                                                          | 0                               | 0  | 0  | 0  | 1             | 0                              | 1  | 1  | 0B  |
| 1 <sup>st</sup> parameter | -                                                                                                                                                             | D7                              | D6 | 0  | 0  | D3            | D2                             | 0  | 0  | -   |
| Description               | This command indicates the current status of the display as described in the table below:                                                                     |                                 |    |    |    |               |                                |    |    |     |
|                           | Bit                                                                                                                                                           | Description                     |    |    |    |               | Value                          |    |    |     |
|                           | D7                                                                                                                                                            | Page Address Order (MY)         |    |    |    |               | '0'=Increment, '1' = Decrement |    |    |     |
|                           | D6                                                                                                                                                            | Column Address Order (MX)       |    |    |    |               | '0'=Increment, '1' = Decrement |    |    |     |
|                           | D5                                                                                                                                                            | Page/Column Exchange (MV)       |    |    |    |               | Set to '0'                     |    |    |     |
|                           | D4                                                                                                                                                            | Line Address Order (ML)         |    |    |    |               | Set to '0'                     |    |    |     |
|                           | D3                                                                                                                                                            | RGB/BGR Order                   |    |    |    |               | '0'=Increment, '1' = Decrement |    |    |     |
|                           | D2                                                                                                                                                            | Display Data Latch Order (MH)   |    |    |    |               | '0'=Increment, '1' = Decrement |    |    |     |
|                           | D1                                                                                                                                                            | Flip horizontal                 |    |    |    |               | Set to '0'                     |    |    |     |
|                           | D0                                                                                                                                                            | Flip vertical                   |    |    |    |               | Set to '0'                     |    |    |     |
| Restriction               | -                                                                                                                                                             |                                 |    |    |    |               |                                |    |    |     |
| Register Availability     | Status                                                                                                                                                        |                                 |    |    |    | Availability  |                                |    |    |     |
|                           | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                      |                                 |    |    |    | Yes           |                                |    |    |     |
|                           | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                       |                                 |    |    |    | Yes           |                                |    |    |     |
|                           | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                     |                                 |    |    |    | Yes           |                                |    |    |     |
|                           | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                      |                                 |    |    |    | Yes           |                                |    |    |     |
|                           | Sleep In or Booster Off                                                                                                                                       |                                 |    |    |    | Yes           |                                |    |    |     |
| Default                   | Status                                                                                                                                                        |                                 |    |    |    | Default Value |                                |    |    |     |
|                           | Power On Sequence                                                                                                                                             |                                 |    |    |    | 00h           |                                |    |    |     |
|                           | S/W Reset                                                                                                                                                     |                                 |    |    |    | 00h           |                                |    |    |     |
|                           | H/W Reset                                                                                                                                                     |                                 |    |    |    | 00h           |                                |    |    |     |
| Flow Chart                | <div><div><div>RDDMADCTL (0BH)</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div>                                                      |                                 |    |    |    |               |                                |    |    |     |
|                           | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                 |    |    |    |               |                                |    |    |     |

## 6.2.6 Read Display Pixel Format (0Ch)

| 0C H                      |                                                                                                                                                                                                                                                | RDDCOLMOD (Read Display COLMOD) |    |    |    |               |                     |    |    |     |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----|----|----|---------------|---------------------|----|----|-----|
|                           | R/W                                                                                                                                                                                                                                            | D7                              | D6 | D5 | D4 | D3            | D2                  | D1 | D0 | HEX |
| Command                   | Read                                                                                                                                                                                                                                           | 0                               | 0  | 0  | 0  | 1             | 1                   | 0  | 0  | 0C  |
| 1 <sup>st</sup> parameter | -                                                                                                                                                                                                                                              | D7                              | D6 | 0  | 0  | D3            | D2                  | 0  | 0  | -   |
| Description               | This command indicates the current status of the display as described in the table below:                                                                                                                                                      |                                 |    |    |    |               |                     |    |    |     |
|                           | Bit                                                                                                                                                                                                                                            | Description                     |    |    |    |               | Value               |    |    |     |
|                           | D7                                                                                                                                                                                                                                             | Reserved                        |    |    |    |               | Set to '0'          |    |    |     |
|                           | D6                                                                                                                                                                                                                                             | RGB Interface Pixel format      |    |    |    |               | '101' =16-bit/pixel |    |    |     |
|                           | D5                                                                                                                                                                                                                                             |                                 |    |    |    |               | '110' =18-bit/pixel |    |    |     |
|                           | D4                                                                                                                                                                                                                                             |                                 |    |    |    |               | '111' =24-bit/pixel |    |    |     |
|                           | D3                                                                                                                                                                                                                                             | Reserved                        |    |    |    |               | Set to '0'          |    |    |     |
|                           | D2                                                                                                                                                                                                                                             | DBI Interface Pixel format      |    |    |    |               | Set to '0'          |    |    |     |
|                           | D1                                                                                                                                                                                                                                             |                                 |    |    |    |               | Set to '0'          |    |    |     |
|                           | D0                                                                                                                                                                                                                                             |                                 |    |    |    |               | Set to '0'          |    |    |     |
| Restriction               | -                                                                                                                                                                                                                                              |                                 |    |    |    |               |                     |    |    |     |
| Register Availability     | Status                                                                                                                                                                                                                                         |                                 |    |    |    | Availability  |                     |    |    |     |
|                           | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                       |                                 |    |    |    | Yes           |                     |    |    |     |
|                           | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                        |                                 |    |    |    | Yes           |                     |    |    |     |
|                           | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                      |                                 |    |    |    | Yes           |                     |    |    |     |
|                           | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                       |                                 |    |    |    | Yes           |                     |    |    |     |
|                           | Sleep In or Booster Off                                                                                                                                                                                                                        |                                 |    |    |    | Yes           |                     |    |    |     |
| Default                   | Status                                                                                                                                                                                                                                         |                                 |    |    |    | Default Value |                     |    |    |     |
|                           | Power On Sequence                                                                                                                                                                                                                              |                                 |    |    |    | 70h           |                     |    |    |     |
|                           | S/W Reset                                                                                                                                                                                                                                      |                                 |    |    |    | No Change     |                     |    |    |     |
|                           | H/W Reset                                                                                                                                                                                                                                      |                                 |    |    |    | 70h           |                     |    |    |     |
| Flow Chart                | <div><div><div>RDDCOLMOD (0CH)</div><div>Host Driver</div><div>Send D[7:0]</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                 |    |    |    |               |                     |    |    |     |

## 6.2.7 Read Display Image Mode (0Dh)

| 0D H                      | RDDIM (Read Display Image Mode)                                                                                                                     |                             |    |    |               |                                           |    |    |    |     |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----|----|---------------|-------------------------------------------|----|----|----|-----|
|                           | R/W                                                                                                                                                 | D7                          | D6 | D5 | D4            | D3                                        | D2 | D1 | D0 | HEX |
| Command                   | Read                                                                                                                                                | 0                           | 0  | 0  | 0             | 1                                         | 1  | 0  | 1  | 0D  |
| 1 <sup>st</sup> parameter | -                                                                                                                                                   | D7                          | D6 | 0  | 0             | D3                                        | D2 | 0  | 0  | -   |
| Description               | This command indicates the current status of the display as described in the table below:                                                           |                             |    |    |               |                                           |    |    |    |     |
|                           | Bit                                                                                                                                                 | Description                 |    |    |               | Value                                     |    |    |    |     |
|                           | D7                                                                                                                                                  | Vertical Scrolling On/Off   |    |    |               | Set to '0'                                |    |    |    |     |
|                           | D6                                                                                                                                                  | Horizontal Scrolling On/off |    |    |               | Set to '0'                                |    |    |    |     |
|                           | D5                                                                                                                                                  | Inversion On/off            |    |    |               | '1' = Inversion On, '0' = Inversion Off   |    |    |    |     |
|                           | D4                                                                                                                                                  | All Pixel On                |    |    |               | '1' = White display, '0' = Normal display |    |    |    |     |
|                           | D3                                                                                                                                                  | All Pixel Off               |    |    |               | '1' = Black display, '0' = Normal display |    |    |    |     |
|                           | D2                                                                                                                                                  | Gamma Curve Selection       |    |    |               | '000'=GC0, '001'=GC1                      |    |    |    |     |
|                           | D1                                                                                                                                                  |                             |    |    |               | '010'=GC2, '011'=GC3                      |    |    |    |     |
|                           | D0                                                                                                                                                  |                             |    |    |               | '100' to '111' = not define               |    |    |    |     |
|                           |                                                                                                                                                     |                             |    |    |               |                                           |    |    |    |     |
| Restriction               | -                                                                                                                                                   |                             |    |    |               |                                           |    |    |    |     |
| Register Availability     | Status                                                                                                                                              |                             |    |    | Availability  |                                           |    |    |    |     |
|                           | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                            |                             |    |    | Yes           |                                           |    |    |    |     |
|                           | Normal Mode On, Idle Mode On, Sleep Out                                                                                                             |                             |    |    | Yes           |                                           |    |    |    |     |
|                           | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                           |                             |    |    | Yes           |                                           |    |    |    |     |
|                           | Partial Mode On, Idle Mode On, Sleep Out                                                                                                            |                             |    |    | Yes           |                                           |    |    |    |     |
|                           | Sleep In or Booster Off                                                                                                                             |                             |    |    | Yes           |                                           |    |    |    |     |
| Default                   | Status                                                                                                                                              |                             |    |    | Default Value |                                           |    |    |    |     |
|                           | Power On Sequence                                                                                                                                   |                             |    |    | 00h           |                                           |    |    |    |     |
|                           | S/W Reset                                                                                                                                           |                             |    |    | 00h           |                                           |    |    |    |     |
|                           | H/W Reset                                                                                                                                           |                             |    |    | 00h           |                                           |    |    |    |     |
| Flow Chart                | <div><div>RDDIM (0DH)</div><div>↓</div><div>Send D[7:0]</div></div> <div>Host Driver</div>                                                          |                             |    |    |               |                                           |    |    |    |     |
|                           | <div>Legend</div> <div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div> |                             |    |    |               |                                           |    |    |    |     |

## 6.2.8 Read Display Signal Mode (0EH)

| 0E H                      |                                                                                                                                                               | RDDSM (Read Display Signal Mode) |    |    |    |               |                                                |    |    |     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----|----|----|---------------|------------------------------------------------|----|----|-----|
|                           | R/W                                                                                                                                                           | D7                               | D6 | D5 | D4 | D3            | D2                                             | D1 | D0 | HEX |
| Command                   | Read                                                                                                                                                          | 0                                | 0  | 0  | 0  | 1             | 1                                              | 1  | 0  | 0E  |
| 1 <sup>st</sup> parameter | -                                                                                                                                                             | D7                               | D6 | 0  | 0  | D3            | D2                                             | 0  | 0  | -   |
| Description               | This command indicates the current status of the display as described in the table below:                                                                     |                                  |    |    |    |               |                                                |    |    |     |
|                           | Bit                                                                                                                                                           | Description                      |    |    |    |               | Value                                          |    |    |     |
|                           | D7                                                                                                                                                            | Tearing Effect Line On/Off       |    |    |    |               | Set to '0'                                     |    |    |     |
|                           | D6                                                                                                                                                            | Tearing Effect Line Mode         |    |    |    |               | Set to '0'                                     |    |    |     |
|                           | D5                                                                                                                                                            | Horizontal Sync. On/Off          |    |    |    |               | '1' = HS bit is '1', '0' = HS bit is '0'       |    |    |     |
|                           | D4                                                                                                                                                            | Vertical Sync. On/Off            |    |    |    |               | '1' = VS bit is '1', '0' = VS bit is '0'       |    |    |     |
|                           | D3                                                                                                                                                            | Pixel Clock. On/Off              |    |    |    |               | '1' = PCLK line is '1', '0' = PCLK line is '0' |    |    |     |
|                           | D2                                                                                                                                                            | Data Enable On/Off               |    |    |    |               | '1' = DE bit is '1', '0' = DE bit is '0'       |    |    |     |
|                           | D1                                                                                                                                                            | Not Define                       |    |    |    |               | Set to '0'                                     |    |    |     |
| Restriction               | -                                                                                                                                                             |                                  |    |    |    |               |                                                |    |    |     |
| Register Availability     | Status                                                                                                                                                        |                                  |    |    |    | Availability  |                                                |    |    |     |
|                           | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                      |                                  |    |    |    | Yes           |                                                |    |    |     |
|                           | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                       |                                  |    |    |    | Yes           |                                                |    |    |     |
|                           | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                     |                                  |    |    |    | Yes           |                                                |    |    |     |
|                           | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                      |                                  |    |    |    | Yes           |                                                |    |    |     |
|                           | Sleep In or Booster Off                                                                                                                                       |                                  |    |    |    | Yes           |                                                |    |    |     |
| Default                   | Status                                                                                                                                                        |                                  |    |    |    | Default Value |                                                |    |    |     |
|                           | Power On Sequence                                                                                                                                             |                                  |    |    |    | 00h           |                                                |    |    |     |
|                           | S/W Reset                                                                                                                                                     |                                  |    |    |    | 00h           |                                                |    |    |     |
|                           | H/W Reset                                                                                                                                                     |                                  |    |    |    | 00h           |                                                |    |    |     |
| Flow Chart                | <div><div><div>RDDSM (0EH)</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div>                                                          |                                  |    |    |    |               |                                                |    |    |     |
|                           | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                  |    |    |    |               |                                                |    |    |     |

## 6.2.9 Read Display Self-Diagnostic Result (0Fh)

| 0F H                      |                                                                                                                                                               | RDDSDR (Read Display Self-Diagnostic Result) |    |    |    |               |                                  |    |    |     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----|----|----|---------------|----------------------------------|----|----|-----|
|                           | R/W                                                                                                                                                           | D7                                           | D6 | D5 | D4 | D3            | D2                               | D1 | D0 | HEX |
| Command                   | Read                                                                                                                                                          | 0                                            | 0  | 0  | 0  | 1             | 1                                | 1  | 1  | 0F  |
| 1 <sup>st</sup> parameter | -                                                                                                                                                             | D7                                           | D6 | 0  | 0  | 0             | 0                                | 0  | 0  | -   |
| Description               | This command indicates the current status of the display as described in the table below:                                                                     |                                              |    |    |    |               |                                  |    |    |     |
|                           | Bit                                                                                                                                                           | Description                                  |    |    |    |               | Value                            |    |    |     |
|                           | D7                                                                                                                                                            | Register Loading Detection                   |    |    |    |               | See Section 5.5.1                |    |    |     |
|                           | D6                                                                                                                                                            | Functionality Detection                      |    |    |    |               | See Section 5.5.2                |    |    |     |
|                           | D5                                                                                                                                                            | Chip Attachment Detection                    |    |    |    |               | Set to '0'                       |    |    |     |
|                           | D4                                                                                                                                                            | Display Glass Break Detection                |    |    |    |               | Set to '0'                       |    |    |     |
|                           | D3                                                                                                                                                            | Not Define                                   |    |    |    |               | Set to '0'                       |    |    |     |
|                           | D2                                                                                                                                                            | Not Define                                   |    |    |    |               | Set to '0'                       |    |    |     |
|                           | D1                                                                                                                                                            | Not Define                                   |    |    |    |               | Set to '0'                       |    |    |     |
|                           | D0                                                                                                                                                            | Checksums Comparison                         |    |    |    |               | '1' = Checksums are not the same |    |    |     |
| Restriction               | -                                                                                                                                                             |                                              |    |    |    |               |                                  |    |    |     |
| Register Availability     | Status                                                                                                                                                        |                                              |    |    |    | Availability  |                                  |    |    |     |
|                           | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                      |                                              |    |    |    | Yes           |                                  |    |    |     |
|                           | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                       |                                              |    |    |    | Yes           |                                  |    |    |     |
|                           | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                     |                                              |    |    |    | Yes           |                                  |    |    |     |
|                           | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                      |                                              |    |    |    | Yes           |                                  |    |    |     |
|                           | Sleep In or Booster Off                                                                                                                                       |                                              |    |    |    | Yes           |                                  |    |    |     |
| Default                   | Status                                                                                                                                                        |                                              |    |    |    | Default Value |                                  |    |    |     |
|                           | Power On Sequence                                                                                                                                             |                                              |    |    |    | 00h           |                                  |    |    |     |
|                           | S/W Reset                                                                                                                                                     |                                              |    |    |    | 00h           |                                  |    |    |     |
|                           | H/W Reset                                                                                                                                                     |                                              |    |    |    | 00h           |                                  |    |    |     |
| Flow Chart                | <div><div><div>RDDSDR</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div>                                                               |                                              |    |    |    |               |                                  |    |    |     |
|                           | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                              |    |    |    |               |                                  |    |    |     |

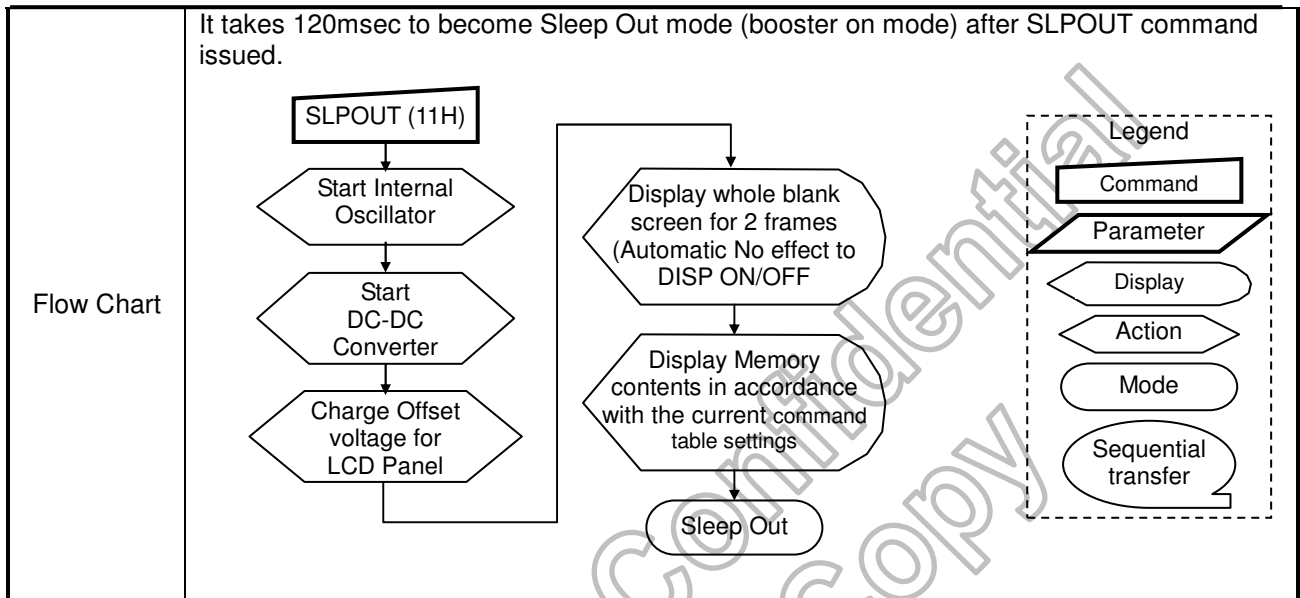
### 6.2.10 Sleep In (10h)

| 10 H                                      | SLPIN (Sleep In)                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|---------------|-----------------------------------------|---------------|-------------------------------------------|---------------|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 10  |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Parameter                                 | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Description                               | <p>This command causes the LCD module to enter the minimum power consumption mode. In this mode the DC/DC converter is stopped, Internal display oscillator is stopped, and panel scanning is stopped.</p> <div><p>Sleep In</p><p>Gate Output</p><p>Source Output</p><p>VCOM Output</p><p>Internal Oscillator</p><p>DC/DC Converter</p><p>STOP</p><p>0V</p><p>0V</p><p>STOP</p><p>OFF</p><p>0V or VDD</p></div>                                                                      |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Restriction                               | <p>This command has no effect when module is already in sleep in mode. Sleep In Mode can only be exit by the Sleep Out Command (11H). It will be necessary to wait 5msec before sending next command; this is to allow time for the supply voltages and clock circuits to stabilize. It will be necessary to wait 120msec after sending Sleep Out command (when in Sleep In Mode) before Sleep In command can be sent.</p>                                                           |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></tbody></table>                                              |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes           | Normal Mode On, Idle Mode On, Sleep Out | Yes           | Partial Mode On, Idle Mode Off, Sleep Out | Yes           | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Default                                   | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>Sleep In mode</td></tr><tr><td>S/W Reset</td><td>Sleep In mode</td></tr><tr><td>H/W Reset</td><td>Sleep In mode</td></tr></tbody></table>                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Sleep In mode | S/W Reset                               | Sleep In mode | H/W Reset                                 | Sleep In mode |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Power On Sequence                         | Sleep In mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| S/W Reset                                 | Sleep In mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| H/W Reset                                 | Sleep In mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Flow Chart                                | <p>It takes about 120msec to get into Sleep In mode after SLPIN command issued.</p> <div><p>SLPIN (10H)</p><p>Display whole blank screen (Automatic No effect to DISP ON/OFF Command)</p><p>Drain charge from LCD panel</p><p>Stop DC/DC Converter</p><p>Stop Internal Oscillator</p><p>Sleep In</p></div> <div><p>Legend</p><ul style="list-style-type: none"><li>Command</li><li>Parameter</li><li>Display</li><li>Action</li><li>Mode</li><li>Sequential transfer</li></ul></div> |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |



### 6.2.11 Sleep Out (11h)

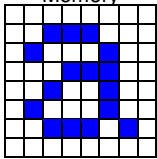
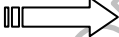
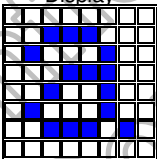
| 11 H                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SLPOUT (Sleep Out) |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----|----|----|----|----|----|----|-----|--------|--|---------------|------------------------------------------|--|---------------|-----------------------------------------|--|---------------|-------------------------------------------|--|---------------|------------------------------------------|--|-----|-------------------------|--|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | D7                 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0                  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 11  |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Parameter                                 | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                    |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Description                               | <p>This command turns off sleep mode. In this mode the DC/DC converter is enabled, Internal display oscillator is started, and panel scanning is started.</p> <p>Blanking display (over 1 frame display) *</p> <p>If DISPON 29H is set</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Restriction                               | <p>This command has no effect when module is already in sleep out mode. Sleep Out Mode can only be left by the Sleep In Command (10h). It will be necessary to wait 15msec before sending next command, this is to allow time for the supply voltages and clock circuits to stabilize. The display module loads all display supplier's factory default values to the registers during this 15msec and there cannot be any abnormal visual effect on the display image if factory default and register values are same when this load is done and when the display module is already Sleep Out –mode. The display module is doing self-diagnostic functions during this 15msec. It will be necessary to wait 120msec after sending Sleep In command (when in Sleep Out mode) before Sleep Out command can be sent.</p> |                    |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Register Availability                     | <table><tr><th colspan="2">Status</th><th>Availability</th></tr><tr><td colspan="2">Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td colspan="2">Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td colspan="2">Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td colspan="2">Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td colspan="2">Sleep In or Booster Off</td><td>Yes</td></tr></table>                                                                                                                                                                                                                                                                                                                                     |                    |    |    |    |    |    |    |    |     | Status |  | Availability  | Normal Mode On, Idle Mode Off, Sleep Out |  | Yes           | Normal Mode On, Idle Mode On, Sleep Out |  | Yes           | Partial Mode On, Idle Mode Off, Sleep Out |  | Yes           | Partial Mode On, Idle Mode On, Sleep Out |  | Yes | Sleep In or Booster Off |  | Yes |
| Status                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Availability       |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Normal Mode On, Idle Mode Off, Sleep Out  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Yes                |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Normal Mode On, Idle Mode On, Sleep Out   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Yes                |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Partial Mode On, Idle Mode Off, Sleep Out |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Yes                |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Partial Mode On, Idle Mode On, Sleep Out  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Yes                |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Sleep In or Booster Off                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Yes                |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Default                                   | <table><tr><th colspan="2">Status</th><th>Default Value</th></tr><tr><td colspan="2">Power On Sequence</td><td>Sleep In mode</td></tr><tr><td colspan="2">S/W Reset</td><td>Sleep In mode</td></tr><tr><td colspan="2">H/W Reset</td><td>Sleep In mode</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                    |    |    |    |    |    |    |    |     | Status |  | Default Value | Power On Sequence                        |  | Sleep In mode | S/W Reset                               |  | Sleep In mode | H/W Reset                                 |  | Sleep In mode |                                          |  |     |                         |  |     |
| Status                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Default Value      |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| Power On Sequence                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sleep In mode      |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| S/W Reset                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sleep In mode      |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |
| H/W Reset                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sleep In mode      |    |    |    |    |    |    |    |     |        |  |               |                                          |  |               |                                         |  |               |                                           |  |               |                                          |  |     |                         |  |     |



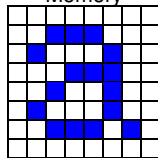
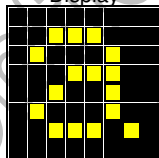
## 6.2.12 Normal Display Mode On (13h)

| 13 H                  | NORMON (Normal Display Mode On)                                                                                  |    |    |    |    |                |    |    |    |     |
|-----------------------|------------------------------------------------------------------------------------------------------------------|----|----|----|----|----------------|----|----|----|-----|
|                       | R/W                                                                                                              | D7 | D6 | D5 | D4 | D3             | D2 | D1 | D0 | HEX |
| Command               | Write                                                                                                            | 0  | 0  | 0  | 1  | 0              | 0  | 1  | 1  | 13  |
| Parameter             | No Parameter                                                                                                     |    |    |    |    |                |    |    |    |     |
| Description           | This command returns the display to normal mode. Normal display mode on means Partial mode off, Scroll mode Off. |    |    |    |    |                |    |    |    |     |
| Restriction           | This command has no effect when module is already in Normal Display mode.                                        |    |    |    |    |                |    |    |    |     |
| Register Availability | Status                                                                                                           |    |    |    |    | Availability   |    |    |    |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                         |    |    |    |    | Yes            |    |    |    |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                          |    |    |    |    | Yes            |    |    |    |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                        |    |    |    |    | Yes            |    |    |    |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                         |    |    |    |    | Yes            |    |    |    |     |
|                       | Sleep In or Booster Off                                                                                          |    |    |    |    | Yes            |    |    |    |     |
| Default               | Status                                                                                                           |    |    |    |    | Default Value  |    |    |    |     |
|                       | Power On Sequence                                                                                                |    |    |    |    | Normal mode ON |    |    |    |     |
|                       | S/W Reset                                                                                                        |    |    |    |    | Normal mode ON |    |    |    |     |
|                       | H/W Reset                                                                                                        |    |    |    |    | Normal mode ON |    |    |    |     |

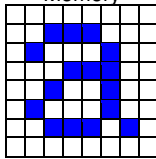
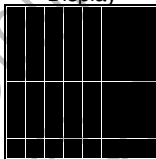
### 6.2.13 Display Inversion Off (20h)

| 20 H                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | INVOFF (Display Inversion Off) |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----------------------|-----------------------------------------|-----------------------|-------------------------------------------|-----------------------|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | D7                             | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0                              | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 20  |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Parameter                                 | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Description                               | <p>This command is used to recover from display inversion mode.<br/>This command makes no change of contents of frame memory.<br/>This command does not change any other status.</p> <p>(Example)</p> <div><div><p>Memory</p></div><div></div><div><p>Display</p></div></div> |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Restriction                               | This command has no effect when module is already in inversion off mode                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table>                                                                                                                         |                                |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes                   | Normal Mode On, Idle Mode On, Sleep Out | Yes                   | Partial Mode On, Idle Mode Off, Sleep Out | Yes                   | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Inversion off</td></tr><tr><td>S/W Reset</td><td>Display Inversion off</td></tr><tr><td>H/W Reset</td><td>Display Inversion off</td></tr></table>                                                                                                                                                                                                                                                                                  |                                |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Inversion off | S/W Reset                               | Display Inversion off | H/W Reset                                 | Display Inversion off |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Power On Sequence                         | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| S/W Reset                                 | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| H/W Reset                                 | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Flow Chart                                | <div><div><p>Display Inversion On Mode</p><p>↓</p><p>INVOFF (20H)</p><p>↓</p><p>Display Inversion OFF Mode</p></div><div><p>Legend</p><ul style="list-style-type: none"><li>Command</li><li>Parameter</li><li>Display</li><li>Action</li><li>Mode</li><li>Sequential transfer</li></ul></div></div>                                                                                                                                                                                                                               |                                |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |

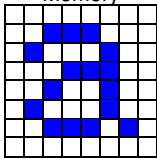
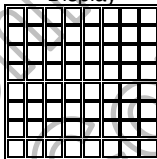
## 6.2.14 Display Inversion On (21h)

| 21 H                                      |                                                                                                                                                                                                                                                                                                                                                                                                                               | INVON (Display Inversion On) |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----------------------|-----------------------------------------|-----------------------|-------------------------------------------|-----------------------|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                           | D7                           | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                                         | 0                            | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 21  |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Parameter                                 | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                  |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Description                               | <p>This command is used to recover from display inversion mode. This command makes no change of contents of frame memory. This command does not change any other status.</p> <p>(Example)</p> <div><div><p>Memory</p></div><div><p>Display</p></div></div> |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Restriction                               | This command has no effect when module is already in inversion on mode                                                                                                                                                                                                                                                                                                                                                        |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table>                     |                              |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes                   | Normal Mode On, Idle Mode On, Sleep Out | Yes                   | Partial Mode On, Idle Mode Off, Sleep Out | Yes                   | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                  |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Inversion off</td></tr><tr><td>S/W Reset</td><td>Display Inversion off</td></tr><tr><td>H/W Reset</td><td>Display Inversion off</td></tr></table>                                                                                                                                                                              |                              |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Inversion off | S/W Reset                               | Display Inversion off | H/W Reset                                 | Display Inversion off |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                 |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Power On Sequence                         | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                                         |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| S/W Reset                                 | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                                         |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| H/W Reset                                 | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                                         |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |
| Flow Chart                                | <div><div><p>Display Inversion OFF Mode</p><p>↓</p><p>INVON (21H)</p><p>↓</p><p>Display Inversion ON Mode</p></div><div><p>Legend</p><p>Command</p><p>Parameter</p><p>Display</p><p>Action</p><p>Mode</p><p>Sequential transfer</p></div></div>                                                                                                                                                                               |                              |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |                         |     |

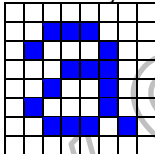
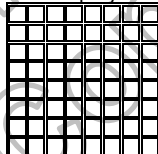
## 6.2.15 All Pixel Off (22h)

| 22 H                                      | ALLPOFF (All Pixel Off)                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|---------------|-----------------------------------------|---------------|-------------------------------------------|---------------|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 0  | 22  |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Parameter                                 | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Description                               | <p>This command turns the display panel black in Sleep Out mode and a status of the Display On/Off register can be on or off.<br/>This command makes no change of contents of frame memory.<br/>This command does not change any other status.</p> <p>(Example)</p> <div><div>Memory</div><div>Display</div></div> |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Restriction                               | This command has no effect when module is already in all pixel off mode                                                                                                                                                                                                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></tbody></table>                                               |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes           | Normal Mode On, Idle Mode On, Sleep Out | Yes           | Partial Mode On, Idle Mode Off, Sleep Out | Yes           | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Default                                   | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>All pixel off</td></tr><tr><td>S/W Reset</td><td>All pixel off</td></tr><tr><td>H/W Reset</td><td>All pixel off</td></tr></tbody></table>                                                                                                                                                                                                                                |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | All pixel off | S/W Reset                               | All pixel off | H/W Reset                                 | All pixel off |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Power On Sequence                         | All pixel off                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| S/W Reset                                 | All pixel off                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| H/W Reset                                 | All pixel off                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Flow Chart                                | <div><div>Normal Display Mode ON</div><div>ALLPOFF</div><div>Black Display</div></div> <div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div>                                                                                                                                                                                                                                             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |

## 6.2.16 All Pixel On (23h)

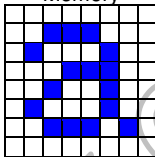
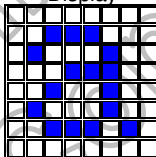
| 23 H                                      | ALLPON (All Pixel On)                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|---------------|-----------------------------------------|---------------|-------------------------------------------|---------------|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 1  | 23  |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Parameter                                 | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Description                               | <p>This command turns the display panel white in Sleep Out mode and a status of the Display On/Off –register can be on or off.<br/>This command makes no change of contents of frame memory.<br/>This command does not change any other status.</p> <p>(Example)</p> <div><div>Memory</div><div>Display</div><div></div></div> |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Restriction                               | This command has no effect when module is already in all pixel on mode                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table>                                                                                         |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes           | Normal Mode On, Idle Mode On, Sleep Out | Yes           | Partial Mode On, Idle Mode Off, Sleep Out | Yes           | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>All pixel off</td></tr><tr><td>S/W Reset</td><td>All pixel off</td></tr><tr><td>H/W Reset</td><td>All pixel off</td></tr></table>                                                                                                                                                                                                                                                                          |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | All pixel off | S/W Reset                               | All pixel off | H/W Reset                                 | All pixel off |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Power On Sequence                         | All pixel off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| S/W Reset                                 | All pixel off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| H/W Reset                                 | All pixel off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |
| Flow Chart                                | <div><div>Normal Display Mode ON</div><div>ALLPON</div><div>White Display</div></div> <div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div>                                                                                                                                                                                                                                                          |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |                         |     |

## 6.2.17 Display Off (28h)

| 28 H                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | DISPOFF (Display off) |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-------------|-----------------------------------------|-------------|-------------------------------------------|-------------|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | D7                    | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0                     | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 28  |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Parameter                                 | NO Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Description                               | <p>This command is used to enter into DISPLAY OFF mode. In this mode, the output from Frame Memory is disabled and blank page inserted.</p> <p>This command makes no change of contents of frame memory.</p> <p>This command does not change any other status.</p> <p>There will be no abnormal visible effect on the display.</p> <p>(Example)</p> <div><div>Memory</div><div>Display</div></div> |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Restriction                               | This command has no effect when module is already in display off mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></tbody></table>                                                                                                                               |                       |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes         | Normal Mode On, Idle Mode On, Sleep Out | Yes         | Partial Mode On, Idle Mode Off, Sleep Out | Yes         | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Default                                   | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>Display Off</td></tr><tr><td>S/W Reset</td><td>Display Off</td></tr><tr><td>H/W Reset</td><td>Display Off</td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                      |                       |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Off | S/W Reset                               | Display Off | H/W Reset                                 | Display Off |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Power On Sequence                         | Display Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| S/W Reset                                 | Display Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| H/W Reset                                 | Display Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Flow Chart                                | <div><div>Display On Mode</div><div>DISPOFF</div><div>Display OFF Mode</div></div> <div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div>                                                                                                                                                                                                                                                                                                                                 |                       |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |



## 6.2.18 Display On (29h)

| 29 H                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DISPON (Display on) |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-------------|-----------------------------------------|-------------|-------------------------------------------|-------------|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | D7                  | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0                   | 0  | 1  | 0  | 1  | 0  | 0  | 1  | 29  |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Parameter                                 | NO Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Description                               | <p>This command is used to recover from DISPLAY OFF mode. Output from the Frame Memory is enabled.</p> <p>This command makes no change of contents of frame memory.</p> <p>This command does not change any other status.</p> <p>(Example)</p> <div><div><p>Memory</p></div><div></div><div><p>Display</p></div></div> |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Restriction                               | This command has no effect when module is already in display on mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table>                                                                                                                                                                  |                     |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes         | Normal Mode On, Idle Mode On, Sleep Out | Yes         | Partial Mode On, Idle Mode Off, Sleep Out | Yes         | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Off</td></tr><tr><td>S/W Reset</td><td>Display Off</td></tr><tr><td>H/W Reset</td><td>Display Off</td></tr></table>                                                                                                                                                                                                                                                                                                                                                         |                     |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Off | S/W Reset                               | Display Off | H/W Reset                                 | Display Off |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Power On Sequence                         | Display Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| S/W Reset                                 | Display Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| H/W Reset                                 | Display Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |
| Flow Chart                                | <div><div><div>Display Off Mode</div><div>↓</div><div>DISPON</div><div>↓</div><div>Display ON Mode</div></div><div><p>Legend</p><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                                                                                                                                                                                                                                                                              |                     |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |                         |     |

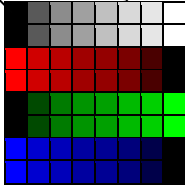
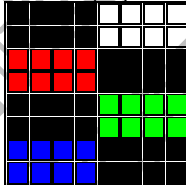
## 6.2.19 Memory Access Control (36h)

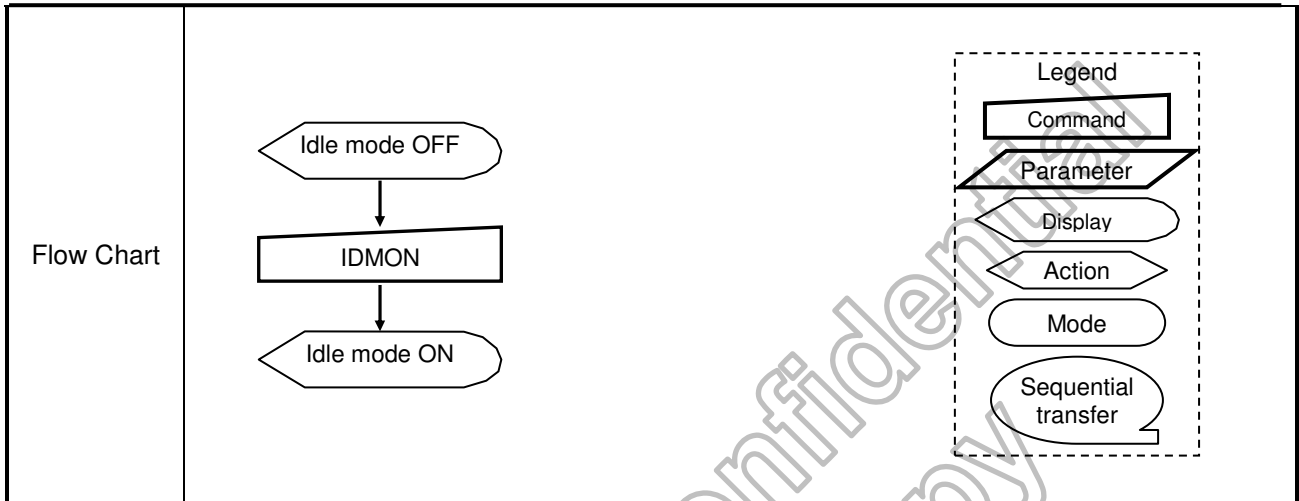
| 36 H                  | MADCTR (Memory Access Control)                                                                                                                                |                           |    |    |                                          |                                                                             |    |    |    |     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----|----|------------------------------------------|-----------------------------------------------------------------------------|----|----|----|-----|
|                       | R/W                                                                                                                                                           | D7                        | D6 | D5 | D4                                       | D3                                                                          | D2 | D1 | D0 | HEX |
| Command               | Write                                                                                                                                                         | 0                         | 0  | 1  | 1                                        | 0                                                                           | 1  | 1  | 0  | 36  |
| Parameter             | -                                                                                                                                                             | MY                        | MX | -  | -                                        | RGB                                                                         | SS | -  | -  | -   |
| Description           | This command defines read/write scanning direction of frame memory.<br>This command makes no change on the other driver status.                               |                           |    |    |                                          |                                                                             |    |    |    |     |
|                       | Bit Assignment                                                                                                                                                |                           |    |    |                                          |                                                                             |    |    |    |     |
|                       | BIT                                                                                                                                                           | NAME                      |    |    |                                          | DESCRIPTION                                                                 |    |    |    |     |
|                       | D7                                                                                                                                                            | PAGE ADDRESS ORDER (MY)   |    |    |                                          | These 2 bits controls interface to display direction                        |    |    |    |     |
|                       | D6                                                                                                                                                            | COLUMN ADDRESS ORDER (MX) |    |    |                                          |                                                                             |    |    |    |     |
|                       | D3                                                                                                                                                            | RGB-BGR ORDER (BGR)       |    |    |                                          | Color selector switch control<br>0=RGB color sequence, 1=BGR color sequence |    |    |    |     |
| D2                    | Horizontal ORDER (SS)                                                                                                                                         |                           |    |    | LCD horizontal refresh direction control |                                                                             |    |    |    |     |
| Restriction           | -                                                                                                                                                             |                           |    |    |                                          |                                                                             |    |    |    |     |
| Register Availability | Status                                                                                                                                                        |                           |    |    |                                          | Availability                                                                |    |    |    |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                      |                           |    |    |                                          | Yes                                                                         |    |    |    |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                       |                           |    |    |                                          | Yes                                                                         |    |    |    |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                     |                           |    |    |                                          | Yes                                                                         |    |    |    |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                      |                           |    |    |                                          | Yes                                                                         |    |    |    |     |
|                       | Sleep In or Booster Off                                                                                                                                       |                           |    |    |                                          | Yes                                                                         |    |    |    |     |
| Default               | Status                                                                                                                                                        |                           |    |    |                                          | Default Value                                                               |    |    |    |     |
|                       | Power On Sequence                                                                                                                                             |                           |    |    |                                          | 00h                                                                         |    |    |    |     |
|                       | S/W Reset                                                                                                                                                     |                           |    |    |                                          | No Change                                                                   |    |    |    |     |
|                       | H/W Reset                                                                                                                                                     |                           |    |    |                                          | 00h                                                                         |    |    |    |     |
| Flow Chart            | <div><div>MADCTR</div><div>↓</div><div>Parameter</div></div>                                                                                                  |                           |    |    |                                          |                                                                             |    |    |    |     |
|                       | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                           |    |    |                                          |                                                                             |    |    |    |     |

### 6.2.20 Idle Mode Off (38h)

| 38 H                  | IDMOFF (Idle Mode Off)                                                                                                                                                                                                                        |    |    |    |    |    |               |    |    |     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|---------------|----|----|-----|
|                       | R/W                                                                                                                                                                                                                                           | D7 | D6 | D5 | D4 | D3 | D2            | D1 | D0 | HEX |
| Command               | Write                                                                                                                                                                                                                                         | 0  | 0  | 1  | 1  | 1  | 0             | 0  | 0  | 38  |
| Parameter             | No Parameter                                                                                                                                                                                                                                  |    |    |    |    |    |               |    |    |     |
| Description           | This command is used to recover from idle mode on<br>In the idle off mode, display panel can display maximum 16.7M colors                                                                                                                     |    |    |    |    |    |               |    |    |     |
| Restriction           | This command has no effect when module is already in idle off mode.                                                                                                                                                                           |    |    |    |    |    |               |    |    |     |
| Register Availability | Status                                                                                                                                                                                                                                        |    |    |    |    |    | Availability  |    |    |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                      |    |    |    |    |    | Yes           |    |    |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                       |    |    |    |    |    | Yes           |    |    |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                     |    |    |    |    |    | Yes           |    |    |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                      |    |    |    |    |    | Yes           |    |    |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                       |    |    |    |    |    | Yes           |    |    |     |
| Default               | Status                                                                                                                                                                                                                                        |    |    |    |    |    | Default Value |    |    |     |
|                       | Power On Sequence                                                                                                                                                                                                                             |    |    |    |    |    | Idle Mode Off |    |    |     |
|                       | S/W Reset                                                                                                                                                                                                                                     |    |    |    |    |    | Idle Mode Off |    |    |     |
|                       | H/W Reset                                                                                                                                                                                                                                     |    |    |    |    |    | Idle Mode Off |    |    |     |
| Flow Chart            | <div><div><div>Idle mode ON</div><div>IDMOFF (38)</div><div>Idle mode OFF</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |    |    |    |    |    |               |    |    |     |

## 6.2.21 Idle Mode On (39h)

| 39 H                                      | IDMON (Idle Mode on)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|-----|--------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------|-------------------------------------------|---------------|------------------------------------------|-----------|-------------------------|-----------|-----|-----------|-----------|-----------|---------|-----------|-----------|-----------|-------|-----------|-----------|-----------|------|-----------|-----------|-----------|--------|-----------|-----------|-----------|-------|-----------|-----------|-----------|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | D7                                                                                                                      | D6                                                                                                                      | D5                                                                                                                      | D4 | D3 | D2 | D1 | D0 | HEX |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                                                       | 0                                                                                                                       | 1                                                                                                                       | 1  | 1  | 0  | 0  | 1  | 39  |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Parameter                                 | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Description                               | <p>This command is used to enter idle mode on.<br/>In the idle on mode, color expression is reduced. The primary and the secondary colors using MSB of each R, G and B in the Frame Memory, 8 color depth data is displayed.</p> <p>(Example)</p> <div><div>Top-Left (0,0)</div><div>Memory</div></div> <div><div>Display</div></div> <table><thead><tr><th>Color</th><th>R<sub>7</sub> R<sub>6</sub> R<sub>5</sub> R<sub>4</sub> R<sub>3</sub> R<sub>2</sub> R<sub>1</sub> R<sub>0</sub></th><th>G<sub>7</sub> G<sub>6</sub> G<sub>5</sub> G<sub>4</sub> G<sub>3</sub> G<sub>2</sub> G<sub>1</sub> G<sub>0</sub></th><th>B<sub>7</sub> B<sub>6</sub> B<sub>5</sub> B<sub>4</sub> B<sub>3</sub> B<sub>2</sub> B<sub>1</sub> B<sub>0</sub></th></tr></thead><tbody><tr><td>Black</td><td>0xxx xxxx</td><td>0xxx xxxx</td><td>0xxx xxxx</td></tr><tr><td>Blue</td><td>0xxx xxxx</td><td>0xxx xxxx</td><td>1xxx xxxx</td></tr><tr><td>Red</td><td>1xxx xxxx</td><td>0xxx xxxx</td><td>0xxx xxxx</td></tr><tr><td>Magenta</td><td>1xxx xxxx</td><td>0xxx xxxx</td><td>1xxx xxxx</td></tr><tr><td>Green</td><td>0xxx xxxx</td><td>1xxx xxxx</td><td>0xxx xxxx</td></tr><tr><td>Cyan</td><td>0xxx xxxx</td><td>1xxx xxxx</td><td>1xxx xxxx</td></tr><tr><td>Yellow</td><td>1xxx xxxx</td><td>1xxx xxxx</td><td>0xxx xxxx</td></tr><tr><td>White</td><td>1xxx xxxx</td><td>1xxx xxxx</td><td>1xxx xxxx</td></tr></tbody></table> |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     | Color  | R <sub>7</sub> R <sub>6</sub> R <sub>5</sub> R <sub>4</sub> R <sub>3</sub> R <sub>2</sub> R <sub>1</sub> R <sub>0</sub> | G <sub>7</sub> G <sub>6</sub> G <sub>5</sub> G <sub>4</sub> G <sub>3</sub> G <sub>2</sub> G <sub>1</sub> G <sub>0</sub> | B <sub>7</sub> B <sub>6</sub> B <sub>5</sub> B <sub>4</sub> B <sub>3</sub> B <sub>2</sub> B <sub>1</sub> B <sub>0</sub> | Black                                   | 0xxx xxxx     | 0xxx xxxx                                 | 0xxx xxxx     | Blue                                     | 0xxx xxxx | 0xxx xxxx               | 1xxx xxxx | Red | 1xxx xxxx | 0xxx xxxx | 0xxx xxxx | Magenta | 1xxx xxxx | 0xxx xxxx | 1xxx xxxx | Green | 0xxx xxxx | 1xxx xxxx | 0xxx xxxx | Cyan | 0xxx xxxx | 1xxx xxxx | 1xxx xxxx | Yellow | 1xxx xxxx | 1xxx xxxx | 0xxx xxxx | White | 1xxx xxxx | 1xxx xxxx | 1xxx xxxx |
|                                           | Color                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R <sub>7</sub> R <sub>6</sub> R <sub>5</sub> R <sub>4</sub> R <sub>3</sub> R <sub>2</sub> R <sub>1</sub> R <sub>0</sub> | G <sub>7</sub> G <sub>6</sub> G <sub>5</sub> G <sub>4</sub> G <sub>3</sub> G <sub>2</sub> G <sub>1</sub> G <sub>0</sub> | B <sub>7</sub> B <sub>6</sub> B <sub>5</sub> B <sub>4</sub> B <sub>3</sub> B <sub>2</sub> B <sub>1</sub> B <sub>0</sub> |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
|                                           | Black                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0xxx xxxx                                                                                                               | 0xxx xxxx                                                                                                               | 0xxx xxxx                                                                                                               |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Blue                                      | 0xxx xxxx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0xxx xxxx                                                                                                               | 1xxx xxxx                                                                                                               |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Red                                       | 1xxx xxxx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0xxx xxxx                                                                                                               | 0xxx xxxx                                                                                                               |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Magenta                                   | 1xxx xxxx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0xxx xxxx                                                                                                               | 1xxx xxxx                                                                                                               |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Green                                     | 0xxx xxxx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1xxx xxxx                                                                                                               | 0xxx xxxx                                                                                                               |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Cyan                                      | 0xxx xxxx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1xxx xxxx                                                                                                               | 1xxx xxxx                                                                                                               |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Yellow                                    | 1xxx xxxx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1xxx xxxx                                                                                                               | 0xxx xxxx                                                                                                               |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| White                                     | 1xxx xxxx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1xxx xxxx                                                                                                               | 1xxx xxxx                                                                                                               |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Restriction                               | This command has no effect when module is already in idle on mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     | Status | Availability                                                                                                            | Normal Mode On, Idle Mode Off, Sleep Out                                                                                | Yes                                                                                                                     | Normal Mode On, Idle Mode On, Sleep Out | Yes           | Partial Mode On, Idle Mode Off, Sleep Out | Yes           | Partial Mode On, Idle Mode On, Sleep Out | Yes       | Sleep In or Booster Off | Yes       |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Default                                   | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>Idle Mode Off</td></tr><tr><td>S/W Reset</td><td>Idle Mode Off</td></tr><tr><td>H/W Reset</td><td>Idle Mode Off</td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     | Status | Default Value                                                                                                           | Power On Sequence                                                                                                       | Idle Mode Off                                                                                                           | S/W Reset                               | Idle Mode Off | H/W Reset                                 | Idle Mode Off |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| Power On Sequence                         | Idle Mode Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| S/W Reset                                 | Idle Mode Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |
| H/W Reset                                 | Idle Mode Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                         |                                                                                                                         |                                                                                                                         |    |    |    |    |    |     |        |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                         |               |                                           |               |                                          |           |                         |           |     |           |           |           |         |           |           |           |       |           |           |           |      |           |           |           |        |           |           |           |       |           |           |           |



## 6.2.22 Write Display Brightness (51h)

| 51 H                  | WRDISBV (Write Display Brightness)                                                                                                                                      |      |                  |      |      |               |      |      |      |     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------|------|------|---------------|------|------|------|-----|
|                       | R/W                                                                                                                                                                     | D7   | D6               | D5   | D4   | D3            | D2   | D1   | D0   | HEX |
| Command               | Write                                                                                                                                                                   | 0    | 1                | 0    | 1    | 0             | 0    | 0    | 1    | 51  |
| Parameter             | Write                                                                                                                                                                   | DBV7 | DBV6             | DBV5 | DBV4 | DBV3          | DBV2 | DBV1 | DBV1 | -   |
| Description           | This command is used to adjust brightness value.<br>In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness. |      |                  |      |      |               |      |      |      |     |
|                       | DBV[7:0]                                                                                                                                                                |      | Brightness Ratio |      |      | Brightness %  |      |      |      |     |
|                       | 00h                                                                                                                                                                     |      | 0/256            |      |      | 0%            |      |      |      |     |
|                       | 01h                                                                                                                                                                     |      | 2/256            |      |      | 0.78%         |      |      |      |     |
|                       | ⋮                                                                                                                                                                       |      | ⋮                |      |      | ⋮             |      |      |      |     |
|                       | FEh                                                                                                                                                                     |      | 255/256          |      |      | 99.6%         |      |      |      |     |
|                       | FFH                                                                                                                                                                     |      | 256/256          |      |      | 100%          |      |      |      |     |
| Restriction           | The display supplier cannot use this command for tuning (e.g. factory tuning, etc.).                                                                                    |      |                  |      |      |               |      |      |      |     |
| Register Availability | Status                                                                                                                                                                  |      |                  |      |      | Availability  |      |      |      |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                |      |                  |      |      | Yes           |      |      |      |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                 |      |                  |      |      | Yes           |      |      |      |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                               |      |                  |      |      | Yes           |      |      |      |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                |      |                  |      |      | Yes           |      |      |      |     |
|                       | Sleep In or Booster Off                                                                                                                                                 |      |                  |      |      | Yes           |      |      |      |     |
| Default               | Status                                                                                                                                                                  |      |                  |      |      | Default Value |      |      |      |     |
|                       | Power On Sequence                                                                                                                                                       |      |                  |      |      | 00h           |      |      |      |     |
|                       | S/W Reset                                                                                                                                                               |      |                  |      |      | 00h           |      |      |      |     |
|                       | H/W Reset                                                                                                                                                               |      |                  |      |      | 00h           |      |      |      |     |
| Flow Chart            | <div><div>WRDISBV</div><div>↓</div><div>DBV[7:0]</div><div>↓</div><div>New Brightness Loaded</div></div>                                                                |      |                  |      |      |               |      |      |      |     |
|                       | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>           |      |                  |      |      |               |      |      |      |     |

### 6.2.23 Read Display Brightness (52h)

| 52 H                  |                                                                                                                                                                                                                                                   | RDISBV (Read Display Brightness) |      |      |      |               |      |      |      |     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------|------|------|---------------|------|------|------|-----|
|                       | R/W                                                                                                                                                                                                                                               | D7                               | D6   | D5   | D4   | D3            | D2   | D1   | D0   | HEX |
| Command               | Read                                                                                                                                                                                                                                              | 0                                | 1    | 0    | 1    | 0             | 0    | 1    | 0    | 52  |
| Parameter             | Read                                                                                                                                                                                                                                              | DBV7                             | DBV6 | DBV5 | DBV4 | DBV3          | DBV2 | DBV1 | DBV1 | -   |
| Description           | This command is used to read brightness value.<br>In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.                                                                             |                                  |      |      |      |               |      |      |      |     |
| Restriction           | -                                                                                                                                                                                                                                                 |                                  |      |      |      |               |      |      |      |     |
| Register Availability | Status                                                                                                                                                                                                                                            |                                  |      |      |      | Availability  |      |      |      |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                          |                                  |      |      |      | Yes           |      |      |      |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                           |                                  |      |      |      | Yes           |      |      |      |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                         |                                  |      |      |      | Yes           |      |      |      |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                          |                                  |      |      |      | Yes           |      |      |      |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                           |                                  |      |      |      | Yes           |      |      |      |     |
| Default               | Status                                                                                                                                                                                                                                            |                                  |      |      |      | Default Value |      |      |      |     |
|                       | Power On Sequence                                                                                                                                                                                                                                 |                                  |      |      |      | 00h           |      |      |      |     |
|                       | S/W Reset                                                                                                                                                                                                                                         |                                  |      |      |      | 00h           |      |      |      |     |
|                       | H/W Reset                                                                                                                                                                                                                                         |                                  |      |      |      | 00h           |      |      |      |     |
| Flow Chart            | <div><div><div>RDISBV</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                  |      |      |      |               |      |      |      |     |

## 6.2.24 Write CTRL Display (53h)

| 53 H                  |                                                                                                                                                               | WRCTRLD (Write CTRL Display Value)                                               |    |       |    |               |    |    |    |     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----|-------|----|---------------|----|----|----|-----|
|                       | R/W                                                                                                                                                           | D7                                                                               | D6 | D5    | D4 | D3            | D2 | D1 | D0 | HEX |
| Command               | Write                                                                                                                                                         | 0                                                                                | 1  | 0     | 1  | 0             | 0  | 1  | 1  | 53  |
| Parameter             | Write                                                                                                                                                         | 0                                                                                | 0  | BCTRL | 0  | DD            | BL | 0  | 0  | -   |
| Description           | This command is used to control display brightness.                                                                                                           |                                                                                  |    |       |    |               |    |    |    |     |
|                       | Bits                                                                                                                                                          | Value                                                                            |    |       |    |               |    |    |    |     |
|                       | BCTRL                                                                                                                                                         | '0' off, DBV[7:0] is 00h; PWM keep low<br>'1' On, DBV[7:0] is active; PWM output |    |       |    |               |    |    |    |     |
|                       | DD                                                                                                                                                            | '0' Brightness dimming is off<br>'1' Brightness dimming is on                    |    |       |    |               |    |    |    |     |
|                       | BL                                                                                                                                                            | '0' Backlight Control is off<br>'1' Backlight Control is on                      |    |       |    |               |    |    |    |     |
| Restriction           | -                                                                                                                                                             |                                                                                  |    |       |    |               |    |    |    |     |
| Register Availability | Status                                                                                                                                                        |                                                                                  |    |       |    | Availability  |    |    |    |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                      |                                                                                  |    |       |    | Yes           |    |    |    |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                       |                                                                                  |    |       |    | Yes           |    |    |    |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                     |                                                                                  |    |       |    | Yes           |    |    |    |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                      |                                                                                  |    |       |    | Yes           |    |    |    |     |
|                       | Sleep In or Booster Off                                                                                                                                       |                                                                                  |    |       |    | Yes           |    |    |    |     |
| Default               | Status                                                                                                                                                        |                                                                                  |    |       |    | Default Value |    |    |    |     |
|                       | Power On Sequence                                                                                                                                             |                                                                                  |    |       |    | 00h           |    |    |    |     |
|                       | S/W Reset                                                                                                                                                     |                                                                                  |    |       |    | 00h           |    |    |    |     |
|                       | H/W Reset                                                                                                                                                     |                                                                                  |    |       |    | 00h           |    |    |    |     |
| Flow Chart            | <div><div>WRCTRLD</div><div>BCTRL, DD, BL</div><div>New Control Value Loaded</div></div>                                                                      |                                                                                  |    |       |    |               |    |    |    |     |
|                       | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                                                                  |    |       |    |               |    |    |    |     |



### 6.2.25 Read CTRL Display Value (54h)

| 54 H                                      | RDCTRLD (Read CTRL Display Value)                                                                                                                                                                                                                                                                                                                                                                         |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-------|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                       | D7 | D6 | D5    | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Command                                   | Read                                                                                                                                                                                                                                                                                                                                                                                                      | 0  | 1  | 0     | 1  | 0  | 1  | 0  | 0  | 54  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Parameter                                 | Read                                                                                                                                                                                                                                                                                                                                                                                                      | 0  | 0  | BCTRL | 0  | DD | BL | 0  | 0  | -   |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Description                               | This command is used to read brightness value.<br>In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.                                                                                                                                                                                                                                     |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table> |    |    |       |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                              |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                                |    |    |       |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | 00h | S/W Reset                               | 00h | H/W Reset                                 | 00h |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                             |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| S/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| H/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Flow Chart                                | <div><div><div>RDCTRLD</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                                        |    |    |       |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |

## 6.2.26 Write Content Adaptive Brightness Control (55h)

| 55 H                  |                                                                                                                                                    | WRCABC (Write Content Adaptive Brightness Control) |                                    |    |    |               |    |    |    |     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------|----|----|---------------|----|----|----|-----|
|                       | R/W                                                                                                                                                | D7                                                 | D6                                 | D5 | D4 | D3            | D2 | D1 | D0 | HEX |
| Command               | Write                                                                                                                                              | 0                                                  | 1                                  | 0  | 1  | 0             | 1  | 0  | 1  | 55  |
| Parameter             | Write                                                                                                                                              | 0                                                  | 0                                  | 0  | 0  | 0             | 0  | C1 | C0 | -   |
| Description           | This command is used to set parameters for image content based adaptive brightness control functionality.                                          |                                                    |                                    |    |    |               |    |    |    |     |
|                       | C1                                                                                                                                                 | C0                                                 | Function                           |    |    |               |    |    |    |     |
|                       | 0                                                                                                                                                  | 0                                                  | Off                                |    |    |               |    |    |    |     |
|                       | 0                                                                                                                                                  | 1                                                  | User Interface Image (UI mode)     |    |    |               |    |    |    |     |
|                       | 1                                                                                                                                                  | 0                                                  | Still Picture Image (Still mode)   |    |    |               |    |    |    |     |
|                       | 1                                                                                                                                                  | 1                                                  | Moving Picture Image (Moving mode) |    |    |               |    |    |    |     |
| Restriction           | This register is synchronized with V-sync by internal circuit.                                                                                     |                                                    |                                    |    |    |               |    |    |    |     |
| Register Availability | Status                                                                                                                                             |                                                    |                                    |    |    | Availability  |    |    |    |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                           |                                                    |                                    |    |    | Yes           |    |    |    |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                            |                                                    |                                    |    |    | Yes           |    |    |    |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                          |                                                    |                                    |    |    | Yes           |    |    |    |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                           |                                                    |                                    |    |    | Yes           |    |    |    |     |
|                       | Sleep In or Booster Off                                                                                                                            |                                                    |                                    |    |    | Yes           |    |    |    |     |
| Default               | Status                                                                                                                                             |                                                    |                                    |    |    | Default Value |    |    |    |     |
|                       | Power On Sequence                                                                                                                                  |                                                    |                                    |    |    | 00h           |    |    |    |     |
|                       | S/W Reset                                                                                                                                          |                                                    |                                    |    |    | 00h           |    |    |    |     |
|                       | H/W Reset                                                                                                                                          |                                                    |                                    |    |    | 00h           |    |    |    |     |
| Flow Chart            | <div><div>WRCABC</div><div>C[1:0]</div><div>New Adaptive Image Mode</div></div>                                                                    |                                                    |                                    |    |    |               |    |    |    |     |
|                       | <div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div> |                                                    |                                    |    |    |               |    |    |    |     |

## 6.2.27 Read Content Adaptive Brightness Control (56h)

| 56 H                  | RDCABC (Read Content Adaptive Brightness Control)                                                                                                             |    |                                    |    |    |               |    |    |    |     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------------------|----|----|---------------|----|----|----|-----|
|                       | R/W                                                                                                                                                           | D7 | D6                                 | D5 | D4 | D3            | D2 | D1 | D0 | HEX |
| Command               | Read                                                                                                                                                          | 0  | 1                                  | 0  | 1  | 1             | 1  | 1  | 0  | 56  |
| Parameter             | Read                                                                                                                                                          | 0  | 0                                  | 0  | 0  | 0             | 0  | C1 | C0 | -   |
| Description           | This command is used to read parameters for image content based adaptive brightness control functionality.                                                    |    |                                    |    |    |               |    |    |    |     |
|                       | C1                                                                                                                                                            | C0 | Function                           |    |    |               |    |    |    |     |
|                       | 0                                                                                                                                                             | 0  | Off                                |    |    |               |    |    |    |     |
|                       | 0                                                                                                                                                             | 1  | User Interface Image (UI mode)     |    |    |               |    |    |    |     |
|                       | 1                                                                                                                                                             | 0  | Still Picture Image (Still mode)   |    |    |               |    |    |    |     |
|                       | 1                                                                                                                                                             | 1  | Moving Picture Image (Moving mode) |    |    |               |    |    |    |     |
| Restriction           | -                                                                                                                                                             |    |                                    |    |    |               |    |    |    |     |
| Register Availability | Status                                                                                                                                                        |    |                                    |    |    | Availability  |    |    |    |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                      |    |                                    |    |    | Yes           |    |    |    |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                       |    |                                    |    |    | Yes           |    |    |    |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                     |    |                                    |    |    | Yes           |    |    |    |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                      |    |                                    |    |    | Yes           |    |    |    |     |
|                       | Sleep In or Booster Off                                                                                                                                       |    |                                    |    |    | Yes           |    |    |    |     |
| Default               | Status                                                                                                                                                        |    |                                    |    |    | Default Value |    |    |    |     |
|                       | Power On Sequence                                                                                                                                             |    |                                    |    |    | 00h           |    |    |    |     |
|                       | S/W Reset                                                                                                                                                     |    |                                    |    |    | 00h           |    |    |    |     |
|                       | H/W Reset                                                                                                                                                     |    |                                    |    |    | 00h           |    |    |    |     |
| Flow Chart            | <div><div><div>RDCABC</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div>                                                               |    |                                    |    |    |               |    |    |    |     |
|                       | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |    |                                    |    |    |               |    |    |    |     |

## 6.2.28 Write CABC Minimum Brightness (5Eh)

| 5E H                                      |                                                                                                                                                                                                                                                                                                                                                                                                           | WRCABCMB (Write CABC minimum brightness) |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                       | D7                                       | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Command                                   | Write                                                                                                                                                                                                                                                                                                                                                                                                     | 0                                        | 1    | 0    | 1    | 1    | 1    | 1    | 0    | 5E  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Parameter                                 | Write                                                                                                                                                                                                                                                                                                                                                                                                     | CMB7                                     | CMB6 | CMB5 | CMB4 | CMB3 | CMB2 | CMB1 | CMB0 | -   |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Description                               | This command is used to set the minimum brightness value of the display for CABC function<br>In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the highest brightness for CABC.                                                                                                                                                                        |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                         |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table> |                                          |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                              |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                                |                                          |      |      |      |      |      |      |      |     | Status | Default Value | Power On Sequence                        | 00h | S/W Reset                               | 00h | H/W Reset                                 | 00h |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                             |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| S/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| H/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Flow Chart                                | <div><div><div>WRCABCMB</div><div>↓</div><div>CMB[7:0]</div><div>↓</div><div>New Display Luminance Value Loaded</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                       |                                          |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |

## 6.2.29 Read CABC minimum brightness (5Fh)

| 5F H                                      | RDCABCMB (Read CABC minimum brightness)                                                                                                                                                                                                                                                                                                                                                                   |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|-------------------------|-----|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                       | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Command                                   | Read                                                                                                                                                                                                                                                                                                                                                                                                      | 0    | 1    | 0    | 1    | 1    | 1    | 1    | 1    | 5F  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Parameter                                 | Read                                                                                                                                                                                                                                                                                                                                                                                                      | CMB7 | CMB6 | CMB5 | CMB4 | CMB3 | CMB2 | CMB1 | CMB0 | -   |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Description                               | This command is used to read the minimum brightness value of the display for CABC function<br>In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the highest brightness for CABC.                                                                                                                                                                       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table> |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                              |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                                |      |      |      |      |      |      |      |      |     | Status | Default Value | Power On Sequence                        | 00h | S/W Reset                               | 00h | H/W Reset                                 | 00h |                                          |     |                         |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                             |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| S/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| H/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |
| Flow Chart                                | <div><div><div>RDCABCMB</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |                         |     |

### 6.2.30 Read Black/White Low Bits (70h)

| 70 H                                      |                                                                                                                                                                                                                                                                                                                                                                                                           | RDBWLB (Read Black White Low Bits) |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------|------|------|-----|-----|-----|-----|-----|--------|---------------|------------------------------------------|------------|-----------------------------------------|-------------------|-------------------------------------------|-----------|------------------------------------------|-----|-------------------------|-----------|-----|-----------|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                       | D7                                 | D6   | D5   | D4   | D3  | D2  | D1  | D0  | HEX |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Command                                   | Read                                                                                                                                                                                                                                                                                                                                                                                                      | 0                                  | 1    | 1    | 1    | 0   | 0   | 0   | 0   | 70  |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Parameter                                 | Read                                                                                                                                                                                                                                                                                                                                                                                                      | Bkx1                               | Bkx0 | Bky1 | Bky0 | Wx1 | Wx0 | Wy1 | Wy0 | -   |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Description                               | This command returns the lowest bits of black and white color characteristic.<br>Black: Bkx and Bky<br>White: Wx and Wy                                                                                                                                                                                                                                                                                   |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                         |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table> |                                    |      |      |      |     |     |     |     |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes        | Normal Mode On, Idle Mode On, Sleep Out | Yes               | Partial Mode On, Idle Mode Off, Sleep Out | Yes       | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes       |     |           |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                              |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Default                                   | <table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>Before OTP</th><th>After OTP</th></tr><tr><td>Power On Sequence</td><td>00h</td><td>OTP Value</td></tr><tr><td>S/W Reset</td><td>00h</td><td>OTP Value</td></tr><tr><td>H/W Reset</td><td>00h</td><td>OTP Value</td></tr></table>                                                                                    |                                    |      |      |      |     |     |     |     |     | Status | Default Value |                                          | Before OTP | After OTP                               | Power On Sequence | 00h                                       | OTP Value | S/W Reset                                | 00h | OTP Value               | H/W Reset | 00h | OTP Value |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                             |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
|                                           | Before OTP                                                                                                                                                                                                                                                                                                                                                                                                | After OTP                          |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                                       | OTP Value                          |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| S/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       | OTP Value                          |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| H/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       | OTP Value                          |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Flow Chart                                | <div><div><div>RDBWLB</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                                         |                                    |      |      |      |     |     |     |     |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |

### 6.2.31 Read Bkx (71h)

| 71 H                  | RDBkx (Read Bkx)                                                                                                                                                                                                                                             |      |      |      |      |               |      |           |      |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|---------------|------|-----------|------|-----|
|                       | R/W                                                                                                                                                                                                                                                          | D7   | D6   | D5   | D4   | D3            | D2   | D1        | D0   | HEX |
| Command               | Read                                                                                                                                                                                                                                                         | 0    | 1    | 1    | 1    | 0             | 0    | 0         | 1    | 71  |
| Parameter             | Read                                                                                                                                                                                                                                                         | Bkx9 | Bkx8 | Bkx7 | Bkx6 | Bkx5          | Bkx4 | Bkx3      | Bkx2 | -   |
| Description           | This command returns the Bkx bits of black color characteristic.                                                                                                                                                                                             |      |      |      |      |               |      |           |      |     |
| Restriction           | -                                                                                                                                                                                                                                                            |      |      |      |      |               |      |           |      |     |
| Register Availability | Status                                                                                                                                                                                                                                                       |      |      |      |      | Availability  |      |           |      |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                     |      |      |      |      | Yes           |      |           |      |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                      |      |      |      |      | Yes           |      |           |      |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                    |      |      |      |      | Yes           |      |           |      |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                     |      |      |      |      | Yes           |      |           |      |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                      |      |      |      |      | Yes           |      |           |      |     |
| Default               | Status                                                                                                                                                                                                                                                       |      |      |      |      | Default Value |      |           |      |     |
|                       |                                                                                                                                                                                                                                                              |      |      |      |      | Before OTP    |      | After OTP |      |     |
|                       | Power On Sequence                                                                                                                                                                                                                                            |      |      |      |      | 00h           |      | OTP Value |      |     |
|                       | S/W Reset                                                                                                                                                                                                                                                    |      |      |      |      | 00h           |      | OTP Value |      |     |
|                       | H/W Reset                                                                                                                                                                                                                                                    |      |      |      |      | 00h           |      | OTP Value |      |     |
| Flow Chart            | <div><div><div>RDBkx</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |      |      |      |      |               |      |           |      |     |

### 6.2.32 Read Bky (72h)

| 72 H                  | RDBky (Read Bky)                                                                                                                                                                                                                                             |      |      |      |      |               |      |           |      |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|---------------|------|-----------|------|-----|
|                       | R/W                                                                                                                                                                                                                                                          | D7   | D6   | D5   | D4   | D3            | D2   | D1        | D0   | HEX |
| Command               | Read                                                                                                                                                                                                                                                         | 0    | 1    | 1    | 1    | 0             | 0    | 1         | 0    | 72  |
| Parameter             | Read                                                                                                                                                                                                                                                         | Bky9 | Bky8 | Bky7 | Bky6 | Bky5          | Bky4 | Bky3      | Bky2 | -   |
| Description           | This command returns the Bky bits of black color characteristic.                                                                                                                                                                                             |      |      |      |      |               |      |           |      |     |
| Restriction           | -                                                                                                                                                                                                                                                            |      |      |      |      |               |      |           |      |     |
| Register Availability | Status                                                                                                                                                                                                                                                       |      |      |      |      | Availability  |      |           |      |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                     |      |      |      |      | Yes           |      |           |      |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                      |      |      |      |      | Yes           |      |           |      |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                    |      |      |      |      | Yes           |      |           |      |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                     |      |      |      |      | Yes           |      |           |      |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                      |      |      |      |      | Yes           |      |           |      |     |
| Default               | Status                                                                                                                                                                                                                                                       |      |      |      |      | Default Value |      |           |      |     |
|                       |                                                                                                                                                                                                                                                              |      |      |      |      | Before OTP    |      | After OTP |      |     |
|                       | Power On Sequence                                                                                                                                                                                                                                            |      |      |      |      | 00h           |      | OTP Value |      |     |
|                       | S/W Reset                                                                                                                                                                                                                                                    |      |      |      |      | 00h           |      | OTP Value |      |     |
|                       | H/W Reset                                                                                                                                                                                                                                                    |      |      |      |      | 00h           |      | OTP Value |      |     |
| Flow Chart            | <div><div><div>RDBky</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |      |      |      |      |               |      |           |      |     |



### 6.2.33 Read Wx (73h)

| 73 H                  | RDWx (Read Wx)                                                                                                                                                                                                                                  |     |     |     |     |               |     |           |     |     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|---------------|-----|-----------|-----|-----|
|                       | R/W                                                                                                                                                                                                                                             | D7  | D6  | D5  | D4  | D3            | D2  | D1        | D0  | HEX |
| Command               | Read                                                                                                                                                                                                                                            | 0   | 1   | 1   | 1   | 0             | 0   | 1         | 1   | 73  |
| Parameter             | Read                                                                                                                                                                                                                                            | Wx9 | Wx8 | Wx7 | Wx6 | Wx5           | Wx4 | Wx3       | Wx2 | -   |
| Description           | This command returns the Wx bits of White color characteristic.                                                                                                                                                                                 |     |     |     |     |               |     |           |     |     |
| Restriction           | -                                                                                                                                                                                                                                               |     |     |     |     |               |     |           |     |     |
| Register Availability | Status                                                                                                                                                                                                                                          |     |     |     |     | Availability  |     |           |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                        |     |     |     |     | Yes           |     |           |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                         |     |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                       |     |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                        |     |     |     |     | Yes           |     |           |     |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                         |     |     |     |     | Yes           |     |           |     |     |
| Default               | Status                                                                                                                                                                                                                                          |     |     |     |     | Default Value |     |           |     |     |
|                       |                                                                                                                                                                                                                                                 |     |     |     |     | Before OTP    |     | After OTP |     |     |
|                       | Power On Sequence                                                                                                                                                                                                                               |     |     |     |     | 00h           |     | OTP Value |     |     |
|                       | S/W Reset                                                                                                                                                                                                                                       |     |     |     |     | 00h           |     | OTP Value |     |     |
|                       | H/W Reset                                                                                                                                                                                                                                       |     |     |     |     | 00h           |     | OTP Value |     |     |
| Flow Chart            | <div><div><div>RDWx</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |     |     |     |     |               |     |           |     |     |

### 6.2.34 Read Wy (74h)

| 74 H                  |                                                                                                                                                                                                                                                 | RDWy (Read Wy) |     |     |     |               |     |           |     |     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|-----|-----|---------------|-----|-----------|-----|-----|
|                       | R/W                                                                                                                                                                                                                                             | D7             | D6  | D5  | D4  | D3            | D2  | D1        | D0  | HEX |
| Command               | Read                                                                                                                                                                                                                                            | 0              | 1   | 1   | 1   | 0             | 1   | 0         | 0   | 74  |
| Parameter             | Read                                                                                                                                                                                                                                            | Wy9            | Wy8 | Wy7 | Wy6 | Wy5           | Wy4 | Wy3       | Wy2 | -   |
| Description           | This command returns the Wy bits of White color characteristic.                                                                                                                                                                                 |                |     |     |     |               |     |           |     |     |
| Restriction           | -                                                                                                                                                                                                                                               |                |     |     |     |               |     |           |     |     |
| Register Availability | Status                                                                                                                                                                                                                                          |                |     |     |     | Availability  |     |           |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                        |                |     |     |     | Yes           |     |           |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                         |                |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                       |                |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                        |                |     |     |     | Yes           |     |           |     |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                         |                |     |     |     | Yes           |     |           |     |     |
| Default               | Status                                                                                                                                                                                                                                          |                |     |     |     | Default Value |     |           |     |     |
|                       |                                                                                                                                                                                                                                                 |                |     |     |     | Before OTP    |     | After OTP |     |     |
|                       | Power On Sequence                                                                                                                                                                                                                               |                |     |     |     | 00h           |     | OTP Value |     |     |
|                       | S/W Reset                                                                                                                                                                                                                                       |                |     |     |     | 00h           |     | OTP Value |     |     |
|                       | H/W Reset                                                                                                                                                                                                                                       |                |     |     |     | 00h           |     | OTP Value |     |     |
| Flow Chart            | <div><div><div>RDWy</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                |     |     |     |               |     |           |     |     |

### 6.2.35 Read Red/Green Low Bits (75h)

| 75 H                  |                                                                                                                                                                                                                                                               | RDRGLB (Read Red Green Low Bits) |     |     |     |               |     |           |     |     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----|-----|-----|---------------|-----|-----------|-----|-----|
|                       | R/W                                                                                                                                                                                                                                                           | D7                               | D6  | D5  | D4  | D3            | D2  | D1        | D0  | HEX |
| Command               | Read                                                                                                                                                                                                                                                          | 0                                | 1   | 1   | 1   | 0             | 1   | 0         | 1   | 75  |
| Parameter             | Read                                                                                                                                                                                                                                                          | Rx1                              | Rx0 | Ry1 | Ry0 | Gx1           | Gx0 | Gy1       | Gy0 | -   |
| Description           | This command returns the lowest bits of Red and Green color characteristic.<br>Red: Rx and Ry<br>Green: Gx and Gy                                                                                                                                             |                                  |     |     |     |               |     |           |     |     |
| Restriction           | -                                                                                                                                                                                                                                                             |                                  |     |     |     |               |     |           |     |     |
| Register Availability | Status                                                                                                                                                                                                                                                        |                                  |     |     |     | Availability  |     |           |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                      |                                  |     |     |     | Yes           |     |           |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                       |                                  |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                     |                                  |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                      |                                  |     |     |     | Yes           |     |           |     |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                       |                                  |     |     |     | Yes           |     |           |     |     |
| Default               | Status                                                                                                                                                                                                                                                        |                                  |     |     |     | Default Value |     |           |     |     |
|                       |                                                                                                                                                                                                                                                               |                                  |     |     |     | Before OTP    |     | After OTP |     |     |
|                       | Power On Sequence                                                                                                                                                                                                                                             |                                  |     |     |     | 00h           |     | OTP Value |     |     |
|                       | S/W Reset                                                                                                                                                                                                                                                     |                                  |     |     |     | 00h           |     | OTP Value |     |     |
|                       | H/W Reset                                                                                                                                                                                                                                                     |                                  |     |     |     | 00h           |     | OTP Value |     |     |
| Flow Chart            | <div><div><div>RDRGLB</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                  |     |     |     |               |     |           |     |     |

### 6.2.36 Read Rx (76h)

| 76 H                  |                                                                                                                                                                                                                                                             | RDRx (Read Rx) |     |     |     |               |     |           |     |     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|-----|-----|---------------|-----|-----------|-----|-----|
|                       | R/W                                                                                                                                                                                                                                                         | D7             | D6  | D5  | D4  | D3            | D2  | D1        | D0  | HEX |
| Command               | Read                                                                                                                                                                                                                                                        | 0              | 1   | 1   | 1   | 0             | 1   | 1         | 0   | 76  |
| Dummy Cycle           |                                                                                                                                                                                                                                                             |                |     |     |     |               |     |           |     |     |
| Parameter             | Read                                                                                                                                                                                                                                                        | Rx9            | Rx8 | Rx7 | Rx6 | Rx5           | Rx4 | Rx3       | Rx2 | -   |
| Description           | This command returns the Rx bits of Red color characteristic.                                                                                                                                                                                               |                |     |     |     |               |     |           |     |     |
| Restriction           | -                                                                                                                                                                                                                                                           |                |     |     |     |               |     |           |     |     |
| Register Availability | Status                                                                                                                                                                                                                                                      |                |     |     |     | Availability  |     |           |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                    |                |     |     |     | Yes           |     |           |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                     |                |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                   |                |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                    |                |     |     |     | Yes           |     |           |     |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                     |                |     |     |     | Yes           |     |           |     |     |
| Default               | Status                                                                                                                                                                                                                                                      |                |     |     |     | Default Value |     |           |     |     |
|                       |                                                                                                                                                                                                                                                             |                |     |     |     | Before OTP    |     | After OTP |     |     |
|                       | Power On Sequence                                                                                                                                                                                                                                           |                |     |     |     | 00h           |     | OTP Value |     |     |
|                       | S/W Reset                                                                                                                                                                                                                                                   |                |     |     |     | 00h           |     | OTP Value |     |     |
|                       | H/W Reset                                                                                                                                                                                                                                                   |                |     |     |     | 00h           |     | OTP Value |     |     |
| Flow Chart            | <div><div><div>RDRx</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                |     |     |     |               |     |           |     |     |

### 6.2.37 Read Ry (77h)

| 77 H                  |                                                                                                                                                                                                                                                             | RDRy (Read Ry) |     |     |     |               |     |           |     |     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|-----|-----|---------------|-----|-----------|-----|-----|
|                       | R/W                                                                                                                                                                                                                                                         | D7             | D6  | D5  | D4  | D3            | D2  | D1        | D0  | HEX |
| Command               | Read                                                                                                                                                                                                                                                        | 0              | 1   | 1   | 1   | 0             | 1   | 1         | 1   | 77  |
| Dummy Cycle           |                                                                                                                                                                                                                                                             |                |     |     |     |               |     |           |     |     |
| Parameter             | Read                                                                                                                                                                                                                                                        | Ry9            | Ry8 | Ry7 | Ry6 | Ry5           | Ry4 | Ry3       | Ry2 | -   |
| Description           | This command returns the Ry bits of Red color characteristic.                                                                                                                                                                                               |                |     |     |     |               |     |           |     |     |
| Restriction           | -                                                                                                                                                                                                                                                           |                |     |     |     |               |     |           |     |     |
| Register Availability | Status                                                                                                                                                                                                                                                      |                |     |     |     | Availability  |     |           |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                    |                |     |     |     | Yes           |     |           |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                     |                |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                   |                |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                    |                |     |     |     | Yes           |     |           |     |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                     |                |     |     |     | Yes           |     |           |     |     |
| Default               | Status                                                                                                                                                                                                                                                      |                |     |     |     | Default Value |     |           |     |     |
|                       |                                                                                                                                                                                                                                                             |                |     |     |     | Before OTP    |     | After OTP |     |     |
|                       | Power On Sequence                                                                                                                                                                                                                                           |                |     |     |     | 00h           |     | OTP Value |     |     |
|                       | S/W Reset                                                                                                                                                                                                                                                   |                |     |     |     | 00h           |     | OTP Value |     |     |
|                       | H/W Reset                                                                                                                                                                                                                                                   |                |     |     |     | 00h           |     | OTP Value |     |     |
| Flow Chart            | <div><div><div>RDRy</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                |     |     |     |               |     |           |     |     |

### 6.2.38 Read Gx (78h)

| 78 H                  | RDGx (Read Gx)                                                                                                                                                |     |     |     |               |              |           |     |     |     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|---------------|--------------|-----------|-----|-----|-----|
|                       | R/W                                                                                                                                                           | D7  | D6  | D5  | D4            | D3           | D2        | D1  | D0  | HEX |
| Command               | Read                                                                                                                                                          | 0   | 1   | 1   | 1             | 1            | 0         | 0   | 0   | 78  |
| Dummy Cycle           |                                                                                                                                                               |     |     |     |               |              |           |     |     |     |
| Parameter             | Read                                                                                                                                                          | Gx9 | Gx8 | Gx7 | Gx6           | Gx5          | Gx4       | Gx3 | Gx2 | -   |
| Description           | This command returns the Gx bits of Green color characteristic.                                                                                               |     |     |     |               |              |           |     |     |     |
| Restriction           | -                                                                                                                                                             |     |     |     |               |              |           |     |     |     |
| Register Availability | Status                                                                                                                                                        |     |     |     |               | Availability |           |     |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                      |     |     |     |               | Yes          |           |     |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                       |     |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                     |     |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                      |     |     |     |               | Yes          |           |     |     |     |
|                       | Sleep In or Booster Off                                                                                                                                       |     |     |     |               | Yes          |           |     |     |     |
| Default               | Status                                                                                                                                                        |     |     |     | Default Value |              |           |     |     |     |
|                       |                                                                                                                                                               |     |     |     | Before OTP    |              | After OTP |     |     |     |
|                       | Power On Sequence                                                                                                                                             |     |     |     | 00h           |              | OTP Value |     |     |     |
|                       | S/W Reset                                                                                                                                                     |     |     |     | 00h           |              | OTP Value |     |     |     |
|                       | H/W Reset                                                                                                                                                     |     |     |     | 00h           |              | OTP Value |     |     |     |
| Flow Chart            | <div><div>RDGx</div><div>Send D[7:0]</div><div>Host Driver</div></div>                                                                                        |     |     |     |               |              |           |     |     |     |
|                       | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |     |     |     |               |              |           |     |     |     |

### 6.2.39 Read Gy (79h)

| 79 H                  |                                                                                                                                                    | RDGy (Read Gy) |     |     |               |              |           |     |     |     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|-----|---------------|--------------|-----------|-----|-----|-----|
|                       | R/W                                                                                                                                                | D7             | D6  | D5  | D4            | D3           | D2        | D1  | D0  | HEX |
| Command               | Read                                                                                                                                               | 0              | 1   | 1   | 1             | 1            | 0         | 0   | 1   | 79  |
| Dummy Cycle           |                                                                                                                                                    |                |     |     |               |              |           |     |     |     |
| Parameter             | Read                                                                                                                                               | Gy9            | Gy8 | Gy7 | Gy6           | Gy5          | Gy4       | Gy3 | Gy2 | -   |
| Description           | This command returns the Gy bits of Green color characteristic.                                                                                    |                |     |     |               |              |           |     |     |     |
| Restriction           | -                                                                                                                                                  |                |     |     |               |              |           |     |     |     |
| Register Availability | Status                                                                                                                                             |                |     |     |               | Availability |           |     |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                           |                |     |     |               | Yes          |           |     |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                            |                |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                          |                |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                           |                |     |     |               | Yes          |           |     |     |     |
|                       | Sleep In or Booster Off                                                                                                                            |                |     |     |               | Yes          |           |     |     |     |
| Default               | Status                                                                                                                                             |                |     |     | Default Value |              |           |     |     |     |
|                       |                                                                                                                                                    |                |     |     | Before OTP    |              | After OTP |     |     |     |
|                       | Power On Sequence                                                                                                                                  |                |     |     | 00h           |              | OTP Value |     |     |     |
|                       | S/W Reset                                                                                                                                          |                |     |     | 00h           |              | OTP Value |     |     |     |
|                       | H/W Reset                                                                                                                                          |                |     |     | 00h           |              | OTP Value |     |     |     |
| Flow Chart            | <div><div>RDGy</div><div>Send D[7:0]</div><div>Host Driver</div></div>                                                                             |                |     |     |               |              |           |     |     |     |
|                       | <div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div> |                |     |     |               |              |           |     |     |     |

#### 6.2.40 Read Blue/AColor Low Bits (7Ah)

| 7A H                  |                                                                                                                                                                                                                                                               | RDBALB (Read Blue AColor Low Bits) |     |     |     |               |     |           |     |     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----|-----|-----|---------------|-----|-----------|-----|-----|
|                       | R/W                                                                                                                                                                                                                                                           | D7                                 | D6  | D5  | D4  | D3            | D2  | D1        | D0  | HEX |
| Command               | Read                                                                                                                                                                                                                                                          | 0                                  | 1   | 1   | 1   | 1             | 0   | 1         | 0   | 7A  |
| Dummy Cycle           |                                                                                                                                                                                                                                                               |                                    |     |     |     |               |     |           |     |     |
| Parameter             | Read                                                                                                                                                                                                                                                          | Bx1                                | Bx0 | By1 | By0 | Ax1           | Ax0 | Ay1       | Ay0 | -   |
| Description           | This command returns the lowest bits of Blue and AColor color characteristic.<br>Blue: Bx and By<br>A: Ax and Ay                                                                                                                                              |                                    |     |     |     |               |     |           |     |     |
| Restriction           | -                                                                                                                                                                                                                                                             |                                    |     |     |     |               |     |           |     |     |
| Register Availability | Status                                                                                                                                                                                                                                                        |                                    |     |     |     | Availability  |     |           |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                      |                                    |     |     |     | Yes           |     |           |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                       |                                    |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                     |                                    |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                      |                                    |     |     |     | Yes           |     |           |     |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                       |                                    |     |     |     | Yes           |     |           |     |     |
| Default               | Status                                                                                                                                                                                                                                                        |                                    |     |     |     | Default Value |     |           |     |     |
|                       |                                                                                                                                                                                                                                                               |                                    |     |     |     | Before OTP    |     | After OTP |     |     |
|                       | Power On Sequence                                                                                                                                                                                                                                             |                                    |     |     |     | 00h           |     | OTP Value |     |     |
|                       | S/W Reset                                                                                                                                                                                                                                                     |                                    |     |     |     | 00h           |     | OTP Value |     |     |
|                       | H/W Reset                                                                                                                                                                                                                                                     |                                    |     |     |     | 00h           |     | OTP Value |     |     |
| Flow Chart            | <div><div><div>RDBALB</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                    |     |     |     |               |     |           |     |     |



### 6.2.41 Read Bx (7Bh)

| 7B H                  |                                                                                                                                                                                                                                                             | RDBx (Read Bx) |     |     |               |              |           |     |     |     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|-----|---------------|--------------|-----------|-----|-----|-----|
|                       | R/W                                                                                                                                                                                                                                                         | D7             | D6  | D5  | D4            | D3           | D2        | D1  | D0  | HEX |
| Command               | Read                                                                                                                                                                                                                                                        | 0              | 1   | 1   | 1             | 1            | 0         | 1   | 1   | 7B  |
| Dummy Cycle           |                                                                                                                                                                                                                                                             |                |     |     |               |              |           |     |     |     |
| Parameter             | Read                                                                                                                                                                                                                                                        | Bx9            | Bx8 | Bx7 | Bx6           | Bx5          | Bx4       | Bx3 | Bx2 | -   |
| Description           | This command returns the Bx bits of Blue color characteristic.                                                                                                                                                                                              |                |     |     |               |              |           |     |     |     |
| Restriction           | -                                                                                                                                                                                                                                                           |                |     |     |               |              |           |     |     |     |
| Register Availability | Status                                                                                                                                                                                                                                                      |                |     |     |               | Availability |           |     |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                    |                |     |     |               | Yes          |           |     |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                     |                |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                   |                |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                    |                |     |     |               | Yes          |           |     |     |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                     |                |     |     |               | Yes          |           |     |     |     |
| Default               | Status                                                                                                                                                                                                                                                      |                |     |     | Default Value |              |           |     |     |     |
|                       |                                                                                                                                                                                                                                                             |                |     |     | Before OTP    |              | After OTP |     |     |     |
|                       | Power On Sequence                                                                                                                                                                                                                                           |                |     |     | 00h           |              | OTP Value |     |     |     |
|                       | S/W Reset                                                                                                                                                                                                                                                   |                |     |     | 00h           |              | OTP Value |     |     |     |
|                       | H/W Reset                                                                                                                                                                                                                                                   |                |     |     | 00h           |              | OTP Value |     |     |     |
| Flow Chart            | <div><div><div>RDBx</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                |     |     |               |              |           |     |     |     |

## 6.2.42 Read By (7Ch)

| 7C H                  | RDBy (Read By)                                                                                                                                                                                                                                              |     |     |     |               |              |           |     |     |     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|---------------|--------------|-----------|-----|-----|-----|
|                       | R/W                                                                                                                                                                                                                                                         | D7  | D6  | D5  | D4            | D3           | D2        | D1  | D0  | HEX |
| Command               | Read                                                                                                                                                                                                                                                        | 0   | 1   | 1   | 1             | 1            | 1         | 0   | 0   | 7C  |
| Dummy Cycle           |                                                                                                                                                                                                                                                             |     |     |     |               |              |           |     |     |     |
| Parameter             | Read                                                                                                                                                                                                                                                        | By9 | By8 | By7 | By6           | By5          | By4       | By3 | By2 | -   |
| Description           | This command returns the By bits of Blue color characteristic.                                                                                                                                                                                              |     |     |     |               |              |           |     |     |     |
| Restriction           | -                                                                                                                                                                                                                                                           |     |     |     |               |              |           |     |     |     |
| Register Availability | Status                                                                                                                                                                                                                                                      |     |     |     |               | Availability |           |     |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                    |     |     |     |               | Yes          |           |     |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                     |     |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                   |     |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                    |     |     |     |               | Yes          |           |     |     |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                     |     |     |     |               | Yes          |           |     |     |     |
| Default               | Status                                                                                                                                                                                                                                                      |     |     |     | Default Value |              |           |     |     |     |
|                       |                                                                                                                                                                                                                                                             |     |     |     | Before OTP    |              | After OTP |     |     |     |
|                       | Power On Sequence                                                                                                                                                                                                                                           |     |     |     | 00h           |              | OTP Value |     |     |     |
|                       | S/W Reset                                                                                                                                                                                                                                                   |     |     |     | 00h           |              | OTP Value |     |     |     |
|                       | H/W Reset                                                                                                                                                                                                                                                   |     |     |     | 00h           |              | OTP Value |     |     |     |
| Flow Chart            | <div><div><div>RDBy</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |     |     |     |               |              |           |     |     |     |

### 6.2.43 Read Ax (7Dh)

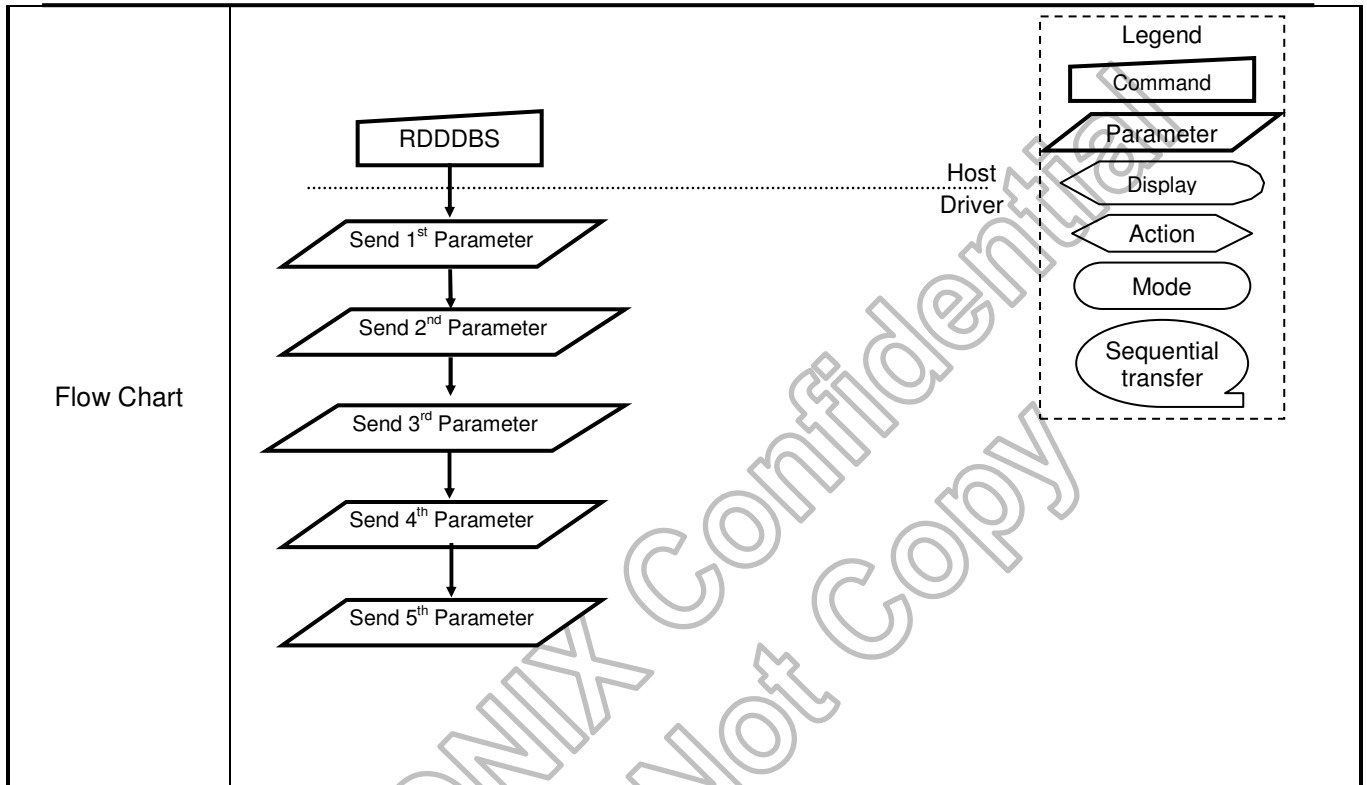
| 7D H                  |                                                                                                                                                               | RD Ax (Read Ax) |     |     |     |               |     |           |     |     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-----|---------------|-----|-----------|-----|-----|
|                       | R/W                                                                                                                                                           | D7              | D6  | D5  | D4  | D3            | D2  | D1        | D0  | HEX |
| Command               | Read                                                                                                                                                          | 0               | 1   | 1   | 1   | 1             | 1   | 0         | 1   | 7D  |
| Dummy Cycle           |                                                                                                                                                               |                 |     |     |     |               |     |           |     |     |
| Parameter             | Read                                                                                                                                                          | Ax9             | Ax8 | Ax7 | Ax6 | Ax5           | Ax4 | Ax3       | Ax2 | -   |
| Description           | This command returns the Ax bits of A color characteristic.                                                                                                   |                 |     |     |     |               |     |           |     |     |
| Restriction           | -                                                                                                                                                             |                 |     |     |     |               |     |           |     |     |
| Register Availability | Status                                                                                                                                                        |                 |     |     |     | Availability  |     |           |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                      |                 |     |     |     | Yes           |     |           |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                       |                 |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                     |                 |     |     |     | Yes           |     |           |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                      |                 |     |     |     | Yes           |     |           |     |     |
|                       | Sleep In or Booster Off                                                                                                                                       |                 |     |     |     | Yes           |     |           |     |     |
| Default               | Status                                                                                                                                                        |                 |     |     |     | Default Value |     |           |     |     |
|                       |                                                                                                                                                               |                 |     |     |     | Before OTP    |     | After OTP |     |     |
|                       | Power On Sequence                                                                                                                                             |                 |     |     |     | 00h           |     | OTP Value |     |     |
|                       | S/W Reset                                                                                                                                                     |                 |     |     |     | 00h           |     | OTP Value |     |     |
|                       | H/W Reset                                                                                                                                                     |                 |     |     |     | 00h           |     | OTP Value |     |     |
| Flow Chart            | <div><div><div>RD Ax</div><div>↓</div><div>Send D[7:0]</div></div><div>Host Driver</div></div>                                                                |                 |     |     |     |               |     |           |     |     |
|                       | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                 |     |     |     |               |     |           |     |     |

#### 6.2.44 Read Ay (7Eh)

| 7E H                  |                                                                                                                                                                                                                                                            | RD <sub>Ay</sub> (Read <sub>Ay</sub> ) |     |     |               |              |           |     |     |     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----|-----|---------------|--------------|-----------|-----|-----|-----|
|                       | R/W                                                                                                                                                                                                                                                        | D7                                     | D6  | D5  | D4            | D3           | D2        | D1  | D0  | HEX |
| Command               | Read                                                                                                                                                                                                                                                       | 0                                      | 1   | 1   | 1             | 1            | 1         | 1   | 0   | 7E  |
| Dummy Cycle           |                                                                                                                                                                                                                                                            |                                        |     |     |               |              |           |     |     |     |
| Parameter             | Read                                                                                                                                                                                                                                                       | Ay9                                    | Ay8 | Ay7 | Ay6           | Ay5          | Ay4       | Ay3 | Ay2 | -   |
| Description           | This command returns the Ay bits of A color characteristic.                                                                                                                                                                                                |                                        |     |     |               |              |           |     |     |     |
| Restriction           | -                                                                                                                                                                                                                                                          |                                        |     |     |               |              |           |     |     |     |
| Register Availability | Status                                                                                                                                                                                                                                                     |                                        |     |     |               | Availability |           |     |     |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                   |                                        |     |     |               | Yes          |           |     |     |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                    |                                        |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                  |                                        |     |     |               | Yes          |           |     |     |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                   |                                        |     |     |               | Yes          |           |     |     |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                    |                                        |     |     |               | Yes          |           |     |     |     |
| Default               | Status                                                                                                                                                                                                                                                     |                                        |     |     | Default Value |              |           |     |     |     |
|                       |                                                                                                                                                                                                                                                            |                                        |     |     | Before OTP    |              | After OTP |     |     |     |
|                       | Power On Sequence                                                                                                                                                                                                                                          |                                        |     |     | 00h           |              | OTP Value |     |     |     |
|                       | S/W Reset                                                                                                                                                                                                                                                  |                                        |     |     | 00h           |              | OTP Value |     |     |     |
|                       | H/W Reset                                                                                                                                                                                                                                                  |                                        |     |     | 00h           |              | OTP Value |     |     |     |
| Flow Chart            | <div><div><div>RD<sub>Ay</sub></div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                        |     |     |               |              |           |     |     |     |

### 6.2.45 Read DDB Start (A1h)

| A1 H                                      | RDDDBS (Read DDB Start)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|------------|-----------------------------------------|-------------------|-------------------------------------------|-----------|------------------------------------------|-----|-------------------------|-----------|-----|-----------|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Command                                   | Read                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1         | 0  | 1  | 0  | 0  | 0  | 0  | 1  | A1  |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 1 <sup>st</sup> Parameter                 | Read                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | x         | x  | x  | x  | x  | x  | x  | x  | -   |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 2 <sup>nd</sup> Parameter                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | x         | x  | x  | x  | x  | x  | x  | x  |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 3 <sup>rd</sup> Parameter                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | x         | x  | x  | x  | x  | x  | x  | x  |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 4 <sup>th</sup> parameter                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | x         | x  | x  | x  | x  | x  | x  | x  |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 5 <sup>th</sup> parameter                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1         | 1  | 1  | 1  | 1  | 1  | 1  | 1  |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Description                               | <p>This command returns the supplier identification and display module mode/revision information.</p> <p>Note: This information is not the same what “Read ID1 (DAh)”, “Read ID2 (DBh)” and “Read ID3 (DCh)” commands are returning.</p> <p>Note: Parameter 0xFF is an “Exit Code”, this means that there is no more data in the DDB block.</p> <p>This read sequence can be interrupted by any command and it can be continued by “Read DDB Continue (A8h)” command when the first parameter, what has been transferred, is the parameter, which has not been sent e.g. RDDDBS =&gt; 1st parameter has been sent =&gt; 2nd parameter has been sent=&gt; interrupt =&gt; RDDDBC =&gt; 3rd parameter of the RDDDBS has been sent.</p> |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table>                                                                                                                                                                                                                                                                                                                            |           |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes        | Normal Mode On, Idle Mode On, Sleep Out | Yes               | Partial Mode On, Idle Mode Off, Sleep Out | Yes       | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes       |     |           |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Default                                   | <table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>Before OTP</th><th>After OTP</th></tr><tr><td>Power On Sequence</td><td>00h</td><td>OTP Value</td></tr><tr><td>S/W Reset</td><td>00h</td><td>OTP Value</td></tr><tr><td>H/W Reset</td><td>00h</td><td>OTP Value</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                               |           |    |    |    |    |    |    |    |     | Status | Default Value |                                          | Before OTP | After OTP                               | Power On Sequence | 00h                                       | OTP Value | S/W Reset                                | 00h | OTP Value               | H/W Reset | 00h | OTP Value |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
|                                           | Before OTP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | After OTP |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | OTP Value |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| S/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | OTP Value |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| H/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | OTP Value |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |



## 6.2.46 Read DDB Continue (A8h)

| A8 H                                      |                                                                                                                                                                                                                                                                                                                                                                                                           | RDDDBC (Read DDB Continue) |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|------------|-----------------------------------------|-------------------|-------------------------------------------|-----------|------------------------------------------|-----|-------------------------|-----------|-----|-----------|
|                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                       | D7                         | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Command                                   | Read                                                                                                                                                                                                                                                                                                                                                                                                      | 1                          | 0  | 1  | 0  | 1  | 0  | 0  | 0  | A8  |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 1 <sup>st</sup> Parameter                 | Read                                                                                                                                                                                                                                                                                                                                                                                                      | x                          | x  | x  | x  | x  | x  | x  | x  | -   |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 2 <sup>nd</sup> Parameter                 |                                                                                                                                                                                                                                                                                                                                                                                                           | x                          | x  | x  | x  | x  | x  | x  | x  |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 3 <sup>rd</sup> Parameter                 |                                                                                                                                                                                                                                                                                                                                                                                                           | x                          | x  | x  | x  | x  | x  | x  | x  |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 4 <sup>th</sup> parameter                 |                                                                                                                                                                                                                                                                                                                                                                                                           | x                          | x  | x  | x  | x  | x  | x  | x  |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| 5 <sup>th</sup> parameter                 |                                                                                                                                                                                                                                                                                                                                                                                                           | 1                          | 1  | 1  | 1  | 1  | 1  | 1  | 1  |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Description                               | This command returns the supplier identification and display module mode/revision information from the point where RDDDBS command was interrupted by an other command.<br>Note: Parameter 0xFF is an “Exit Code”, this means that there is no more data in the DDB block.                                                                                                                                 |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                         |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table> |                            |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes        | Normal Mode On, Idle Mode On, Sleep Out | Yes               | Partial Mode On, Idle Mode Off, Sleep Out | Yes       | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In or Booster Off | Yes       |     |           |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                              |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Sleep In or Booster Off                   | Yes                                                                                                                                                                                                                                                                                                                                                                                                       |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Default                                   | <table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>Before OTP</th><th>After OTP</th></tr><tr><td>Power On Sequence</td><td>00h</td><td>OTP Value</td></tr><tr><td>S/W Reset</td><td>00h</td><td>OTP Value</td></tr><tr><td>H/W Reset</td><td>00h</td><td>OTP Value</td></tr></table>                                                                                    |                            |    |    |    |    |    |    |    |     | Status | Default Value |                                          | Before OTP | After OTP                               | Power On Sequence | 00h                                       | OTP Value | S/W Reset                                | 00h | OTP Value               | H/W Reset | 00h | OTP Value |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                             |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
|                                           | Before OTP                                                                                                                                                                                                                                                                                                                                                                                                | After OTP                  |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                                       | OTP Value                  |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| S/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       | OTP Value                  |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| H/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                       | OTP Value                  |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |
| Flow Chart                                | <div><div><div>RDDDBC</div><div></div><div>RDDDBS Data<br/>D1[7:0],<br/>D2[7:0]...Dn[7:0]..</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                         |                            |    |    |    |    |    |    |    |     |        |               |                                          |            |                                         |                   |                                           |           |                                          |     |                         |           |     |           |

## 6.2.47 Read ID1 Value (DAh)

| DA H                  | RDID1 (Read ID1 Value)                                                                                                                                                                                                                               |      |      |      |      |               |              |           |      |     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|---------------|--------------|-----------|------|-----|
|                       | R/W                                                                                                                                                                                                                                                  | D7   | D6   | D5   | D4   | D3            | D2           | D1        | D0   | HEX |
| Command               | Read                                                                                                                                                                                                                                                 | 1    | 1    | 0    | 1    | 1             | 0            | 1         | 0    | DA  |
| Parameter             | Read                                                                                                                                                                                                                                                 | ID17 | ID16 | ID15 | ID14 | ID13          | ID12         | ID11      | ID10 | -   |
| Description           | This read byte identifies the LCD module's manufacturer.                                                                                                                                                                                             |      |      |      |      |               |              |           |      |     |
| Restriction           | -                                                                                                                                                                                                                                                    |      |      |      |      |               |              |           |      |     |
| Register Availability | Status                                                                                                                                                                                                                                               |      |      |      |      |               | Availability |           |      |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                             |      |      |      |      |               | Yes          |           |      |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                              |      |      |      |      |               | Yes          |           |      |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                            |      |      |      |      |               | Yes          |           |      |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                             |      |      |      |      |               | Yes          |           |      |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                              |      |      |      |      |               | Yes          |           |      |     |
| Default               | Status                                                                                                                                                                                                                                               |      |      |      |      | Default Value |              |           |      |     |
|                       |                                                                                                                                                                                                                                                      |      |      |      |      | Before OTP    |              | After OTP |      |     |
|                       | Power On Sequence                                                                                                                                                                                                                                    |      |      |      |      | 38h           |              | OTP Value |      |     |
|                       | S/W Reset                                                                                                                                                                                                                                            |      |      |      |      | 38h           |              | OTP Value |      |     |
|                       | H/W Reset                                                                                                                                                                                                                                            |      |      |      |      | 38h           |              | OTP Value |      |     |
| Flow Chart            | <div><div>RDID1</div><div>↓</div><div>Send ID1[7:0]</div></div> <div>Host Driver</div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |      |      |      |      |               |              |           |      |     |



## 6.2.48 Read ID2 Value (DBh)

| DB H                  | RDID2 (Read ID2 Value)                                                                                                                                                                                                                                         |      |      |      |      |               |              |           |      |     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|---------------|--------------|-----------|------|-----|
|                       | R/W                                                                                                                                                                                                                                                            | D7   | D6   | D5   | D4   | D3            | D2           | D1        | D0   | HEX |
| Command               | Read                                                                                                                                                                                                                                                           | 1    | 1    | 0    | 1    | 1             | 0            | 1         | 1    | DB  |
| Parameter             | Read                                                                                                                                                                                                                                                           | ID27 | ID26 | ID25 | ID24 | ID23          | ID22         | ID21      | ID20 | -   |
| Description           | This read byte identifies the LCD module version.                                                                                                                                                                                                              |      |      |      |      |               |              |           |      |     |
| Restriction           | -                                                                                                                                                                                                                                                              |      |      |      |      |               |              |           |      |     |
| Register Availability | Status                                                                                                                                                                                                                                                         |      |      |      |      |               | Availability |           |      |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                       |      |      |      |      |               | Yes          |           |      |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                        |      |      |      |      |               | Yes          |           |      |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                      |      |      |      |      |               | Yes          |           |      |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                       |      |      |      |      |               | Yes          |           |      |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                                        |      |      |      |      |               | Yes          |           |      |     |
| Default               | Status                                                                                                                                                                                                                                                         |      |      |      |      | Default Value |              |           |      |     |
|                       |                                                                                                                                                                                                                                                                |      |      |      |      | Before OTP    |              | After OTP |      |     |
|                       | Power On Sequence                                                                                                                                                                                                                                              |      |      |      |      | 21h           |              | OTP Value |      |     |
|                       | S/W Reset                                                                                                                                                                                                                                                      |      |      |      |      | 21h           |              | OTP Value |      |     |
|                       | H/W Reset                                                                                                                                                                                                                                                      |      |      |      |      | 21h           |              | OTP Value |      |     |
| Flow Chart            | <div><div><div>RDID1</div><div>↓</div><div>Send ID2[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |      |      |      |      |               |              |           |      |     |

## 6.2.49 Read ID3 Value (DCh)

| DC H                  |                                                                                                                                                                                                                                                      | RDID3 (Read ID3 Value) |      |      |      |               |      |           |      |     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------|------|------|---------------|------|-----------|------|-----|
|                       | R/W                                                                                                                                                                                                                                                  | D7                     | D6   | D5   | D4   | D3            | D2   | D1        | D0   | HEX |
| Command               | Read                                                                                                                                                                                                                                                 | 1                      | 1    | 0    | 1    | 1             | 1    | 0         | 0    | DC  |
| Dummy Cycle           |                                                                                                                                                                                                                                                      |                        |      |      |      |               |      |           |      |     |
| Parameter             | Read                                                                                                                                                                                                                                                 | ID37                   | ID36 | ID35 | ID34 | ID33          | ID32 | ID31      | ID30 | -   |
| Description           | This read byte identifies the LCD module/driver.                                                                                                                                                                                                     |                        |      |      |      |               |      |           |      |     |
| Restriction           | -                                                                                                                                                                                                                                                    |                        |      |      |      |               |      |           |      |     |
| Register Availability | Status                                                                                                                                                                                                                                               |                        |      |      |      | Availability  |      |           |      |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                             |                        |      |      |      | Yes           |      |           |      |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                              |                        |      |      |      | Yes           |      |           |      |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                            |                        |      |      |      | Yes           |      |           |      |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                             |                        |      |      |      | Yes           |      |           |      |     |
|                       | Sleep In or Booster Off                                                                                                                                                                                                                              |                        |      |      |      | Yes           |      |           |      |     |
| Default               | Status                                                                                                                                                                                                                                               |                        |      |      |      | Default Value |      |           |      |     |
|                       |                                                                                                                                                                                                                                                      |                        |      |      |      | Before OTP    |      | After OTP |      |     |
|                       | Power On Sequence                                                                                                                                                                                                                                    |                        |      |      |      | 1Fh           |      | OTP Value |      |     |
|                       | S/W Reset                                                                                                                                                                                                                                            |                        |      |      |      | 1Fh           |      | OTP Value |      |     |
|                       | H/W Reset                                                                                                                                                                                                                                            |                        |      |      |      | 1Fh           |      | OTP Value |      |     |
| Flow Chart            | <div><div>RDID1</div><div>↓</div><div>Send ID3[7:0]</div></div> <div>Host Driver</div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                        |      |      |      |               |      |           |      |     |

### 6.2.50 SETEXTC (B9h)

| B9 H          | SETEXC                                                                                                                                                |          |          |          |          |          |          |          |          |     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|-----|
|               | R/W                                                                                                                                                   | D7       | D6       | D5       | D4       | D3       | D2       | D1       | D0       | HEX |
| Command       | W                                                                                                                                                     | 1        | 0        | 1        | 1        | 1        | 0        | 0        | 1        | B9  |
| Parameter 1st | W                                                                                                                                                     | EXTC1[7] | EXTC1[6] | EXTC1[5] | EXTC1[4] | EXTC1[3] | EXTC1[2] | EXTC1[1] | EXTC1[0] | 00  |
| Parameter 2nd | W                                                                                                                                                     | EXTC2[7] | EXTC2[6] | EXTC2[5] | EXTC2[4] | EXTC2[3] | EXTC2[2] | EXTC2[1] | EXTC2[0] | 00  |
| Parameter 3rd | W                                                                                                                                                     | EXTC3[7] | EXTC3[6] | EXTC3[5] | EXTC3[4] | EXTC3[3] | EXTC3[2] | EXTC3[1] | EXTC3[0] | 00  |
| Description   | <p>This command is used to enable USER Command.</p> <p>Enable User command: Set B9H=F1h, 12h, 83h<br/>Disable User command: Set B9H=00h, 00h, 00h</p> |          |          |          |          |          |          |          |          |     |

### 6.2.51 SETDISP (B2h)

| B2 H                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                           | SETDISP      |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------|--------------------|--------------------|-------|-------------|-------------|-------------|------------------------|----------------|---|---------------|---|---|----------------|--------|---|--------|---|--------|------|--------|--------|-------------------|
|                                                                                                                                                                                                                                                                                                                                     | R/W                                                                                                                                                                                                                                                                       | D7           | D6                 | D5                 | D4                 | D3    | D2          | D1          | D0          | HEX                    |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| Command                                                                                                                                                                                                                                                                                                                             | W                                                                                                                                                                                                                                                                         | 1            | 0                  | 1                  | 1                  | 0     | 0           | 1           | 0           | B2                     |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| Parameter 1st                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                       | NL[7]        | NL[6]              | NL[5]              | NL[4]              | NL[3] | NL[2]       | NL[1]       | NL[0]       | C8                     |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| Parameter 2nd                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                       | RES_V_LSB[1] | RES_V_LSB[0]       | BLK_CON[1]         | BLK_CON[0]         | x     | RESO_SEL[2] | RESO_SEL[1] | RESO_SEL[0] | 02                     |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| Parameter 3rd                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                       | WHITE_GND_EN | WHITE_FRAME_SEL[2] | WHITE_FRAME_SEL[1] | WHITE_FRAME_SEL[0] | x     | ISC[2]      | ISC[1]      | ISC[0]      | A0                     |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| Description                                                                                                                                                                                                                                                                                                                         | This command is used to control the display resolution.                                                                                                                                                                                                                   |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>• <b>NL[7:0] / RES_V_LSB[1:0]</b> : Specify the gate number of vertical direction.<ul style="list-style-type: none"><li>- Gate Number = 480 + NL*4 + RES_V_LSB</li></ul></li></ul>                                                  |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>• <b>BLK_CON[1:0]</b> : Non-display area source output control.</li></ul>                                                                                                                                                           |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     | <table><tr><th colspan="2">BLK_CON</th><th>Source output</th></tr><tr><td>0</td><td>0</td><td>Keep last data</td></tr><tr><td>0</td><td>1</td><td>VSSD</td></tr><tr><td>1</td><td>0</td><td>Hi-Z</td></tr><tr><td>1</td><td>1</td><td>Setting inhibited</td></tr></table> |              |                    |                    |                    |       |             |             |             |                        | BLK_CON        |   | Source output | 0 | 0 | Keep last data | 0      | 1 | VSSD   | 1 | 0      | Hi-Z | 1      | 1      | Setting inhibited |
|                                                                                                                                                                                                                                                                                                                                     | BLK_CON                                                                                                                                                                                                                                                                   |              | Source output      |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     | 0                                                                                                                                                                                                                                                                         | 0            | Keep last data     |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     | 0                                                                                                                                                                                                                                                                         | 1            | VSSD               |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                         | 0            | Hi-Z               |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                         | 1            | Setting inhibited  |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>• <b>RESO_SEL[2:0]</b> : Specify the channel number of source direction.</li></ul>                                                                                                                                                  |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| <table><tr><th>RESO_SEL[2:0]<br/>(Hex)</th><th>Channel number</th></tr><tr><td>0</td><td>-</td></tr><tr><td>1</td><td>-</td></tr><tr><td>2</td><td>720RGB</td></tr><tr><td>3</td><td>640RGB</td></tr><tr><td>4</td><td>600RGB</td></tr><tr><td>5</td><td>540RGB</td></tr><tr><td>others</td><td>Setting inhibited</td></tr></table> |                                                                                                                                                                                                                                                                           |              |                    |                    |                    |       |             |             |             | RESO_SEL[2:0]<br>(Hex) | Channel number | 0 | -             | 1 | - | 2              | 720RGB | 3 | 640RGB | 4 | 600RGB | 5    | 540RGB | others | Setting inhibited |
| RESO_SEL[2:0]<br>(Hex)                                                                                                                                                                                                                                                                                                              | Channel number                                                                                                                                                                                                                                                            |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| 0                                                                                                                                                                                                                                                                                                                                   | -                                                                                                                                                                                                                                                                         |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| 1                                                                                                                                                                                                                                                                                                                                   | -                                                                                                                                                                                                                                                                         |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| 2                                                                                                                                                                                                                                                                                                                                   | 720RGB                                                                                                                                                                                                                                                                    |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| 3                                                                                                                                                                                                                                                                                                                                   | 640RGB                                                                                                                                                                                                                                                                    |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| 4                                                                                                                                                                                                                                                                                                                                   | 600RGB                                                                                                                                                                                                                                                                    |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| 5                                                                                                                                                                                                                                                                                                                                   | 540RGB                                                                                                                                                                                                                                                                    |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| others                                                                                                                                                                                                                                                                                                                              | Setting inhibited                                                                                                                                                                                                                                                         |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
| <ul style="list-style-type: none"><li>• <b>WHITE_GND_EN</b> : Determine source voltage during Blanking Time when accessing Sleep-Out / Sleep-In command.<ul style="list-style-type: none"><li>- 1 : Source Voltage = GND</li><li>- 0 : Source Voltage = Lowest Voltage</li></ul></li></ul>                                          |                                                                                                                                                                                                                                                                           |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                           |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                           |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |
|                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                           |              |                    |                    |                    |       |             |             |             |                        |                |   |               |   |   |                |        |   |        |   |        |      |        |        |                   |

- **WHITE\_FRAME\_SEL[2:0]** : Blank timing control when access sleep out command.

| WHITE_FRAME |   |   | Blank Frame Period |
|-------------|---|---|--------------------|
| 0           | 0 | 0 | 0 frame            |
| 0           | 0 | 1 | 1 frame            |
| 0           | 1 | 0 | 2 frames           |
| 0           | 1 | 1 | 3 frames           |
| 1           | 0 | 0 | 4 frames           |
| 1           | 0 | 1 | 5 frames           |
| 1           | 1 | 0 | 6 frames           |
| 1           | 1 | 1 | 7 frames           |

- **ISC[2:0]** : Source output refresh control.

| ISC |   |   | Refresh Period |
|-----|---|---|----------------|
| 0   | 0 | 0 | 0 frame        |
| 0   | 0 | 1 | 1 frame        |
| 0   | 1 | 0 | 2 frames       |
| :   | : | : | :              |
| 1   | 1 | 1 | 7 frames       |

## 6.2.52 SETRGBIF (B3h)

| B3 H                                                                          |                                                                                             | SETRGBIF                |                         |                         |                         |                         |                         |                         |                         |     |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-----|
|                                                                               | R/W                                                                                         | D7                      | D6                      | D5                      | D4                      | D3                      | D2                      | D1                      | D0                      | HEX |
| Command                                                                       | W                                                                                           | 1                       | 0                       | 1                       | 1                       | 0                       | 0                       | 1                       | 1                       | B3  |
| Parameter 1st                                                                 | R/W                                                                                         | VBP_RG<br>B_GEN[7<br>]  | VBP_RG<br>B_GEN[6<br>]  | VBP_RG<br>B_GEN[5<br>]  | VBP_RG<br>B_GEN[4<br>]  | VBP_RG<br>B_GEN[3<br>]  | VBP_RG<br>B_GEN[2<br>]  | VBP_RG<br>B_GEN[1<br>]  | VBP_RG<br>B_GEN[0<br>]  | 28  |
| Parameter 2nd                                                                 | R/W                                                                                         | VFP_RG<br>B_GEN[7<br>]  | VFP_RG<br>B_GEN[6<br>]  | VFP_RG<br>B_GEN[5<br>]  | VFP_RG<br>B_GEN[4<br>]  | VFP_RG<br>B_GEN[3<br>]  | VFP_RG<br>B_GEN[2<br>]  | VFP_RG<br>B_GEN[1<br>]  | VFP_RG<br>B_GEN[0<br>]  | 28  |
| Parameter 3rd                                                                 | R/W                                                                                         | DE_BP<br>RGB_GE<br>N[7] | DE_BP<br>RGB_GE<br>N[6] | DE_BP<br>RGB_GE<br>N[5] | DE_BP<br>RGB_GE<br>N[4] | DE_BP<br>RGB_GE<br>N[3] | DE_BP<br>RGB_GE<br>N[2] | DE_BP<br>RGB_GE<br>N[1] | DE_BP<br>RGB_GE<br>N[0] | 28  |
| Parameter 4th                                                                 | R/W                                                                                         | DE_FP<br>RGB_GE<br>N[7] | DE_FP<br>RGB_GE<br>N[6] | DE_FP<br>RGB_GE<br>N[5] | DE_FP<br>RGB_GE<br>N[4] | DE_FP<br>RGB_GE<br>N[3] | DE_FP<br>RGB_GE<br>N[2] | DE_FP<br>RGB_GE<br>N[1] | DE_FP<br>RGB_GE<br>N[0] | 28  |
| Description                                                                   | This command is used to control RGB I/F porch timing for internal use.                      |                         |                         |                         |                         |                         |                         |                         |                         |     |
|                                                                               | • <b>VBP_RGB_GEN[7:0]</b> : Vertical back porch HS number selection in Blank Frame Period.  |                         |                         |                         |                         |                         |                         |                         |                         |     |
|                                                                               | • <b>VFP_RGB_GEN[7:0]</b> : Vertical front porch HS number selection in Blank Frame Period. |                         |                         |                         |                         |                         |                         |                         |                         |     |
|                                                                               | <b>VBP_RGB_GEN[7:0]<br/>VFP_RGB_GEN[7:0]<br/>(Hex)</b>                                      |                         | <b>Hsync number</b>     |                         |                         |                         |                         |                         |                         |     |
|                                                                               | 0                                                                                           |                         | Setting inhibited       |                         |                         |                         |                         |                         |                         |     |
|                                                                               | 1                                                                                           |                         | Setting inhibited       |                         |                         |                         |                         |                         |                         |     |
|                                                                               | 2                                                                                           |                         | 2                       |                         |                         |                         |                         |                         |                         |     |
|                                                                               | :                                                                                           |                         | :                       |                         |                         |                         |                         |                         |                         |     |
|                                                                               | FE                                                                                          |                         | 254                     |                         |                         |                         |                         |                         |                         |     |
|                                                                               | FF                                                                                          |                         | Setting inhibited       |                         |                         |                         |                         |                         |                         |     |
| • <b>DE_BP_RGB_GEN[7:0]</b> : HBP OSC number selection in Blank Frame Period. |                                                                                             |                         |                         |                         |                         |                         |                         |                         |                         |     |
| • <b>DE_FP_RGB_GEN[7:0]</b> : HFP OSC number selection in Blank Frame Period. |                                                                                             |                         |                         |                         |                         |                         |                         |                         |                         |     |
| <b>DE_BP_RGB_GEN[7:0]<br/>DE_FP_RGB_GEN[7:0]<br/>(Hex)</b>                    |                                                                                             | <b>OSC number</b>       |                         |                         |                         |                         |                         |                         |                         |     |
| 0                                                                             |                                                                                             | Setting inhibited       |                         |                         |                         |                         |                         |                         |                         |     |
| 1                                                                             |                                                                                             | Setting inhibited       |                         |                         |                         |                         |                         |                         |                         |     |
| 2                                                                             |                                                                                             | 2                       |                         |                         |                         |                         |                         |                         |                         |     |
| :                                                                             |                                                                                             | :                       |                         |                         |                         |                         |                         |                         |                         |     |
| FE                                                                            |                                                                                             | 254                     |                         |                         |                         |                         |                         |                         |                         |     |
| FF                                                                            |                                                                                             | Setting inhibited       |                         |                         |                         |                         |                         |                         |                         |     |

### 6.2.53 SETCYC (B4h)

| B4 H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | SETCYC        |                   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|------------------------|----------|----|---------|---------|---------|------------------------|----------|---------------|----------------|------------------------|---|---|------------------|--------|---|---|-----------------|--------|---|---|-------------------|--------|---|---|-----------------|--------|---|---|-----------------|--------|---|---|-----------------|--------|---|---|-----------------|--------|---|---|-----------------|--------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D7            | D6                | D5                     | D4       | D3 | D2      | D1      | D0      | HEX                    |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| Command                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1             | 0                 | 1                      | 1        | 0  | 1       | 0       | 0       | B4                     |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| Parameter 1st                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ZINV_S2401_EN | ZINV_G_EVEN_EN    | ZINV_EN                | ZINV2_EN | x  | N_NW[2] | N_NW[1] | N_NW[0] | 80                     |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| Parameter 2nd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | x             | x                 | x                      | x        | x  | I_NW[2] | I_NW[1] | I_NW[0] | 00                     |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | This command is used to control display inversion type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |                   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"><li>• <b>ZINV_S2401_EN</b> : Specify extra source for Zig-Zag Inversion.<ul style="list-style-type: none"><li>- 1 : S2401 for use</li><li>- 0 : S0 for use</li></ul></li><li>• <b>ZINV_G_EVEN_EN</b> : Specify in which row source data dislocates.<ul style="list-style-type: none"><li>- 1 : Odd row</li><li>- 0 : Even row</li></ul></li><li>• <b>ZINV_EN</b> : Enable Zig-Zag inversion.<ul style="list-style-type: none"><li>- 1 : Enable Zig-Zag Inversion</li><li>- 0 : Disable Zig-Zag Inversion.</li></ul></li><li>• <b>ZINV2_EN</b> : Specify Zig-Zag inversion selection.<ul style="list-style-type: none"><li>- 1 : Enable Zig-Zag2 Inversion</li><li>- 0 : Enable Zig-Zag1 Inversion.</li></ul></li></ul> <p>◎ For Zig-Zag2 Inversion Type Definition, please refer to the following table.</p> |               |                   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <table><tr><th>ZINV2_EN</th><th>ZINV_S2401_EN</th><th>ZINV_G_EVEN_EN</th><th>Zig-Zag Inversion Type</th></tr><tr><td>0</td><td>0</td><td>1</td><td>Type A</td></tr><tr><td>0</td><td>0</td><td>0</td><td>Type B</td></tr><tr><td>0</td><td>1</td><td>0</td><td>Type C</td></tr><tr><td>0</td><td>1</td><td>1</td><td>Type D</td></tr><tr><td>1</td><td>0</td><td>1</td><td>Type E</td></tr><tr><td>1</td><td>0</td><td>0</td><td>Type F</td></tr><tr><td>1</td><td>1</td><td>0</td><td>Type G</td></tr><tr><td>1</td><td>1</td><td>1</td><td>Type H</td></tr></table>                                                                                                                                                                                                                                                                          |               |                   |                        |          |    |         |         |         |                        | ZINV2_EN | ZINV_S2401_EN | ZINV_G_EVEN_EN | Zig-Zag Inversion Type | 0 | 0 | 1                | Type A | 0 | 0 | 0               | Type B | 0 | 1 | 0                 | Type C | 0 | 1 | 1               | Type D | 1 | 0 | 1               | Type E | 1 | 0 | 0               | Type F | 1 | 1 | 0               | Type G | 1 | 1 | 1               | Type H |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ZINV2_EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ZINV_S2401_EN | ZINV_G_EVEN_EN    | Zig-Zag Inversion Type |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | 1                 | Type A                 |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | 0                 | Type B                 |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1             | 0                 | Type C                 |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1             | 1                 | Type D                 |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | 1                 | Type E                 |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | 0                 | Type F                 |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | Type G            |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1             | Type H            |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| <ul style="list-style-type: none"><li>• <b>N_NW[2:0]</b>: Normal mode inversion type selection.</li><li>• <b>I_NW[2:0]</b> : IDLE mode inversion type selection.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| <table><tr><th colspan="3">N_NW[2:0]<br/>I_NW[2:0]</th><th>Inversion Type</th></tr><tr><td>0</td><td>0</td><td>0</td><td>Column inversion</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1-dot inversion</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1+2-dot inversion</td></tr><tr><td>0</td><td>1</td><td>1</td><td>3-dot inversion</td></tr><tr><td>1</td><td>0</td><td>0</td><td>4-dot inversion</td></tr><tr><td>1</td><td>0</td><td>1</td><td>8-dot inversion</td></tr><tr><td>1</td><td>1</td><td>0</td><td>2-dot inversion</td></tr><tr><td>1</td><td>1</td><td>1</td><td>Setting invalid</td></tr></table> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                   |                        |          |    |         |         |         | N_NW[2:0]<br>I_NW[2:0] |          |               | Inversion Type | 0                      | 0 | 0 | Column inversion | 0      | 0 | 1 | 1-dot inversion | 0      | 1 | 0 | 1+2-dot inversion | 0      | 1 | 1 | 3-dot inversion | 1      | 0 | 0 | 4-dot inversion | 1      | 0 | 1 | 8-dot inversion | 1      | 1 | 0 | 2-dot inversion | 1      | 1 | 1 | Setting invalid |        |
| N_NW[2:0]<br>I_NW[2:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               | Inversion Type    |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | Column inversion  |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1             | 1-dot inversion   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | 1+2-dot inversion |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1             | 3-dot inversion   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | 4-dot inversion   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1             | 8-dot inversion   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | 2-dot inversion   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1             | Setting invalid   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                   |                        |          |    |         |         |         |                        |          |               |                |                        |   |   |                  |        |   |   |                 |        |   |   |                   |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |   |   |                 |        |

## 6.2.54 SETBGP (B5h)

| B5 H          | SETBGP                                                             |    |                        |    |                                          |              |                        |              |              |     |
|---------------|--------------------------------------------------------------------|----|------------------------|----|------------------------------------------|--------------|------------------------|--------------|--------------|-----|
|               | R/W                                                                | D7 | D6                     | D5 | D4                                       | D3           | D2                     | D1           | D0           | HEX |
| Command       | W                                                                  | 1  | 0                      | 1  | 1                                        | 0            | 1                      | 0            | 1            | B5  |
| Parameter 1st | R/W                                                                | x  | x                      | x  | VREF_SEL[4]                              | VREF_SEL[3]  | VREF_SEL[2]            | VREF_SEL[1]  | VREF_SEL[0]  | 09  |
| Parameter 2nd | R/W                                                                | x  | x                      | x  | NVREF_SEL[4]                             | NVREF_SEL[3] | NVREF_SEL[2]           | NVREF_SEL[1] | NVREF_SEL[0] | 09  |
| Description   | This command is used for internal reference voltage setting.       |    |                        |    |                                          |              |                        |              |              |     |
|               | • VREF_SEL[4:0] / NVREF_SEL[4:0] : Specify the VREF/NVREF voltage. |    |                        |    |                                          |              |                        |              |              |     |
|               | VREF_SEL[4:0]<br>NVREF_SEL[4:0]<br>(Hex)                           |    | VREF<br> NVREF <br>(V) |    | VREF_SEL[4:0]<br>NVREF_SEL[4:0]<br>(Hex) |              | VREF<br> NVREF <br>(V) |              |              |     |
|               | 0                                                                  |    | 3.5                    |    | 10                                       |              | 5.1                    |              |              |     |
|               | 1                                                                  |    | 3.6                    |    | 11                                       |              | 5.2                    |              |              |     |
|               | 2                                                                  |    | 3.7                    |    | 12                                       |              | 5.3                    |              |              |     |
|               | 3                                                                  |    | 3.8                    |    | 13                                       |              | 5.4                    |              |              |     |
|               | 4                                                                  |    | 3.9                    |    | 14                                       |              | 5.45                   |              |              |     |
|               | 5                                                                  |    | 4.0                    |    | 15                                       |              | 5.5                    |              |              |     |
|               | 6                                                                  |    | 4.1                    |    | 16                                       |              | 5.55                   |              |              |     |
|               | 7                                                                  |    | 4.2                    |    | 17                                       |              | 5.6                    |              |              |     |
|               | 8                                                                  |    | 4.3                    |    | 18                                       |              | 5.65                   |              |              |     |
|               | 9                                                                  |    | 4.4                    |    | 19                                       |              | 5.7                    |              |              |     |
|               | A                                                                  |    | 4.5                    |    | 1A                                       |              | 5.75                   |              |              |     |
|               | B                                                                  |    | 4.6                    |    | 1B                                       |              | 5.8                    |              |              |     |
|               | C                                                                  |    | 4.7                    |    | 1C                                       |              | 5.85                   |              |              |     |
|               | D                                                                  |    | 4.8                    |    | 1D                                       |              | 5.9                    |              |              |     |
|               | E                                                                  |    | 4.9                    |    | 1E                                       |              | 5.95                   |              |              |     |
|               | F                                                                  |    | 5.0                    |    | 1F                                       |              | 6.0                    |              |              |     |



## 6.2.55 SETVCOM (B6h)

| B6 H          |                                                                                                                                                                                                   | SETVCOM         |                 |                                         |                 |                 |                                         |                          |                          |     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------------------------------|-----------------|-----------------|-----------------------------------------|--------------------------|--------------------------|-----|
|               | R/W                                                                                                                                                                                               | D7              | D6              | D5                                      | D4              | D3              | D2                                      | D1                       | D0                       | HEX |
| Command       | W                                                                                                                                                                                                 | 1               | 0               | 1                                       | 1               | 0               | 1                                       | 1                        | 0                        | B6  |
| Parameter 1st | R/W                                                                                                                                                                                               | VCOMD<br>C_F[7] | VCOMD<br>C_F[6] | VCOMD<br>C_F[5]                         | VCOMD<br>C_F[4] | VCOMD<br>C_F[3] | VCOMD<br>C_F[2]                         | VCOMD<br>C_F[1]          | VCOMD<br>C_F[0]          | 4D  |
| Parameter 2nd | R/W                                                                                                                                                                                               | VCOMD<br>C_B[7] | VCOMD<br>C_B[6] | VCOMD<br>C_B[5]                         | VCOMD<br>C_B[4] | VCOMD<br>C_B[3] | VCOMD<br>C_B[2]                         | VCOMD<br>C_B[1]          | VCOMD<br>C_B[0]          | 4D  |
| Parameter 3rd | R                                                                                                                                                                                                 | x               | x               | x                                       | x               | x               | VCOM_O<br>TP_TIME<br>[2]                | VCOM_O<br>TP_TIME<br>[1] | VCOM_O<br>TP_TIME<br>[0] | 00  |
| Description   | This command is used to set VCOM Voltage.                                                                                                                                                         |                 |                 |                                         |                 |                 |                                         |                          |                          |     |
|               | <ul style="list-style-type: none"><li>• <b>VCOMDC_F[7:0]</b> : Specify the VCOMDC voltage at GS_PANEL="0".</li><li>• <b>VCOMDC_B[7:0]</b> : Specify the VCOMDC voltage at GS_PANEL="1".</li></ul> |                 |                 |                                         |                 |                 |                                         |                          |                          |     |
|               | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex)                                                                                                                                                           |                 | VCOMDC<br>(V)   | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) |                 | VCOMDC<br>(V)   | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) |                          | VCOMDC<br>(V)            |     |
|               | 0                                                                                                                                                                                                 |                 | 0               | 10                                      |                 | -0.36           | 20                                      |                          | -0.55                    |     |
|               | 1                                                                                                                                                                                                 |                 | -0.025          | 11                                      |                 | -0.375          | 21                                      |                          | -0.56                    |     |
|               | 2                                                                                                                                                                                                 |                 | -0.05           | 12                                      |                 | -0.39           | 22                                      |                          | -0.57                    |     |
|               | 3                                                                                                                                                                                                 |                 | -0.075          | 13                                      |                 | -0.405          | 23                                      |                          | -0.58                    |     |
|               | 4                                                                                                                                                                                                 |                 | -0.1            | 14                                      |                 | -0.42           | 24                                      |                          | -0.59                    |     |
|               | 5                                                                                                                                                                                                 |                 | -0.125          | 15                                      |                 | -0.435          | 25                                      |                          | -0.6                     |     |
|               | 6                                                                                                                                                                                                 |                 | -0.15           | 16                                      |                 | -0.45           | 26                                      |                          | -0.61                    |     |
|               | 7                                                                                                                                                                                                 |                 | -0.175          | 17                                      |                 | -0.46           | 27                                      |                          | -0.62                    |     |
|               | 8                                                                                                                                                                                                 |                 | -0.2            | 18                                      |                 | -0.47           | 28                                      |                          | -0.63                    |     |
|               | 9                                                                                                                                                                                                 |                 | -0.225          | 19                                      |                 | -0.48           | 29                                      |                          | -0.64                    |     |
|               | A                                                                                                                                                                                                 |                 | -0.25           | 1A                                      |                 | -0.49           | 2A                                      |                          | -0.65                    |     |
|               | B                                                                                                                                                                                                 |                 | -0.275          | 1B                                      |                 | -0.5            | 2B                                      |                          | -0.66                    |     |
|               | C                                                                                                                                                                                                 |                 | -0.3            | 1C                                      |                 | -0.51           | 2C                                      |                          | -0.67                    |     |
|               | D                                                                                                                                                                                                 |                 | -0.315          | 1D                                      |                 | -0.52           | 2D                                      |                          | -0.68                    |     |
|               | E                                                                                                                                                                                                 |                 | -0.33           | 1E                                      |                 | -0.53           | 2E                                      |                          | -0.69                    |     |
|               | F                                                                                                                                                                                                 |                 | -0.345          | 1F                                      |                 | -0.54           | 2F                                      |                          | -0.7                     |     |
|               | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex)                                                                                                                                                           |                 | VCOMDC<br>(V)   | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) |                 | VCOMDC<br>(V)   | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) |                          | VCOMDC<br>(V)            |     |
|               | 30                                                                                                                                                                                                |                 | -0.71           | 40                                      |                 | -0.87           | 50                                      |                          | -1.03                    |     |
|               | 31                                                                                                                                                                                                |                 | -0.72           | 41                                      |                 | -0.88           | 51                                      |                          | -1.04                    |     |
|               | 32                                                                                                                                                                                                |                 | -0.73           | 42                                      |                 | -0.89           | 52                                      |                          | -1.05                    |     |
|               | 33                                                                                                                                                                                                |                 | -0.74           | 43                                      |                 | -0.9            | 53                                      |                          | -1.06                    |     |
|               | 34                                                                                                                                                                                                |                 | -0.75           | 44                                      |                 | -0.91           | 54                                      |                          | -1.07                    |     |
|               | 35                                                                                                                                                                                                |                 | -0.76           | 45                                      |                 | -0.92           | 55                                      |                          | -1.08                    |     |
|               | 36                                                                                                                                                                                                |                 | -0.77           | 46                                      |                 | -0.93           | 56                                      |                          | -1.09                    |     |
| 37            |                                                                                                                                                                                                   | -0.78           | 47              |                                         | -0.94           | 57              |                                         | -1.1                     |                          |     |
| 38            |                                                                                                                                                                                                   | -0.79           | 48              |                                         | -0.95           | 58              |                                         | -1.11                    |                          |     |
| 39            |                                                                                                                                                                                                   | -0.8            | 49              |                                         | -0.96           | 59              |                                         | -1.12                    |                          |     |
| 3A            |                                                                                                                                                                                                   | -0.81           | 4A              |                                         | -0.97           | 5A              |                                         | -1.13                    |                          |     |
| 3B            |                                                                                                                                                                                                   | -0.82           | 4B              |                                         | -0.98           | 5B              |                                         | -1.14                    |                          |     |
| 3C            |                                                                                                                                                                                                   | -0.83           | 4C              |                                         | -0.99           | 5C              |                                         | -1.15                    |                          |     |
| 3D            |                                                                                                                                                                                                   | -0.84           | 4D              |                                         | -1              | 5D              |                                         | -1.16                    |                          |     |
| 3E            |                                                                                                                                                                                                   | -0.85           | 4E              |                                         | -1.01           | 5E              |                                         | -1.17                    |                          |     |
| 3F            |                                                                                                                                                                                                   | -0.86           | 4F              |                                         | -1.02           | 5F              |                                         | -1.18                    |                          |     |

| VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) |
|-----------------------------------------|---------------|-----------------------------------------|---------------|-----------------------------------------|---------------|
| 60                                      | -1.19         | 70                                      | -1.35         | 80                                      | -1.51         |
| 61                                      | -1.2          | 71                                      | -1.36         | 81                                      | -1.52         |
| 62                                      | -1.21         | 72                                      | -1.37         | 82                                      | -1.53         |
| 63                                      | -1.22         | 73                                      | -1.38         | 83                                      | -1.54         |
| 64                                      | -1.23         | 74                                      | -1.39         | 84                                      | -1.55         |
| 65                                      | -1.24         | 75                                      | -1.4          | 85                                      | -1.56         |
| 66                                      | -1.25         | 76                                      | -1.41         | 86                                      | -1.57         |
| 67                                      | -1.26         | 77                                      | -1.42         | 87                                      | -1.58         |
| 68                                      | -1.27         | 78                                      | -1.43         | 88                                      | -1.59         |
| 69                                      | -1.28         | 79                                      | -1.44         | 89                                      | -1.6          |
| 6A                                      | -1.29         | 7A                                      | -1.45         | 8A                                      | -1.61         |
| 6B                                      | -1.3          | 7B                                      | -1.46         | 8B                                      | -1.62         |
| 6C                                      | -1.31         | 7C                                      | -1.47         | 8C                                      | -1.63         |
| 6D                                      | -1.32         | 7D                                      | -1.48         | 8D                                      | -1.64         |
| 6E                                      | -1.33         | 7E                                      | -1.49         | 8E                                      | -1.65         |
| 6F                                      | -1.34         | 7F                                      | -1.5          | 8F                                      | -1.66         |

| VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) |
|-----------------------------------------|---------------|-----------------------------------------|---------------|-----------------------------------------|---------------|
| 90                                      | -1.67         | A0                                      | -1.845        | B0                                      | -2.085        |
| 91                                      | -1.68         | A1                                      | -1.86         | B1                                      | -2.1          |
| 92                                      | -1.69         | A2                                      | -1.875        | B2                                      | -2.115        |
| 93                                      | -1.7          | A3                                      | -1.89         | B3                                      | -2.13         |
| 94                                      | -1.71         | A4                                      | -1.905        | B4                                      | -2.145        |
| 95                                      | -1.72         | A5                                      | -1.92         | B5                                      | -2.16         |
| 96                                      | -1.73         | A6                                      | -1.935        | B6                                      | -2.175        |
| 97                                      | -1.74         | A7                                      | -1.95         | B7                                      | -2.19         |
| 98                                      | -1.75         | A8                                      | -1.965        | B8                                      | -2.205        |
| 99                                      | -1.76         | A9                                      | -1.98         | B9                                      | -2.22         |
| 9A                                      | -1.77         | AA                                      | -1.995        | BA                                      | -2.235        |
| 9B                                      | -1.78         | AB                                      | -2.01         | BB                                      | -2.25         |
| 9C                                      | -1.79         | AC                                      | -2.025        | BC                                      | -2.265        |
| 9D                                      | -1.8          | AD                                      | -2.04         | BD                                      | -2.28         |
| 9E                                      | -1.815        | AE                                      | -2.055        | BE                                      | -2.295        |
| 9F                                      | -1.83         | AF                                      | -2.07         | BF                                      | -2.31         |

| VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) | VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) |
|-----------------------------------------|---------------|-----------------------------------------|---------------|-----------------------------------------|---------------|
| C0                                      | -2.325        | D0                                      | -2.565        | E0                                      | -2.88         |
| C1                                      | -2.34         | D1                                      | -2.58         | E1                                      | -2.9          |
| C2                                      | -2.355        | D2                                      | -2.6          | E2                                      | -2.92         |
| C3                                      | -2.37         | D3                                      | -2.62         | E3                                      | -2.94         |
| C4                                      | -2.385        | D4                                      | -2.64         | E4                                      | -2.96         |
| C5                                      | -2.4          | D5                                      | -2.66         | E5                                      | -2.98         |
| C6                                      | -2.415        | D6                                      | -2.68         | E6                                      | -3            |
| C7                                      | -2.43         | D7                                      | -2.7          | E7                                      | -3.02         |
| C8                                      | -2.445        | D8                                      | -2.72         | E8                                      | -3.04         |
| C9                                      | -2.46         | D9                                      | -2.74         | E9                                      | -3.06         |
| CA                                      | -2.475        | DA                                      | -2.76         | EA                                      | -3.08         |
| CB                                      | -2.49         | DB                                      | -2.78         | EB                                      | -3.1          |
| CC                                      | -2.505        | DC                                      | -2.8          | EC                                      | -3.12         |
| CD                                      | -2.52         | DD                                      | -2.82         | ED                                      | -3.14         |
| CE                                      | -2.535        | DE                                      | -2.84         | EE                                      | -3.16         |
| CF                                      | -2.55         | DF                                      | -2.86         | EF                                      | -3.18         |

| VCOMDC_F[7:0]<br>VCOMDC_B[7:0]<br>(Hex) | VCOMDC<br>(V) |
|-----------------------------------------|---------------|
| F0                                      | -3.2          |
| F1                                      | -3.22         |
| F2                                      | -3.24         |
| F3                                      | -3.26         |
| F4                                      | -3.28         |
| F5                                      | -3.3          |
| F6                                      | -3.32         |
| F7                                      | -3.34         |
| F8                                      | -3.36         |
| F9                                      | -3.38         |
| FA                                      | -3.4          |
| FB                                      | -3.42         |
| FC                                      | -3.44         |
| FD                                      | -3.46         |
| FE                                      | -3.48         |
| FF                                      | -3.5          |

• **VCOM\_OTP\_TIME[2:0]** : Read VCOM OTP programming times.

| VCOM_OTP_TIME[2:0] | OTP Programming times |
|--------------------|-----------------------|
| 0                  | NO OTPED              |
| 1                  | 1                     |
| 2                  | 2                     |
| 3                  | 3                     |
| 4                  | 4                     |

## 6.2.56 SETOTP (B7h)

| B7 H                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                  | SETOTP           |                        |                  |                  |                  |                  |                  |                  |     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------|------------------|------------------|------------------|------------------|------------------|------------------|-----|
|                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                                              | D7               | D6                     | D5               | D4               | D3               | D2               | D1               | D0               | HEX |
| Command                                                                                                                                                                                                                                               | W                                                                                                                                                                                                                                                                                                | 1                | 0                      | 1                | 1                | 0                | 1                | 1                | 1                | B7  |
| Parameter 1st                                                                                                                                                                                                                                         | R/W                                                                                                                                                                                                                                                                                              | OTP_KEY<br>Y[7]  | OTP_KEY<br>Y[6]        | OTP_KEY<br>Y[5]  | OTP_KEY<br>Y[4]  | OTP_KEY<br>Y[3]  | OTP_KEY<br>Y[2]  | OTP_KEY<br>Y[1]  | OTP_KEY<br>Y[0]  | FF  |
| Parameter 2nd                                                                                                                                                                                                                                         | R/W                                                                                                                                                                                                                                                                                              | OTP_MA<br>SK[7]  | OTP_MA<br>SK[6]        | OTP_MA<br>SK[5]  | OTP_MA<br>SK[4]  | OTP_MA<br>SK[3]  | OTP_MA<br>SK[2]  | OTP_MA<br>SK[1]  | OTP_MA<br>SK[0]  | 00  |
| Parameter 3rd                                                                                                                                                                                                                                         | R/W                                                                                                                                                                                                                                                                                              | x                | VPP_SE<br>L[2]         | VPP_SE<br>L[1]   | VPP_SE<br>L[0]   | x                | x                | x                | OTP_IND<br>EX[8] | 30  |
| Parameter 4th                                                                                                                                                                                                                                         | R/W                                                                                                                                                                                                                                                                                              | OTP_IND<br>EX[7] | OTP_IND<br>EX[6]       | OTP_IND<br>EX[5] | OTP_IND<br>EX[4] | OTP_IND<br>EX[3] | OTP_IND<br>EX[2] | OTP_IND<br>EX[1] | OTP_IND<br>EX[0] | 00  |
| Parameter 5th                                                                                                                                                                                                                                         | R/W                                                                                                                                                                                                                                                                                              | Load_DI<br>S     | VPP_EN                 | OTP_SE<br>L      | OTP_PW<br>E      | OTP_PT<br>M[1]   | OTP_PT<br>M[0]   | OTP_PO<br>R      | OTP_PO<br>RG     | 00  |
| Parameter 6th                                                                                                                                                                                                                                         | R                                                                                                                                                                                                                                                                                                | OTP_DA<br>TA[7]  | OTP_DA<br>TA[6]        | OTP_DA<br>TA[5]  | OTP_DA<br>TA[4]  | OTP_DA<br>TA[3]  | OTP_DA<br>TA[2]  | OTP_DA<br>TA[1]  | OTP_DA<br>TA[0]  | 00  |
| Description                                                                                                                                                                                                                                           | This command is used to set OTP related setting.                                                                                                                                                                                                                                                 |                  |                        |                  |                  |                  |                  |                  |                  |     |
|                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"><li>• <b>OTP_KEY[7:0]</b> : Enable OTP function key.<ul style="list-style-type: none"><li>- OTP_KEY = “5A”, enable OTP function.</li><li>- OTP_KEY = “00”, disable OTP function.</li></ul></li></ul>                                                           |                  |                        |                  |                  |                  |                  |                  |                  |     |
|                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"><li>• <b>OTP_MASK[7:0]</b> : Mask function for OTP programming.<ul style="list-style-type: none"><li>- 1 : Means this bit will not be OTP in all programming process.</li><li>- 0 : Means this bit will be OTP in all programming process.</li></ul></li></ul> |                  |                        |                  |                  |                  |                  |                  |                  |     |
|                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"><li>• <b>VPPSEL[2:0]</b> : Specify VPP Voltage selection.</li></ul>                                                                                                                                                                                            |                  |                        |                  |                  |                  |                  |                  |                  |     |
|                                                                                                                                                                                                                                                       | <b>VPPSEL[2:0]</b>                                                                                                                                                                                                                                                                               |                  | <b>VPP Voltage (V)</b> |                  |                  |                  |                  |                  |                  |     |
|                                                                                                                                                                                                                                                       | 0                                                                                                                                                                                                                                                                                                |                  | 7.2                    |                  |                  |                  |                  |                  |                  |     |
|                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                |                  | 7.3                    |                  |                  |                  |                  |                  |                  |     |
|                                                                                                                                                                                                                                                       | 2                                                                                                                                                                                                                                                                                                |                  | 7.4                    |                  |                  |                  |                  |                  |                  |     |
|                                                                                                                                                                                                                                                       | 3                                                                                                                                                                                                                                                                                                |                  | 7.5                    |                  |                  |                  |                  |                  |                  |     |
|                                                                                                                                                                                                                                                       | 4                                                                                                                                                                                                                                                                                                |                  | 7.6                    |                  |                  |                  |                  |                  |                  |     |
| 5                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                  | 7.7              |                        |                  |                  |                  |                  |                  |                  |     |
| 6                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                  | 7.8              |                        |                  |                  |                  |                  |                  |                  |     |
| 7                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                  | External Power   |                        |                  |                  |                  |                  |                  |                  |     |
| <ul style="list-style-type: none"><li>• <b>OTP_INDEX[8:0]</b> : Set OTP Index of target register or byte.</li></ul>                                                                                                                                   |                                                                                                                                                                                                                                                                                                  |                  |                        |                  |                  |                  |                  |                  |                  |     |
| <ul style="list-style-type: none"><li>• <b>Load_DIS</b> : Disable OTP load function.<ul style="list-style-type: none"><li>- 1 : Disable OTP load function.</li><li>- 0 : Enable OTP load function.</li></ul></li></ul>                                |                                                                                                                                                                                                                                                                                                  |                  |                        |                  |                  |                  |                  |                  |                  |     |
| <ul style="list-style-type: none"><li>• <b>VPP_EN</b> : Enable VPP Power.<ul style="list-style-type: none"><li>- 1 : Enable VPP Power.</li><li>- 0 : Disable VPP Power.</li></ul></li></ul>                                                           |                                                                                                                                                                                                                                                                                                  |                  |                        |                  |                  |                  |                  |                  |                  |     |
| <ul style="list-style-type: none"><li>• <b>OTP_SEL</b> : OTP functional block selection.<ul style="list-style-type: none"><li>- Automatically set by IC in auto OTP.</li><li>- Manually set by user in manual OTP of DG register.</li></ul></li></ul> |                                                                                                                                                                                                                                                                                                  |                  |                        |                  |                  |                  |                  |                  |                  |     |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | <ul style="list-style-type: none"> <li>• <b>OTP_PWE</b> : OTP Write Signal. <ul style="list-style-type: none"> <li>- 1 : OTP Write</li> <li>- 0 : OTP don't Write</li> </ul> </li> <li>• <b>OTP_PTM[1:0]</b> : For testing OTP robustness in test mode. <ul style="list-style-type: none"> <li>- For CP usage.</li> </ul> </li> <li>• <b>OTP_POR</b> : OTP Read Signal. <ul style="list-style-type: none"> <li>- 1 : OTP Read</li> <li>- 0 : OTP don't Read</li> </ul> </li> <li>• <b>OTP_PORG</b> : OTP Auto programming with related register. <ul style="list-style-type: none"> <li>- 1 : OTP auto programming start.</li> <li>- 0 : OTP auto programming stop.</li> </ul> </li> <li>• <b>OTP_DATA[7:0]</b> : OTP Data read back when VPP_POR="1".</li> </ul> |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 6.2.57 SETPOWER\_EXT (B8h)

| B8 H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | SETPOWER_EXT    |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|----------|----------|---------------|---------------|---------------|---------------|-----------------|----------------|-----------------|----------------|-------|---------|-----|------------|---|---------|---|------------|-----|------------|------------|------------|---|---------|-------|------------|------------|------------|---|------------|---|-----------|-----|------------|--------|-----------|---|------------|-------|-----------|------------|-----------------|---|---|---|-------|-----|------------|------------|---|---|---|-------|-----|--------------|--------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D7              | D6              | D5       | D4       | D3            | D2            | D1            | D0            | HEX             |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| Command                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1               | 0               | 1        | 1        | 1             | 0             | 0             | 0             | B8              |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| Parameter 1st                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | x               | PCCS[2]         | PCCS[1]  | PCCS[0]  | ECP_DC_DIV[3] | ECP_DC_DIV[2] | ECP_DC_DIV[1] | ECP_DC_DIV[0] | 34              |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| Parameter 2nd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | x               | SINGLE_PUMP     | DT[1]    | DT[0]    | SYNC_STB      | XDK_ECP[1]    | XDK_ECP[0]    | x             | 22              |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| Parameter 3rd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | x               | x               | x        | x        | x             | PFM_DC_DIV[2] | PFM_DC_DIV[1] | PFM_DC_DIV[0] | 00              |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| Parameter 4th                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | x               | x               | x        | x        | x             | x             | ECP_SYNC_EN   | VGX_SYNC_EN   | 03              |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| Parameter 5th                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | A_DC[1]         | A_DC[0]         | A_DTP[2] | A_DTP[1] | A_DTP[0]      | A_DTN[2]      | A_DTN[1]      | A_DTN[0]      | ED              |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| Parameter 6th                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DCS[1]          | DCS[0]          | DTPS[2]  | DTPS[1]  | DTPS[0]       | DTNS[2]       | DTNS[1]       | DTNS[0]       | 00              |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | This command is used to set display related register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | • <b>PCCS[2:0]</b> : External power IC or PFM selection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <table><tr><th colspan="3">PCCS[2:0]</th><th>IOVCC</th><th>VCI</th><th>VSP</th><th>VSN</th></tr><tr><td>0</td><td>0</td><td>0</td><td>IOVCC</td><td>VCI</td><td>PFM TYPE-A</td><td>PFM TYPE-A</td></tr><tr><td>0</td><td>0</td><td>1</td><td>IOVCC</td><td>VCI</td><td>PFM TYPE-C</td><td>PFM TYPE-C</td></tr><tr><td>0</td><td>1</td><td>0</td><td>IOVCC</td><td>VCI</td><td>FL1002</td><td>FL1002</td></tr><tr><td>0</td><td>1</td><td>1</td><td>IOVCC</td><td>VCI</td><td>PFM TYPE-B</td><td>PFM TYPE-B</td></tr><tr><td>1</td><td>0</td><td>0</td><td>IOVCC</td><td>VCI</td><td>PFM TYPE-A</td><td>PFM TYPE-A</td></tr><tr><td>1</td><td>1</td><td>1</td><td>IOVCC</td><td>VCI</td><td>External VSP</td><td>External VSN</td></tr></table> |                 |                 |          |          |               |               |               |               |                 | PCCS[2:0]      |                 |                | IOVCC | VCI     | VSP | VSN        | 0 | 0       | 0 | IOVCC      | VCI | PFM TYPE-A | PFM TYPE-A | 0          | 0 | 1       | IOVCC | VCI        | PFM TYPE-C | PFM TYPE-C | 0 | 1          | 0 | IOVCC     | VCI | FL1002     | FL1002 | 0         | 1 | 1          | IOVCC | VCI       | PFM TYPE-B | PFM TYPE-B      | 1 | 0 | 0 | IOVCC | VCI | PFM TYPE-A | PFM TYPE-A | 1 | 1 | 1 | IOVCC | VCI | External VSP | External VSN |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PCCS[2:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |                 | IOVCC    | VCI      | VSP           | VSN           |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0               | 0               | IOVCC    | VCI      | PFM TYPE-A    | PFM TYPE-A    |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0               | 1               | IOVCC    | VCI      | PFM TYPE-C    | PFM TYPE-C    |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1               | 0               | IOVCC    | VCI      | FL1002        | FL1002        |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1               | 1               | IOVCC    | VCI      | PFM TYPE-B    | PFM TYPE-B    |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0               | 0               | IOVCC    | VCI      | PFM TYPE-A    | PFM TYPE-A    |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1               | 1               | IOVCC    | VCI      | External VSP  | External VSN  |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| • <b>ECP_DC_DIV[3:0]</b> : Specify VCSW1 / VCSW2 Frequency for Pumping VSP / VSN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| <table><tr><th>ECP_DC_DIV[3:0]</th><th>VCSW1/2 Period</th><th>ECP_DC_DIV[3:0]</th><th>VCSW1/2 Period</th></tr><tr><td>0</td><td>8 Hsync</td><td>8</td><td>1/12 Hsync</td></tr><tr><td>1</td><td>4 Hsync</td><td>9</td><td>1/16 Hsync</td></tr><tr><td>2</td><td>2 Hsync</td><td>A</td><td>1/20 Hsync</td></tr><tr><td>3</td><td>1 Hsync</td><td>B</td><td>1/25 Hsync</td></tr><tr><td>4</td><td>1/2 Hsync</td><td>C</td><td>1/36 Hsync</td></tr><tr><td>5</td><td>1/4 Hsync</td><td>D</td><td>1/56 Hsync</td></tr><tr><td>6</td><td>1/6 Hsync</td><td>E</td><td>1/72 Hsync</td></tr><tr><td>7</td><td>1/8 Hsync</td><td>F</td><td>Setting invalid</td></tr></table> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                 |          |          |               |               |               |               | ECP_DC_DIV[3:0] | VCSW1/2 Period | ECP_DC_DIV[3:0] | VCSW1/2 Period | 0     | 8 Hsync | 8   | 1/12 Hsync | 1 | 4 Hsync | 9 | 1/16 Hsync | 2   | 2 Hsync    | A          | 1/20 Hsync | 3 | 1 Hsync | B     | 1/25 Hsync | 4          | 1/2 Hsync  | C | 1/36 Hsync | 5 | 1/4 Hsync | D   | 1/56 Hsync | 6      | 1/6 Hsync | E | 1/72 Hsync | 7     | 1/8 Hsync | F          | Setting invalid |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| ECP_DC_DIV[3:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | VCSW1/2 Period                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ECP_DC_DIV[3:0] | VCSW1/2 Period  |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 8 Hsync                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 8               | 1/12 Hsync      |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 Hsync                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 9               | 1/16 Hsync      |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2 Hsync                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | A               | 1/20 Hsync      |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1 Hsync                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | B               | 1/25 Hsync      |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1/2 Hsync                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | C               | 1/36 Hsync      |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1/4 Hsync                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | D               | 1/56 Hsync      |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1/6 Hsync                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | E               | 1/72 Hsync      |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1/8 Hsync                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | F               | Setting invalid |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| • <b>SINGLE_PUMP</b> : Specify signal pump state.<br>- 1 : VCSW1 Toggle ; VSCW2 GND.<br>- 0 : VCSW1 / VCSW2 Toggle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| • <b>DT[1:0]</b> : Specify VCSW1/VCSW2 soft start time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| <table><tr><th>DT[1:0]</th><th>Period (ms)</th></tr><tr><td>0</td><td>5</td></tr><tr><td>1</td><td>10</td></tr><tr><td>2</td><td>15</td></tr><tr><td>3</td><td>20</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                 |          |          |               |               |               |               | DT[1:0]         | Period (ms)    | 0               | 5              | 1     | 10      | 2   | 15         | 3 | 20      |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| DT[1:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Period (ms)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |
| • <b>SYNC_STB</b> : Enter synchronous standby mode.<br>- 1 : All power signal synchronous turn off.<br>- 0 : All power signal sequence turn off.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                 |          |          |               |               |               |               |                 |                |                 |                |       |         |     |            |   |         |   |            |     |            |            |            |   |         |       |            |            |            |   |            |   |           |     |            |        |           |   |            |       |           |            |                 |   |   |   |       |     |            |            |   |   |   |       |     |              |              |

Description

- **XDK\_ECP[1:0]** : Specify Pumping ratio of VSP / VSN with VCI.

| XDK_ECP[1:0] | Ratio             |
|--------------|-------------------|
| 0            | X1.5              |
| 1            | X2                |
| 2            | X3                |
| 3            | Setting Inhibited |

- **PFM\_DC\_DIV[2:0]** : Specify PFM operation frequency FoscD.

| PFM_DC_DIV[2:0] | FoscD  |
|-----------------|--------|
| 0               | Fosc/1 |
| 1               | Fosc/2 |
| 2               | Fosc/3 |
| 3               | Fosc/4 |
| 4               | Fosc/5 |
| 5               | Fosc/6 |
| 6               | Fosc/7 |
| 7               | Fosc/8 |

- **ECP\_SYNC\_EN** : Enable power IC pumping frequency synchronization.

- 1 : Synchronize with external Hsync.
- 0 : Synchronize with internal Hsync.

- **VGX\_SYNC\_EN** : Enable VGH/VGL pumping frequency synchronization.

- 1 : Synchronize with external Hsync.
- 0 : Synchronize with internal Hsync.

- **A\_DC[1:0]** : Specify PFM operation half period.

| A_DC[1:0] | Period   |
|-----------|----------|
| 0         | 4/FoscD  |
| 1         | 6/FoscD  |
| 2         | 8/FoscD  |
| 3         | 12/FoscD |

- **A\_DTP[2:0]** : Specify PFM period of VSP.

- **A\_DTN[2:0]** : Specify PFM period of VSN.

- **DTPS[2:0]** : Specify PFM period of VSP at soft start stage.

- **DTNS[2:0]** : Specify PFM period of VSN at soft start stage.

| DTN[2:0]<br>DTP[2:0]<br>DTPS[2:0]<br>DTNS[2:0] | Period  |
|------------------------------------------------|---------|
| 0                                              | 1/FoscD |
| 1                                              | 2/FoscD |
| 2                                              | 3/FoscD |
| 3                                              | 4/FoscD |
| 4                                              | 5/FoscD |
| 5                                              | 6/FoscD |
| 6                                              | 7/FoscD |
| 7                                              | 8/FoscD |

- **DCS[1:0]** : Specify PFM operation half period at soft start stage.

| DCS[1:0] | Period   |
|----------|----------|
| 0        | 12/FoscD |
| 1        | 24/FoscD |
| 2        | 48/FoscD |
| 3        | 60/FoscD |

## 6.2.58 SETMIPI (BAh)

| BA H          | SETMIPI |                    |                    |                    |                |                   |                   |                    |                    |     |
|---------------|---------|--------------------|--------------------|--------------------|----------------|-------------------|-------------------|--------------------|--------------------|-----|
|               | R/W     | D7                 | D6                 | D5                 | D4             | D3                | D2                | D1                 | D0                 | HEX |
| Command       | W       | 1                  | 0                  | 1                  | 1              | 1                 | 0                 | 1                  | 0                  | BA  |
| Parameter 1st | R/W     | x                  | x                  | x                  | x              | VC_Main<br>[1]    | VC_Main<br>[0]    | Lane_Nu<br>mber[1] | Lane_Nu<br>mber[0] | 33  |
| Parameter 2nd | R/W     | DSI_LDO<br>_SEL[2] | DSI_LDO<br>_SEL[1] | DSI_LDO<br>_SEL[0] | x              | x                 | x                 | RTERM[<br>1]       | RTERM[<br>0]       | 61  |
| Parameter 3rd | R/W     | x                  | x                  | x                  | x              | IHSRX[3]          | IHSRX[2]          | IHSRX[1]           | IHSRX[0]           | 06  |
| Parameter 4th | R/W     | x                  | x                  | x                  | x              | Tx_clk_s<br>el[1] | Tx_clk_s<br>el[0] | x                  | x                  | F9  |
| Parameter 5th | R/W     | HFP_OS<br>C[7]     | HFP_OS<br>C[6]     | HFP_OS<br>C[5]     | HFP_OS<br>C[4] | HFP_OS<br>C[3]    | HFP_OS<br>C[2]    | HFP_OS<br>C[1]     | HFP_OS<br>C[0]     | FF  |
| Parameter 6th | R/W     | HBP_OS<br>C[7]     | HBP_OS<br>C[6]     | HBP_OS<br>C[5]     | HBP_OS<br>C[4] | HBP_OS<br>C[3]    | HBP_OS<br>C[2]    | HBP_OS<br>C[1]     | HBP_OS<br>C[0]     | 0A  |



Description

This command is used to set MIPI related register.

• **VC\_Main [1:0]** : Specify the virtual channel ID for multiple slave ID.

• **Lane\_Number [1:0]** : Specify the lane number selection.

| Lane[1:0] | MIPI Lane |
|-----------|-----------|
| 0         | 1 lane    |
| 1         | 2 lanes   |
| 2         | 3 lanes   |
| 3         | 4 lanes   |

• **DSI\_LDO\_SEL[2:0]** : Specify the MIPI LDO voltage selection.

| DSI_LDO_SEL[2:0] | LDO voltage (V) | DSI_LDO_SEL[2:0] | LDO voltage (V) |
|------------------|-----------------|------------------|-----------------|
| 0                | 1.4             | 4                | 1.7             |
| 1                | 1.45            | 5                | 1.8             |
| 2                | 1.5             | 6                | 1.9             |
| 3                | 1.6             | 7                | 2.0             |

• **RTERM[1:0]** : Specify the terminal resistance

| RTERM [1:0] | Resistance ( $\Omega$ ) |
|-------------|-------------------------|
| 0           | 80                      |
| 1           | 90                      |
| 2           | 100                     |
| 3           | 110                     |

• **IHSRX[3:0]** : Specify the MIPI Low High Speed driving ability .

| IHSRX [3:0] | Driving Ability | IHSRX [3:0] | Driving Ability |
|-------------|-----------------|-------------|-----------------|
| 0           | X1              | 8           | X9              |
| 1           | X2              | 9           | X10             |
| 2           | X3              | 10          | X11             |
| 3           | X4              | 11          | X12             |
| 4           | X5              | 12          | X13             |
| 5           | X6              | 13          | X14             |
| 6           | X7              | 14          | X15             |
| 7           | X8              | 15          | X16             |

• **Tx\_clk[1:0]** : Specify TXCLK speed in DSI LP mode for transmission.

| Tx_clk[1:0] | TXCLK      |
|-------------|------------|
| 0           | fDSICLK/4  |
| 1           | fDSICLK/8  |
| 2           | fDSICLK/16 |
| 3           | fDSICLK/32 |

• **HFP\_OSC[7:0]** : Specify the min. HFP number in DSI mode.

• **HBP\_OSC[7:0]** : Specify the min. HBP number in DSI mode.

### 6.2.59 SETVDC (BCh)

| BC H                                                  | SETVDC                                                  |          |              |              |              |    |             |             |             |     |
|-------------------------------------------------------|---------------------------------------------------------|----------|--------------|--------------|--------------|----|-------------|-------------|-------------|-----|
|                                                       | R/W                                                     | D7       | D6           | D5           | D4           | D3 | D2          | D1          | D0          | HEX |
| Command                                               | W                                                       | 1        | 0            | 1            | 1            | 1  | 1           | 0           | 0           | BC  |
| Parameter 1st                                         | R/W                                                     | x        | NVDDD_SEL[2] | NVDDD_SEL[1] | NVDDD_SEL[0] | x  | VDDD_SEL[2] | VDDD_SEL[1] | VDDD_SEL[0] | 47  |
| Description                                           | This command is used to control NVDDD/VDDD Voltage.     |          |              |              |              |    |             |             |             |     |
|                                                       | • NVDDD_SEL[2:0] : Specify the NVDDD voltage selection. |          |              |              |              |    |             |             |             |     |
|                                                       | NVDDD_SEL[2:0]                                          |          | NVDDD (V)    |              |              |    |             |             |             |     |
|                                                       | 0                                                       |          | -1.5         |              |              |    |             |             |             |     |
|                                                       | 1                                                       |          | -1.6         |              |              |    |             |             |             |     |
| 2                                                     |                                                         | -1.7     |              |              |              |    |             |             |             |     |
| 3                                                     |                                                         | -1.75    |              |              |              |    |             |             |             |     |
| 4                                                     |                                                         | -1.8     |              |              |              |    |             |             |             |     |
| 5                                                     |                                                         | -1.85    |              |              |              |    |             |             |             |     |
| 6                                                     |                                                         | -1.9     |              |              |              |    |             |             |             |     |
| 7                                                     |                                                         | -2.0     |              |              |              |    |             |             |             |     |
| • VDDD_SEL[2:0] : Specify the VDDD voltage selection. |                                                         |          |              |              |              |    |             |             |             |     |
| VDDD_SEL[2:0]                                         |                                                         | VDDD (V) |              |              |              |    |             |             |             |     |
| 0                                                     |                                                         | 1.5      |              |              |              |    |             |             |             |     |
| 1                                                     |                                                         | 1.6      |              |              |              |    |             |             |             |     |
| 2                                                     |                                                         | 1.7      |              |              |              |    |             |             |             |     |
| 3                                                     |                                                         | 1.75     |              |              |              |    |             |             |             |     |
| 4                                                     |                                                         | 1.8      |              |              |              |    |             |             |             |     |
| 5                                                     |                                                         | 1.85     |              |              |              |    |             |             |             |     |
| 6                                                     |                                                         | 1.9      |              |              |              |    |             |             |             |     |
| 7                                                     |                                                         | 2.0      |              |              |              |    |             |             |             |     |

## 6.2.60 SETSCR (C0h)

| C0 H          | SETSCR |             |             |             |             |             |             |             |             |     |
|---------------|--------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Command       | R/W    | D7          | D6          | D5          | D4          | D3          | D2          | D1          | D0          | HEX |
| Command       | W      | 1           | 1           | 0           | 0           | 0           | 0           | 0           | 0           | C0  |
| Parameter 1st | R/W    | N_POPO N[7] | N_POPO N[6] | N_POPO N[5] | N_POPO N[4] | N_POPO N[3] | N_POPO N[2] | N_POPO N[1] | N_POPO N[0] | 73  |
| Parameter 2nd | R/W    | N_NOPO N[7] | N_NOPO N[6] | N_NOPO N[5] | N_NOPO N[4] | N_NOPO N[3] | N_NOPO N[2] | N_NOPO N[1] | N_NOPO N[0] | 73  |
| Parameter 3rd | R/W    | I_POPO N[7] | I_POPO N[6] | I_POPO N[5] | I_POPO N[4] | I_POPO N[3] | I_POPO N[2] | I_POPO N[1] | I_POPO N[0] | 50  |
| Parameter 4th | R/W    | I_NOPO N[7] | I_NOPO N[6] | I_NOPO N[5] | I_NOPO N[4] | I_NOPO N[3] | I_NOPO N[2] | I_NOPO N[1] | I_NOPO N[0] | 50  |
| Parameter 5th | R/W    | SCR[31]     | SCR[30]     | SCR[29]     | SCR[28]     | SCR[27]     | SCR[26]     | SCR[25]     | SCR[24]     | C0  |
| Parameter 6th | R/W    | SCR[23]     | SCR[22]     | SCR[21]     | SCR[20]     | SCR[19]     | SCR[18]     | SCR[17]     | SCR[16]     | 00  |
| Parameter 7th | R/W    | SCR[15]     | SCR[14]     | SCR[13]     | SCR[12]     | SCR[11]     | SCR[10]     | SCR[9]      | SCR[8]      | 08  |
| Parameter 8th | R/W    | SCR[7]      | SCR[6]      | SCR[5]      | SCR[4]      | SCR[3]      | SCR[2]      | SCR[1]      | SCR[0]      | 70  |

This command is used to set related setting of Source driving.

**Description**

- **N\_POPO [7:0]**  
- Specify the source OP Amp driving period for positive polarity in Normal Mode.
- **N\_NOPON [7:0]**  
- Specify the source OP Amp driving period for negative polarity in Normal Mode.
- **I\_POPO [7:0]**  
- Specify the source OP Amp driving period for positive polarity in Idle mode.
- **I\_NOPON [7:0]**  
- Specify the source OP Amp driving period for negative polarity in Idle Mode.

| OPON[7:0] | Source OP Period         |
|-----------|--------------------------|
| 0         | $1 \times 4 / F_{osc}$   |
| 1         | $1 \times 4 / F_{osc}$   |
| 2         | $2 \times 4 / F_{osc}$   |
| 3         | $3 \times 4 / F_{osc}$   |
| :         | :                        |
| FD        | $253 \times 4 / F_{osc}$ |
| FE        | $254 \times 4 / F_{osc}$ |
| FF        | $255 \times 4 / F_{osc}$ |

Description

- **SCR [3:0]** : Source and Gamma bias current core tune.

| SCR[3:0] | lbias | SCR[3:0] | lbias                |
|----------|-------|----------|----------------------|
| 0        | 1     | C        | 3.0                  |
| 1        | 1.25  | D        | 3.25                 |
| 2        | 1.5   | E        | 3.5                  |
| 3        | 1.75  | F        | 3.75                 |
| 4        | 2.0   | others   | Setting<br>Inhibited |
| 5        | 2.25  |          |                      |
| 6        | 2.5   |          |                      |
| 7        | 2.75  |          |                      |

- **SCR [8:4]** : Source bias current fine tune.
- **SCR [13:9]** : Gamma bias current fine tune.

| SCR[8:4]<br>SCR[13:9]<br>(Hex) | Current<br>xlbias | SCR[8:4]<br>SCR[13:9] | Current<br>xlbias |
|--------------------------------|-------------------|-----------------------|-------------------|
| 0                              | Setting Inhibited | 0                     | 16                |
| 1                              | 1                 | 1                     | 17                |
| 2                              | 2                 | 2                     | 18                |
| 3                              | 3                 | 3                     | 19                |
| 4                              | 4                 | 4                     | 20                |
| 5                              | 5                 | 5                     | 21                |
| 6                              | 6                 | 6                     | 22                |
| 7                              | 7                 | 7                     | 23                |
| 8                              | 8                 | 8                     | 24                |
| 9                              | 9                 | 9                     | 25                |
| A                              | 10                | A                     | 26                |
| B                              | 11                | B                     | 27                |
| C                              | 12                | C                     | 28                |
| D                              | 13                | D                     | 29                |
| E                              | 14                | E                     | 30                |
| F                              | 15                | F                     | 31                |

## 6.2.61 SETPOWER (C1h)

| C1 H                                                                                                                                                                                                      | SETPOWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |                     |                     |                     |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|--------------------|------------------|------------------|------------------|-----|---------------------|--------------------|---------------------|--------------------|---------------------|--------------------|---|----|---|-----|---|-----|---|----|---|-----|---|-----|---|----|---|-----|---|-----|---|-----|---|-----|-----|-----|
|                                                                                                                                                                                                           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D7                  | D6                  | D5                  | D4                  | D3                 | D2               | D1               | D0               | HEX |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Command                                                                                                                                                                                                   | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1                   | 1                   | 0                   | 0                   | 0                  | 0                | 0                | 1                | C1  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 1st                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VBTHS[3<br>]        | VBTHS[2<br>]        | VBTHS[1<br>]        | VBTHS[0<br>]        | VBTLS[3]           | VBTLS[2]         | VBTLS[1]         | VBTLS[0]         | 53  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 2nd                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | FBOFF_VGH           | FBOFF_VGL           | x                   | x                   | x                  | x                | x                | x                | C0  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 3rd                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | x                   | x                   | VRP[5]              | VRP[4]              | VRP[3]             | VRP[2]           | VRP[1]           | VRP[0]           | 16  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 4th                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | x                   | x                   | VRN[5]              | VRN[4]              | VRN[3]             | VRN[2]           | VRN[1]           | VRN[0]           | 16  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 5th                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | x                   | x                   | x                   | x                   | x                  | x                | x                | x                | 00  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 6th                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGL_DE<br>T_EN      | VGH_DE<br>T_EN      | VGL_TU<br>RBO       | VGH_TU<br>RBO       | x                  | APS[2]           | APS[1]           | APS[0]           | D1  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 7th                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH1_L<br>DIV[3]    | VGH1_L<br>DIV[2]    | VGH1_L<br>DIV[1]    | VGH1_L<br>DIV[0]    | VGL1_L<br>DIV[3]   | VGL1_L<br>DIV[2] | VGL1_L<br>DIV[1] | VGL1_L<br>DIV[0] | CC  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 8th                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH1_R<br>DIV[3]    | VGH1_R<br>DIV[2]    | VGH1_R<br>DIV[1]    | VGH1_R<br>DIV[0]    | VGL1_R<br>DIV[3]   | VGL1_R<br>DIV[2] | VGL1_R<br>DIV[1] | VGL1_R<br>DIV[0] | DD  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 9th                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH2_L<br>DIV[3]    | VGH2_L<br>DIV[2]    | VGH2_L<br>DIV[1]    | VGH2_L<br>DIV[0]    | VGL2_L<br>DIV[3]   | VGL2_L<br>DIV[2] | VGL2_L<br>DIV[1] | VGL2_L<br>DIV[0] | 67  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 10th                                                                                                                                                                                            | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH2_R<br>DIV[3]    | VGH2_R<br>DIV[2]    | VGH2_R<br>DIV[1]    | VGH2_R<br>DIV[0]    | VGL2_R<br>DIV[3]   | VGL2_R<br>DIV[2] | VGL2_R<br>DIV[1] | VGL2_R<br>DIV[0] | 77  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 11h                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH3_L<br>DIV[3]    | VGH3_L<br>DIV[2]    | VGH3_L<br>DIV[1]    | VGH3_L<br>DIV[0]    | VGL3_L<br>DIV[3]   | VGL3_L<br>DIV[2] | VGL3_L<br>DIV[1] | VGL3_L<br>DIV[0] | 33  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 12h                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH3_R<br>DIV[3]    | VGH3_R<br>DIV[2]    | VGH3_R<br>DIV[1]    | VGH3_R<br>DIV[0]    | VGL3_R<br>DIV[3]   | VGL3_R<br>DIV[2] | VGL3_R<br>DIV[1] | VGL3_R<br>DIV[0] | 33  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 13h                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH4_L<br>DIV[3]    | VGH4_L<br>DIV[2]    | VGH4_L<br>DIV[1]    | VGH4_L<br>DIV[0]    | VGL4_L<br>DIV[3]   | VGL4_L<br>DIV[2] | VGL4_L<br>DIV[1] | VGL4_L<br>DIV[0] | 11  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 14h                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH4_R<br>DIV[3]    | VGH4_R<br>DIV[2]    | VGH4_R<br>DIV[1]    | VGH4_R<br>DIV[0]    | VGL4_R<br>DIV[3]   | VGL4_R<br>DIV[2] | VGL4_R<br>DIV[1] | VGL4_R<br>DIV[0] | 11  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 15h                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH5_L<br>DIV[3]    | VGH5_L<br>DIV[2]    | VGH5_L<br>DIV[1]    | VGH5_L<br>DIV[0]    | VGL5_L<br>DIV[3]   | VGL5_L<br>DIV[2] | VGL5_L<br>DIV[1] | VGL5_L<br>DIV[0] | 00  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Parameter 16h                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH5_R<br>DIV[3]    | VGH5_R<br>DIV[2]    | VGH5_R<br>DIV[1]    | VGH5_R<br>DIV[0]    | VGL5_R<br>DIV[3]   | VGL5_R<br>DIV[2] | VGL5_R<br>DIV[1] | VGL5_R<br>DIV[0] | 00  |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| Description                                                                                                                                                                                               | This command is used to set related setting of power.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |                     |                     |                     |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
|                                                                                                                                                                                                           | • <b>VBTHS[3:0]</b> : The voltage adjustment of VGH.<br>- FBOFF_VGH must be 0                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |                     |                     |                     |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
|                                                                                                                                                                                                           | <table><tr><th>VBTHS[3:0]<br/>(Hex)</th><th>VGH Voltage<br/>(V)</th><th>VBTHS[3:0]<br/>(Hex)</th><th>VGH Voltage<br/>(V)</th><th>VBTHS[3:0]<br/>(Hex)</th><th>VGH Voltage<br/>(V)</th></tr><tr><td>0</td><td>10</td><td>4</td><td>14</td><td>8</td><td>18</td></tr><tr><td>1</td><td>11</td><td>5</td><td>15</td><td>9</td><td>18</td></tr><tr><td>2</td><td>12</td><td>6</td><td>16</td><td>A</td><td>18</td></tr><tr><td>3</td><td>13</td><td>7</td><td>17</td><td>B~F</td><td>18</td></tr></table>          |                     |                     |                     |                     |                    |                  |                  |                  |     | VBTHS[3:0]<br>(Hex) | VGH Voltage<br>(V) | VBTHS[3:0]<br>(Hex) | VGH Voltage<br>(V) | VBTHS[3:0]<br>(Hex) | VGH Voltage<br>(V) | 0 | 10 | 4 | 14  | 8 | 18  | 1 | 11 | 5 | 15  | 9 | 18  | 2 | 12 | 6 | 16  | A | 18  | 3 | 13  | 7 | 17  | B~F | 18  |
|                                                                                                                                                                                                           | VBTHS[3:0]<br>(Hex)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH Voltage<br>(V)  | VBTHS[3:0]<br>(Hex) | VGH Voltage<br>(V)  | VBTHS[3:0]<br>(Hex) | VGH Voltage<br>(V) |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
|                                                                                                                                                                                                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 10                  | 4                   | 14                  | 8                   | 18                 |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
|                                                                                                                                                                                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 11                  | 5                   | 15                  | 9                   | 18                 |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
|                                                                                                                                                                                                           | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 12                  | 6                   | 16                  | A                   | 18                 |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
|                                                                                                                                                                                                           | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 13                  | 7                   | 17                  | B~F                 | 18                 |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
|                                                                                                                                                                                                           | • <b>VBTLS[3:0]</b> : The voltage adjustment of VGL.<br>- FBOFF_VGL must be 0                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |                     |                     |                     |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
|                                                                                                                                                                                                           | <table><tr><th>VBTLS[3:0]<br/>(Hex)</th><th>VGL Voltage<br/>(V)</th><th>VBTLS[3:0]<br/>(Hex)</th><th>VGL Voltage<br/>(V)</th><th>VBTLS[3:0]<br/>(Hex)</th><th>VGL Voltage<br/>(V)</th></tr><tr><td>0</td><td>-7</td><td>4</td><td>-11</td><td>8</td><td>-15</td></tr><tr><td>1</td><td>-8</td><td>5</td><td>-12</td><td>9</td><td>-16</td></tr><tr><td>2</td><td>-9</td><td>6</td><td>-13</td><td>A</td><td>-17</td></tr><tr><td>3</td><td>-10</td><td>7</td><td>-14</td><td>B~F</td><td>-18</td></tr></table> |                     |                     |                     |                     |                    |                  |                  |                  |     | VBTLS[3:0]<br>(Hex) | VGL Voltage<br>(V) | VBTLS[3:0]<br>(Hex) | VGL Voltage<br>(V) | VBTLS[3:0]<br>(Hex) | VGL Voltage<br>(V) | 0 | -7 | 4 | -11 | 8 | -15 | 1 | -8 | 5 | -12 | 9 | -16 | 2 | -9 | 6 | -13 | A | -17 | 3 | -10 | 7 | -14 | B~F | -18 |
| VBTLS[3:0]<br>(Hex)                                                                                                                                                                                       | VGL Voltage<br>(V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | VBTLS[3:0]<br>(Hex) | VGL Voltage<br>(V)  | VBTLS[3:0]<br>(Hex) | VGL Voltage<br>(V)  |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| 0                                                                                                                                                                                                         | -7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4                   | -11                 | 8                   | -15                 |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| 1                                                                                                                                                                                                         | -8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5                   | -12                 | 9                   | -16                 |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| 2                                                                                                                                                                                                         | -9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 6                   | -13                 | A                   | -17                 |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| 3                                                                                                                                                                                                         | -10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 7                   | -14                 | B~F                 | -18                 |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| • <b>FBOFF_VGH</b> : Enable VGH feedback detect function.<br>- 1 : Disable VGH feedback voltage detection. Output voltage = 15V<br>- 0 : Enable VGH feedback voltage detection. Output voltage = VBTHS    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |                     |                     |                     |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |
| • <b>FBOFF_VGL</b> : Enable VGL feedback detect function.<br>- 1 : Disable VGL feedback voltage detection. Output voltage = -10V.<br>- 0 : Enable VGL feedback voltage detection. Output voltage = VBTL5. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |                     |                     |                     |                    |                  |                  |                  |     |                     |                    |                     |                    |                     |                    |   |    |   |     |   |     |   |    |   |     |   |     |   |    |   |     |   |     |   |     |   |     |     |     |

Description

- **VRP[5:0]/VRN[5:0]** : Specify the VSPROUT/VSNROUT voltage.
  - VSPROUT= {Decimal (VRH [5:0]) x0.05+3.3} x (VREF/4.8) if VREF [4] =0.
  - VSNROUT= {Decimal (VRH [5:0]) x0.05+3.3} x (VREF/5.6) if VREF [4] =1.

| VRP[5:0]<br>VRN[5:0]<br>(Hex) | VSPROUT<br> VSNROUT <br>(V) | VRP[5:0]<br>VRN[5:0]<br>(Hex) | VSPROUT<br> VSNROUT <br>(V) | VRP[5:0]<br>VRN[5:0]<br>(Hex) | VSPROUT<br> VSNROUT <br>(V) |
|-------------------------------|-----------------------------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 0                             | 3.23                        | E                             | 3.92                        | 1C                            | 4.60                        |
| 1                             | 3.28                        | F                             | 3.97                        | 1D                            | 4.65                        |
| 2                             | 3.33                        | 10                            | 4.01                        | 1E                            | 4.70                        |
| 3                             | 3.38                        | 11                            | 4.06                        | 1F*                           | 4.70                        |
| 4                             | 3.43                        | 12                            | 4.11                        | 20*                           | 5.2                         |
| 5                             | 3.48                        | 13                            | 4.16                        | 21*                           | 5.3                         |
| 6                             | 3.53                        | 14                            | 4.21                        | 22*                           | 5.4                         |
| 7                             | 3.57                        | 15                            | 4.26                        | 23*                           | 5.5                         |
| 8                             | 3.62                        | 16                            | 4.31                        | 24*                           | 5.55                        |
| 9                             | 3.67                        | 17                            | 4.36                        | 25*                           | 5.6                         |
| A                             | 3.72                        | 18                            | 4.41                        | 26*                           | 5.65                        |
| B                             | 3.77                        | 19                            | 4.46                        | 27*                           | 5.7                         |
| C                             | 3.82                        | 1A                            | 4.50                        | 28                            | STOP                        |
| D                             | 3.87                        | 1B                            | 4.55                        | 29~3F                         | STOP                        |

Note: When VREF [4] =0, the max voltage of VSPROUT/VSNROUT will be 4.7V/-4.7V.

- **VGL\_DET\_EN** : Enable VGL voltage Detect Function.
  - 09h\_01[3] = 1: VGL voltage Abnormal.
  - 09h\_01[3] = 0: VGL voltage Normal.
- **VGH\_DET\_EN** : Enable VGH voltage Detect Function.
  - 09h\_01[0] = 1 : VGH voltage Abnormal.
  - 09h\_01[0] = 0 : VGH voltage Normal.
- **VGL\_TURBO** : Enlarge VGL Voltage at FBOFF\_VGL=1.
  - 1 : VGL=-15V
  - 0 : VGL=-10V
- **VGH\_TURBO** : Enlarge VGH Voltage at FBOFF\_VGH=1.
  - 1 : VGH=20V
  - 0 : VGH=15V

Description

- **APS[2:0]** : Adjust power bias current of all regulator voltage.

| APS[2:0] |   |   | Bias Current    |
|----------|---|---|-----------------|
| 0        | 0 | 0 | Stop            |
| 0        | 0 | 1 | Small           |
| 0        | 1 | 0 | Medium Small    |
| 0        | 1 | 1 | Medium Large    |
| 1        | 0 | 0 | Large           |
| others   |   |   | Setting Invalid |

- **VGH1\_L\_DIV[3:0]** : Specify left side VGH stage 1 pumping frequency.
- **VGH1\_R\_DIV[3:0]** : Specify right side VGH stage 1 pumping frequency.
- **VGH2\_L\_DIV[3:0]** : Specify left side VGH stage 2 pumping frequency.
- **VGH2\_R\_DIV[3:0]** : Specify right side VGH stage 2 pumping frequency.
- **VGH3\_L\_DIV[3:0]** : Specify left side VGH stage 3 pumping frequency.
- **VGH3\_R\_DIV[3:0]** : Specify right side VGH stage 3 pumping frequency.
- **VGH4\_L\_DIV[3:0]** : Specify left side VGH stage 4 pumping frequency.
- **VGH4\_R\_DIV[3:0]** : Specify right side VGH stage 4 pumping frequency.
- **VGH5\_L\_DIV[3:0]** : Specify left side VGH stage 5 pumping frequency.
- **VGH5\_R\_DIV[3:0]** : Specify right side VGH stage 5 pumping frequency.
- **VGL1\_L\_DIV[3:0]** : Specify left side VGL stage 1 pumping frequency.
- **VGL1\_R\_DIV[3:0]** : Specify right side VGL stage 1 pumping frequency.
- **VGL2\_L\_DIV[3:0]** : Specify left side VGL stage 2 pumping frequency.
- **VGL2\_R\_DIV[3:0]** : Specify right side VGL stage 2 pumping frequency.
- **VGL3\_L\_DIV[3:0]** : Specify left side VGL stage 3 pumping frequency.
- **VGL3\_R\_DIV[3:0]** : Specify right side VGL stage 3 pumping frequency.
- **VGL4\_L\_DIV[3:0]** : Specify left side VGL stage 4 pumping frequency.
- **VGL4\_R\_DIV[3:0]** : Specify right side VGL stage 4 pumping frequency.
- **VGL5\_L\_DIV[3:0]** : Specify left side VGL stage 5 pumping frequency.
- **VGL5\_R\_DIV[3:0]** : Specify right side VGL stage 5 pumping frequency.

| VGH1/2/3/4/5_DIV[3:0]<br>VGL1/2/3/4/5_DIV[3:0] |   |   |   | Frequency (MHz) |
|------------------------------------------------|---|---|---|-----------------|
| 0                                              | 0 | 0 | 0 | 36              |
| 0                                              | 0 | 0 | 1 | 18              |
| 0                                              | 0 | 1 | 0 | 12              |
| 0                                              | 0 | 1 | 1 | 9               |
| 0                                              | 1 | 0 | 0 | 7.2             |
| 0                                              | 1 | 0 | 1 | 6               |
| 0                                              | 1 | 1 | 0 | 5.1             |
| 0                                              | 1 | 1 | 1 | 4.5             |
| 1                                              | 0 | 0 | 0 | 4               |
| 1                                              | 0 | 0 | 1 | 3.6             |
| 1                                              | 0 | 1 | 0 | 3.3             |
| 1                                              | 0 | 1 | 1 | 3               |
| 1                                              | 1 | 0 | 0 | 2.6             |
| 1                                              | 1 | 0 | 1 | 2.1             |
| 1                                              | 1 | 1 | 0 | 1.8             |
| 1                                              | 1 | 1 | 1 | 1.5             |

### 6.2.62 SETID (C3h)

| C3 H          | SETID                                                                                                                        |        |                              |        |        |        |            |            |            |     |
|---------------|------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------|--------|--------|--------|------------|------------|------------|-----|
|               | R/W                                                                                                                          | D7     | D6                           | D5     | D4     | D3     | D2         | D1         | D0         | HEX |
| Command       | W                                                                                                                            | 1      | 1                            | 0      | 0      | 0      | 0          | 1          | 1          | C3  |
| Parameter 1st | R/W                                                                                                                          | ID1[7] | ID1[6]                       | ID1[5] | ID1[4] | ID1[3] | ID1[2]     | ID1[1]     | ID1[0]     | 38  |
| Parameter 2nd | R/W                                                                                                                          | ID2[7] | ID2[6]                       | ID2[5] | ID2[4] | ID2[3] | ID2[2]     | ID2[1]     | ID2[0]     | 21  |
| Parameter 3rd | R/W                                                                                                                          | ID3[7] | ID3[6]                       | ID3[5] | ID3[4] | ID3[3] | ID3[2]     | ID3[1]     | ID3[0]     | 1F  |
| Parameter 4th | R                                                                                                                            | x      | x                            | x      | x      | x      | ID_TIME[2] | ID_TIME[1] | ID_TIME[0] | 00  |
| Description   | This command is used for setting ID.                                                                                         |        |                              |        |        |        |            |            |            |     |
|               | • <b>ID1[7:0] / ID2[7:0] / ID3[7:0]</b> : Set ID1 / ID2 / ID3 respectively<br>- 04h / DAh / DBh / DEh change correspondingly |        |                              |        |        |        |            |            |            |     |
|               | • <b>ID_TIME[2:0]</b> : Read ID OTP programming times.                                                                       |        |                              |        |        |        |            |            |            |     |
|               | <b>ID_TIME[2:0]</b>                                                                                                          |        | <b>OTP Programming times</b> |        |        |        |            |            |            |     |
|               | 0                                                                                                                            |        | NO OTPED                     |        |        |        |            |            |            |     |
|               | 1                                                                                                                            |        | 1                            |        |        |        |            |            |            |     |
|               | 2                                                                                                                            |        | 2                            |        |        |        |            |            |            |     |
|               | 3                                                                                                                            |        | 3                            |        |        |        |            |            |            |     |
| 4             |                                                                                                                              | 4      |                              |        |        |        |            |            |            |     |

### 6.2.63 SETDDB (C4h)

| C4 H          | SETDDB                                                                                                                                                   |         |         |         |         |         |         |         |         |     |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|-----|
|               | R/W                                                                                                                                                      | D7      | D6      | D5      | D4      | D3      | D2      | D1      | D0      | HEX |
| Command       | W                                                                                                                                                        | 1       | 1       | 0       | 0       | 0       | 1       | 0       | 0       | C4  |
| Parameter 1st | R/W                                                                                                                                                      | DDB1[7] | DDB1[6] | DDB1[5] | DDB1[4] | DDB1[3] | DDB1[2] | DDB1[1] | DDB1[0] | 00  |
| Parameter 2nd | R/W                                                                                                                                                      | DDB2[7] | DDB2[6] | DDB2[5] | DDB2[4] | DDB2[3] | DDB2[2] | DDB2[1] | DDB2[0] | 00  |
| Parameter 3rd | R/W                                                                                                                                                      | DDB3[7] | DDB3[6] | DDB3[5] | DDB3[4] | DDB3[3] | DDB3[2] | DDB3[1] | DDB3[0] | 00  |
| Parameter 4th | R/W                                                                                                                                                      | DDB4[7] | DDB4[6] | DDB4[5] | DDB4[4] | DDB4[3] | DDB4[2] | DDB4[1] | DDB4[0] | 00  |
| Description   | <p>This command is used to set CMD DDB1~4 value.</p> <ul style="list-style-type: none"> <li>• <b>DDB1[7:0] ~ DDB4[7:0]</b> : Set DDB1 ~ DDB4.</li> </ul> |         |         |         |         |         |         |         |         |     |



## 6.2.64 SETIO (C7h)

| C7 H                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                     | SETIO           |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|----------------|----------------|-------------|----------------|----------------|----------------|--------------|-----------------|-----------------|---|---|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|
|                                                                                                                                                                                                                                                                                                                                     | R/W                                                                                                                                                                                                                                                                                                                                 | D7              | D6             | D5             | D4             | D3          | D2             | D1             | D0             | HEX          |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| Command                                                                                                                                                                                                                                                                                                                             | W                                                                                                                                                                                                                                                                                                                                   | 1               | 1              | 0              | 0              | 0           | 1              | 1              | 1              | C7           |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| Parameter 1st                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                                                                                 | PWM_O<br>E      | x              | INVPWM         | VOUT_O<br>E    | HOUT_O<br>E | x              | x              | x              | 00           |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| Parameter 2nd                                                                                                                                                                                                                                                                                                                       | R/W                                                                                                                                                                                                                                                                                                                                 | x               | V_DELA<br>Y[2] | V_DELA<br>Y[1] | V_DELA<br>Y[0] | x           | H_DELA<br>Y[2] | H_DELA<br>Y[1] | H_DELA<br>Y[0] | 00           |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| Description                                                                                                                                                                                                                                                                                                                         | This command is used to set I/O related register.                                                                                                                                                                                                                                                                                   |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>• <b>PWM_OE</b> : PWM output CABC PWM signal.<ul style="list-style-type: none"><li>- 1 : Enable CABC PWM signal.</li><li>- 0 : Disable CABC PWM signal.</li></ul></li></ul>                                                                                                                   |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>• <b>INVPWM</b> : Inverse polarity of CABC PWM signal.<ul style="list-style-type: none"><li>- 1 : CABC PWM signal is inversed.</li><li>- 0 : Original CABC PWM signal.</li></ul></li></ul>                                                                                                    |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>• <b>VOUT_OE</b> : VOUT pin output frame synchronization signal.</li></ul>                                                                                                                                                                                                                    |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>• <b>HOUT_OE</b> : VOUT pin output horizontal synchronization signal.</li></ul>                                                                                                                                                                                                               |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>• <b>V_DELAY[2:0]</b> : Specify Vertical Sync. delay time.</li></ul>                                                                                                                                                                                                                          |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|                                                                                                                                                                                                                                                                                                                                     | <table><tr><th>V_DELAY[2:0]</th><th>Delay time (ns)</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1666</td></tr><tr><td>2</td><td>3332</td></tr><tr><td>3</td><td>4998</td></tr><tr><td>4</td><td>6664</td></tr><tr><td>5</td><td>8330</td></tr><tr><td>6</td><td>9996</td></tr><tr><td>7</td><td>11662</td></tr></table> |                 |                |                |                |             |                |                |                |              | V_DELAY[2:0]    | Delay time (ns) | 0 | 0 | 1    | 1666 | 2    | 3332 | 3    | 4998 | 4    | 6664 | 5    | 8330 | 6    | 9996 | 7     | 11662 |
|                                                                                                                                                                                                                                                                                                                                     | V_DELAY[2:0]                                                                                                                                                                                                                                                                                                                        | Delay time (ns) |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|                                                                                                                                                                                                                                                                                                                                     | 0                                                                                                                                                                                                                                                                                                                                   | 0               |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|                                                                                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                                                                                   | 1666            |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 2                                                                                                                                                                                                                                                                                                                                   | 3332                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 3                                                                                                                                                                                                                                                                                                                                   | 4998                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 4                                                                                                                                                                                                                                                                                                                                   | 6664                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 5                                                                                                                                                                                                                                                                                                                                   | 8330                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 6                                                                                                                                                                                                                                                                                                                                   | 9996                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 7                                                                                                                                                                                                                                                                                                                                   | 11662                                                                                                                                                                                                                                                                                                                               |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| <ul style="list-style-type: none"><li>• <b>H_DELAY[2:0]</b> : Specify horizontal sync delay time.</li></ul>                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                     |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| <table><tr><th>H_DELAY[2:0]</th><th>Delay time (ns)</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1666</td></tr><tr><td>2</td><td>3332</td></tr><tr><td>3</td><td>4998</td></tr><tr><td>4</td><td>6664</td></tr><tr><td>5</td><td>8330</td></tr><tr><td>6</td><td>9996</td></tr><tr><td>7</td><td>11662</td></tr></table> |                                                                                                                                                                                                                                                                                                                                     |                 |                |                |                |             |                |                |                | H_DELAY[2:0] | Delay time (ns) | 0               | 0 | 1 | 1666 | 2    | 3332 | 3    | 4998 | 4    | 6664 | 5    | 8330 | 6    | 9996 | 7    | 11662 |       |
| H_DELAY[2:0]                                                                                                                                                                                                                                                                                                                        | Delay time (ns)                                                                                                                                                                                                                                                                                                                     |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 0                                                                                                                                                                                                                                                                                                                                   | 0                                                                                                                                                                                                                                                                                                                                   |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 1                                                                                                                                                                                                                                                                                                                                   | 1666                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 2                                                                                                                                                                                                                                                                                                                                   | 3332                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 3                                                                                                                                                                                                                                                                                                                                   | 4998                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 4                                                                                                                                                                                                                                                                                                                                   | 6664                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 5                                                                                                                                                                                                                                                                                                                                   | 8330                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 6                                                                                                                                                                                                                                                                                                                                   | 9996                                                                                                                                                                                                                                                                                                                                |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
| 7                                                                                                                                                                                                                                                                                                                                   | 11662                                                                                                                                                                                                                                                                                                                               |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |
|                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                     |                 |                |                |                |             |                |                |                |              |                 |                 |   |   |      |      |      |      |      |      |      |      |      |      |      |      |       |       |

### 6.2.65 SETCABC (C8h)

| C8 H                                                         | SETCABC                                                 |                      |                   |                   |                   |                   |                   |                   |                   |     |
|--------------------------------------------------------------|---------------------------------------------------------|----------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-----|
|                                                              | R/W                                                     | D7                   | D6                | D5                | D4                | D3                | D2                | D1                | D0                | HEX |
| Command                                                      | W                                                       | 1                    | 1                 | 0                 | 0                 | 1                 | 0                 | 0                 | 0                 | C8  |
| Parameter 1st                                                | R/W                                                     | x                    | PWMDIV<br>[2]     | PWMDIV<br>[1]     | PWMDIV<br>[0]     | x                 | x                 | x                 | x                 | 10  |
| Parameter 2nd                                                | R/W                                                     | PWM_PE<br>RIOD[7]    | PWM_PE<br>RIOD[6] | PWM_PE<br>RIOD[5] | PWM_PE<br>RIOD[4] | PWM_PE<br>RIOD[3] | PWM_PE<br>RIOD[2] | PWM_PE<br>RIOD[1] | PWM_PE<br>RIOD[0] | 40  |
| Description                                                  | This command is used to set CABC related register.      |                      |                   |                   |                   |                   |                   |                   |                   |     |
|                                                              | • <b>PWMDIV[2:0]</b> : CABC PWM Frequency Fpwm control. |                      |                   |                   |                   |                   |                   |                   |                   |     |
|                                                              | <b>PWMDIV[2:0]</b>                                      |                      | <b>PWM Fpwm</b>   |                   |                   |                   |                   |                   |                   |     |
|                                                              | 0                                                       |                      | Fosc/1            |                   |                   |                   |                   |                   |                   |     |
|                                                              | 1                                                       |                      | Fosc/2            |                   |                   |                   |                   |                   |                   |     |
| 2                                                            |                                                         | Fosc/4               |                   |                   |                   |                   |                   |                   |                   |     |
| 3                                                            |                                                         | Fosc/8               |                   |                   |                   |                   |                   |                   |                   |     |
| 4                                                            |                                                         | Fosc/16              |                   |                   |                   |                   |                   |                   |                   |     |
|                                                              |                                                         |                      |                   |                   |                   |                   |                   |                   |                   |     |
| • <b>PWMPERIOD[7:0]</b> : CABC PWM output frequency control. |                                                         |                      |                   |                   |                   |                   |                   |                   |                   |     |
| <b>PWMPERIOD[7:0]</b>                                        |                                                         | <b>PWM frequency</b> |                   |                   |                   |                   |                   |                   |                   |     |
| 0                                                            |                                                         | Fpwm/1               |                   |                   |                   |                   |                   |                   |                   |     |
| 1                                                            |                                                         | Fpwm/1               |                   |                   |                   |                   |                   |                   |                   |     |
| 2                                                            |                                                         | Fpwm/2               |                   |                   |                   |                   |                   |                   |                   |     |
| :                                                            |                                                         | :                    |                   |                   |                   |                   |                   |                   |                   |     |
| 3F                                                           |                                                         | Fpwm/127             |                   |                   |                   |                   |                   |                   |                   |     |

### 6.2.66 SETPANEL (CCh)

| C8 H          | SETPANEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |              |              |               |               |     |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|--------------|--------------|---------------|---------------|-----|
|               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D7 | D6 | D5 | D4 | D3           | D2           | D1            | D0            | HEX |
| Command       | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1  | 1  | 0  | 0  | 1            | 1            | 0             | 0             | CC  |
| Parameter 1st | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    | x  | x  | x  | SS_PAN<br>EL | GS_PAN<br>EL | REV_PA<br>NEL | BGR_PA<br>NEL | 00  |
| Description   | <p>This command is used to set display related register.</p> <ul style="list-style-type: none"> <li>• <b>SS_PANEL</b> : Reverse the source scan direction. <ul style="list-style-type: none"> <li>- 1 : enable reverse the source scan direction.</li> <li>- 0 : normal scan direction from S1-&gt;S1440.</li> </ul> </li> <li>• <b>GS_PANEL</b> : Reverse the vertical scan direction. <ul style="list-style-type: none"> <li>- 1 : Enable reverse the vertical scan direction.</li> <li>- 0 : Normal vertical scan direction.</li> </ul> </li> <li>• <b>REV_PANEL</b> : Reverse the source polarity. <ul style="list-style-type: none"> <li>- 1 : Normally black panel</li> <li>- 0 : Normally white panel.</li> </ul> </li> <li>• <b>BGR_PANEL</b>: reverse the source mapping direction with RGB. <ul style="list-style-type: none"> <li>- 1 : S1:S2:S3 = "B" : "G" : "R"</li> <li>- 0 : S1:S2:S3 = "R" : "G" : "B"</li> </ul> </li> </ul> |    |    |    |    |              |              |               |               |     |

## 6.2.67 DGC\_R (CDh)

| CD H           | DGC_R                                                                                                                                                                                                                                                                                                                                                                                               |                    |                    |                    |                    |                    |                    |                    |                    |     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-----|
|                | R/W                                                                                                                                                                                                                                                                                                                                                                                                 | D7                 | D6                 | D5                 | D4                 | D3                 | D2                 | D1                 | D0                 | HEX |
| Command        | W                                                                                                                                                                                                                                                                                                                                                                                                   | 1                  | 1                  | 0                  | 0                  | 1                  | 1                  | 0                  | 1                  | CD  |
| Parameter 1st  | R/W                                                                                                                                                                                                                                                                                                                                                                                                 | x                  | x                  | x                  | x                  | x                  | x                  | x                  | DGC_EN             | 00  |
| Parameter 2nd  | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R00[7] | DGC_LU<br>T_R00[6] | DGC_LU<br>T_R00[5] | DGC_LU<br>T_R00[4] | DGC_LU<br>T_R00[3] | DGC_LU<br>T_R00[2] | DGC_LU<br>T_R00[1] | DGC_LU<br>T_R00[0] | 80  |
| Parameter 3rd  | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R01[7] | DGC_LU<br>T_R01[6] | DGC_LU<br>T_R01[5] | DGC_LU<br>T_R01[4] | DGC_LU<br>T_R01[3] | DGC_LU<br>T_R01[2] | DGC_LU<br>T_R01[1] | DGC_LU<br>T_R01[0] | 80  |
| Parameter 4th  | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R02[7] | DGC_LU<br>T_R02[6] | DGC_LU<br>T_R02[5] | DGC_LU<br>T_R02[4] | DGC_LU<br>T_R02[3] | DGC_LU<br>T_R02[2] | DGC_LU<br>T_R02[1] | DGC_LU<br>T_R02[0] | 80  |
| :              | :                                                                                                                                                                                                                                                                                                                                                                                                   | :                  | :                  | :                  | :                  | :                  | :                  | :                  | :                  |     |
| Parameter 10th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R08[7] | DGC_LU<br>T_R08[6] | DGC_LU<br>T_R08[5] | DGC_LU<br>T_R08[4] | DGC_LU<br>T_R08[3] | DGC_LU<br>T_R08[2] | DGC_LU<br>T_R08[1] | DGC_LU<br>T_R08[0] | 80  |
| Parameter 11th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R09[7] | DGC_LU<br>T_R09[6] | DGC_LU<br>T_R09[5] | DGC_LU<br>T_R09[4] | DGC_LU<br>T_R09[3] | DGC_LU<br>T_R09[2] | DGC_LU<br>T_R09[1] | DGC_LU<br>T_R09[0] | 80  |
| Parameter 12th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R10[7] | DGC_LU<br>T_R10[6] | DGC_LU<br>T_R10[5] | DGC_LU<br>T_R10[4] | DGC_LU<br>T_R10[3] | DGC_LU<br>T_R10[2] | DGC_LU<br>T_R10[1] | DGC_LU<br>T_R10[0] | 80  |
| Parameter 13th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R11[7] | DGC_LU<br>T_R11[6] | DGC_LU<br>T_R11[5] | DGC_LU<br>T_R11[4] | DGC_LU<br>T_R11[3] | DGC_LU<br>T_R11[2] | DGC_LU<br>T_R11[1] | DGC_LU<br>T_R11[0] | 80  |
| :              | :                                                                                                                                                                                                                                                                                                                                                                                                   | :                  | :                  | :                  | :                  | :                  | :                  | :                  | :                  |     |
| Parameter 20th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R18[7] | DGC_LU<br>T_R18[6] | DGC_LU<br>T_R18[5] | DGC_LU<br>T_R18[4] | DGC_LU<br>T_R18[3] | DGC_LU<br>T_R18[2] | DGC_LU<br>T_R18[1] | DGC_LU<br>T_R18[0] | 80  |
| Parameter 21th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R19[7] | DGC_LU<br>T_R19[6] | DGC_LU<br>T_R19[5] | DGC_LU<br>T_R19[4] | DGC_LU<br>T_R19[3] | DGC_LU<br>T_R19[2] | DGC_LU<br>T_R19[1] | DGC_LU<br>T_R19[0] | 80  |
| Parameter 22th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R20[7] | DGC_LU<br>T_R20[6] | DGC_LU<br>T_R20[5] | DGC_LU<br>T_R20[4] | DGC_LU<br>T_R20[3] | DGC_LU<br>T_R20[2] | DGC_LU<br>T_R20[1] | DGC_LU<br>T_R20[0] | 80  |
| Parameter 23th | W:                                                                                                                                                                                                                                                                                                                                                                                                  | DGC_LU<br>T_R21[7] | DGC_LU<br>T_R21[6] | DGC_LU<br>T_R21[5] | DGC_LU<br>T_R21[4] | DGC_LU<br>T_R21[3] | DGC_LU<br>T_R21[2] | DGC_LU<br>T_R21[1] | DGC_LU<br>T_R21[0] | 80  |
| :              | :                                                                                                                                                                                                                                                                                                                                                                                                   | :                  | :                  | :                  | :                  | :                  | :                  | :                  | :                  |     |
| Parameter 30th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R28[7] | DGC_LU<br>T_R28[6] | DGC_LU<br>T_R28[5] | DGC_LU<br>T_R28[4] | DGC_LU<br>T_R28[3] | DGC_LU<br>T_R28[2] | DGC_LU<br>T_R28[1] | DGC_LU<br>T_R28[0] | 80  |
| Parameter 31th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R29[7] | DGC_LU<br>T_R29[6] | DGC_LU<br>T_R29[5] | DGC_LU<br>T_R29[4] | DGC_LU<br>T_R29[3] | DGC_LU<br>T_R29[2] | DGC_LU<br>T_R29[1] | DGC_LU<br>T_R29[0] | 80  |
| Parameter 32th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R30[7] | DGC_LU<br>T_R30[6] | DGC_LU<br>T_R30[5] | DGC_LU<br>T_R30[4] | DGC_LU<br>T_R30[3] | DGC_LU<br>T_R30[2] | DGC_LU<br>T_R30[1] | DGC_LU<br>T_R30[0] | 80  |
| Parameter 33th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R31[7] | DGC_LU<br>T_R31[6] | DGC_LU<br>T_R31[5] | DGC_LU<br>T_R31[4] | DGC_LU<br>T_R31[3] | DGC_LU<br>T_R31[2] | DGC_LU<br>T_R31[1] | DGC_LU<br>T_R31[0] | 80  |
| Parameter 34th | W                                                                                                                                                                                                                                                                                                                                                                                                   | DGC_LU<br>T_R32[7] | DGC_LU<br>T_R32[6] | DGC_LU<br>T_R32[5] | DGC_LU<br>T_R32[4] | DGC_LU<br>T_R32[3] | DGC_LU<br>T_R32[2] | DGC_LU<br>T_R32[1] | DGC_LU<br>T_R32[0] | 80  |
| Description    | <p>This command is used to set Red color digital gamma.</p> <ul style="list-style-type: none"> <li>• <b>DGC_EN</b> : Control digital gamma function. <ul style="list-style-type: none"> <li>- 1 : Enable digital gamma function.</li> <li>- 0 : Disable digital gamma function.</li> </ul> </li> <li>• <b>DGC_LUT_R00 [7:0] ~ DGC_LUT_R32 [7:0]</b> : Red color digital gamma LUT table.</li> </ul> |                    |                    |                    |                    |                    |                    |                    |                    |     |

## 6.2.68 DGC\_G (CEh)

| CE H           | DGC_G                                                                                                                                                      |                    |                    |                    |                    |                    |                    |                    |                    |     |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-----|
|                | R/W                                                                                                                                                        | D7                 | D6                 | D5                 | D4                 | D3                 | D2                 | D1                 | D0                 | HEX |
| Command        | W                                                                                                                                                          | 1                  | 1                  | 0                  | 0                  | 1                  | 1                  | 1                  | 0                  | CE  |
| Parameter 1st  | W                                                                                                                                                          | DGC_LU<br>T_G00[7] | DGC_LU<br>T_G00[6] | DGC_LU<br>T_G00[5] | DGC_LU<br>T_G00[4] | DGC_LU<br>T_G00[3] | DGC_LU<br>T_G00[2] | DGC_LU<br>T_G00[1] | DGC_LU<br>T_G00[0] | 80  |
| Parameter 2nd  | W                                                                                                                                                          | DGC_LU<br>T_G01[7] | DGC_LU<br>T_G01[6] | DGC_LU<br>T_G01[5] | DGC_LU<br>T_G01[4] | DGC_LU<br>T_G01[3] | DGC_LU<br>T_G01[2] | DGC_LU<br>T_G01[1] | DGC_LU<br>T_G01[0] | 80  |
| Parameter 3rd  | W                                                                                                                                                          | DGC_LU<br>T_G02[7] | DGC_LU<br>T_G02[6] | DGC_LU<br>T_G02[5] | DGC_LU<br>T_G02[4] | DGC_LU<br>T_G02[3] | DGC_LU<br>T_G02[2] | DGC_LU<br>T_G02[1] | DGC_LU<br>T_G02[0] | 80  |
| Parameter 4th  | W                                                                                                                                                          | DGC_LU<br>T_G03[7] | DGC_LU<br>T_G03[6] | DGC_LU<br>T_G03[5] | DGC_LU<br>T_G03[4] | DGC_LU<br>T_G03[3] | DGC_LU<br>T_G03[2] | DGC_LU<br>T_G03[1] | DGC_LU<br>T_G03[0] | 80  |
| :              | :                                                                                                                                                          | :                  | :                  | :                  | :                  | :                  | :                  | :                  | :                  |     |
| Parameter 10th | W                                                                                                                                                          | DGC_LU<br>T_G09[7] | DGC_LU<br>T_G09[6] | DGC_LU<br>T_G09[5] | DGC_LU<br>T_G09[4] | DGC_LU<br>T_G09[3] | DGC_LU<br>T_G09[2] | DGC_LU<br>T_G09[1] | DGC_LU<br>T_G09[0] | 80  |
| Parameter 11th | W                                                                                                                                                          | DGC_LU<br>T_G10[7] | DGC_LU<br>T_G10[6] | DGC_LU<br>T_G10[5] | DGC_LU<br>T_G10[4] | DGC_LU<br>T_G10[3] | DGC_LU<br>T_G10[2] | DGC_LU<br>T_G10[1] | DGC_LU<br>T_G10[0] | 80  |
| Parameter 12th | W                                                                                                                                                          | DGC_LU<br>T_G11[7] | DGC_LU<br>T_G11[6] | DGC_LU<br>T_G11[5] | DGC_LU<br>T_G11[4] | DGC_LU<br>T_G11[3] | DGC_LU<br>T_G11[2] | DGC_LU<br>T_G11[1] | DGC_LU<br>T_G11[0] | 80  |
| Parameter 13th | W                                                                                                                                                          | DGC_LU<br>T_G12[7] | DGC_LU<br>T_G12[6] | DGC_LU<br>T_G12[5] | DGC_LU<br>T_G12[4] | DGC_LU<br>T_G12[3] | DGC_LU<br>T_G12[2] | DGC_LU<br>T_G12[1] | DGC_LU<br>T_G12[0] | 80  |
| :              | :                                                                                                                                                          | :                  | :                  | :                  | :                  | :                  | :                  | :                  | :                  |     |
| Parameter 19th | W                                                                                                                                                          | DGC_LU<br>T_G18[7] | DGC_LU<br>T_G18[6] | DGC_LU<br>T_G18[5] | DGC_LU<br>T_G18[4] | DGC_LU<br>T_G18[3] | DGC_LU<br>T_G18[2] | DGC_LU<br>T_G18[1] | DGC_LU<br>T_G18[0] | 80  |
| Parameter 20th | W                                                                                                                                                          | DGC_LU<br>T_G19[7] | DGC_LU<br>T_G19[6] | DGC_LU<br>T_G19[5] | DGC_LU<br>T_G19[4] | DGC_LU<br>T_G19[3] | DGC_LU<br>T_G19[2] | DGC_LU<br>T_G19[1] | DGC_LU<br>T_G19[0] | 80  |
| Parameter 21th | W                                                                                                                                                          | DGC_LU<br>T_G20[7] | DGC_LU<br>T_G20[6] | DGC_LU<br>T_G20[5] | DGC_LU<br>T_G20[4] | DGC_LU<br>T_G20[3] | DGC_LU<br>T_G20[2] | DGC_LU<br>T_G20[1] | DGC_LU<br>T_G20[0] | 80  |
| Parameter 22th | W                                                                                                                                                          | DGC_LU<br>T_G21[7] | DGC_LU<br>T_G21[6] | DGC_LU<br>T_G21[5] | DGC_LU<br>T_G21[4] | DGC_LU<br>T_G21[3] | DGC_LU<br>T_G21[2] | DGC_LU<br>T_G21[1] | DGC_LU<br>T_G21[0] | 80  |
| :              | :                                                                                                                                                          | :                  | :                  | :                  | :                  | :                  | :                  | :                  | :                  |     |
| Parameter 30th | W                                                                                                                                                          | DGC_LU<br>T_G29[7] | DGC_LU<br>T_G29[6] | DGC_LU<br>T_G29[5] | DGC_LU<br>T_G29[4] | DGC_LU<br>T_G29[3] | DGC_LU<br>T_G29[2] | DGC_LU<br>T_G29[1] | DGC_LU<br>T_G29[0] | 80  |
| Parameter 31th | W                                                                                                                                                          | DGC_LU<br>T_G30[7] | DGC_LU<br>T_G30[6] | DGC_LU<br>T_G30[5] | DGC_LU<br>T_G30[4] | DGC_LU<br>T_G30[3] | DGC_LU<br>T_G30[2] | DGC_LU<br>T_G30[1] | DGC_LU<br>T_G30[0] | 80  |
| Parameter 32th | W                                                                                                                                                          | DGC_LU<br>T_G31[7] | DGC_LU<br>T_G31[6] | DGC_LU<br>T_G31[5] | DGC_LU<br>T_G31[4] | DGC_LU<br>T_G31[3] | DGC_LU<br>T_G31[2] | DGC_LU<br>T_G31[1] | DGC_LU<br>T_G31[0] | 80  |
| Parameter 33th | W                                                                                                                                                          | DGC_LU<br>T_G32[7] | DGC_LU<br>T_G32[6] | DGC_LU<br>T_G32[5] | DGC_LU<br>T_G32[4] | DGC_LU<br>T_G32[3] | DGC_LU<br>T_G32[2] | DGC_LU<br>T_G32[1] | DGC_LU<br>T_G32[0] | 80  |
| Description    | <p>This command is used to set Green color digital gamma.</p> <p>• <b>DGC_LUT_G00 [7:0] ~ DGC_LUT_G32 [7:0]</b> : Green color digital gamma LUT table.</p> |                    |                    |                    |                    |                    |                    |                    |                    |     |

## 6.2.69 DGC\_B (CFh)

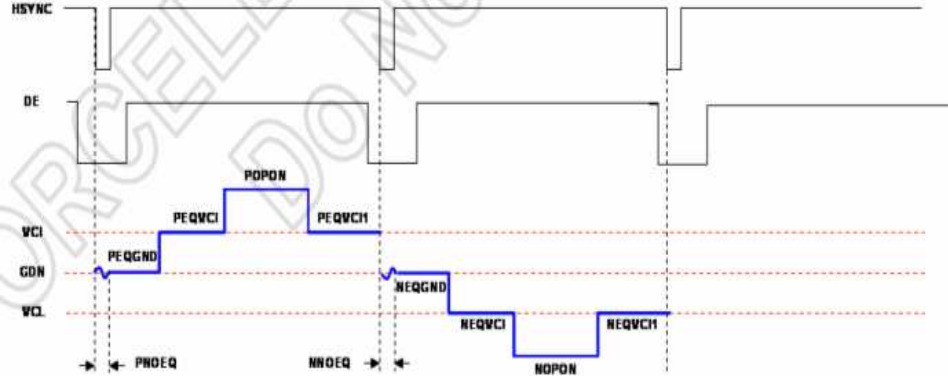
| CF H           | DGC_B                                                                                                                                                    |                    |                    |                    |                    |                    |                    |                    |                    |     |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-----|
|                | R/W                                                                                                                                                      | D7                 | D6                 | D5                 | D4                 | D3                 | D2                 | D1                 | D0                 | HEX |
| Command        | W                                                                                                                                                        | 1                  | 1                  | 0                  | 0                  | 1                  | 1                  | 1                  | 1                  | CF  |
| Parameter 1st  | W                                                                                                                                                        | DGC_LU<br>T_B00[7] | DGC_LU<br>T_B00[6] | DGC_LU<br>T_B00[5] | DGC_LU<br>T_B00[4] | DGC_LU<br>T_B00[3] | DGC_LU<br>T_B00[2] | DGC_LU<br>T_B00[1] | DGC_LU<br>T_B00[0] | 80  |
| Parameter 2nd  | W                                                                                                                                                        | DGC_LU<br>T_B01[7] | DGC_LU<br>T_B01[6] | DGC_LU<br>T_B01[5] | DGC_LU<br>T_B01[4] | DGC_LU<br>T_B01[3] | DGC_LU<br>T_B01[2] | DGC_LU<br>T_B01[1] | DGC_LU<br>T_B01[0] | 80  |
| Parameter 3rd  | W                                                                                                                                                        | DGC_LU<br>T_B02[7] | DGC_LU<br>T_B02[6] | DGC_LU<br>T_B02[5] | DGC_LU<br>T_B02[4] | DGC_LU<br>T_B02[3] | DGC_LU<br>T_B02[2] | DGC_LU<br>T_B02[1] | DGC_LU<br>T_B02[0] | 80  |
| Parameter 4th  | W                                                                                                                                                        | DGC_LU<br>T_B03[7] | DGC_LU<br>T_B03[6] | DGC_LU<br>T_B03[5] | DGC_LU<br>T_B03[4] | DGC_LU<br>T_B03[3] | DGC_LU<br>T_B03[2] | DGC_LU<br>T_B03[1] | DGC_LU<br>T_B03[0] | 80  |
| :              | :                                                                                                                                                        | :                  | :                  | :                  | :                  | :                  | :                  | :                  | :                  |     |
| Parameter 10th | W                                                                                                                                                        | DGC_LU<br>T_B09[7] | DGC_LU<br>T_B09[6] | DGC_LU<br>T_B09[5] | DGC_LU<br>T_B09[4] | DGC_LU<br>T_B09[3] | DGC_LU<br>T_B09[2] | DGC_LU<br>T_B09[1] | DGC_LU<br>T_B09[0] | 80  |
| Parameter 11th | W                                                                                                                                                        | DGC_LU<br>T_B10[7] | DGC_LU<br>T_B10[6] | DGC_LU<br>T_B10[5] | DGC_LU<br>T_B10[4] | DGC_LU<br>T_B10[3] | DGC_LU<br>T_B10[2] | DGC_LU<br>T_B10[1] | DGC_LU<br>T_B10[0] | 80  |
| Parameter 12th | W                                                                                                                                                        | DGC_LU<br>T_B11[7] | DGC_LU<br>T_B11[6] | DGC_LU<br>T_B11[5] | DGC_LU<br>T_B11[4] | DGC_LU<br>T_B11[3] | DGC_LU<br>T_B11[2] | DGC_LU<br>T_B11[1] | DGC_LU<br>T_B11[0] | 80  |
| Parameter 13th | :W                                                                                                                                                       | DGC_LU<br>T_B12[7] | DGC_LU<br>T_B12[6] | DGC_LU<br>T_B12[5] | DGC_LU<br>T_B12[4] | DGC_LU<br>T_B12[3] | DGC_LU<br>T_B12[2] | DGC_LU<br>T_B12[1] | DGC_LU<br>T_B12[0] | 80  |
| :              | :                                                                                                                                                        | :                  | :                  | :                  | :                  | :                  | :                  | :                  | :                  |     |
| Parameter 19th | W                                                                                                                                                        | DGC_LU<br>T_B18[7] | DGC_LU<br>T_B18[6] | DGC_LU<br>T_B18[5] | DGC_LU<br>T_B18[4] | DGC_LU<br>T_B18[3] | DGC_LU<br>T_B18[2] | DGC_LU<br>T_B18[1] | DGC_LU<br>T_B18[0] | 80  |
| Parameter 20th | W                                                                                                                                                        | DGC_LU<br>T_B19[7] | DGC_LU<br>T_B19[6] | DGC_LU<br>T_B19[5] | DGC_LU<br>T_B19[4] | DGC_LU<br>T_B19[3] | DGC_LU<br>T_B19[2] | DGC_LU<br>T_B19[1] | DGC_LU<br>T_B19[0] | 80  |
| Parameter 21th | W                                                                                                                                                        | DGC_LU<br>T_B20[7] | DGC_LU<br>T_B20[6] | DGC_LU<br>T_B20[5] | DGC_LU<br>T_B20[4] | DGC_LU<br>T_B20[3] | DGC_LU<br>T_B20[2] | DGC_LU<br>T_B20[1] | DGC_LU<br>T_B20[0] | 80  |
| Parameter 22th | W                                                                                                                                                        | DGC_LU<br>T_B21[7] | DGC_LU<br>T_B21[6] | DGC_LU<br>T_B21[5] | DGC_LU<br>T_B21[4] | DGC_LU<br>T_B21[3] | DGC_LU<br>T_B21[2] | DGC_LU<br>T_B21[1] | DGC_LU<br>T_B21[0] | 80  |
| :              | :                                                                                                                                                        | :                  | :                  | :                  | :                  | :                  | :                  | :                  | :                  |     |
| Parameter 30th | W                                                                                                                                                        | DGC_LU<br>T_B29[7] | DGC_LU<br>T_B29[6] | DGC_LU<br>T_B29[5] | DGC_LU<br>T_B29[4] | DGC_LU<br>T_B29[3] | DGC_LU<br>T_B29[2] | DGC_LU<br>T_B29[1] | DGC_LU<br>T_B29[0] | 80  |
| Parameter 31th | W                                                                                                                                                        | DGC_LU<br>T_B30[7] | DGC_LU<br>T_B30[6] | DGC_LU<br>T_B30[5] | DGC_LU<br>T_B30[4] | DGC_LU<br>T_B30[3] | DGC_LU<br>T_B30[2] | DGC_LU<br>T_B30[1] | DGC_LU<br>T_B30[0] | 80  |
| Parameter 32th | W                                                                                                                                                        | DGC_LU<br>T_B31[7] | DGC_LU<br>T_B31[6] | DGC_LU<br>T_B31[5] | DGC_LU<br>T_B31[4] | DGC_LU<br>T_B31[3] | DGC_LU<br>T_B31[2] | DGC_LU<br>T_B31[1] | DGC_LU<br>T_B31[0] | 80  |
| Parameter 33th | W                                                                                                                                                        | DGC_LU<br>T_B32[7] | DGC_LU<br>T_B32[6] | DGC_LU<br>T_B32[5] | DGC_LU<br>T_B32[4] | DGC_LU<br>T_B32[3] | DGC_LU<br>T_B32[2] | DGC_LU<br>T_B32[1] | DGC_LU<br>T_B32[0] | 80  |
| Description    | <p>This command is used to set Blue color digital gamma.</p> <p>• <b>DGC_LUT_B00 [7:0] ~ DGC_LUT_B32 [7:0]</b> : Blue color digital gamma LUT table.</p> |                    |                    |                    |                    |                    |                    |                    |                    |     |

## 6.2.70 SETGAMMA (E0h)

| E0 H           | SETGAMMA                                                                                                 |    |         |         |         |         |         |         |         |     |
|----------------|----------------------------------------------------------------------------------------------------------|----|---------|---------|---------|---------|---------|---------|---------|-----|
|                | R/W                                                                                                      | D7 | D6      | D5      | D4      | D3      | D2      | D1      | D0      | HEX |
| Command        | W                                                                                                        | 1  | 1       | 1       | 0       | 0       | 0       | 0       | 0       | E0  |
| Parameter 1st  | R/W                                                                                                      | x  | x       | PVR0[5] | PVR0[4] | PVR0[3] | PVR0[2] | PVR0[1] | PVR0[0] | 0B  |
| Parameter 2nd  | R/W                                                                                                      | x  | x       | PVR1[5] | PVR1[4] | PVR1[3] | PVR1[2] | PVR1[1] | PVR1[0] | 0C  |
| Parameter 3rd  | R/W                                                                                                      | x  | x       | PVR2[5] | PVR2[4] | PVR2[3] | PVR2[2] | PVR2[1] | PVR2[0] | 11  |
| Parameter 4th  | R/W                                                                                                      | x  | x       | PVR3[5] | PVR3[4] | PVR3[3] | PVR3[2] | PVR3[1] | PVR3[0] | 1D  |
| Parameter 5th  | R/W                                                                                                      | x  | x       | PVR4[5] | PVR4[4] | PVR4[3] | PVR4[2] | PVR4[1] | PVR4[0] | 25  |
| Parameter 6th  | R/W                                                                                                      | x  | x       | PVR5[5] | PVR5[4] | PVR5[3] | PVR5[2] | PVR5[1] | PVR5[0] | 37  |
| Parameter 7th  | R/W                                                                                                      | x  | PPR0[6] | PPR0[5] | PPR0[4] | PPR0[3] | PPR0[2] | PPR0[1] | PPR0[0] | 43  |
| Parameter 8th  | R/W                                                                                                      | x  | PPR1[6] | PPR1[5] | PPR1[4] | PPR1[3] | PPR1[2] | PPR1[1] | PPR1[0] | 4B  |
| Parameter 9th  | R/W                                                                                                      | x  | x       | x       | PPK0[4] | PPK0[3] | PPK0[2] | PPK0[1] | PPK0[0] | 4E  |
| Parameter 10th | R/W                                                                                                      | x  | x       | x       | PPK1[4] | PPK1[3] | PPK1[2] | PPK1[1] | PPK1[0] | 47  |
| Parameter 11th | R/W                                                                                                      | x  | x       | x       | PPK2[4] | PPK2[3] | PPK2[2] | PPK2[1] | PPK2[0] | 41  |
| Parameter 12th | R/W                                                                                                      | x  | x       | x       | PPK3[4] | PPK3[3] | PPK3[2] | PPK3[1] | PPK3[0] | 39  |
| Parameter 13th | R/W                                                                                                      | x  | x       | x       | PPK4[4] | PPK4[3] | PPK4[2] | PPK4[1] | PPK4[0] | 35  |
| Parameter 14th | R/W                                                                                                      | x  | x       | x       | PPK5[4] | PPK5[3] | PPK5[2] | PPK5[1] | PPK5[0] | 31  |
| Parameter 15th | R/W                                                                                                      | x  | x       | x       | PPK6[4] | PPK6[3] | PPK6[2] | PPK6[1] | PPK6[0] | 2E  |
| Parameter 16th | R/W                                                                                                      | x  | x       | x       | PPK7[4] | PPK7[3] | PPK7[2] | PPK7[1] | PPK7[0] | 21  |
| Parameter 17th | R/W                                                                                                      | x  | x       | x       | PPK8[4] | PPK8[3] | PPK8[2] | PPK8[1] | PPK8[0] | 1C  |
| Parameter 18th | R/W                                                                                                      | x  | x       | NVR0[5] | NVR0[4] | NVR0[3] | NVR0[2] | NVR0[1] | NVR0[0] | 1D  |
| Parameter 19th | R/W                                                                                                      | x  | x       | NVR1[5] | NVR1[4] | NVR1[3] | NVR1[2] | NVR1[1] | NVR1[0] | 1D  |
| Parameter 20th | R/W                                                                                                      | x  | x       | NVR2[5] | NVR2[4] | NVR2[3] | NVR2[2] | NVR2[1] | NVR2[0] | 26  |
| Parameter 21st | R/W                                                                                                      | x  | x       | NVR3[5] | NVR3[4] | NVR3[3] | NVR3[2] | NVR3[1] | NVR3[0] | 31  |
| Parameter 22nd | R/W                                                                                                      | x  | x       | NVR4[5] | NVR4[4] | NVR4[3] | NVR4[2] | NVR4[1] | NVR4[0] | 44  |
| Parameter 23rd | R/W                                                                                                      | x  | x       | NVR5[5] | NVR5[4] | NVR5[3] | NVR5[2] | NVR5[1] | NVR5[0] | 4E  |
| Parameter 24th | R/W                                                                                                      | x  | NPR0[6] | NPR0[5] | NPR0[4] | NPR0[3] | NPR0[2] | NPR0[1] | NPR0[0] | 56  |
| Parameter 25th | R/W                                                                                                      | x  | NPR1[6] | NPR1[5] | NPR1[4] | NPR1[3] | NPR1[2] | NPR1[1] | NPR1[0] | 44  |
| Parameter 26th | R/W                                                                                                      | x  | x       | x       | NPK0[4] | NPK0[3] | NPK0[2] | NPK0[1] | NPK0[0] | 3F  |
| Parameter 27th | R/W                                                                                                      | x  | x       | x       | NPK1[4] | NPK1[3] | NPK1[2] | NPK1[1] | NPK1[0] | 39  |
| Parameter 28th | R/W                                                                                                      | x  | x       | x       | NPK2[4] | NPK2[3] | NPK2[2] | NPK2[1] | NPK2[0] | 33  |
| Parameter 29th | R/W                                                                                                      | x  | x       | x       | NPK3[4] | NPK3[3] | NPK3[2] | NPK3[1] | NPK3[0] | 31  |
| Parameter 30th | R/W                                                                                                      | x  | x       | x       | NPK4[4] | NPK4[3] | NPK4[2] | NPK4[1] | NPK4[0] | 2E  |
| Parameter 31st | R/W                                                                                                      | x  | x       | x       | NPK5[4] | NPK5[3] | NPK5[2] | NPK5[1] | NPK5[0] | 28  |
| Parameter 32nd | R/W                                                                                                      | x  | x       | x       | NPK6[4] | NPK6[3] | NPK6[2] | NPK6[1] | NPK6[0] | 1D  |
| Parameter 33rd | R/W                                                                                                      | x  | x       | x       | NPK7[4] | NPK7[3] | NPK7[2] | NPK7[1] | NPK7[0] | E0  |
| Parameter 34th | R/W                                                                                                      | x  | x       | x       | NPK8[4] | NPK8[3] | NPK8[2] | NPK8[1] | NPK8[0] | 00  |
| Description    | This command is used to set the gray scale voltage to adjust the gamma characteristics of the TFT panel. |    |         |         |         |         |         |         |         |     |



### 6.2.71 SETEQ (E3h)

| E3 H           | SETEQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |               |            |               |                    |                   |                      |                      |     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|------------|---------------|--------------------|-------------------|----------------------|----------------------|-----|
| Command        | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | D7                  | D6            | D5         | D4            | D3                 | D2                | D1                   | D0                   | HEX |
| Command        | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                   | 1             | 1          | 0             | 0                  | 0                 | 1                    | 1                    | E3  |
| Parameter 1st  | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PNOEQ[7]            | PNOEQ[6]      | PNOEQ[5]   | PNOEQ[4]      | PNOEQ[3]           | PNOEQ[2]          | PNOEQ[1]             | PNOEQ[0]             | 00  |
| Parameter 2nd  | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | NNOEQ[7]            | NNOEQ[6]      | PEOEQ[5]   | NNOEQ[4]      | NNOEQ[3]           | NNOEQ[2]          | NNOEQ[1]             | NNOEQ[0]             | 00  |
| Parameter 3rd  | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PEQGND[7]           | PEQGND[6]     | PEQGND[5]  | PEQGND[4]     | PEQGND[3]          | PEQGND[2]         | PEQGND[1]            | PEQGND[0]            | 03  |
| Parameter 4th  | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | NEQGN D[7]          | NEQGN D[6]    | NEQGN D[5] | NEQGN D[4]    | NEQGN D[3]         | NEQGN D[2]        | NEQGN D[1]           | NEQGN D[0]           | 03  |
| Parameter 5th  | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PEQVCI[7]           | PEQVCI[6]     | PEQVCI[5]  | PEQVCI[4]     | PEQVCI[3]          | PEQVCI[2]         | PEQVCI[1]            | PEQVCI[0]            | 03  |
| Parameter 6th  | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | NEQVCI[7]           | NEQVCI[6]     | NEQVCI[5]  | NEQVCI[4]     | NEQVCI[3]          | NEQVCI[2]         | NEQVCI[1]            | NEQVCI[0]            | 03  |
| Parameter 7th  | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PEQVCI1[7]          | PEQVCI1[6]    | PEQVCI1[5] | PEQVCI1[4]    | PEQVCI1[3]         | PEQVCI1[2]        | PEQVCI1[1]           | PEQVCI1[0]           | 03  |
| Parameter 8th  | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | NEQVCI1[7]          | NEQVCI1[6]    | NEQVCI1[5] | NEQVCI1[4]    | NEQVCI1[3]         | NEQVCI1[2]        | NEQVCI1[1]           | NEQVCI1[0]           | 03  |
| Parameter 9th  | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | x                   | x             | x          | x             | x                  | x                 | x                    | x                    | 00  |
| Parameter 10th | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | x                   | x             | x          | x             | x                  | x                 | x                    | x                    | 00  |
| Parameter 11th | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | x                   | x             | x          | x             | x                  | x                 | x                    | x                    | 00  |
| Parameter 12th | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | x                   | x             | x          | x             | x                  | x                 | x                    | x                    | 00  |
| Parameter 13th | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ESD_DE T_DATA WHITE | ESD_WH ITE_EN | x          | x             | x                  | x                 | x                    | x                    | C0  |
| Parameter 14th | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | x                   | x             | x          | SLPIN_O PTION | VEDIO NO_CHE CK_EN | ESD_WH ITE_GND_EN | ESD_DE T_TIME_SEL[1] | ESD_DE T_TIME_SEL[0] | 14  |
| Description    | This command is used to set EQ related register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |               |            |               |                    |                   |                      |                      |     |
|                |  <ul style="list-style-type: none"> <li>• <b>PNOEQ[7:0]</b> : Specify the temporal spacing between HSYNC and PEQGND.</li> <li>• <b>NNOEQ[7:0]</b> : Specify the temporal spacing between HSYNC and NEQGN.</li> <li>• <b>PEQGND[7:0]</b> : Specify the source EQ GND period when Source up to positive voltage.</li> <li>• <b>NEQGN[7:0]</b> : Specify the source EQ GND period when Source down to negative voltage.</li> <li>• <b>PEQVCI[7:0]</b> : Specify the source EQ VCI period when Source up to positive voltage.</li> <li>• <b>NEQVCI[7:0]</b> : Specify the source EQ VCI period when Source down to negative voltage.</li> <li>• <b>PEQVCI1[7:0]</b> : Specify the temporal period of PEQVCI1.</li> <li>• <b>NEQVCI1[7:0]</b> : Specify the temporal period of NEQVCI1.</li> </ul> |                     |               |            |               |                    |                   |                      |                      |     |

Description

|                                                                                                                      |                  |
|----------------------------------------------------------------------------------------------------------------------|------------------|
| PNOEQ[7:0]<br>NNOEQ[7:0]<br>PEQGND[7:0]<br>NEQGND[7:0]<br>PEQVCI[7:0]<br>NEQVCI[7:0]<br>PEQVCI1[7:0]<br>NEQVCI1[7:0] | <b>EQ Period</b> |
| 0                                                                                                                    | 0*4/Fosc         |
| 1                                                                                                                    | 1*4/Fosc         |
| 2                                                                                                                    | 2*4/Fosc         |
| 3                                                                                                                    | 3*4/Fosc         |
| :                                                                                                                    | :                |
| FD                                                                                                                   | 253*4/Fosc       |
| FE                                                                                                                   | 254*4/Fosc       |
| FF                                                                                                                   | 255*4/Fosc       |

- **ESD\_DET\_DATA\_WHITE** : Select white or black pattern to protect GOA glass.
- **ESD\_WHITE\_EN** : Enable ESD detection function to protect GOA glass.
  - At Sleep-Out State, IC automatically displaying white or black pattern when MIPI-DSI is interrupted or not sent.
- **SLPIN\_OPTION**
  - 0 : Need VSYNC (additional frame) after Sleep-In command to display sleep-in blanking frame then into Sleep-In State.
  - 1 : No Need VSYNC (additional frame) after Sleep-In command to display sleep-in blanking frame then into Sleep-In State.
- **VEDIO\_NO\_CHECK\_EN** : Enable vedio function detection.
  - 0 : Enable vedio function detection.
  - 1 : Disable vedio function detection, continually displaying white or black pattern after sleep-out blanking frame without detecting if MIPI video comes or not
- **ESD\_WHITE\_GND\_EN** : Enable ESD white pattern scanning voltage pull ground.  
 "0", Disable scanning voltage pull ground.  
 "1", Enable scanning voltage pull ground.

- **ESD\_DET\_TIME\_SEL[1:0]** : ESD detection function period.

| ESD_TIME_SEL[1:0] | Period   |
|-------------------|----------|
| 0                 | 0 frames |
| 1                 | 1 frames |
| 2                 | 2 frames |
| 3                 | 3 frames |



## 6.2.72 SETCOLOREN (E4h)

| E4 H          | SETCOLOREN                                                                                                                                                                                                                                                         |    |    |    |    |    |    |    |         |     |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|---------|-----|
|               | R/W                                                                                                                                                                                                                                                                | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0      | HEX |
| Command       | W                                                                                                                                                                                                                                                                  | 1  | 1  | 1  | 0  | 0  | 1  | 0  | 0       | E4  |
| Parameter 1st | R/W                                                                                                                                                                                                                                                                | x  | x  | x  | x  | x  | x  | x  | CE_MASK | 01  |
| Description   | <p>This command is to set color enhanment mode.</p> <ul style="list-style-type: none"> <li>• <b>CE_MASK</b> : Enable / Disable the color enhanced mask function. <ul style="list-style-type: none"> <li>- 0 : Disable</li> <li>- 1 : Enable</li> </ul> </li> </ul> |    |    |    |    |    |    |    |         |     |

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### 6.2.73 SETCOLOREN (E5h)

| E5 H                                                 | SETCOLOREN                                        |                 |                        |            |                |                |                |            |            |           |
|------------------------------------------------------|---------------------------------------------------|-----------------|------------------------|------------|----------------|----------------|----------------|------------|------------|-----------|
|                                                      | R/W                                               | D7              | D6                     | D5         | D4             | D3             | D2             | D1         | D0         | HEX       |
| Command                                              | W                                                 | 1               | 1                      | 1          | 0              | 0              | 1              | 0          | 1          | E5        |
| Parameter 1st                                        | R/W                                               | SP_LEV<br>EL[1] | SP_LEV<br>EL[0]        | CE_EN      | x              | x              | x              | x          | x          | 00        |
| Parameter 2nd                                        | R/W                                               | x               | x                      | x          | x              | x              | x              | x          | x          | No<br>use |
| Parameter 3rd                                        | R/W                                               | x               | x                      | x          | Blue_3[1<br>2] | Blue_3[1<br>1] | Blue_3[1<br>0] | Blue_3[9]  | Blue_3[8]  | 08        |
| Parameter 4th                                        | R/W                                               | Blue_3[7]       | Blue_3[6]              | Blue_3[5]  | Blue_3[4]      | Blue_3[3]      | Blue_3[2]      | Blue_3[1]  | Blue_3[0]  | 32        |
| Parameter 5th                                        | R/W                                               | x               | x                      | x          | Blue_2[1<br>2] | Blue_2[1<br>1] | Blue_2[1<br>0] | Blue_2[9]  | Blue_2[8]  | 1C        |
| Parameter 6th                                        | R/W                                               | Blue_2[7]       | Blue_2[6]              | Blue_2[5]  | Blue_2[4]      | Blue_2[3]      | Blue_2[2]      | Blue_2[1]  | Blue_2[0]  | 71        |
| Parameter 7th                                        | R/W                                               | x               | x                      | x          | Blue_1[1<br>2] | Blue_1[1<br>1] | Blue_1[1<br>0] | Blue_1[9]  | Blue_1[8]  | 1F        |
| Parameter 8th                                        | R/W                                               | Blue_1[7]       | Blue_1[6]              | Blue_1[5]  | Blue_1[4]      | Blue_1[3]      | Blue_1[2]      | Blue_1[1]  | Blue_1[0]  | 90        |
| Parameter 9th                                        | R/W                                               | x               | x                      | x          | Green_3[12]    | Green_3[11]    | Green_3[10]    | Green_3[9] | Green_3[8] | 1E        |
| Parameter 10th                                       | R/W                                               | Green_3[7]      | Green_3[6]             | Green_3[5] | Green_3[4]     | Green_3[3]     | Green_3[2]     | Green_3[1] | Green_3[0] | CD        |
| Parameter 11th                                       | R/W                                               | x               | x                      | x          | Green_2[12]    | Green_2[11]    | Green_2[10]    | Green_2[9] | Green_2[8] | 07        |
| Parameter 12th                                       | R/W                                               | Green_2[7]      | Green_2[6]             | Green_2[5] | Green_2[4]     | Green_2[3]     | Green_2[2]     | Green_2[1] | Green_2[0] | 4C        |
| Parameter 13th                                       | R/W                                               | x               | x                      | x          | Green_1[12]    | Green_1[11]    | Green_1[10]    | Green_1[9] | Green_1[8] | 1E        |
| Parameter 14th                                       | R/W                                               | Green_1[7]      | Green_1[6]             | Green_1[5] | Green_1[4]     | Green_1[3]     | Green_1[2]     | Green_1[1] | Green_1[0] | 06        |
| Parameter 15th                                       | R/W                                               | x               | x                      | x          | Red_3[12]      | Red_3[11]      | Red_3[10]      | Red_3[9]   | Red_3[8]   | 1F        |
| Parameter 16th                                       | R/W                                               | Red_3[7]        | Red_3[6]               | Red_3[5]   | Red_3[4]       | Red_3[3]       | Red_3[2]       | Red_3[1]   | Red_3[0]   | 66        |
| Parameter 17th                                       | R/W                                               | x               | x                      | x          | Red_2[12]      | Red_2[11]      | Red_2[10]      | Red_2[9]   | Red_2[8]   | 1E        |
| Parameter 18th                                       | R/W                                               | Red_2[7]        | Red_2[6]               | Red_2[5]   | Red_2[4]       | Red_2[3]       | Red_2[2]       | Red_2[1]   | Red_2[0]   | 2A        |
| Parameter 19th                                       | R/W                                               | x               | x                      | x          | Red_1[12]      | Red_1[11]      | Red_1[10]      | Red_1[9]   | Red_1[8]   | 06        |
| Parameter 20th                                       | R/W                                               | Red_1[7]        | Red_1[6]               | Red_1[5]   | Red_1[4]       | Red_1[3]       | Red_1[2]       | Red_1[1]   | Red_1[0]   | 70        |
| Description                                          | This command is to set color enhanment mode.      |                 |                        |            |                |                |                |            |            |           |
|                                                      | • <b>SP_LEVEL[1:0]</b> : Specify sharpness level. |                 |                        |            |                |                |                |            |            |           |
|                                                      | <b>SP_LEVEL[1:0]</b>                              |                 | <b>Sharpness Level</b> |            |                |                |                |            |            |           |
|                                                      | 0                                                 |                 | No sharpness           |            |                |                |                |            |            |           |
|                                                      | 1                                                 |                 | Small                  |            |                |                |                |            |            |           |
|                                                      | 2                                                 |                 | Medium                 |            |                |                |                |            |            |           |
| 3                                                    |                                                   | Large           |                        |            |                |                |                |            |            |           |
| • <b>CE_EN</b> : Enable the color enhanced function. |                                                   |                 |                        |            |                |                |                |            |            |           |

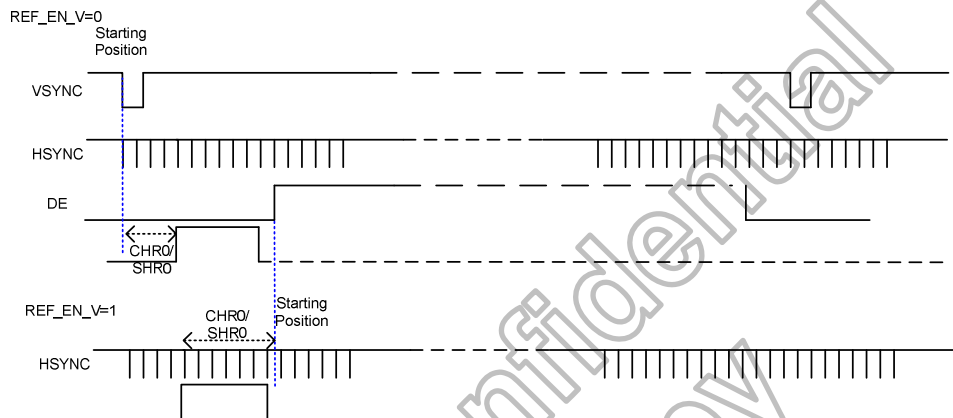
## 6.2.74 SETGIP1 (E9h)

| E9 H           | SETGIP1 |                  |                  |                  |                  |                  |                  |                  |                  |     |
|----------------|---------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|-----|
|                | R/W     | D7               | D6               | D5               | D4               | D3               | D2               | D1               | D0               | HEX |
| Command        | W       | 1                | 1                | 1                | 0                | 1                | 0                | 0                | 1                | E9  |
| Parameter 1st  | R/W     | REF_EN           | x                | x                | x                | PANEL_SEL[3]     | PANEL_SEL[2]     | PANEL_SEL[1]     | PANEL_SEL[0]     | 24  |
| Parameter 2nd  | R/W     | x                | x                | x                | SHR0[12]         | SHR0[11]         | SHR0[10]         | SHR0[9]          | SHR0[8]          | 00  |
| Parameter 3rd  | R/W     | SHR0[7]          | SHR0[6]          | SHR0[5]          | SHR0[4]          | SHR0[3]          | SHR0[2]          | SHR0[1]          | SHR0[0]          | 00  |
| Parameter 4th  | R/W     | x                | x                | x                | SHR0[12]         | SHR1[11]         | SHR1[10]         | SHR1[9]          | SHR1[8]          | 00  |
| Parameter 5th  | R/W     | SHR1[7]          | SHR1[6]          | SHR1[5]          | SHR1[4]          | SHR1[3]          | SHR1[2]          | SHR1[1]          | SHR1[0]          | 00  |
| Parameter 6th  | R/W     | SPON[7]          | SPON[6]          | SPON[5]          | SPON[4]          | SPON[3]          | SPON[2]          | SPON[1]          | SPON[0]          | 00  |
| Parameter 7th  | R/W     | SPOFF[7]         | SPOFF[6]         | SPOFF[5]         | SPOFF[4]         | SPOFF[3]         | SPOFF[2]         | SPOFF[1]         | SPOFF[0]         | 00  |
| Parameter 8th  | R/W     | SHR0_1[3]        | SHR0_1[2]        | SHR0_1[1]        | SHR0_1[0]        | SHR0_2[3]        | SHR0_2[2]        | SHR0_2[1]        | SHR0_2[0]        | 00  |
| Parameter 9th  | R/W     | SHR0_3[3]        | SHR0_3[2]        | SHR0_3[1]        | SHR0_3[0]        | SHR1_1[3]        | SHR1_1[2]        | SHR1_1[1]        | SHR1_1[0]        | 00  |
| Parameter 10th | R/W     | SHR1_2[3]        | SHR1_2[2]        | SHR1_2[1]        | SHR1_2[0]        | SHR1_3[3]        | SHR1_3[2]        | SHR1_3[1]        | SHR1_3[0]        | 00  |
| Parameter 11th | R/W     | SHP[3]           | SHP[2]           | SHP[1]           | SHP[0]           | SCP[3]           | SCP[2]           | SCP[1]           | SCP[0]           | 00  |
| Parameter 12th | R/W     | CHR[7]           | CHR[6]           | CHR[5]           | CHR[4]           | CHR[3]           | CHR[2]           | CHR[1]           | CHR[0]           | 00  |
| Parameter 13th | R/W     | CON[7]           | CON[6]           | CON[5]           | CON[4]           | CON[3]           | CON[2]           | CON[1]           | CON[0]           | 02  |
| Parameter 14th | R/W     | COF[7]           | COF[6]           | COF[5]           | COF[4]           | COF[3]           | COF[2]           | COF[1]           | COF[0]           | 02  |
| Parameter 15th | R/W     | CHP[3]           | CHP[2]           | CHP[1]           | CHP[0]           | CCP[3]           | CCP[2]           | CCP[1]           | CCP[0]           | 00  |
| Parameter 16th | R/W     | USER_GIP_GATE[7] | USER_GIP_GATE[6] | USER_GIP_GATE[5] | USER_GIP_GATE[4] | USER_GIP_GATE[3] | USER_GIP_GATE[2] | USER_GIP_GATE[1] | USER_GIP_GATE[0] | 00  |
| Parameter 17th | R/W     | x                | x                | CGTS_L[21]       | CGTS_L[20]       | CGTS_L[19]       | CGTS_L[18]       | CGTS_L[17]       | CGTS_L[16]       | 00  |
| Parameter 18th | R/W     | CGTS_L[15]       | CGTS_L[14]       | CGTS_L[13]       | CGTS_L[12]       | CGTS_L[11]       | CGTS_L[10]       | CGTS_L[9]        | CGTS_L[8]        | 00  |
| Parameter 19th | R/W     | CGTS_L[7]        | CGTS_L[6]        | CGTS_L[5]        | CGTS_L[4]        | CGTS_L[3]        | CGTS_L[2]        | CGTS_L[1]        | CGTS_L[0]        | 00  |
| Parameter 20th | R/W     | x                | x                | CGTS_IN_V_L[21]  | CGTS_IN_V_L[20]  | CGTS_IN_V_L[19]  | CGTS_IN_V_L[18]  | CGTS_IN_V_L[17]  | CGTS_IN_V_L[16]  | 00  |
| Parameter 21st | R/W     | CGTS_IN_V_L[15]  | CGTS_IN_V_L[14]  | CGTS_IN_V_L[13]  | CGTS_IN_V_L[12]  | CGTS_IN_V_L[11]  | CGTS_IN_V_L[10]  | CGTS_IN_V_L[9]   | CGTS_IN_V_L[8]   | 00  |
| Parameter 22nd | R/W     | CGTS_IN_V_L[7]   | CGTS_IN_V_L[6]   | CGTS_IN_V_L[5]   | CGTS_IN_V_L[4]   | CGTS_IN_V_L[3]   | CGTS_IN_V_L[2]   | CGTS_IN_V_L[1]   | CGTS_IN_V_L[0]   | 00  |
| Parameter 23rd | R/W     | x                | x                | CGTS_R[21]       | CGTS_R[20]       | CGTS_R[19]       | CGTS_R[18]       | CGTS_R[17]       | CGTS_R[16]       | 00  |
| Parameter 24th | R/W     | CGTS_R[15]       | CGTS_R[14]       | CGTS_R[13]       | CGTS_R[12]       | CGTS_R[11]       | CGTS_R[10]       | CGTS_R[9]        | CGTS_R[8]        | 00  |
| Parameter 25th | R/W     | CGTS_R[7]        | CGTS_R[6]        | CGTS_R[5]        | CGTS_R[4]        | CGTS_R[3]        | CGTS_R[2]        | CGTS_R[1]        | CGTS_R[0]        | 00  |
| Parameter 26th | R/W     | x                | x                | CGTS_IN_V_R[21]  | CGTS_IN_V_R[20]  | CGTS_IN_V_R[19]  | CGTS_IN_V_R[18]  | CGTS_IN_V_R[17]  | CGTS_IN_V_R[16]  | 00  |
| Parameter 27th | R/W     | CGTS_IN_V_R[15]  | CGTS_IN_V_R[14]  | CGTS_IN_V_R[13]  | CGTS_IN_V_R[12]  | CGTS_IN_V_R[11]  | CGTS_IN_V_R[10]  | CGTS_IN_V_R[9]   | CGTS_IN_V_R[8]   | 00  |
| Parameter 28th | R/W     | CGTS_IN_V_R[7]   | CGTS_IN_V_R[6]   | CGTS_IN_V_R[5]   | CGTS_IN_V_R[4]   | CGTS_IN_V_R[3]   | CGTS_IN_V_R[2]   | CGTS_IN_V_R[1]   | CGTS_IN_V_R[0]   | 00  |
| Parameter 29th | R/W     | COS1_L[3]        | COS1_L[2]        | COS1_L[1]        | COS1_L[0]        | COS2_L[3]        | COS2_L[2]        | COS2_L[1]        | COS2_L[0]        | 00  |
| Parameter 30th | R/W     | COS3_L[3]        | COS3_L[2]        | COS3_L[1]        | COS3_L[0]        | COS4_L[3]        | COS4_L[2]        | COS4_L[1]        | COS4_L[0]        | 00  |

|                |     |             |             |             |             |             |             |             |             |    |
|----------------|-----|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|----|
| Parameter 31st | R/W | COS5_L[3]   | COS5_L[2]   | COS5_L[1]   | COS5_L[0]   | COS6_L[3]   | COS6_L[2]   | COS6_L[1]   | COS6_L[0]   | 00 |
| :              | :   | :           | :           | :           | :           | :           | :           | :           | :           |    |
| Parameter 37th | R/W | COS17_L[3]  | COS17_L[2]  | COS17_L[1]  | COS17_L[0]  | COS18_L[3]  | COS18_L[2]  | COS18_L[1]  | COS18_L[0]  | 00 |
| Parameter 38th | R/W | COS19_L[3]  | COS19_L[2]  | COS19_L[1]  | COS19_L[0]  | COS20_L[3]  | COS20_L[2]  | COS20_L[1]  | COS20_L[0]  | 00 |
| Parameter 39th | R/W | COS21_L[3]  | COS21_L[2]  | COS21_L[1]  | COS21_L[0]  | COS22_L[3]  | COS22_L[2]  | COS22_L[1]  | COS22_L[0]  | 00 |
| Parameter 40th | R/W | COS1_R[3]   | COS1_R[2]   | COS1_R[1]   | COS1_R[0]   | COS2_R[3]   | COS2_R[2]   | COS2_R[1]   | COS2_R[0]   | 00 |
| Parameter 41st | R/W | COS3_R[3]   | COS3_R[2]   | COS3_R[1]   | COS3_R[0]   | COS4_R[3]   | COS4_R[2]   | COS4_R[1]   | COS4_R[0]   | 00 |
| Parameter 42nd | R/W | COS5_R[3]   | COS5_R[2]   | COS5_R[1]   | COS5_R[0]   | COS6_R[3]   | COS6_R[2]   | COS6_R[1]   | COS6_R[0]   | 00 |
| :              | :   | :           | :           | :           | :           | :           | :           | :           | :           |    |
| Parameter 48th | R/W | COS17_R[3]  | COS17_R[2]  | COS17_R[1]  | COS17_R[0]  | COS18_R[3]  | COS18_R[2]  | COS18_R[1]  | COS18_R[0]  | 00 |
| Parameter 49th | R/W | COS19_R[3]  | COS19_R[2]  | COS19_R[1]  | COS19_R[0]  | COS20_R[3]  | COS20_R[2]  | COS20_R[1]  | COS20_R[0]  | 00 |
| Parameter 50th | R/W | COS21_R[3]  | COS21_R[2]  | COS21_R[1]  | COS21_R[0]  | COS22_R[3]  | COS22_R[2]  | COS22_R[1]  | COS22_R[0]  | 00 |
| Parameter 51st | R/W | TCON_OPT[7] | TCON_OPT[6] | TCON_OPT[5] | TCON_OPT[4] | TCON_OPT[3] | TCON_OPT[2] | TCON_OPT[1] | TCON_OPT[0] | 00 |
| Parameter 52nd | R/W | x           | GIP_OPT[22] | GIP_OPT[21] | GIP_OPT[20] | GIP_OPT[19] | GIP_OPT[18] | GIP_OPT[17] | GIP_OPT[16] | 00 |
| Parameter 53rd | R/W | GIP_OPT[15] | GIP_OPT[14] | GIP_OPT[13] | GIP_OPT[12] | GIP_OPT[11] | GIP_OPT[10] | GIP_OPT[9]  | GIP_OPT[8]  | 00 |
| Parameter 54th | R/W | GIP_OPT[7]  | GIP_OPT[6]  | GIP_OPT[5]  | GIP_OPT[4]  | GIP_OPT[3]  | GIP_OPT[2]  | GIP_OPT[1]  | GIP_OPT[0]  | 00 |
| Parameter 55th | R/W | CHR2[7]     | CHR2[6]     | CHR2[5]     | CHR2[4]     | CHR2[3]     | CHR2[2]     | CHR2[1]     | CHR2[0]     | 00 |
| Parameter 56th | R/W | CON2[7]     | CON2[6]     | CON2[5]     | CON2[4]     | CON2[3]     | CON2[2]     | CON2[1]     | CON2[0]     | 02 |
| Parameter 57th | R/W | COFF2[7]    | COFF2[6]    | COFF2[5]    | COFF2[4]    | COFF2[3]    | COFF2[2]    | COFF2[1]    | COFF2[0]    | 02 |
| Parameter 58th | R/W | CHP2[3]     | CHP2[2]     | CHP2[1]     | CHP2[0]     | CCP2[3]     | CCP2[2]     | CCP2[1]     | CCP2[0]     | 00 |
| Parameter 59th | R/W | x           | x           | CKS[21]     | CKS[20]     | CKS[19]     | CKS[18]     | CKS[17]     | CKS[16]     | 00 |
| Parameter 60th | R/W | CKS[15]     | CKS[14]     | CKS[13]     | CKS[12]     | CKS[11]     | CKS[10]     | CKS[9]      | CKS[8]      | 00 |
| Parameter 61st | R/W | CKS[7]      | CKS[6]      | CKS[5]      | CKS[4]      | CKS[3]      | CKS[2]      | CKS[1]      | CKS[0]      | 00 |
| Parameter 62nd | R/W | COFF[9]     | COFF[8]     | CON[9]      | CON[8]      | SPOFF[9]    | SPOFF[8]    | SPON[9]     | SPON[8]     | 00 |
| Parameter 63rd | R/W | COFF2[9]    | COFF[8]     | CON2[9]     | CON2[8]     | x           | x           | x           | x           | 00 |

Description

This command is to set forward GIP timing.



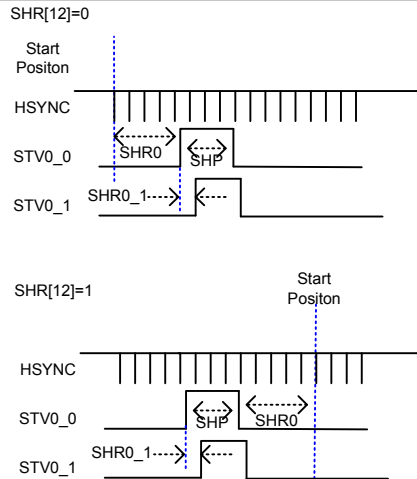
• **REF\_EN :**

- 0 : SHR0, SHR1, CHR, CHR2 refer to Internal VSYNC
- 1 : SHR0, SHR1, CHR, CHR2 refer to Internal DE
- ◎ REF\_EN=1 : SHR0, SHR1, CHR, CHR2 = MSB Sign Bit.

• **PANELI\_SEL[3:0] :** Specify GIP signal at abnormal power off state.

| Panel_sel[3:0] | GIP signal State |
|----------------|------------------|
| 2              | VGH              |
| 4              | VGL              |
| 8              | GND              |

Description



- **SHR0[12:0]** : Specify the starting position of GIP STV group 0 (STV0\_0).
- **SHR1[12:0]** : Specify the starting position of GIP STV group 1 (STV1\_0).

| SHR0[12:0]<br>SHR1[12:0]<br>(Hex) | Starting Position |
|-----------------------------------|-------------------|
| 0                                 | 0 HSYNC           |
| 1                                 | 1 HSYNC           |
| 2                                 | 2 HSYNC           |
| :                                 | :                 |
| FFF                               | 4095 HSYNC        |

- **SPON[9:0]** : Specify the distance of STV rising edge and HYSNC.
- **SPOFF[9:0]** : Specify the distance of STV falling edge and HYSNC.

| SPON[9:0]<br>SPOFF[9:0]<br>(Hex) | Distance   |
|----------------------------------|------------|
| 0                                | 0*2 Fosc   |
| 1                                | 1*2Fosc    |
| 2                                | 2*2Fosc    |
| :                                | :          |
| 3FF                              | 1023*2Fosc |

- **SHR0\_1[3:0] / SHR0\_2[3:0] / SHR0\_3[3:0]**: Specify the STV0\_x distance with STV0\_0.
- **SHR1\_1[3:0] / SHR1\_2[3:0] / SHR1\_3[3:0]**: Specify the STV1\_x distance with STV1\_0.  
(X=1~3)

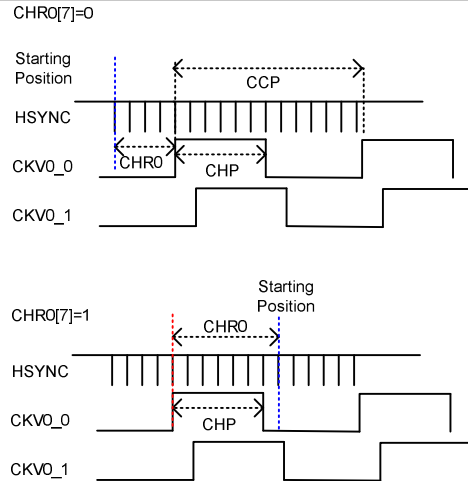
- **SHP[3:0]** : Specify the STV signal high pulse width.

| SHP[3:0]<br>(Hex) | Width    |
|-------------------|----------|
| 0                 | 0 HSYNC  |
| 1                 | 1 HSYNC  |
| 2                 | 2 HSYNC  |
| :                 | :        |
| F                 | 16 HSYNC |

- **SCP[3:0]** : Specify the total number of STV signal.

| SCP[3:0]<br>(Hex) | STV NO. |
|-------------------|---------|
| 0                 | 1       |
| 1                 | 2       |
| 2                 | 3       |
| :                 | :       |
| F                 | 16      |

Description



- **CHR[7:0]** : Specify the starting position of GIP CKV group 0 (CKV0\_0).
- **CHR2[7:0]** : Specify the starting position of GIP CKV group 1 (CKV1\_0).

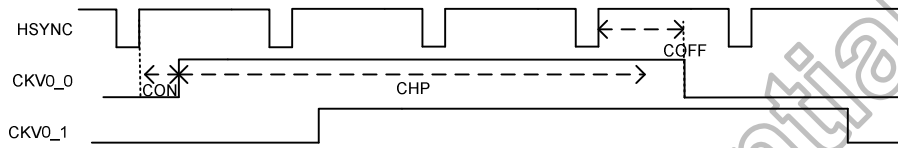
| <b>CHR[7:0]<br/>CHR2[7:0]<br/>(Hex)</b> | <b>Starting Position</b> |
|-----------------------------------------|--------------------------|
| 0                                       | 0 HSYNC                  |
| 1                                       | 1 HSYNC                  |
| 2                                       | 2 HSYNC                  |
| :                                       | :                        |
| FF                                      | 255 HSYNC                |

- **CHP[3:0]** : Specify the CKV signal high pulse width.
- **CHP2[3:0]** : Specify the CKV1 signal high pulse width

| <b>CHP[3:0]<br/>CHP2[3:0]<br/>(Hex)</b> | <b>Width</b> |
|-----------------------------------------|--------------|
| 0                                       | 1 HSYNC      |
| 1                                       | 2 HSYNC      |
| 2                                       | 3 HSYNC      |
| :                                       | :            |
| F                                       | 16 HSYNC     |

- **CCP[3:0]** : Specify the total period cycle of CKV signal.
- **CCP2[3:0]** : Specify the total period cycle of CKV1 signal.

| <b>CCP[3:0]<br/>CCP2[3:0]<br/>(Hex)</b> | <b>CKV Period.</b> |
|-----------------------------------------|--------------------|
| 0                                       | 1 HSYNC            |
| 1                                       | 2 HSYNC            |
| 2                                       | 3 HSYNC            |
| :                                       | :                  |
| F                                       | 16 HSYNC           |



- **CON[9:0]** : Specify the distance of CKV rising edge and HYSNC.
- **COFF[9:0]** : Specify the distance of CKV falling edge and HYSNC.
- **CON2[9:0]** : Specify the distance of CKV1 rising edge and HYSNC.
- **COFF2[9:0]** : Specify the distance of CKV1 falling edge and HYSNC.

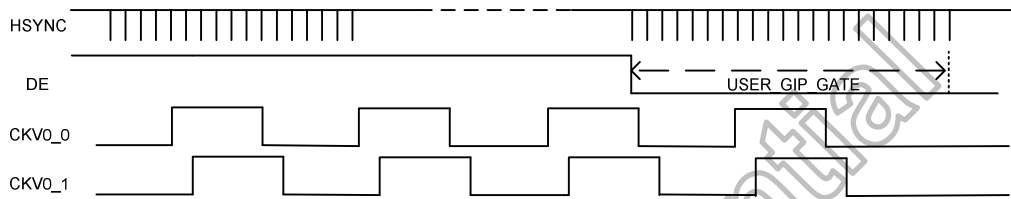
| CON[9:0]<br>COFF[9:0]<br>CON2[9:0]<br>COFF2[9:0]<br>(Hex) | Distance              |
|-----------------------------------------------------------|-----------------------|
| 0                                                         | $0 \times 2F_{osc}$   |
| 1                                                         | $1 \times 2F_{osc}$   |
| 2                                                         | $2 \times 2F_{osc}$   |
| .                                                         | .                     |
| FF                                                        | $255 \times 2F_{osc}$ |

Description

- **CGTS\_L[21:0]** : Select Left side GIP output pad signal.
  - 1 : Output pad assign as STV.
  - 0 : Output pad assign as CKV.
- **CGTS\_R[21:0]** : Select Right side GIP output pad signal.
  - 1 : Output pad assign as STV.
  - 0 : Output pad assign as CKV.
- **CGTS\_INV\_L[21:0]** : Reverse the polarity of Left side GIP output pad signal.
  - 1 : Reverse the polarity.
  - 0 : Normal polarity.
- **CGTS\_INV\_R[21:0]** : Reverse the polarity of Right side GIP output pad signal.
  - 1 : Reverse the polarity.
  - 0 : Normal polarity.



Description



- **USER\_GIP\_GATE[7:0]** : Specify the extra gate counter at blanking area.

| <b>USER_GIP_GATE[7:0]<br/>(Hex)</b> | <b>Gate number</b> |
|-------------------------------------|--------------------|
| 0                                   | 0                  |
| 1                                   | 1                  |
| 2                                   | 2                  |
| :                                   | :                  |
| FF                                  | 255                |

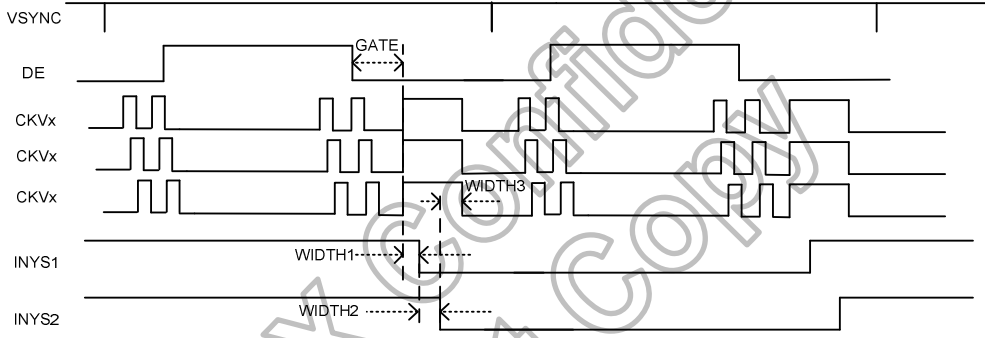
- **COSx\_L[3:0]** : Specify Left side GIP output pad signal.
- **COSx\_R[3:0]** : Specify Right side GIP output pad signal.  
(x=1~22)

| <b>COS<br/>(Hex)</b> | <b>GIP OUTPUT</b> |                 |                |
|----------------------|-------------------|-----------------|----------------|
|                      | <b>CGTS="1"</b>   | <b>CGTS="0"</b> |                |
|                      |                   | <b>CKS="0"</b>  | <b>CKS="1"</b> |
| 0                    | STV0_0            | CKV0_0          | CKV1_0         |
| 1                    | STV0_1            | CKV0_1          | CKV1_1         |
| 2                    | STV0_2            | CKV0_2          | CKV1_2         |
| 3                    | STV0_3            | CKV0_3          | CKV1_3         |
| 4                    | STV1_0            | CKV0_4          | CKV1_4         |
| 5                    | STV1_1            | CKV0_5          | CKV1_5         |
| 6                    | STV1_2            | CKV0_6          | CKV1_6         |
| 7                    | STV1_3            | CKV0_7          | CKV1_7         |
| 8                    | Always Low        | Always Low      |                |
| 9                    | STV2_0            | CKV All On      |                |
| A                    | STV2_1            | YS1_Signal      |                |
| B                    | STV2_2            | YS2_Signal      |                |
| C                    | STV2_3            | CKV All On_2    |                |
| D                    | -                 | YS1_Signal_2    |                |
| E                    | -                 | YS2_Signal_2    |                |
| F                    | Always High       | Always High     |                |

- **GIP\_OPT[22:0]** : Specify GIP optional setting for GIP signal.

## 6.2.75 SETGIP2 (EAH)

| EA H           | SETGIP2 |                             |                             |                               |                               |                               |                               |                               |                               |     |
|----------------|---------|-----------------------------|-----------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-----|
|                | R/W     | D7                          | D6                          | D5                            | D4                            | D3                            | D2                            | D1                            | D0                            | HEX |
| Command        | W       | 1                           | 1                           | 1                             | 0                             | 1                             | 0                             | 1                             | 0                             | EA  |
| Parameter 1st  | R/W     | YS2_SE<br>L[1]              | YS2_SEL<br>[0]              | YS1_SEL<br>[1]                | YS1_SEL<br>[0]                | YS2_XO<br>R                   | YS1_XO<br>R                   | YS_FLG<br>EN                  | ALL_ON<br>EN                  | 94  |
| Parameter 2nd  | R/W     | GATE[7]                     | GATE[6]                     | GATE[5]                       | GATE[4]                       | GATE[3]                       | GATE[2]                       | GATE[1]                       | GATE[0]                       | 00  |
| Parameter 3rd  | R/W     | CK_ALL_<br>ON_EN            | STV_ALL_<br>ON_EN           | CK_ALL_<br>ON_WID<br>TH1[5]   | CK_ALL_<br>ON_WID<br>TH1[4]   | CK_ALL_<br>ON_WID<br>TH1[3]   | CK_ALL_<br>ON_WID<br>TH1[2]   | CK_ALL_<br>ON_WID<br>TH1[1]   | CK_ALL_<br>ON_WID<br>TH1[0]   | 00  |
| Parameter 4th  | R/W     | x                           | x                           | CK_ALL_<br>ON_WID<br>TH2[5]   | CK_ALL_<br>ON_WID<br>TH2[4]   | CK_ALL_<br>ON_WID<br>TH2[3]   | CK_ALL_<br>ON_WID<br>TH2[2]   | CK_ALL_<br>ON_WID<br>TH2[1]   | CK_ALL_<br>ON_WID<br>TH2[0]   | 00  |
| Parameter 5th  | R/W     | x                           | x                           | CK_ALL_<br>ON_WID<br>TH3[5]   | CK_ALL_<br>ON_WID<br>TH3[4]   | CK_ALL_<br>ON_WID<br>TH3[3]   | CK_ALL_<br>ON_WID<br>TH3[2]   | CK_ALL_<br>ON_WID<br>TH3[1]   | CK_ALL_<br>ON_WID<br>TH3[0]   | 00  |
| Parameter 6th  | R/W     | YS_FLG<br>G_PERI<br>OD[7]   | YS_FLG<br>G_PERI<br>OD[6]   | YS_FLG<br>G_PERI<br>OD[5]     | YS_FLG<br>G_PERI<br>OD[4]     | YS_FLG<br>G_PERI<br>OD[3]     | YS_FLG<br>G_PERI<br>OD[2]     | YS_FLG<br>G_PERI<br>OD[1]     | YS_FLG<br>G_PERI<br>OD[0]     | 00  |
| Parameter 7th  | R/W     | YS2_SE<br>L_2[1]            | YS2_SEL<br>_2[0]            | YS1_SEL<br>_2[1]              | YS1_SEL<br>_2[0]              | YS2_XO<br>R_2                 | YS1_XO<br>R_2                 | YS_FLG<br>EN_2                | ALL_ON<br>EN_2                | 94  |
| Parameter 8th  | R/W     | USER_G<br>IP_GATE<br>1_2[7] | USER_G<br>IP_GATE<br>1_2[6] | USER_G<br>IP_GATE<br>1_2[5]   | USER_G<br>IP_GATE<br>1_2[4]   | USER_G<br>IP_GATE<br>1_2[3]   | USER_G<br>IP_GATE<br>1_2[2]   | USER_G<br>IP_GATE<br>1_2[1]   | USER_G<br>IP_GATE<br>1_2[0]   | 00  |
| Parameter 9th  | R/W     | CK_ALL_<br>ON_EN<br>_2      | STV_ALL_<br>ON_EN<br>_2     | CK_ALL_<br>ON_WID<br>TH1_2[5] | CK_ALL_<br>ON_WID<br>TH1_2[4] | CK_ALL_<br>ON_WID<br>TH1_2[3] | CK_ALL_<br>ON_WID<br>TH1_2[2] | CK_ALL_<br>ON_WID<br>TH1_2[1] | CK_ALL_<br>ON_WID<br>TH1_2[0] | 00  |
| Parameter 10th | R/W     | x                           | x                           | CK_ALL_<br>ON_WID<br>TH2_2[5] | CK_ALL_<br>ON_WID<br>TH2_2[4] | CK_ALL_<br>ON_WID<br>TH2_2[3] | CK_ALL_<br>ON_WID<br>TH2_2[2] | CK_ALL_<br>ON_WID<br>TH2_2[1] | CK_ALL_<br>ON_WID<br>TH2_2[0] | 00  |
| Parameter 11th | R/W     | x                           | x                           | CK_ALL_<br>ON_WID<br>TH3_2[5] | CK_ALL_<br>ON_WID<br>TH3_2[4] | CK_ALL_<br>ON_WID<br>TH3_2[3] | CK_ALL_<br>ON_WID<br>TH3_2[2] | CK_ALL_<br>ON_WID<br>TH3_2[1] | CK_ALL_<br>ON_WID<br>TH3_2[0] | 00  |
| Parameter 12th | R/W     | YS_FLG<br>G_PERI<br>OD_2[7] | YS_FLG<br>G_PERI<br>OD_2[6] | YS_FLG<br>G_PERI<br>OD_2[5]   | YS_FLG<br>G_PERI<br>OD_2[4]   | YS_FLG<br>G_PERI<br>OD_2[3]   | YS_FLG<br>G_PERI<br>OD_2[2]   | YS_FLG<br>G_PERI<br>OD_2[1]   | YS_FLG<br>G_PERI<br>OD_2[0]   | 00  |
| Parameter 13th | R/W     | COS1_L<br>GS[3]             | COS1_L<br>GS[2]             | COS1_L<br>GS[1]               | COS1_L<br>GS[0]               | COS2_L<br>GS[3]               | COS2_L<br>GS[2]               | COS2_L<br>GS[1]               | COS2_L<br>GS[0]               | 00  |
| Parameter 14th | R/W     | COS3_L<br>GS[3]             | COS3_L<br>GS[2]             | COS3_L<br>GS[1]               | COS3_L<br>GS[0]               | COS4_L<br>GS[3]               | COS4_L<br>GS[2]               | COS4_L<br>GS[1]               | COS4_L<br>GS[0]               | 00  |
| Parameter 15th | R/W     | COS5_L<br>GS[3]             | COS5_L<br>GS[2]             | COS5_L<br>GS[1]               | COS5_L<br>GS[0]               | COS6_L<br>GS[3]               | COS6_L<br>GS[2]               | COS6_L<br>GS[1]               | COS6_L<br>GS[0]               | 00  |
| :              | :       | :                           | :                           | :                             | :                             | :                             | :                             | :                             | :                             | :   |
| Parameter 21th | R/W     | COS17_<br>L_GS[3]           | COS17_<br>L_GS[2]           | COS17_<br>L_GS[1]             | COS17_<br>L_GS[0]             | COS18_<br>L_GS[3]             | COS18_<br>L_GS[2]             | COS18_<br>L_GS[1]             | COS18_<br>L_GS[0]             | 00  |
| Parameter 22th | R/W     | COS19_<br>L_GS[3]           | COS19_<br>L_GS[2]           | COS19_<br>L_GS[1]             | COS19_<br>L_GS[0]             | COS20_<br>L_GS[3]             | COS20_<br>L_GS[2]             | COS20_<br>L_GS[1]             | COS20_<br>L_GS[0]             | 00  |
| Parameter 23th | R/W     | COS21_<br>L_GS[3]           | COS21_<br>L_GS[2]           | COS21_<br>L_GS[1]             | COS21_<br>L_GS[0]             | COS22_<br>L_GS[3]             | COS22_<br>L_GS[2]             | COS22_<br>L_GS[1]             | COS22_<br>L_GS[0]             | 00  |
| Parameter 24th | R/W     | COS1_R<br>GS[3]             | COS1_R<br>GS[2]             | COS1_R<br>GS[1]               | COS1_R<br>GS[0]               | COS2_R<br>GS[3]               | COS2_R<br>GS[2]               | COS2_R<br>GS[1]               | COS2_R<br>GS[0]               | 00  |
| Parameter 25th | R/W     | COS3_R<br>GS[3]             | COS3_R<br>GS[2]             | COS3_R<br>GS[1]               | COS3_R<br>GS[0]               | COS4_R<br>GS[3]               | COS4_R<br>GS[2]               | COS4_R<br>GS[1]               | COS4_R<br>GS[0]               | 00  |
| Parameter 26th | R/W     | COS5_R<br>GS[3]             | COS5_R<br>GS[2]             | COS5_R<br>GS[1]               | COS5_R<br>GS[0]               | COS6_R<br>GS[3]               | COS6_R<br>GS[2]               | COS6_R<br>GS[1]               | COS6_R<br>GS[0]               | 00  |
| :              | :       | :                           | :                           | :                             | :                             | :                             | :                             | :                             | :                             | :   |
| Parameter 32th | R/W     | COS17_<br>R_GS[3]           | COS17_<br>R_GS[2]           | COS17_<br>R_GS[1]             | COS17_<br>R_GS[0]             | COS18_<br>R_GS[3]             | COS18_<br>R_GS[2]             | COS18_<br>R_GS[1]             | COS18_<br>R_GS[0]             | 00  |
| Parameter 33th | R/W     | COS19_<br>R_GS[3]           | COS19_<br>R_GS[2]           | COS19_<br>R_GS[1]             | COS19_<br>R_GS[0]             | COS20_<br>R_GS[3]             | COS20_<br>R_GS[2]             | COS20_<br>R_GS[1]             | COS20_<br>R_GS[0]             | 00  |
| Parameter 34th | R/W     | COS21_<br>R_GS[3]           | COS21_<br>R_GS[2]           | COS21_<br>R_GS[1]             | COS21_<br>R_GS[0]             | COS22_<br>R_GS[3]             | COS22_<br>R_GS[2]             | COS22_<br>R_GS[1]             | COS22_<br>R_GS[0]             | 00  |
| Parameter 35th | R/W     | x                           | x                           | EQOPT[1]                      | EQOPT[0]                      | x                             | x                             | EQ_SEL[1]                     | EQ_SEL[0]                     | 30  |
| Parameter 36th | R/W     | EQ_DEL<br>AY[7]             | EQ_DEL<br>AY[6]             | EQ_DEL<br>AY[5]               | EQ_DEL<br>AY[4]               | EQ_DEL<br>AY[3]               | EQ_DEL<br>AY[2]               | EQ_DEL<br>AY[1]               | EQ_DEL<br>AY[0]               | 00  |

| Parameter 37th                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W                 | x                             | x                            | x                            | x                            | EQ_DEL<br>AY_HSY<br>NC[3]    | EQ_DEL<br>AY_HSY<br>NC[2]    | EQ_DEL<br>AY_HSY<br>NC[1]    | EQ_DEL<br>AY_HSY<br>NC[0]    | 00 |                               |                     |   |             |   |                |   |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|----|-------------------------------|---------------------|---|-------------|---|----------------|---|-----------------|
| Parameter 38th                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W                 | x                             | x                            | x                            | x                            | x                            | x                            | HSYNC_<br>TO_CL1<br>CNT10[9] | HSYNC_<br>TO_CL1<br>CNT10[8] | 00 |                               |                     |   |             |   |                |   |                 |
| Parameter 39th                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W                 | HSYNC_<br>TO_CL1<br>_CNT10[7] | HSYNC_<br>TO_CL1<br>CNT10[6] | HSYNC_<br>TO_CL1<br>CNT10[5] | HSYNC_<br>TO_CL1<br>CNT10[4] | HSYNC_<br>TO_CL1<br>CNT10[3] | HSYNC_<br>TO_CL1<br>CNT10[2] | HSYNC_<br>TO_CL1<br>CNT10[1] | HSYNC_<br>TO_CL1<br>CNT10[0] | 02 |                               |                     |   |             |   |                |   |                 |
| <p>This command is to set backward GIP timing.</p> <div></div> <ul style="list-style-type: none"><li>• <b>YS1_SEL[1:0]</b> : Specify YS1 Signal Mode.</li><li>• <b>YS2_SEL[1:0]</b> : Specify YS2 Signal Mode.</li></ul> <table border="1" data-bbox="357 960 796 1140"><tr><th>YS1[1:0]<br/>YS2[1:0]<br/>(Hex)</th><th>YS1/YS2 Signal Mode</th></tr><tr><td>0</td><td>INYS1/INYS2</td></tr><tr><td>1</td><td>INYS1 or INVS2</td></tr><tr><td>2</td><td>INYS1 and INYS2</td></tr></table> <ul style="list-style-type: none"><li>• <b>YS_FLAG_EN</b> : Enable YS signal function.</li><li>• <b>YS1_XOR</b> : Reverse YS1 signal.</li><li>• <b>YS2_XOR</b> : Reverse YS2 signal.</li><li>• <b>ALL_ON_EN</b> : Enable ALL ON function.</li><li>• <b>USER_GIP_GATE[7:0]</b> : Specify the distance of GIP ALL On rising edge and DE.</li><li>• <b>CK_ALL_ON_WIDTH1[5:0] / CK_ALL_ON_WIDTH2[5:0] / CK_ALL_ON_WIDTH3[5:0]</b> : Specify the timing of YS1 and YS2 signal.</li></ul> |                     |                               |                              |                              |                              |                              |                              |                              |                              |    | YS1[1:0]<br>YS2[1:0]<br>(Hex) | YS1/YS2 Signal Mode | 0 | INYS1/INYS2 | 1 | INYS1 or INVS2 | 2 | INYS1 and INYS2 |
| YS1[1:0]<br>YS2[1:0]<br>(Hex)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | YS1/YS2 Signal Mode |                               |                              |                              |                              |                              |                              |                              |                              |    |                               |                     |   |             |   |                |   |                 |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | INYS1/INYS2         |                               |                              |                              |                              |                              |                              |                              |                              |    |                               |                     |   |             |   |                |   |                 |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | INYS1 or INVS2      |                               |                              |                              |                              |                              |                              |                              |                              |    |                               |                     |   |             |   |                |   |                 |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | INYS1 and INYS2     |                               |                              |                              |                              |                              |                              |                              |                              |    |                               |                     |   |             |   |                |   |                 |

Description

- **EQOPT[1:0]** : Specify GIP output EQ signal.

| EQOTP<br>(Hex) | P_EQ | N_EQ |
|----------------|------|------|
| 0              | Yes  | Yes  |
| 1              | No   | Yes  |
| 2              | Yes  | No   |
| 3              | No   | No   |

- **EQ\_SEL[1:0]** : Specify GIP output EQ signal level.

| EQ_SEL<br>(Hex) | P_EQ | N_EQ |
|-----------------|------|------|
| 0               | -    | -    |
| 1               | -    | -    |
| 2               | -    | -    |
| 3               | VSP  | VSP  |

- **EQ\_DELAY[7:0]** : Specify the distance of EQ rising edge and HYSNC.

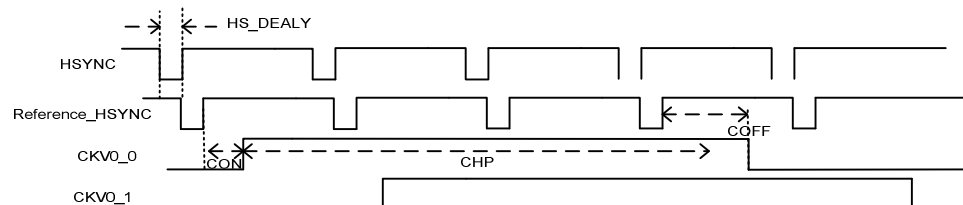
| EQ_DEALY<br>(Hex) | Distance |
|-------------------|----------|
| 0                 | 0 Fosc   |
| 1                 | 1 Fosc   |
| 2                 | 2 Fosc   |
| :                 | :        |
| FF                | 255 Fosc |

- **EQ\_DELAY\_HSYNC[3:0]** : Specify the distance of EQ rising edge and HYSNC.

| EQ_DEALY_HS<br>(Hex) | Distance |
|----------------------|----------|
| 0                    | 0 Hsync  |
| 1                    | 1 Hsync  |
| 2                    | 2 Hsync  |
| :                    | :        |
| 16                   | 16 Hsync |

- **HSYNC\_TO\_CL1\_CNT10[9:0]** : Specify the GIP reference HSYNC between external HSYNC

| HSYNC_TO_CL1_CNT10[9:0]<br>(Hex) | Distance |
|----------------------------------|----------|
| 0                                | 0 Fosc   |
| 1                                | 1 Fosc   |
| 2                                | 2 Fosc   |
| :                                | :        |
| 1FF                              | 511 Fosc |



## 6.2.76 SETCOLOR (EBh)

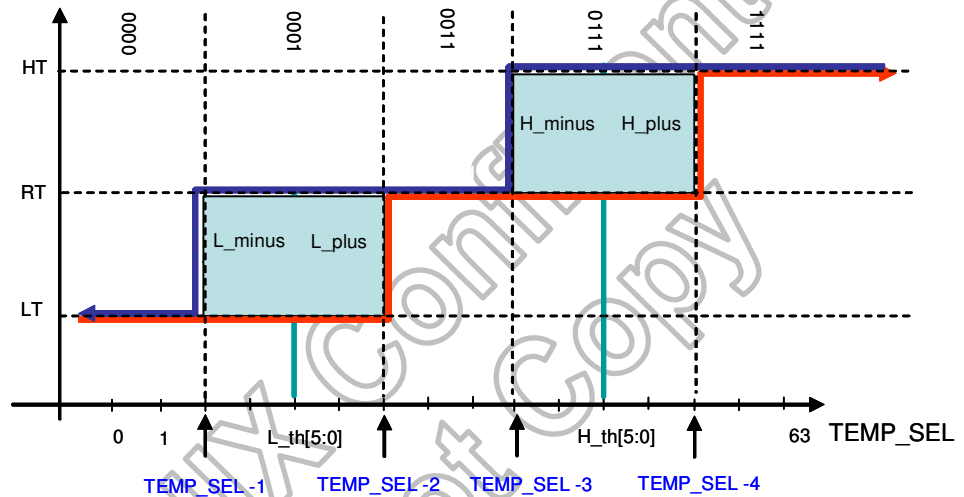
| EB H           | SETCOLOR                                                      |        |        |        |        |        |        |        |        |     |
|----------------|---------------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|-----|
|                | R/W                                                           | D7     | D6     | D5     | D4     | D3     | D2     | D1     | D0     | HEX |
| Command        | W                                                             | 1      | 1      | 1      | 0      | 1      | 0      | 1      | 1      | EB  |
| Parameter 1st  | W                                                             | Bkx[1] | Bkx[0] | Bky[1] | Bky[0] | Wx[1]  | Wx[0]  | Wy[1]  | Wy[0]  | 00  |
| Parameter 2nd  | W                                                             | BKx[9] | BKx[8] | BKx[7] | BKx[6] | BKx[5] | BKx[4] | BKx[3] | BKx[2] | 00  |
| Parameter 3rd  | W                                                             | BKy[9] | BKy[8] | BKy[7] | BKy[6] | BKy[5] | BKy[4] | BKy[3] | BKy[2] | 00  |
| Parameter 4th  | W                                                             | Wx[9]  | Wx[8]  | Wx[7]  | Wx[6]  | Wx[5]  | Wx[4]  | Wx[3]  | Wx[2]  | 00  |
| Parameter 5th  | W                                                             | Wy[9]  | Wy[8]  | Wy[7]  | Wy[6]  | Wy[5]  | Wy[4]  | Wy[3]  | Wy[2]  | 00  |
| Parameter 6th  | W                                                             | Rx[1]  | Rx[0]  | Ry[1]  | Ry[0]  | Gx[1]  | Gx[0]  | Gy[1]  | Gy[0]  | 00  |
| Parameter 7th  | W                                                             | Rx[9]  | Rx[8]  | Rx[7]  | Rx[6]  | Rx[5]  | Rx[4]  | Rx[3]  | Rx[2]  | 00  |
| Parameter 8th  | W                                                             | Ry[9]  | Ry[8]  | Ry[7]  | Ry[6]  | Ry[5]  | Ry[4]  | Ry[3]  | Ry[2]  | 00  |
| Parameter 9th  | W                                                             | Gx[9]  | Gx[8]  | Gx[7]  | Gx[6]  | Gx[5]  | Gx[4]  | Gx[3]  | Gx[2]  | 00  |
| Parameter 10th | W                                                             | Gy[9]  | Gy[8]  | Gy[7]  | Gy[6]  | Gy[5]  | Gy[4]  | Gy[3]  | Gy[2]  | 00  |
| Parameter 11th | W                                                             | Bx[1]  | Bx[0]  | By[1]  | By[0]  | Ax[1]  | Ax[0]  | Ay[1]  | Ay[0]  | 00  |
| Parameter 12th | W                                                             | Bx[9]  | Bx[8]  | Bx[7]  | Bx[6]  | Bx[5]  | Bx[4]  | Bx[3]  | Bx[2]  | 00  |
| Parameter 13th | W                                                             | By[9]  | By[8]  | By[7]  | By[6]  | By[5]  | By[4]  | By[3]  | By[2]  | 00  |
| Parameter 14th | W                                                             | Ax[9]  | Ax[8]  | Ax[7]  | Ax[6]  | Ax[5]  | Ax[4]  | Ax[3]  | Ax[2]  | 00  |
| Parameter 15th | W                                                             | Ay[9]  | Ay[8]  | Ay[7]  | Ay[6]  | Ay[5]  | Ay[4]  | Ay[3]  | Ay[2]  | 00  |
| Description    | This command is used to record the Panel Optical Performance. |        |        |        |        |        |        |        |        |     |

## 6.2.77 TEMP\_SENSOR (F1h)

| F1 H                                                                                         | TEMP_SENSOR                                                                                                                                                     |                                |                     |                 |                 |                  |                  |                  |                  |     |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------|-----------------|-----------------|------------------|------------------|------------------|------------------|-----|
|                                                                                              | R/W                                                                                                                                                             | D7                             | D6                  | D5              | D4              | D3               | D2               | D1               | D0               | HEX |
| Command                                                                                      | W                                                                                                                                                               | 1                              | 1                   | 1               | 1               | 0                | 0                | 0                | 1                | F1  |
| Parameter 1st                                                                                | R/W                                                                                                                                                             | PASSW<br>ORD[7]                | PASSW<br>ORD[6]     | PASSW<br>ORD[5] | PASSW<br>ORD[4] | PASSW<br>ORD[3]  | PASSW<br>ORD[2]  | PASSW<br>ORD[1]  | PASSW<br>ORD[0]  | 00  |
| Parameter 2nd                                                                                | R/W                                                                                                                                                             | TEMP_S<br>T[3]                 | TEMP_S<br>T[2]      | TEMP_S<br>T[1]  | TEMP_S<br>T[0]  | TEMP_F<br>RA [3] | TEMP_F<br>RA [2] | TEMP_F<br>RA [1] | TEMP_F<br>RA [0] | 3F  |
| Parameter 3rd                                                                                | R/W                                                                                                                                                             | x                              | x                   | L_TH[5]         | L_TH[4]         | L_TH[3]          | L_TH[2]          | L_TH[1]          | L_TH[0]          | 14  |
| Parameter 4th                                                                                | R/W                                                                                                                                                             | x                              | x                   | H_TH[5]         | H_TH[4]         | H_TH[3]          | H_TH[2]          | H_TH[1]          | H_TH[0]          | 30  |
| Parameter 5th                                                                                | R/W                                                                                                                                                             | x                              | L_PLUS[<br>2]       | L_PLUS[<br>1]   | L_PLUS[<br>0]   | x                | L_MINUS<br>[2]   | L_MINUS<br>[1]   | L_MINUS<br>[0]   | 55  |
| Parameter 6th                                                                                | R/W                                                                                                                                                             | x                              | H_PLUS[<br>2]       | H_PLUS[<br>1]   | H_PLUS[<br>0]   | x                | H_MINU<br>S[2]   | H_MINU<br>S[1]   | H_MINU<br>S[0]   | 55  |
| Parameter 7th                                                                                | R/W                                                                                                                                                             | TEMP_FI<br>LTER_E<br>N         | x                   | x               | x               | SAME_N<br>UM[3]  | SAME_N<br>UM[2]  | SAME_N<br>UM[1]  | SAME_N<br>UM[0]  | 0F  |
| Description                                                                                  | This command is used to control the temperature function.                                                                                                       |                                |                     |                 |                 |                  |                  |                  |                  |     |
|                                                                                              | • <b>PASSWORD[7:0]</b> : Enable Temp_sensor function key<br>- PASSWORD = “A5”, enable Temp_sensor function.<br>- PASSWORD = “00”, disable Temp_sensor function. |                                |                     |                 |                 |                  |                  |                  |                  |     |
|                                                                                              | • <b>TEMP_FRA[3:0]</b> : Control the temperature sensing rate                                                                                                   |                                |                     |                 |                 |                  |                  |                  |                  |     |
|                                                                                              | <b>TEMP_FRA[3:0]<br/>(Hex)</b>                                                                                                                                  |                                | <b>Sensing Rate</b> |                 |                 |                  |                  |                  |                  |     |
|                                                                                              | 0                                                                                                                                                               |                                | 1 frame             |                 |                 |                  |                  |                  |                  |     |
|                                                                                              | 1                                                                                                                                                               |                                | 2 frame             |                 |                 |                  |                  |                  |                  |     |
|                                                                                              | 2                                                                                                                                                               |                                | 4 frames            |                 |                 |                  |                  |                  |                  |     |
|                                                                                              | :                                                                                                                                                               |                                | :                   |                 |                 |                  |                  |                  |                  |     |
|                                                                                              | E                                                                                                                                                               |                                | 16384 frames        |                 |                 |                  |                  |                  |                  |     |
|                                                                                              | F                                                                                                                                                               |                                | STOP                |                 |                 |                  |                  |                  |                  |     |
| • <b>TEMP_ST [3:0]</b> : Control the temperature sensing period at vertical blanking period. |                                                                                                                                                                 |                                |                     |                 |                 |                  |                  |                  |                  |     |
| <b>TEMP_ST[3:0]<br/>(Hex)</b>                                                                |                                                                                                                                                                 | <b>Sensing Period<br/>(us)</b> |                     |                 |                 |                  |                  |                  |                  |     |
| 0                                                                                            |                                                                                                                                                                 | 25                             |                     |                 |                 |                  |                  |                  |                  |     |
| 1                                                                                            |                                                                                                                                                                 | 50                             |                     |                 |                 |                  |                  |                  |                  |     |
| 2                                                                                            |                                                                                                                                                                 | 75                             |                     |                 |                 |                  |                  |                  |                  |     |
| :                                                                                            |                                                                                                                                                                 | :                              |                     |                 |                 |                  |                  |                  |                  |     |
| E                                                                                            |                                                                                                                                                                 | 200                            |                     |                 |                 |                  |                  |                  |                  |     |
| F                                                                                            |                                                                                                                                                                 | 200                            |                     |                 |                 |                  |                  |                  |                  |     |
| • <b>TEMP_TEST_EN</b> : Enable temperature sensor test mode.                                 |                                                                                                                                                                 |                                |                     |                 |                 |                  |                  |                  |                  |     |
| • <b>T_TEMP_EN</b> : Enable temperature sensor comparator function.                          |                                                                                                                                                                 |                                |                     |                 |                 |                  |                  |                  |                  |     |

Description

- **L\_TH[5:0]** : Define Low temperature threshold voltage.
- **H\_TH[5:0]** : Define High temperature thresholds voltage.
- **L\_PLUS[2:0]** : Specify Low temperature Hysteretic Window plus range.
- **H\_PLUS[2:0]** : Specify High temperature Hysteretic Window plus range.
- **L\_MINUS[2:0]** : Specify Low temperature Hysteretic Window minus range.
- **H\_MINUS[2:0]** : Specify High temperature Hysteretic Window minus range.



- **TEMP\_FILTER\_EN** : Enable temperature sensor filter function.
  - 0 : Disable
  - 1 : Enable
- **SAME\_NUM[3:0]** : The sates machine will change the state if the same output times.

## 6.2.78 TEMP\_VOLTAGE (F2h)

| F2 H           | TEMP_VOLTAGE                                                                               |                |                |                |                |              |              |              |                |     |
|----------------|--------------------------------------------------------------------------------------------|----------------|----------------|----------------|----------------|--------------|--------------|--------------|----------------|-----|
|                | R/W                                                                                        | D7             | D6             | D5             | D4             | D3           | D2           | D1           | D0             | HEX |
| Command        | W                                                                                          | 1              | 1              | 1              | 1              | 0            | 0            | 1            | 0              | F2  |
| Parameter 1st  | R/W                                                                                        | VBTHS1[3]      | VBTHS1[2]      | VBTHS1[1]      | VBTHS1[0]      | VBTLS1[3]    | VBTLS1[2]    | VBTLS1[1]    | VBTLS1[0]      | 53  |
| Parameter 2nd  | R/W                                                                                        | ECP_DC_DIV1[3] | ECP_DC_DIV1[2] | ECP_DC_DIV1[1] | ECP_DC_DIV1[0] | VGLREG1[3]   | VGLREG1[2]   | VGLREG1[1]   | VGLREG1[0]     | 43  |
| Parameter 3rd  | R/W                                                                                        | VCOMDC_F1[7]   | VCOMDC_F1[6]   | VCOMDC_F1[5]   | VCOMDC_F1[4]   | VCOMDC_F1[3] | VCOMDC_F1[2] | VCOMDC_F1[1] | VCOMDC_F1[0]   | 4D  |
| Parameter 4th  | R/W                                                                                        | VCOMDC_B1[7]   | VCOMDC_B1[6]   | VCOMDC_B1[5]   | VCOMDC_B1[4]   | VCOMDC_B1[3] | VCOMDC_B1[2] | VCOMDC_B1[1] | VCOMDC_B1[0]   | 4D  |
| Parameter 5th  | R/W                                                                                        | x              | x              | PVR01[5]       | PVR01[4]       | PVR01[3]     | PVR01[2]     | PVR01[1]     | PVR01[0]       | 00  |
| Parameter 6th  | R/W                                                                                        | x              | x              | PVR51[5]       | PVR51[4]       | PVR51[3]     | PVR51[2]     | PVR51[1]     | PVR51[0]       | 00  |
| Parameter 7th  | R/W                                                                                        | x              | x              | NVR01[5]       | NVR01[4]       | NVR01[3]     | NVR01[2]     | NVR01[1]     | NVR01[0]       | 00  |
| Parameter 8th  | R/W                                                                                        | x              | x              | NVR51[5]       | NVR51[4]       | NVR51[3]     | NVR51[2]     | NVR51[1]     | NVR51[0]       | 00  |
| Parameter 9th  | R/W                                                                                        | x              | x              | x              | x              | x            | N_NW_HT[2]   | N_NW_HT[1]   | N_NW_HT[0]     | 00  |
| Parameter 10th |                                                                                            | VBTHS4[3]      | VBTHS4[2]      | VBTHS4[1]      | VBTHS4[0]      | VBTLS4[3]    | VBTLS4[2]    | VBTLS4[1]    | VBTLS4[0]      | 53  |
| Parameter 11th | R/W                                                                                        | ECP_DC_DIV4[3] | ECP_DC_DIV4[2] | ECP_DC_DIV4[1] | ECP_DC_DIV4[0] | VGLREG4[3]   | VGLREG4[2]   | VGLREG4[1]   | VGLREG4[0]     | 43  |
| Parameter 12th |                                                                                            | VCOMDC_F4[7]   | VCOMDC_F4[6]   | VCOMDC_F4[5]   | VCOMDC_F4[4]   | VCOMDC_F4[3] | VCOMDC_F4[2] | VCOMDC_F4[1] | VCOMDC_F4[0]   | 4D  |
| Parameter 13th | R/W                                                                                        | VCOMDC_B4[7]   | VCOMDC_B4[6]   | VCOMDC_B4[5]   | VCOMDC_B4[4]   | VCOMDC_B4[3] | VCOMDC_B4[2] | VCOMDC_B4[1] | VCOMDC_B4[0]   | 4D  |
| Parameter 14th | R/W                                                                                        | x              | x              | PVR04[5]       | PVR04[4]       | PVR04[3]     | PVR04[2]     | PVR04[1]     | PVR04[0]       | 00  |
| Parameter 15th | R/W                                                                                        | x              | x              | PVR54[5]       | PVR54[4]       | PVR54[3]     | PVR54[2]     | PVR54[1]     | PVR54[0]       | 00  |
| Parameter 16th | R/W                                                                                        | x              | x              | NVR04[5]       | NVR04[4]       | NVR04[3]     | NVR04[2]     | NVR04[1]     | NVR04[0]       | 00  |
| Parameter 17th | R/W                                                                                        | x              | x              | NVR54[5]       | NVR54[4]       | NVR54[3]     | NVR54[2]     | NVR54[1]     | NVR54[0]       | 00  |
| Parameter 18th | R/W                                                                                        | x              | x              | x              | x              | x            | N_NW_LT[2]   | N_NW_LT[1]   | N_NW_LT[0]     | 00  |
| Parameter 19th | R/W                                                                                        | x              | x              | x              | x              | x            | x            | x            | TEMP_SENSOR_ON | 00  |
| Description    | This command is used to control the voltage at 2 different temperature conditions.         |                |                |                |                |              |              |              |                |     |
|                | Voltage                                                                                    |                |                |                | Temperature    |              |              |              |                |     |
|                | VBTHS1/VBTL1S1/ ECP_DC_DIV1<br>VGLREG1/ VCOMDC_F1/ VCOMDC_B1<br>PVR01/ PVR51/ NVR01/ NVR51 |                |                |                | HIGH           |              |              |              |                |     |
|                | VBTHS4/VBTL4S4/ ECP_DC_DIV4<br>VGLREG4/ VCOMDC_F4/ VCOMDC_B4<br>PVR04/ PVR54/ NVR04/ NVR54 |                |                |                | LOW            |              |              |              |                |     |



## 7. Electrical Characteristics

### 7.1 Absolute maximum ratings

| Item                       | Symbol                                               | Unit | Spec. |      |           |
|----------------------------|------------------------------------------------------|------|-------|------|-----------|
|                            |                                                      |      | Min.  | Typ. | Max.      |
| Power Supply Voltage 1     | IOVCC~VSSD                                           | V    | -0.3  | -    | +5.5      |
| Power Supply Voltage 2     | VCI ~ VSSA                                           | V    | -0.3  | -    | +6.6      |
| Power Supply Voltage 3     | VSP ~ VSSA                                           | V    | -0.3  | -    | +6.6      |
| Power Supply Voltage 4     | VSSA ~ VSN                                           | V    | -0.3  | -    | +6.6      |
| Power Supply Voltage 5     | VGH ~ VGL                                            | V    | -0.3  | -    | +35       |
| Logic Input Voltage        | V <sub>IN</sub>                                      | V    | -0.3  | -    | IOVCC+0.3 |
| Logic Output Voltage       | V <sub>O</sub>                                       | V    | -0.3  | -    | IOVCC+0.3 |
| Differential Input Voltage | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P,<br>DSI_D1P/DSI_D1N | V    | -0.3  | -    | 2.0       |
| Operating Temperature      | T <sub>opr</sub>                                     | °C   | -40   | -    | +85       |
| Storage Temperature        | T <sub>stg</sub>                                     | °C   | -55   | -    | +110      |

**Table 7.1: Absolute Maximum Ratings**

## 7.2 DC characteristics

### 7.2.1 Basic Characteristics

| Parameter                       | Symbol  | Conditions             | Spec.    |      |          | Unit |
|---------------------------------|---------|------------------------|----------|------|----------|------|
|                                 |         |                        | Min.     | Typ. | Max.     |      |
| Power & Operating Voltages      |         |                        |          |      |          |      |
| Logic Operating voltage         | IOVCC   | I/O supply voltage     | 1.65     | 1.8  | 2.0      | V    |
| Analog Operating voltage        | VCI     | Operation voltage      | 2.5      | -    | 6.2      |      |
| Input / Output                  |         |                        |          |      |          |      |
| Logic High level input voltage  | VIH     | -                      | 0.7IOVCC | -    | IOVCC    | V    |
| Logic Low level input voltage   | VIL     | -                      | VSSD     | -    | 0.3IOVCC |      |
| Logic High level output voltage | VOH     | IOH = -1.0mA           | 0.8IOVCC | -    | IOVCC    |      |
| Logic Low level output voltage  | VOL     | IOL = +1.0mA           | VSSD     | -    | 0.2IOVCC |      |
| Input leakage current           | IIL     | -                      | -1       | -    | 1        | μA   |
| DC/DC Converter Operation       |         |                        |          |      |          |      |
| VSP booster voltage             | VSP     | IVSP=1mA               | 4.5      |      | 6.2      | V    |
| VSN booster voltage             | VSN     | IVSN=-1mA              | -6.2     |      | -4.5     |      |
| VGH booster voltage             | VGH     | Ivgh=1mA               | 10       |      | 20       |      |
| VGL booster voltage             | VGL     | Ivgl=-1mA              | -15      |      | -7.5     |      |
| VGH and VGL difference          | VGH-VGL | -                      | -        | -    | 32       |      |
| Oscillator tolerance            | OSC     | 25°C                   | -3       |      | 3        | %    |
| Source Driver                   |         |                        |          |      |          |      |
| Gamma reference voltage         | VSPR    | -                      | 3.3      |      | 5.6      | V    |
|                                 | VSNR    | -                      | -5.6     |      | -3.3     |      |
| Output voltage deviation        | DVOS    | VSSD+1.0 ~ VSPROUT-1.0 | -        | -    | +/- 20   | mV   |
|                                 |         | VSSD+0.1V ~ VSSD+1.0   |          |      |          |      |
|                                 |         | VSPR-1.0 ~ VSPR-0.1V   | -        | -    | +/- 50   | mV   |
| Output offset voltage           | Voff    | -                      |          | -    | +/-50    | mV   |

## 7.2.2 DSI DC Characteristics

### LP Mode

| Parameter                       | Symbol          | Conditions            | Spec. |      |      | Unit |
|---------------------------------|-----------------|-----------------------|-------|------|------|------|
|                                 |                 |                       | Min.  | Typ. | Max. |      |
| Logic high level input voltage  | $V_{IHLPCD}$    | LP-CD                 | 450   | -    | 1350 | mV   |
| Logic low level input voltage   | $V_{ILLPCD}$    | LP-CD                 | 0     | -    | 200  | mV   |
| Logic high level input voltage  | $V_{IHLPRX}$    | LP-RX(CLK, D0)        | 880   | -    | 1350 | mV   |
| Logic low level input voltage   | $V_{ILLPRX}$    | LP-RX(CLK, D0)        | 0     | -    | 550  | mV   |
| Logic low level input voltage   | $V_{ILLPRXULP}$ | LP-RX(CLK ULP mode)   | 0     | -    | 300  | mV   |
| Logic high level output voltage | $V_{OHLPTX}$    | LP-TX(D0)             | 1.1   | -    | 1.3  | V    |
| Logic low level output voltage  | $V_{OLLPTX}$    | LP-TX(D0)             | -50   | -    | 50   | mV   |
| Logic high level input current  | $V_{IH}$        | LP-CD, LP-RX          | -     | -    | 10   | uA   |
| Logic low level input current   | $V_{IL}$        | LP-CD, LP-RX          | -10   | -    | -    | uA   |
| Input pulse rejection           | SGD             | DSI-CLK+/-, DSI-D0+/1 | -     | -    | 300  | Vps  |

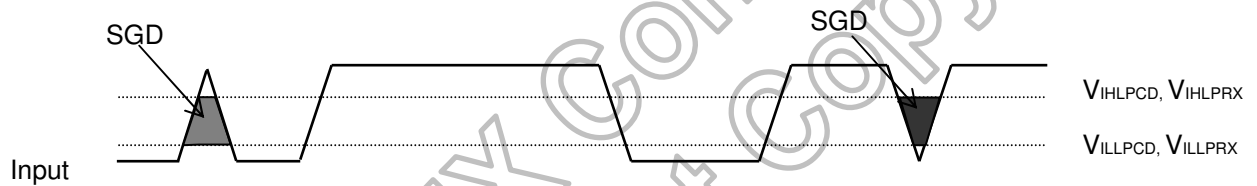
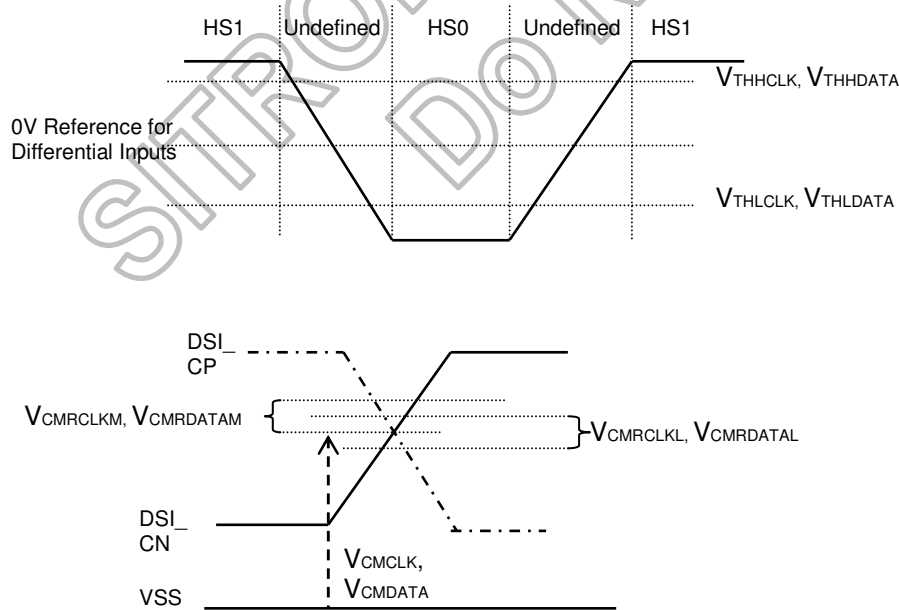


Figure 7.1: Input glitch rejections of low-power receivers

## High Speed Mode

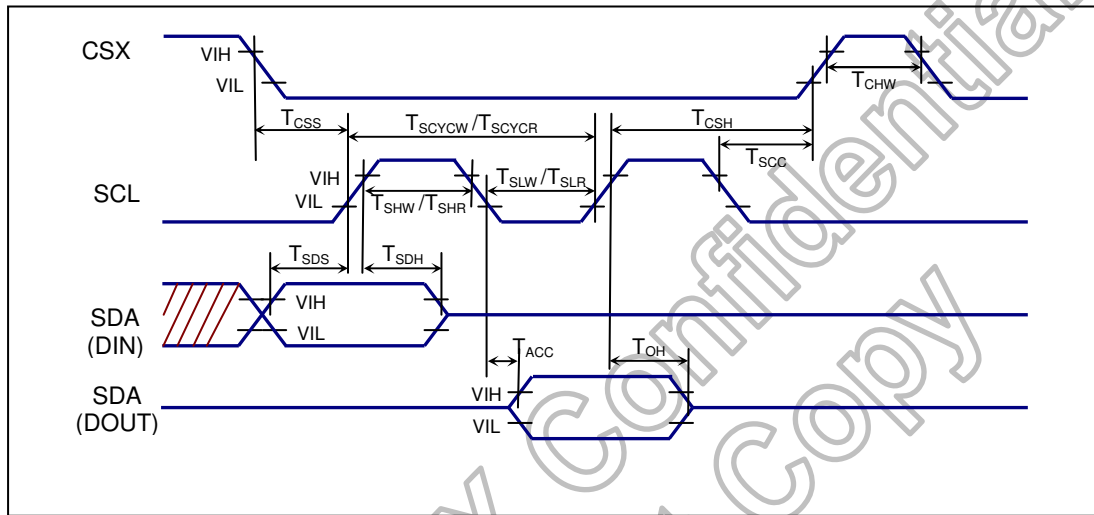
| Parameter                                                | Symbol                          | Conditions                       | Spec. |      |      | Unit     |
|----------------------------------------------------------|---------------------------------|----------------------------------|-------|------|------|----------|
|                                                          |                                 |                                  | Min.  | Typ. | Max. |          |
| Input common mode                                        | $V_{CMCLK}$<br>$V_{CMDATA}$     | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | 70    | -    | 330  | mV       |
| Input common mode variation<br><450 MHz                  | $V_{CMRCLKL}$<br>$V_{CMRDATAL}$ | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | -50   | -    | 50   | mV       |
| Input common mode variation<br>>450 MHz                  | $V_{CMRCLKM}$<br>$V_{CMRDATAM}$ | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | -     | -    | 100  | mV       |
| Low-level differential Input<br>threshold                | $V_{THLCLK}$<br>$V_{THLDATA}$   | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | -70   | -    | -    | mV       |
| High-level differential Input<br>threshold               | $V_{THHCLK}$<br>$V_{THHDATA}$   | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | -     | -    | 70   | mV       |
| Single ended input low voltage                           | $V_{ILHS}$                      | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | -40   | -    | -    | mV       |
| Single ended input high voltage                          | $V_{IHHS}$                      | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | -     | -    | 460  | mV       |
| Differential input termination<br>resistor               | $R_{TERM}$                      | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | 80    | 100  | 125  | $\Omega$ |
| Single-ended threshold voltage<br>for termination enable | $V_{TERMEN}$                    | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | -     | -    | 450  | mV       |
| Termination capacitor                                    | $C_{TERM}$                      | DSI_CP/DSI_CN<br>DSI_D0P/DSI_D0P | -     | -    | -    | pF       |



**Figure 7.2: Differential voltage range and Command mode voltage**

## 7.3 AC characteristics

### 7.3.1 Serial Interface Characteristics



**Figure 7.3: Serial Interface Characteristics**

(VSSA=0V, IOVCC=1.8V, VCI=2.8V, T<sub>A</sub> = 25°C)

| Signal          | Symbol           | Parameter                      | Min. | Max. | Unit | Description                                                         |
|-----------------|------------------|--------------------------------|------|------|------|---------------------------------------------------------------------|
| CSX             | t <sub>CSS</sub> | Chip select setup time (Write) | 15   | -    | ns   | -                                                                   |
|                 | t <sub>CSS</sub> | Chip select setup time (Read)  | 60   | -    |      |                                                                     |
|                 | t <sub>CSH</sub> | Chip select hold time (Write)  | 15   | -    |      |                                                                     |
|                 | t <sub>CSH</sub> | Chip select hold time (Read)   | 65   | -    |      |                                                                     |
| DCX             | t <sub>AST</sub> | Address setup time             | 0    | -    | ns   | -                                                                   |
|                 | t <sub>AHT</sub> | Address hold time (Write/Read) | 10   | -    |      |                                                                     |
| SCL<br>(Write)  | t <sub>WC</sub>  | Write cycle                    | 66   | -    | ns   | -                                                                   |
|                 | t <sub>WRH</sub> | Control pulse "H" duration     | 15   | -    |      |                                                                     |
|                 | t <sub>WRL</sub> | Control pulse "L" duration     | 15   | -    |      |                                                                     |
| SCL<br>(Read)   | t <sub>RC</sub>  | Read cycle                     | 150  | -    | ns   | -                                                                   |
|                 | t <sub>RDH</sub> | Control pulse "H" duration     | 60   | -    |      |                                                                     |
|                 | t <sub>RDL</sub> | Control pulse "L" duration     | 60   | -    |      |                                                                     |
| SDA<br>(Input)  | t <sub>DS</sub>  | Data setup time                | 10   | -    | ns   | For maximum C <sub>L</sub> =30pF<br>For minimum C <sub>L</sub> =8pF |
|                 | t <sub>DH</sub>  | Data hold time                 | 10   | -    |      |                                                                     |
| SDA<br>(Output) | t <sub>ACC</sub> | Read access time               | -    | 100  | ns   |                                                                     |
|                 | t <sub>OH</sub>  | Output disable time            | 10   | -    |      |                                                                     |

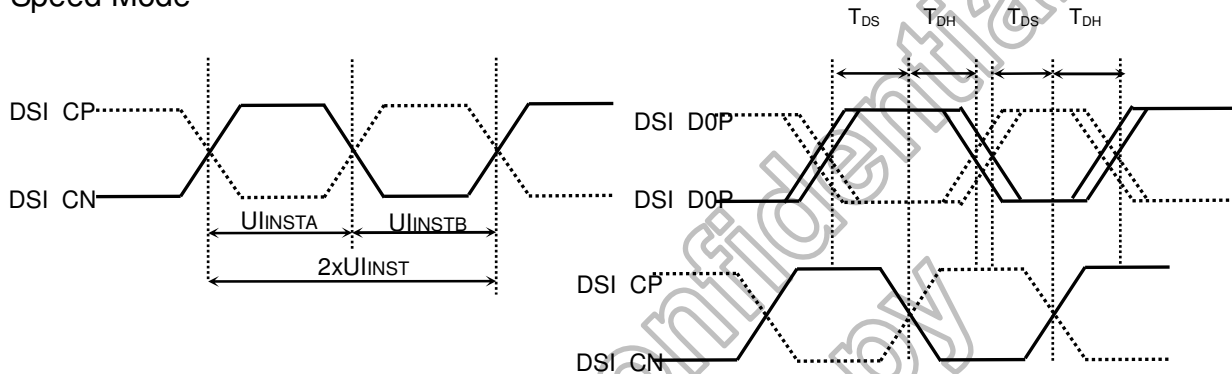
**Note:** The input signal rise time and fall time (tr, tf) is specified at 15 ns or less.

Logic high and low levels are specified as 30% and 70% of IOVCC for Input signals.

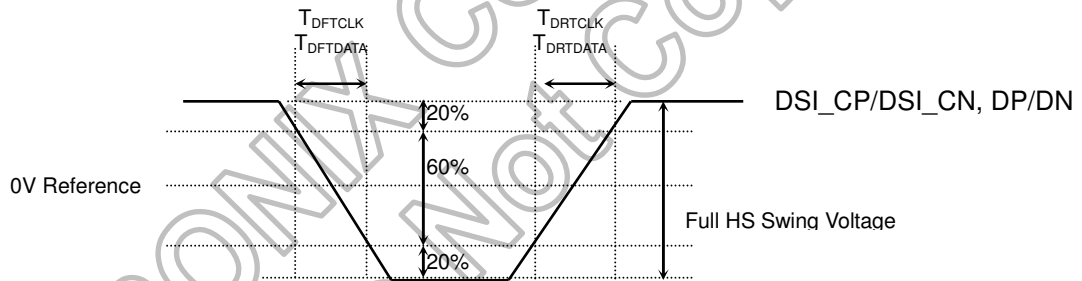
**Table 7.2: Serial Interface Characteristics**

### 7.3.2 DSI Interface Timing Characteristics

#### High Speed Mode



**Figure 7.4: DSI clock timing Characteristics**



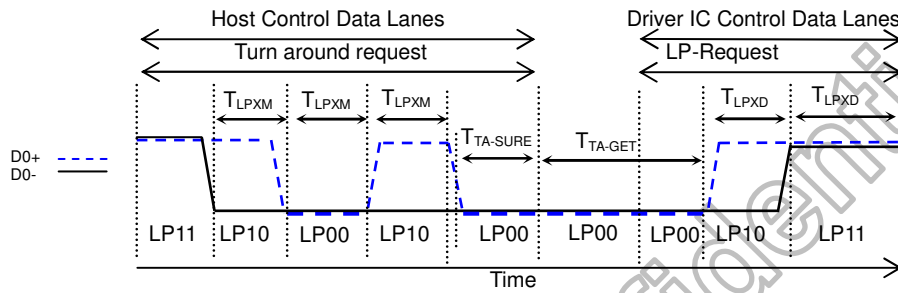
**Figure 7.5: Rising and falling time on clock and data channel**

(VSSA=0V, IOVCC=1.65V to 3.3V, VCI=2.5V to 3.3V,  $T_A = -30$  to  $70^{\circ}\text{C}$ )

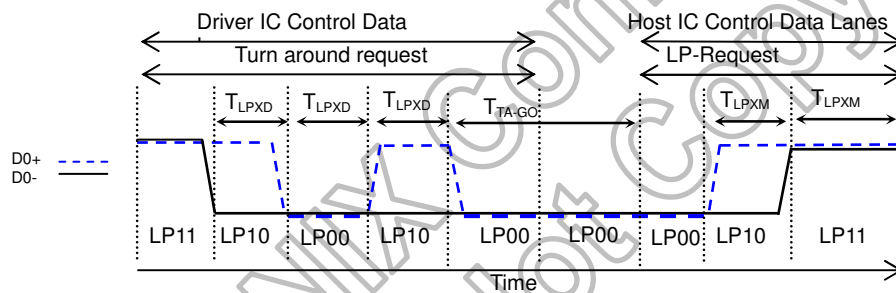
| Signal            | Item                             | Symbol                       | Spec.     |      |         | Unit |
|-------------------|----------------------------------|------------------------------|-----------|------|---------|------|
|                   |                                  |                              | Min.      | Typ. | Max.    |      |
| DSI_CP/<br>DSI_CN | Double UI instantaneous          | $2xUI_{INST}$                | TBD       | -    | 25      | ns   |
|                   | UI instantaneous                 | $UI_{INSTA}$<br>$UI_{INSTB}$ | TBD       | -    | 12.5    | ns   |
| DP/DN             | Data to clock setup time         | $T_{DS}$                     | $0.15xUI$ | -    | -       | ps   |
|                   | Data to clock hold time          | $T_{DH}$                     | $0.15xUI$ | -    | -       | ps   |
| DSI_CP/<br>DSI_CN | Differential rise time for clock | $T_{DRTCLK}$                 | 150       | -    | $0.3UI$ | ps   |
|                   | Differential fall time for clock | $T_{DFTCLK}$                 | 150       | -    | $0.3UI$ | ps   |
| DP/DN             | Differential rise time for data  | $T_{DRTDATA}$                | 150       | -    | $0.3UI$ | ps   |
|                   | Differential fall time for data  | $T_{DFTDATA}$                | 150       | -    | $0.3UI$ | ps   |

**Table 7.3: DSI High Speed Mode Characteristics**

## Low Power Mode



**Figure 7.6: BTA from HOST to Display Module Timing**



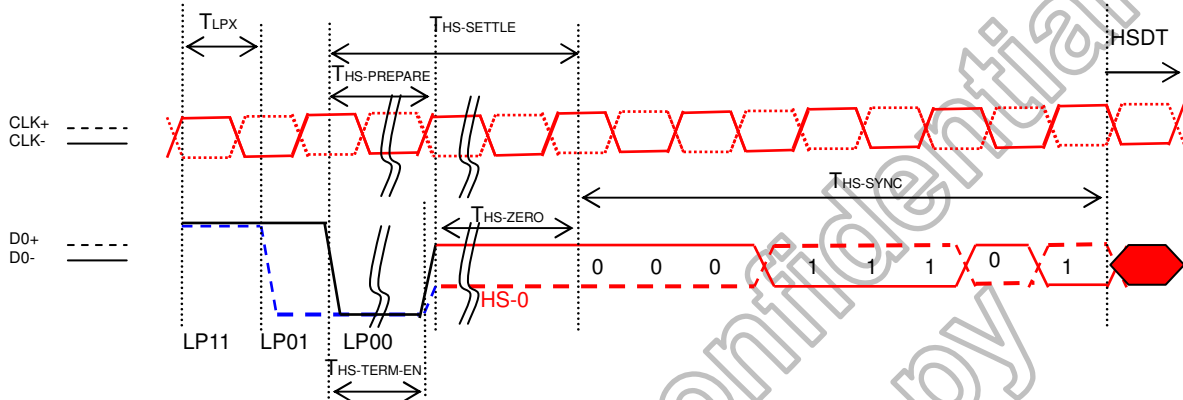
**Figure 7.7: BTA from Display Module Timing to HOST**

(VSSA=0V, IOVCC=1.65V to 3.3V, VCI=2.3V to 3.3V,  $T_A = -30$  to  $70^\circ\text{C}$ )

| Signal              | Item                                                    | Symbol        | Spec.               |      |                     | Unit |
|---------------------|---------------------------------------------------------|---------------|---------------------|------|---------------------|------|
|                     |                                                         |               | Min.                | Typ. | Max.                |      |
| DSI_D0P/<br>DSI_D0P | Length of LP-00/LP01/LP10/LP11<br>Host → Display module | $T_{LPXM}$    | 50                  | -    | -                   | ns   |
|                     | Length of LP-00/LP01/LP10/LP11<br>Display module → Host | $T_{LPXD}$    | 50                  | -    | -                   | ns   |
|                     | Time-out before the MPU start driver                    | $T_{TA-SURE}$ | $T_{LPXD}$          | -    | $2 \times T_{LPXD}$ | ns   |
|                     | Time to drive LP-00 by display module                   | $T_{TA-GET}$  | $5 \times T_{LPXD}$ | -    | -                   | ns   |
|                     | Time to drive LP-00 after turnaround request<br>Host    | $T_{TAGO}$    | $4 \times T_{LPXD}$ | -    | -                   | ns   |

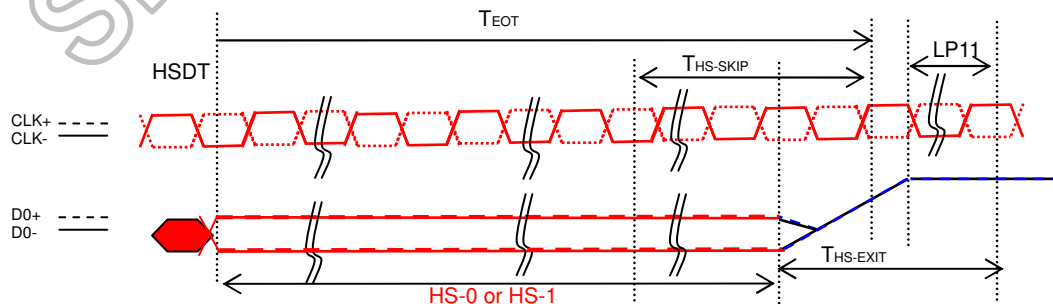
**Table 7.4: DSI Low Power Mode Characteristics**

## DSI BURSTS



| Signal              | Item                                                | Symbol                  | Spec.               |      |         | Unit |
|---------------------|-----------------------------------------------------|-------------------------|---------------------|------|---------|------|
|                     |                                                     |                         | Min.                | Typ. | Max.    |      |
| DSI_D0P/<br>DSI_D0P | Length of LP-00/LP01/LP10/LP11                      | T <sub>LPX</sub>        | 50                  | -    | -       | ns   |
|                     | Time to Driver LP-00 to prepare for HS transmission | T <sub>HS-PREPARE</sub> | 40+4UI              | -    | 85+6UI  | ns   |
|                     | Time to enable data receiver line termination       | T <sub>HS-TERM-EN</sub> | -                   | -    | 35+4xUI | ns   |
|                     | Time to drive LP-00 by display module               | T <sub>TA-GET</sub>     | 5xT <sub>LPXD</sub> | -    | -       | ns   |
|                     | Time to drive LP-00 after turnaround request Host   | T <sub>TAGO</sub>       | 4xT <sub>LPXD</sub> | -    | -       | ns   |

**Table 7.5: DSI Low Power Mode to High Speed Mode Timing**

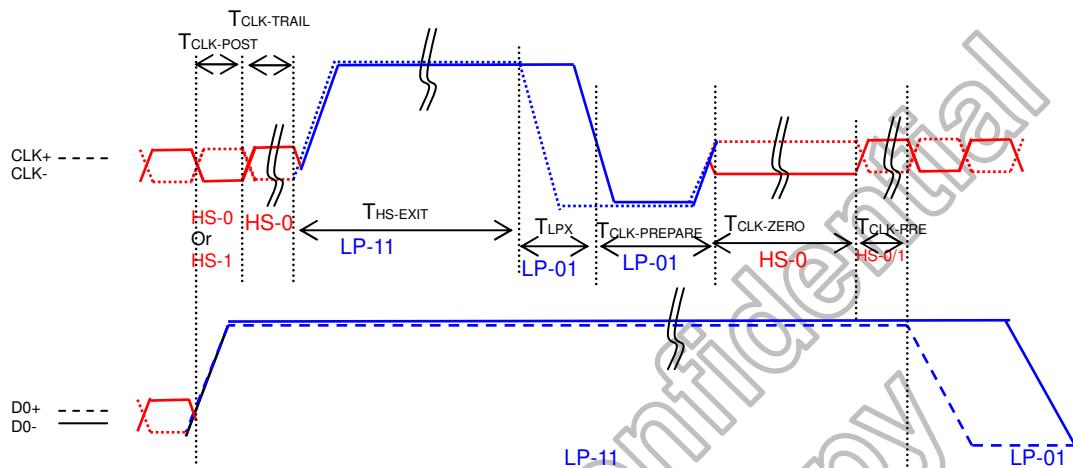


NOTE:  
If the last bit is HS-0, the transmitter changes from HS-0 to HS-1  
If the last bit is HS-0, the transmitter changes from HS-1 to HS-0

| Signal              | Item                                                          | Symbol               | Spec. |      |         | Unit |
|---------------------|---------------------------------------------------------------|----------------------|-------|------|---------|------|
|                     |                                                               |                      | Min.  | Typ. | Max.    |      |
| DSI_D0P/<br>DSI_D0P | Time-Out at Display Module to Ignore Transition Period of EoT | T <sub>HS-SKIP</sub> | 40    | -    | 55+4xUI | ns   |
|                     | Time to Driver LP-11 after HS Burst                           | T <sub>HS-EXIT</sub> | 100   | -    | -       | ns   |

**Table 7.6: DSI Low Power Mode to High Speed Mode Timing**

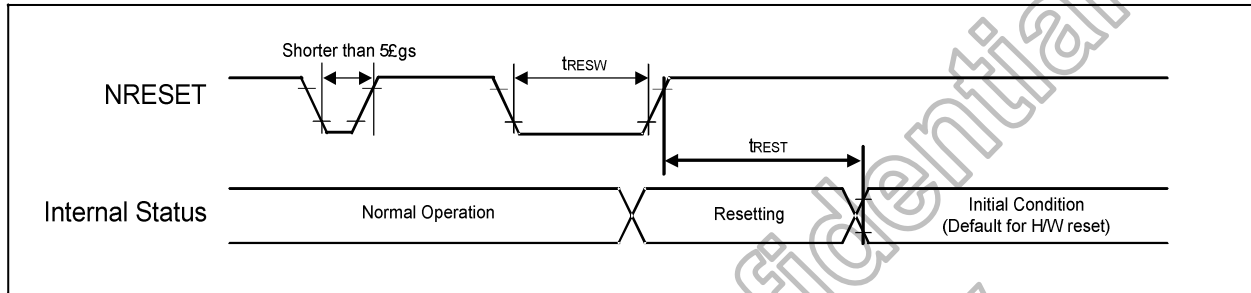




| Signal            | Item                                                                                                                 | Symbol                                           | Spec.    |      |      | Unit |
|-------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------|------|------|------|
|                   |                                                                                                                      |                                                  | Min.     | Typ. | Max. |      |
| DSI_CP/<br>DSI_CN | Time that the MCU shall continue sending HS clock after the last associated Data Lane has transitioned to LP mode    | T <sub>CLK-POST</sub>                            | 60+52xUI | -    | -    | ns   |
|                   | Time to drive HS differential state after last payload clock bit of a HS transmission burst                          | T <sub>CLK-TRAIL</sub>                           | 60       | -    | -    | ns   |
|                   | Time to drive LP-11 after HS burst                                                                                   | T <sub>HS-EXIT</sub>                             | 100      | -    | -    | ns   |
|                   | Time to drive LP-00 to prepare for HS transmission                                                                   | T <sub>CLK-PREPARE</sub>                         | 38       | -    | 95   | ns   |
|                   | Time-out at Clock Lane Display Module to enable HS Termination                                                       | T <sub>CLK-TERM-EN</sub>                         | -        | -    | 38   | ns   |
|                   | Minimum lead HS-0 drive period before starting Clock                                                                 | T <sub>CLK-PREPARE</sub> + T <sub>CLK-ZERO</sub> | 300      | -    | -    | ns   |
|                   | Time that the HS clock shall be driven prior to any associated data Lane beginning the transition from LP to HS mode | T <sub>CLK-PRE</sub>                             | 8xUI     |      |      |      |

**Table 7.7: Clock Lanes High Speed Mode to/from Low Power Mode Timing**

### 7.3.3 Reset input timing



**Figure 7.8: Reset input timing**

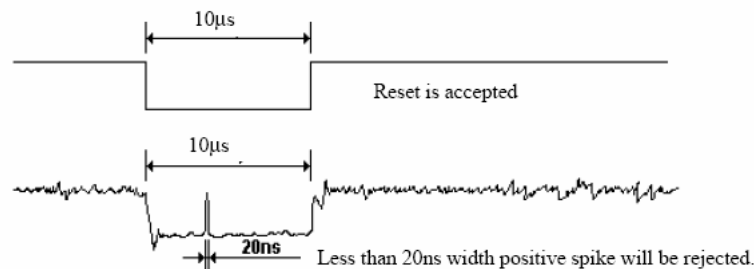
| Symbol | Parameter                            | Related Pins | Spec. |      |      | Note                                  | Unit |
|--------|--------------------------------------|--------------|-------|------|------|---------------------------------------|------|
|        |                                      |              | Min.  | Typ. | Max. |                                       |      |
| tRESW  | Reset low pulse width <sup>(1)</sup> | NRESET       | 10    | -    | -    | -                                     | μs   |
| tREST  | Reset complete time <sup>(2)</sup>   | -            | 15    | -    | -    | When reset applied during SLPIN mode  | ms   |
|        |                                      | -            | 120   | -    | -    | When reset applied during SLPOUT mode | ms   |

**Table 7.8: Reset Input Timing**

**Note:** (1) Spike due to an electrostatic discharge on NRESET line does not cause irregular system reset according to the following table.

| NRESET Pulse           | Action         |
|------------------------|----------------|
| Shorter than 5 μs      | Reset Rejected |
| Longer than 10 μs      | Reset          |
| Between 5 μs and 10 μs | Reset Start    |

- (2) During the resetting period, the display will be blanked (The display is entering blanking sequence, which Maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode) and then return to Default condition for H/W reset.
- (3) During Reset Complete Time, ID and VCOM value in OTP will be latched to internal register during this period. This loading is done every time when there is H/W reset complete time (tREST) within 15ms after a rising edge of NRESET.
- (4) Spike Rejection also applies during a valid reset pulse as shown as below:



- (5) It is necessary to wait 15msec after releasing NRESET before sending commands. Also Sleep Out command cannot be sent for 120msec.

## 8. Reference Application

### 8.1 Interface

The display, which is using DSI interface, is connected to the MPU as it is illustrated below.

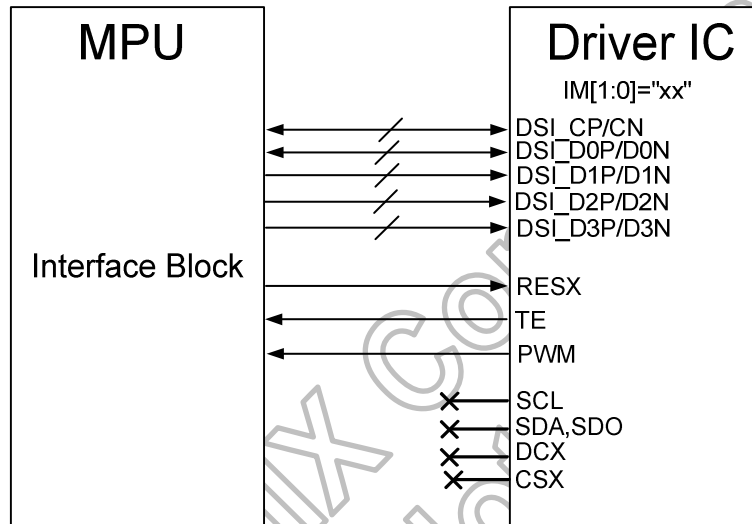


Figure 8.1: Interfacing for MIPI by Setting Lane [1:0]="11"

Notes :

1. When TE is not in use, please let it open.
2. When PWM is not in use, please let it open.
3. Connect DSI\_D3\_P/N to DSI\_VSS in 3 data lanes application (LANE[1:0]="10").
4. Connect DSI\_D3P/N and DSI\_D2P/N to DSI\_VSS in 2 data lanes application (LANE[1:0]="01").

## 8.2 Connections with Panel

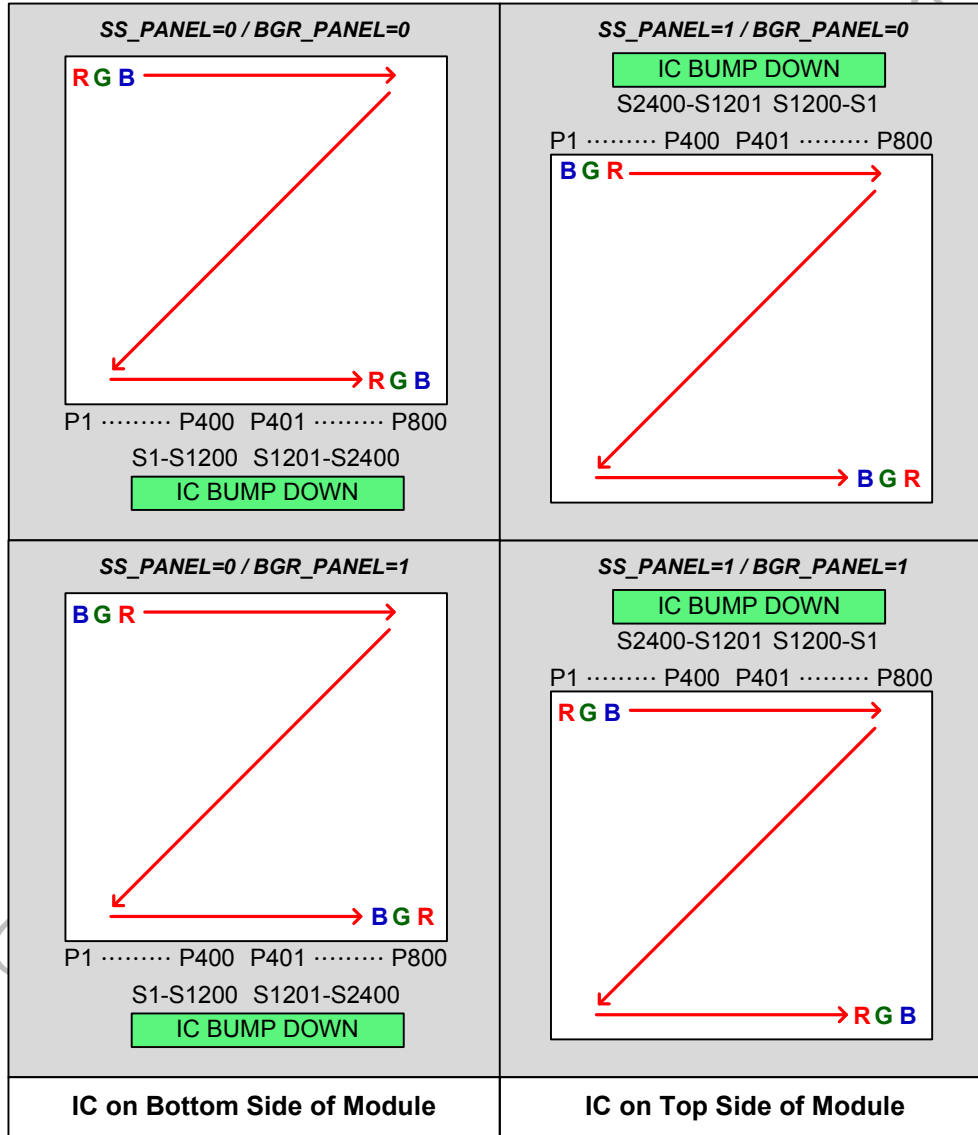


Figure 8.2: Source Pin Connection for Panel

The relationship between Sn output sequence and SS\_PANEL / BGR\_PANEL.

| RESO SEL[2:0] | Resolution | SS PANEL | BGR PANEL | Sn Output Sequence                                                                                                                                                                       | Used Sn Pin              |
|---------------|------------|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 2             | 720RGB     | 0        | 0         | S1 <sub>(R)</sub> → S2 <sub>(G)</sub> → S3 <sub>(B)</sub> → ... → S1080 <sub>(B)</sub> → S1321 <sub>(R)</sub> → ... → S2398 <sub>(R)</sub> → S2399 <sub>(G)</sub> → S2400 <sub>(B)</sub> | S1~S1080 and S1321~S2400 |
|               |            | 0        | 1         | S1 <sub>(B)</sub> → S2 <sub>(G)</sub> → S3 <sub>(R)</sub> → ... → S1080 <sub>(R)</sub> → S1321 <sub>(B)</sub> → ... → S2398 <sub>(B)</sub> → S2399 <sub>(G)</sub> → S2400 <sub>(R)</sub> |                          |
|               |            | 1        | 0         | S2400 <sub>(B)</sub> → S2399 <sub>(G)</sub> → S2398 <sub>(R)</sub> → ... → S1321 <sub>(R)</sub> → S1080 <sub>(B)</sub> → ... → S3 <sub>(B)</sub> → S2 <sub>(G)</sub> → S1 <sub>(R)</sub> |                          |
|               |            | 1        | 1         | S2400 <sub>(R)</sub> → S2399 <sub>(G)</sub> → S2398 <sub>(B)</sub> → ... → S1321 <sub>(B)</sub> → S1080 <sub>(R)</sub> → ... → S3 <sub>(R)</sub> → S2 <sub>(G)</sub> → S1 <sub>(B)</sub> |                          |
| 3             | 640RGB     | 0        | 0         | S1 <sub>(R)</sub> → S2 <sub>(G)</sub> → S3 <sub>(B)</sub> → ... → S960 <sub>(B)</sub> → S1441 <sub>(R)</sub> → ... → S2398 <sub>(R)</sub> → S2399 <sub>(G)</sub> → S2400 <sub>(B)</sub>  | S1~S960 and S1441~S2400  |
|               |            | 0        | 1         | S1 <sub>(B)</sub> → S2 <sub>(G)</sub> → S3 <sub>(R)</sub> → ... → S960 <sub>(R)</sub> → S1441 <sub>(B)</sub> → ... → S2398 <sub>(B)</sub> → S2399 <sub>(G)</sub> → S2400 <sub>(R)</sub>  |                          |
|               |            | 1        | 0         | S2400 <sub>(B)</sub> → S2399 <sub>(G)</sub> → S2398 <sub>(R)</sub> → ... → S1441 <sub>(R)</sub> → S960 <sub>(B)</sub> → ... → S3 <sub>(B)</sub> → S2 <sub>(G)</sub> → S1 <sub>(R)</sub>  |                          |
|               |            | 1        | 1         | S2400 <sub>(R)</sub> → S2399 <sub>(G)</sub> → S2398 <sub>(B)</sub> → ... → S1441 <sub>(B)</sub> → S960 <sub>(R)</sub> → ... → S3 <sub>(R)</sub> → S2 <sub>(G)</sub> → S1 <sub>(B)</sub>  |                          |
| 4             | 600RGB     | 0        | 0         | S1 <sub>(R)</sub> → S2 <sub>(G)</sub> → S3 <sub>(B)</sub> → ... → S900 <sub>(B)</sub> → S1501 <sub>(R)</sub> → ... → S2398 <sub>(R)</sub> → S2399 <sub>(G)</sub> → S2400 <sub>(B)</sub>  | S1~S900 and S1501~S2400  |
|               |            | 0        | 1         | S1 <sub>(B)</sub> → S2 <sub>(G)</sub> → S3 <sub>(R)</sub> → ... → S900 <sub>(R)</sub> → S1501 <sub>(B)</sub> → ... → S2398 <sub>(B)</sub> → S2399 <sub>(G)</sub> → S2400 <sub>(R)</sub>  |                          |
|               |            | 1        | 0         | S2400 <sub>(B)</sub> → S2399 <sub>(G)</sub> → S2398 <sub>(R)</sub> → ... → S1501 <sub>(R)</sub> → S900 <sub>(B)</sub> → ... → S3 <sub>(B)</sub> → S2 <sub>(G)</sub> → S1 <sub>(R)</sub>  |                          |
|               |            | 1        | 1         | S2400 <sub>(R)</sub> → S2399 <sub>(G)</sub> → S2398 <sub>(B)</sub> → ... → S1501 <sub>(B)</sub> → S900 <sub>(R)</sub> → ... → S3 <sub>(R)</sub> → S2 <sub>(G)</sub> → S1 <sub>(B)</sub>  |                          |
| 5             | 540RGB     | 0        | 0         | S1 <sub>(R)</sub> → S2 <sub>(G)</sub> → S3 <sub>(B)</sub> → ... → S810 <sub>(B)</sub> → S1591 <sub>(R)</sub> → ... → S2398 <sub>(R)</sub> → S2399 <sub>(G)</sub> → S2400 <sub>(B)</sub>  | S1~S810 and S1591~S2400  |
|               |            | 0        | 1         | S1 <sub>(B)</sub> → S2 <sub>(G)</sub> → S3 <sub>(R)</sub> → ... → S810 <sub>(R)</sub> → S1591 <sub>(B)</sub> → ... → S2398 <sub>(B)</sub> → S2399 <sub>(G)</sub> → S2400 <sub>(R)</sub>  |                          |
|               |            | 1        | 0         | S2400 <sub>(B)</sub> → S2399 <sub>(G)</sub> → S2398 <sub>(R)</sub> → ... → S1591 <sub>(R)</sub> → S810 <sub>(B)</sub> → ... → S3 <sub>(B)</sub> → S2 <sub>(G)</sub> → S1 <sub>(R)</sub>  |                          |
|               |            | 1        | 1         | S2400 <sub>(R)</sub> → S2399 <sub>(G)</sub> → S2398 <sub>(B)</sub> → ... → S1591 <sub>(B)</sub> → S810 <sub>(R)</sub> → ... → S3 <sub>(R)</sub> → S2 <sub>(G)</sub> → S1 <sub>(B)</sub>  |                          |

### 8.3 DC/DC Converter

#### 8.3.1 External Power IC Mode - FL1002

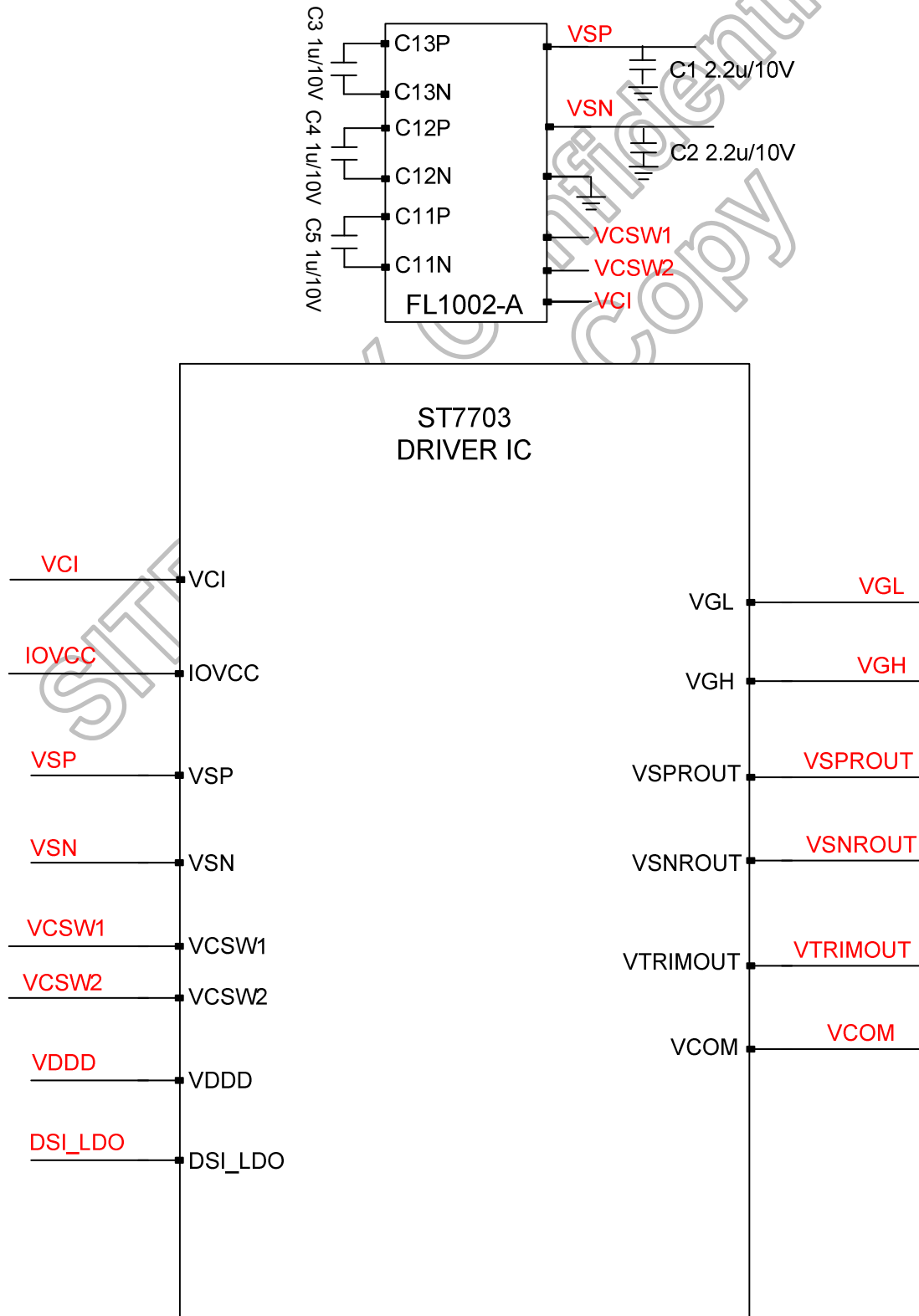


Figure 8.3: Using External Power IC Application

### 8.3.2 PFM Mode - Type C

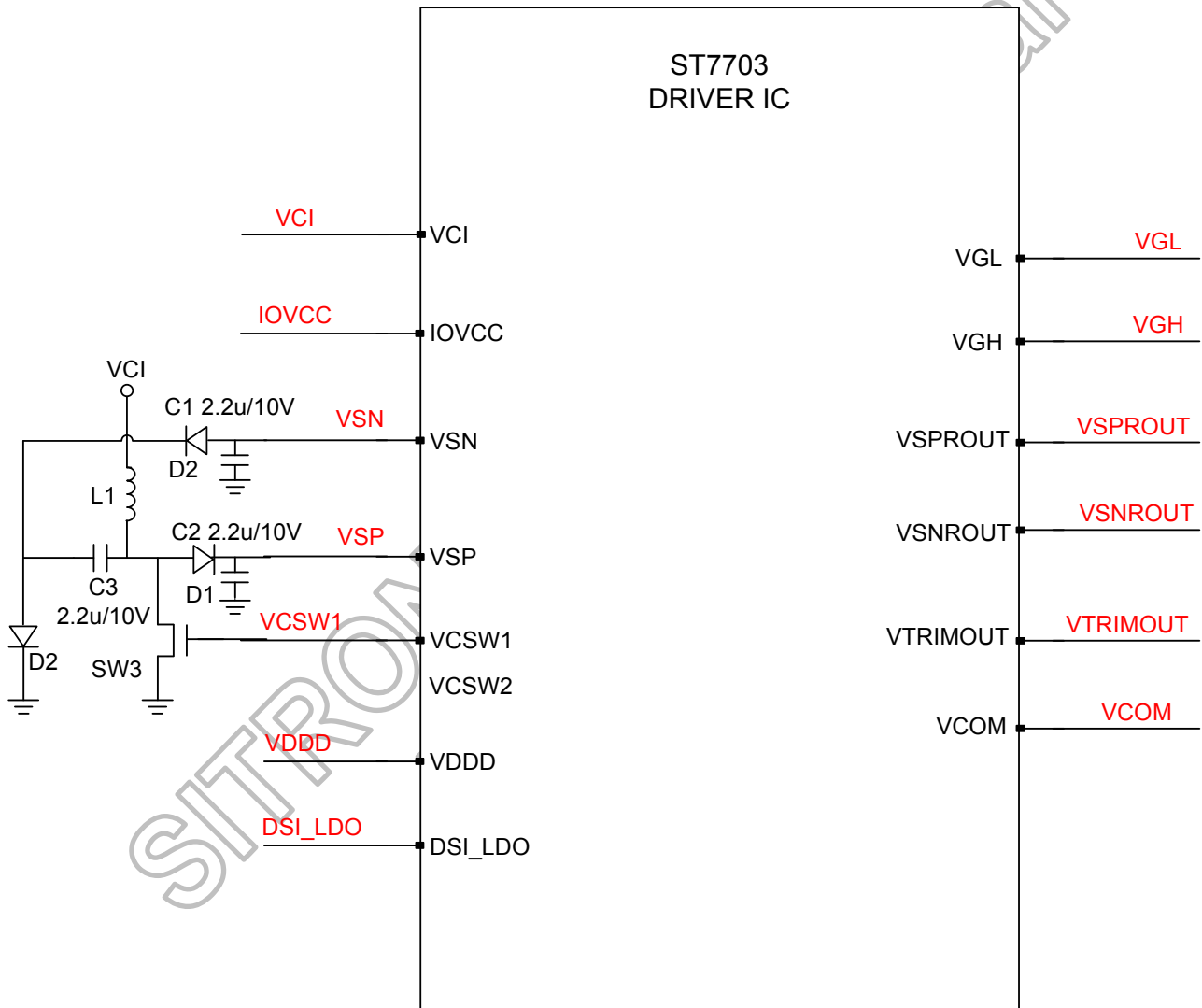
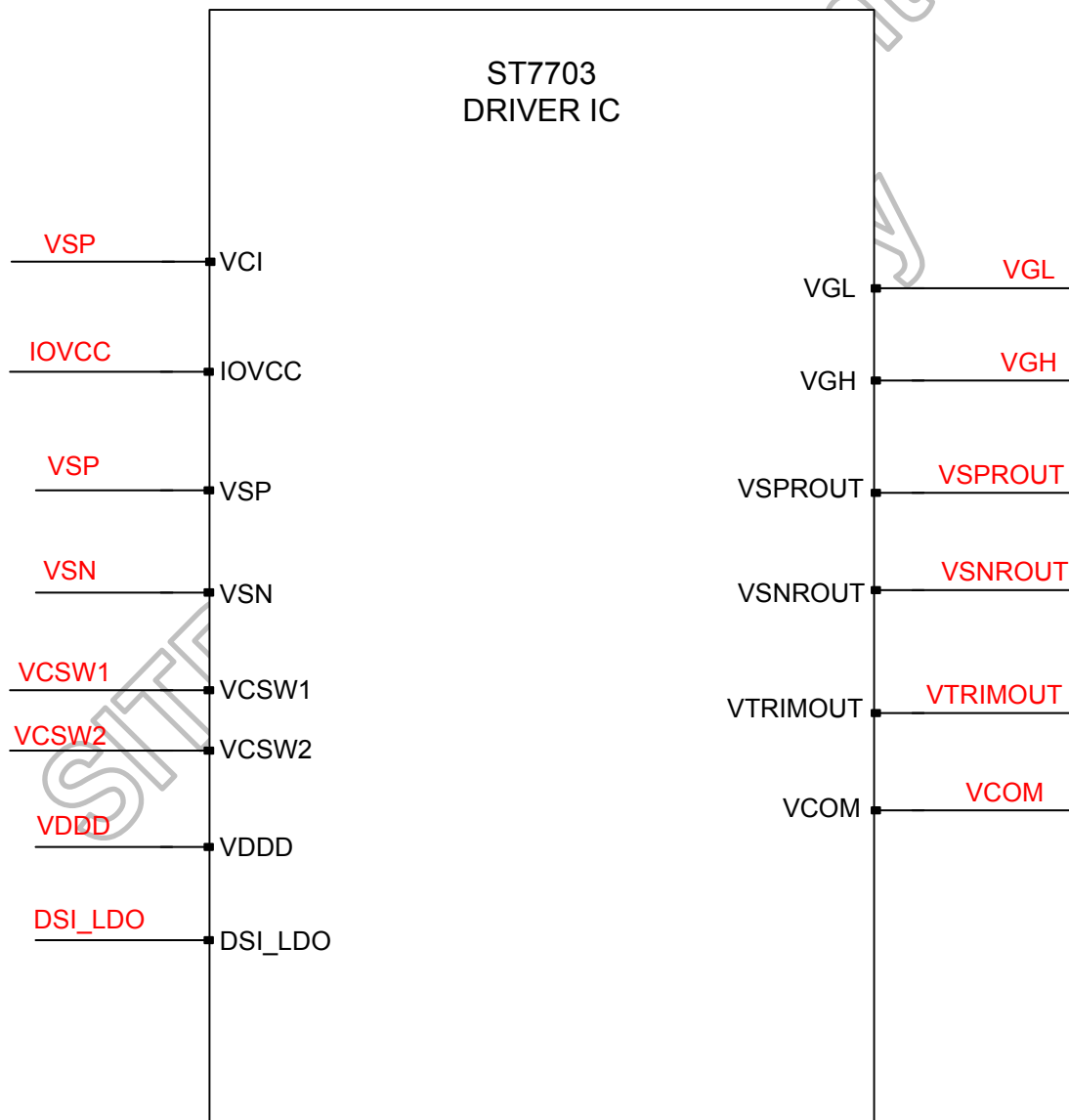


Figure 8.4: Using PFM - Type C Application

### 8.3.3 Three-Power Mode



**Figure 8.5: Using Three Power Mode Application**

## 8.4 DSI Power On/Off Timing

### 8.4.1 Power On Timing of External Power IC

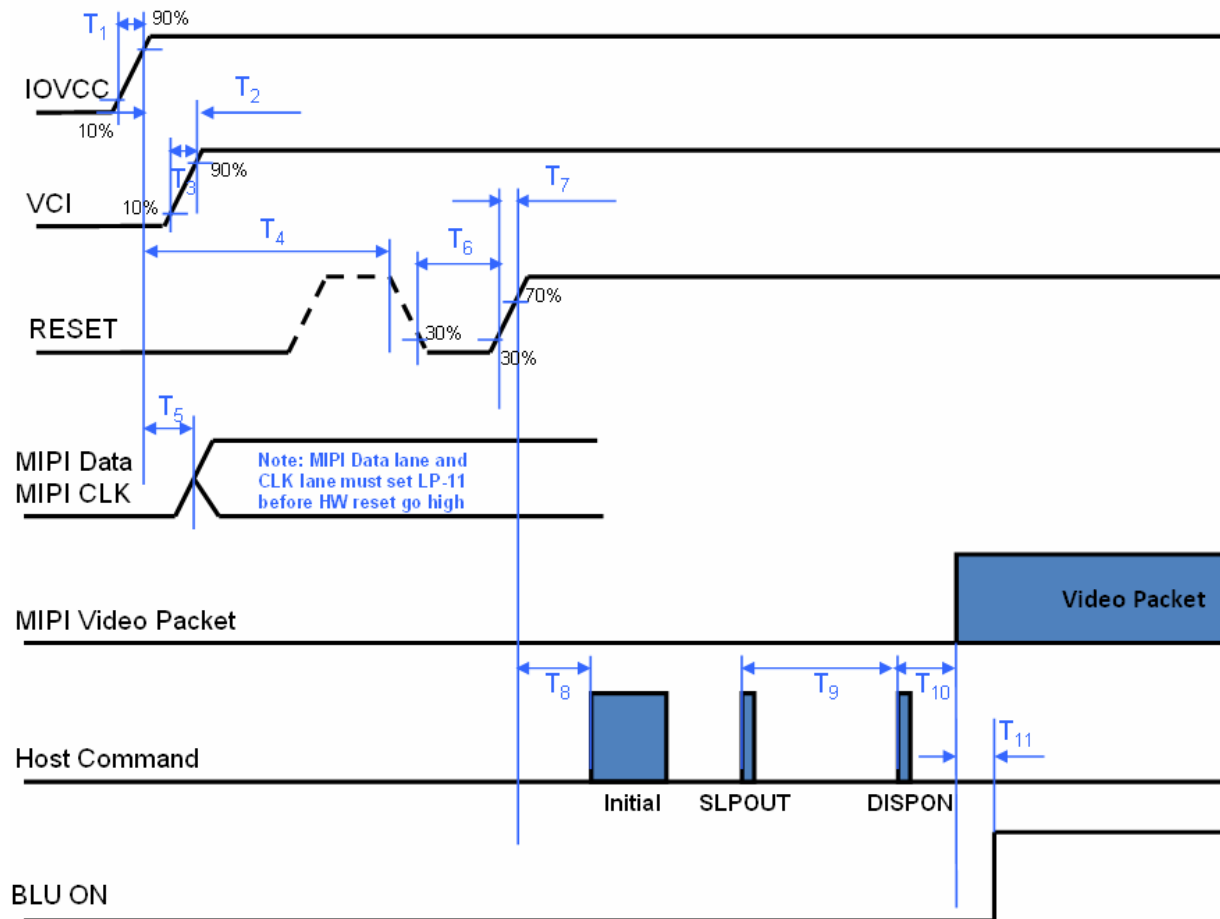


Figure 8-6: DSI Power On Sequence of Power IC Mode

|     | Min.     | Typ. | Max. | Unit |
|-----|----------|------|------|------|
| T1  | 0.01     | -    | 10   | ms   |
| T2  | No Limit |      |      | ms   |
| T3  | 0.01     | -    | 10   | ms   |
| T4  | 1        | -    | -    | ms   |
| T5  | 1        | -    | -    | ms   |
| T6  | 10       | -    | -    | us   |
| T7  | No Limit |      |      | ns   |
| T8  | 15       | -    | -    | ms   |
| T9  | 120      | -    | -    | ms   |
| T10 | No Limit |      |      | ms   |
| T11 | 100      | 150  | -    | ms   |

Table 8-1: DSI Power On Timing of Power IC Mode



#### 8.4.2 Power Off Timing of External Power IC

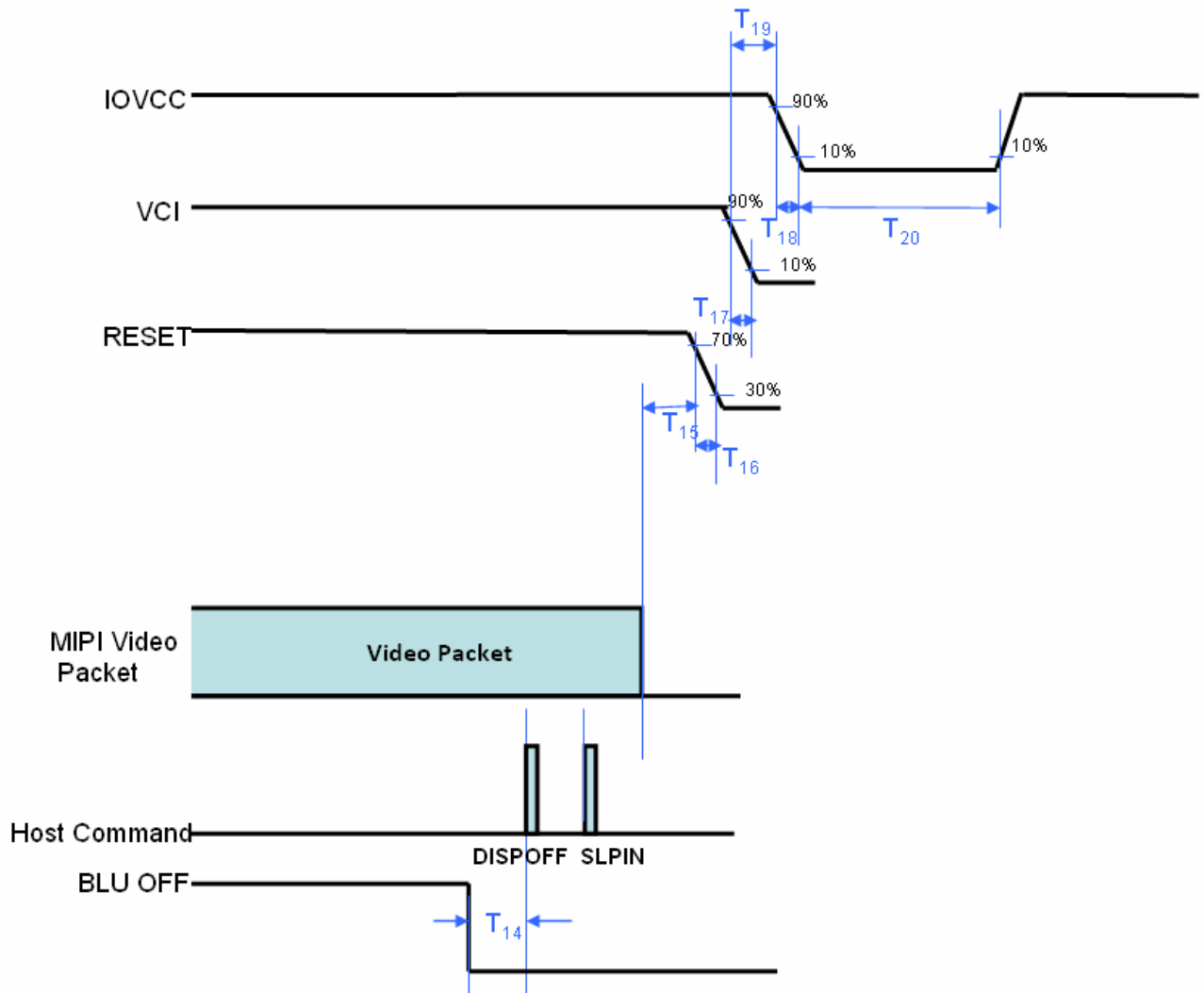


Figure 8-7: DSI Power Off Sequence of Power IC Mode

|     | Min.     | Typ. | Max. | Unit |
|-----|----------|------|------|------|
| T14 | 40       | 100  | -    | ms   |
| T15 | 10       | -    | -    | ms   |
| T16 | No Limit |      |      | ms   |
| T17 | No Limit |      |      | ms   |
| T18 | No Limit |      |      | ms   |
| T19 | No Limit |      |      | ms   |
| T20 | 500      |      |      | ms   |

Table 8-2: DSI Power Off Timing of Power IC Mode

### 8.4.3 Power On Timing of 3-Power Mode

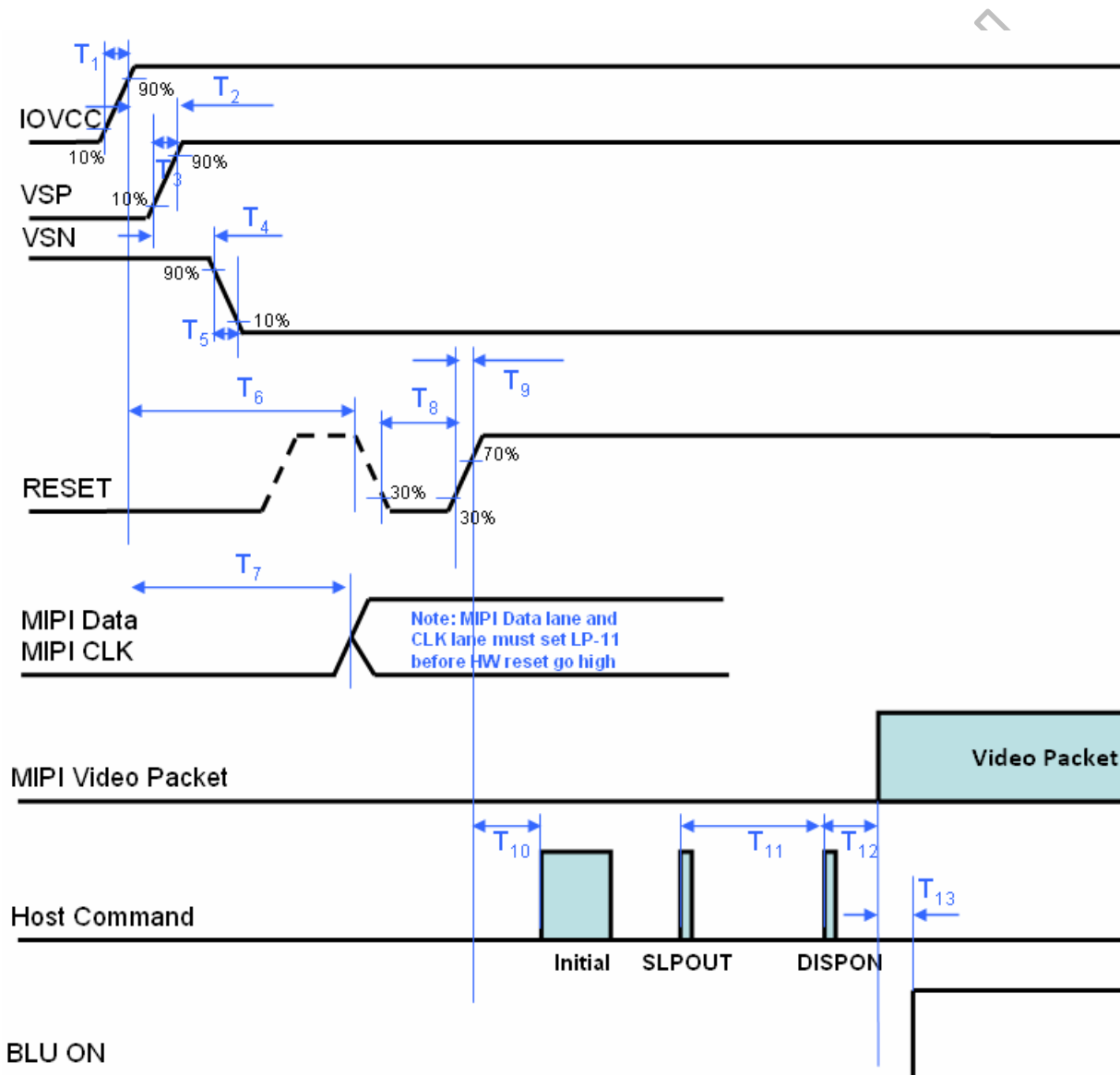


Figure 8-8: DSI Power On Sequence of 3 Power Mode

|     | Min.     | Typ. | Max. | Unit |
|-----|----------|------|------|------|
| T1  | 0.01     | -    | 10   | ms   |
| T2  | No Limit |      |      | ms   |
| T3  | 0.01     | -    | 10   | ms   |
| T4  | No Limit |      |      | ms   |
| T5  | 0.01     | -    | 10   | ms   |
| T6  | 1        | -    | -    | ms   |
| T7  | 1        | -    | -    | ms   |
| T8  | 10       | -    | -    | us   |
| T9  | No Limit |      |      | ns   |
| T10 | 15       | -    | -    | ms   |
| T11 | 120      | -    | -    | ms   |
| T12 | No Limit |      |      | ms   |
| T13 | 100      | 150  | -    | ms   |

Table 8-3: DSI Power On Timing of 3 Power Mode

#### 8.4.4 Power Off Timing of 3-Power Mode

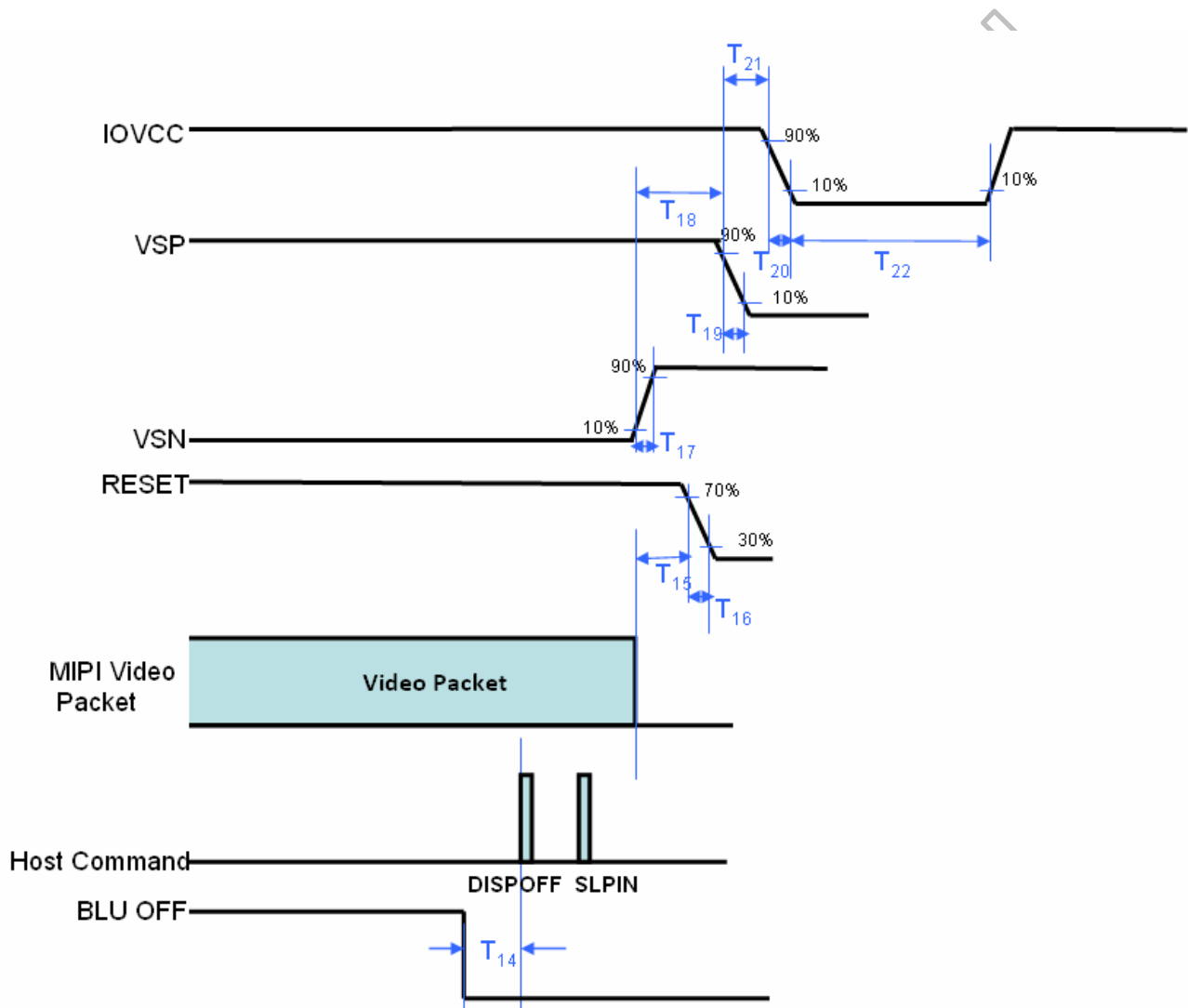


Figure 8-9: DSI Power Off Sequence of 3 Power Mode

|     | Min.     | Typ. | Max. | Unit |
|-----|----------|------|------|------|
| T14 | 40       | 100  | -    | ms   |
| T15 | 10       | -    | -    | ms   |
| T16 | No Limit |      |      | ms   |
| T17 | No Limit |      |      | ms   |
| T18 | No Limit |      |      | ms   |
| T19 | No Limit |      |      | ms   |
| T20 | No Limit |      |      | ms   |
| T21 | No Limit |      |      | ms   |
| T22 | 500      | -    | -    | ms   |

Table 8-4: DSI Power Off Timing of 3 Power Mode

## 8.5 Maximum Layout Resistance

| Name                 | Type         | Maximum layout resistance | Unit     |
|----------------------|--------------|---------------------------|----------|
| IOVCC                | Power supply | 10                        | $\Omega$ |
| VCI                  | Power supply | 10                        | $\Omega$ |
| VSSD                 | Power supply | 10                        | $\Omega$ |
| VSSA                 | Power supply | 10                        | $\Omega$ |
| DSI_VSS              | Power supply | 10                        | $\Omega$ |
| IM0,IM1,LANSEL       | Input        | 100                       | $\Omega$ |
| VCSW1, VCSW2         | Output       | 30                        | $\Omega$ |
| DCX,SCL,CSX,RESX     | Input        | 100                       | $\Omega$ |
| SDA                  | Input/Output | 100                       | $\Omega$ |
| SDO                  | Output       | 100                       | $\Omega$ |
| VOUT,HOUT            | Output       | 100                       | $\Omega$ |
| PWM                  | Output       | 100                       | $\Omega$ |
| VCOM                 | Output       | 10                        | $\Omega$ |
| DSI_D0P              | Input/Output | 8                         | $\Omega$ |
| DSI_D0N              | Input/Output | 8                         | $\Omega$ |
| DSI_CP               | Input        | 8                         | $\Omega$ |
| DSI_CN               | Input        | 8                         | $\Omega$ |
| DSI_D1P              | Input        | 8                         | $\Omega$ |
| DSI_D1N              | Input        | 8                         | $\Omega$ |
| DSI_D2P              | Input        | 8                         | $\Omega$ |
| DSI_D2N              | Input        | 8                         | $\Omega$ |
| DSI_D3P              | Input        | 8                         | $\Omega$ |
| DSI_D3N              | Input        | 8                         | $\Omega$ |
| VDDD                 | Output       | 10                        | $\Omega$ |
| VSP,VSN              | Input/Output | 10                        | $\Omega$ |
| VSPROUT, VSNROUT     | Output       | 10                        | $\Omega$ |
| VTRIMOUT             | Output       | 10                        | $\Omega$ |
| VGL                  | Output       | 10                        | $\Omega$ |
| VGH                  | Output       | 10                        | $\Omega$ |
| DSI_LDO              | Output       | 10                        | $\Omega$ |
| OSC                  | Input        | 100                       | $\Omega$ |
| VTESTOUTP, VTESTOUTN | Output       | 100                       | $\Omega$ |
| CGOUTL_1~22          | Output       | 30                        | $\Omega$ |
| CGOUTR_1~22          | Output       | 30                        | $\Omega$ |

**Table 8-5: Maximum Layout Resistance**